



The Economic Case for Greater LGBTI+ Equality in the United States



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Foreword

The OECD Directorate for Employment, Labour and Social Affairs (ELS) has been supporting the inclusion of increasingly diverse groups in member countries through its work on gender equality, ageing and employment, the labour market integration of youth, the inclusion of people with disability, or the integration of immigrants and visible minorities. Since 2016, following a Call to Action signed in 2014 by 12 member countries, ELS has been leading the OECD's work on the inclusion of LGBTI+ people, i.e. lesbians, gay men, bisexuals, transgender and intersex individuals.

With the 2019 edition of *Society at a Glance* and with *Over the Rainbow? The Road to LGBTI Inclusion* (2020), the OECD previously explored the socio-economic situation of LGBTI+ individuals and the extent to which laws in OECD countries ensure their equal treatment. In 2023, the OECD published the first country review undertaken as part of the OECD work on LGBTI+ inclusion, to explore legal and policy achievements towards LGBTI+ equality in Germany, at both the national and subnational levels. Another significant study was also released in 2023, based on an innovative randomised control trial in France measuring the effectiveness of school-based interventions in reducing homo- and transphobia.

This new report introduces a country review focused on the economic benefits from creating a level playing field for LGBTI+ individuals. It explores whether achieving greater LGBTI+ equality could lead to a meaningful increase in GDP, even in a country like the United States which has shown an above-average performance with respect to legal and social acceptance of LGBTI+ individuals. The report introduces a novel analytical framework with potential for application to other countries in similar reviews.

The analysis of a unique set of nationally representative microdata reveals that LGBTI+ individuals continue to face considerable disparities in areas critical to individual and societal well-being, including education, employment, and health. Consequently, the economic returns of achieving LGBTI+ equality are substantial. Should the United States succeed in closing the unexplained LGBTI+ gaps in employment and labour productivity by 2050, an increase in GDP equal to 2.6% of the baseline GDP could be expected. This corresponds to a yearly increase in GDP of 0.1%. A more ambitious goal of closing the gaps by 2030 would lead to a higher yearly GDP increase, representing not 5%, but about 10% of the average annual US GDP growth observed between 2013 and 2023.

This report was written by Marie-Anne Valfort and Marc Folch under the supervision of Monika Queisser (Head of the Social Policy Division), and under the leadership of Stefano Scarpetta (Director of ELS) and Mark Pearson (Deputy Director of ELS).

We are extremely grateful to all the United States officials, Jennifer Burnszynski (U.S. Department of Health and Human Services – Office of the Assistant Secretary for Planning and Evaluation), Jason Dale (U.S. Department of Labor – Office of International Relations and Economic Research), Rachel Floyd-Nelson (U.S. Department of Labor – Office of International Relations and Economic Research), Sheimaliz Glover (U.S. Department of State), Sarah Morgan (U.S. Department of Labor – Office of International Relations and Economic Research), as well as participants of the 47th meeting of the Working Party on Social Policy for providing thoughtful comments on an earlier version of this report. We warmly thank Jonas Fluchtmann (OECD – ELS) and Yvan Guillemette (OECD – ECO) for their guidance on navigating the

OECD long-term model, and Emily Hewlett, Gaetan Lafortune, and Doron Wijkers (all OECD – ELS) for reviewing our estimates of the benefits of removing the economic and well-being burden of mental health disparities for LGBTI+ Americans. We are also very grateful to Caroline Medina (Whitman-Walker Institute) for her insights on the methodology and findings from the 2022 Center for American Progress survey on LGBTQI+ people. Finally, we thank the Gallup corporation, and notably Jerry Hansen and Kris Hodgins, for their support in accessing the Gallup data. Hanna Varkki prepared the manuscript for publication.

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Executive summary

Ensuring equality of opportunities in the labour market and society for LGBTI+ individuals is a human rights imperative, but it also makes a lot of economic sense. Yet, solid evidence supporting the economic case for greater LGBTI+ equality is still scarce due to challenges in accurately measuring the size and life situation of the LGBTI+ population. This report bridges this gap by using a unique set of microdata from the United States, offering the first extensive nationally representative sample of the LGBTI+ US adult population. The report presents current data on the share of US adults identifying as LGBTI+, their geographic distribution and key demographics, followed by an evaluation of the state of equality for LGBTI+ Americans. Finally, the report provides an estimate of the potential increase in GDP resulting from levelling the playing field for LGBTI+ Americans.

Who are LGBTI+ Americans?

In 2023, more than one in ten (11.2%) American adults self-identified as LGBTI+. This figure is equivalent to nearly 30 million American adults, which is larger than the population of the state of New York or Florida. Among these, 2.3% self-identified as non-cisgender (individuals whose gender identity does not match their sex assigned at birth), while 8.9% as cisgender non-heterosexual. Consistent with the well-documented increase in acceptance of homosexuality, the share of individuals identifying as LGBTI+ has risen significantly, with the percentage of self-declared LGBT adults nearly doubling from 2012 to 2022.

The share of self-identified LGBTI+ adults varies across the US, with higher concentration in areas characterized by greater social and legal acceptance of LGBTI+ individuals. For instance, in 2023, US adults were 24% more likely to identify as LGBTI+ in the West, compared to the South. This pattern reflects both the relocation of LGBTI+ individuals, presumably to avoid discrimination, and the increased likelihood of LGBTI+ adults disclosing their identities in more accepting environments.

Further analysis reveals that LGBTI+ Americans differ on key demographic characteristics compared to their cisgender straight peers:

- Individuals assigned female at birth are more likely to identify as LGBTI+ than those assigned male, with this gap widening over the past decade. This trend may be influenced by greater societal acceptance of women expressing same-sex relationships compared to men, women's more positive attitudes towards LGBTI+ identities, and advancements in gender equality that reduce pressure of heteronormativity and traditional femininity.
- LGBTI+ self-identification decreases with age. Notably, the recent surge in LGBTI+ identification is driven by younger generations. This trend is attributed to young LGBTI+ individuals being raised in more accepting environments, fostering greater comfort in disclosing their identities.
- LGBTI+ self-identification varies by race and ethnicity, with lower rates among non-Hispanic Blacks and Asians (compared to non-Hispanic Whites), possibly due to compounded stigma and more negative attitudes within these groups. Conversely, the higher propensity among Hispanics to identify as LGBTI+ is predominantly due to their younger age. Additionally, individuals from other

racial categories, including Indigenous and mixed-race groups, are more likely to identify as LGBTI+, potentially reflecting more accepting attitudes and a more fluid mindset within these populations.

- LGBTI+ Americans are less likely to be partnered, married, or live in households with children. Although their partnership and marriage rates have increased, this group still faces legal, financial, and discriminatory barriers that restrict access to parenthood.
- Non-cisgender individuals and young cisgender non-heterosexuals (except young cisgender gay men) face educational attainment disadvantages, notably stemming from challenging school environments, exclusionary behaviours from family, and mental health struggles.

Are LGBTI+ Americans discriminated against?

Surveys on attitudes towards LGBTI+ individuals, perceptions of discrimination, and field experiments reveal that anti-LGBTI+ discrimination is still a reality in the United States. This discrimination hampers the hiring of LGBTI+ individuals and often relegates them to lower-skilled positions when employed. Reduced labour market prospects in turn may diminish incentives for labour force participation and lower productivity at work. These patterns are exacerbated by the detrimental impact of discrimination on mental health.

Consistent with these mechanisms, this report reveals significant labour market penalties for LGBTI+ individuals, even after adjusting for location, demographics, as well as sector and occupation for the employed. Overall, LGBTI+ adults are 7% less likely to be employed and those who are employed earn 7% less than their cisgender straight counterparts. These labour market penalties result in an 8% lower household income and are particularly pronounced among non-cisgender adults, possibly due to greater discrimination exposure. These disparities are also wider among younger cohorts, potentially because of a lower non-disclosure bias compared to older generations. Additionally, LGBTI+ adults face significant mental health challenges, being more than 50% more likely to be at risk of generalized anxiety or major depressive disorder, with non-cisgender individuals again experiencing the most severe penalties. Finally, evidence suggests that the COVID-19 pandemic has exacerbated these economic and mental health challenges.

Does LGBTI+ equality pay?

To quantify the economic gains of LGBTI+ inclusion, this report utilizes the OECD long-term model. This model provides a framework for estimating the additional GDP that could result from a gradual convergence of the employment rate and labour productivity of LGBTI+ adults to those of their cisgender straight peers, compared to a baseline scenario where no such efforts are made.

Should the United States succeed in closing the unexplained LGBTI+ gaps in employment and labour productivity by 2050, it could expect an increase in GDP equal to 2.6% of the baseline GDP. This corresponds to a yearly increase in GDP equal to 0.1%, which amounts to 5% of the average annual US GDP growth observed between 2013 and 2023. A more ambitious goal of closing the gaps by 2030 would lead to a higher yearly GDP increase, representing about 10% of the average annual US GDP growth over the past ten years.

There are obviously many more benefits from LGBTI+ equality than just the economic gains. These include improved public finances through increased tax revenues and decreased public expenditures, as well as the potential to tap into a diversity and gender equality dividend. Specifically, LGBTI+ equality can have ripple effects on the emancipation of other groups, particularly women, as it inherently challenges heteronormativity, which confines men and women to rigid roles and obstructs women's empowerment.

1

The economic case for greater LGBTI+ equality in the United States: An overview

This introductory chapter summarises the report's findings on the economic case for greater LGBTI+ equality, drawing from a unique set of nationally representative microdata from the United States. The report first presents the latest data on the percentage of US adults identifying as LGBTI+, their geographic distribution and essential demographics. It reveals that 11.2% (or nearly 30 million) of US adults self-identified as LGBTI+ in 2023, a number that has nearly doubled in the past decade. In addition, self-identified LGBTI+ Americans differ on their geographic location and key demographic characteristics (sex assigned at birth, age, race/ethnicity, family structure, and educational attainment), compared to their cisgender straight peers. The report then focuses on evaluating equality for LGBTI+ Americans, underscoring persistent disparities faced by LGBTI+ Americans, especially in labour market outcomes, household income, and mental health. Finally, the report quantifies the substantial economic returns of levelling the playing field for LGBTI+ Americans.

1.1. Introduction

Ensuring that LGBTI+ people can live as who they are without being discriminated against is a question of human rights, but it also makes a lot of economic sense. Yet, compelling evidence on the economic case for greater LGBTI+ equality is still scarce, primarily due to challenges in accurately measuring the size and life situation of the LGBTI+ population. These difficulties stem from two main reasons:

- Firstly, sexual orientation and gender identity are considered very sensitive personal information which individuals are often reluctant to disclose, especially in surveys involving an interviewer, be it face-to-face or over the phone, but also in self-administered online surveys whenever respondents' names are collected (OECD, 2019^[1]; Valfort, 2017^[2]). In such cases, respondents feel that their responses are not truly anonymous (Robertson et al., 2018^[3]).
- Secondly, there are very few large-scale nationally representative surveys that include questions on both sexual orientation and gender identity. As of 2023, only two OECD countries (Canada and the United Kingdom) have included a question on sexual orientation and/or gender identity in their population censuses (OECD, 2023^[4]).

Against this backdrop, the United States has recently been active in bridging the data gap which has thus far hindered comprehensive estimates of the size of the LGBTI+ population and of their life situation. In particular, the launch of the Census Bureau's Household Pulse Survey (HPS) in 2020 constituted a breakthrough. This is a large-scale, online, fully anonymous, and probability-based survey initially designed to assess the social and economic impact of COVID-19 on American households. Since 21 July 2021, the HPS has been collecting detailed information on sexual orientation, sex assigned at birth and gender identity, thereby offering the first extensive nationally representative sample of the LGBTI+ US adult population, defined as follows:

- **Cisgender non-heterosexual (LGB+) individuals**, composed of those whose gender identity matches their sex assigned at birth and who self-identify as lesbians, gay men, bisexuals, or with another non-heterosexual identity, such as pansexual.
- **Non-cisgender individuals**, composed of those whose gender identity does not match their sex assigned at birth. This includes transgender individuals (persons assigned female at birth but identifying as male, and *vice versa*), as well as other non-cisgender identities such as being non-binary (a person who was assigned either female or male at birth but whose gender identity refers to a combination of the two genders).¹ While the majority of non-cisgender individuals self-identify as non-heterosexual, this population group also includes non-cisgender individuals who self-identify as heterosexual individuals.²

The analysis in this report relies on HPS data collected from 21 July 2021, to 10 July 2023. This dataset contains 1 648 098 respondents of which 155 447 self-identify as LGBTI+, an unprecedented sample size. In addition, to document trends over the past decade, the report also utilises data collected by the Gallup corporation from 2012 to 2022, as part of phone-based surveys which included a question on whether respondents self-identified as LGBT or not. However, compared to the HPS data, the Gallup surveys have two main limitations. Firstly, the phone-based format provides less anonymity compared to the online-based format, confirming the recurring finding that estimates of the LGBTI+ population are higher in surveys where respondents, rather than interviewers, answer the question about LGBTI+ status (OECD, 2019^[1]; Valfort, 2017^[2]). Secondly, in most of the Gallup sample (from 2012 to 2017), respondents were classified in a dichotomous way (being LGBT or not), making it impossible to distinguish between different sexual orientations and gender identities. While the most recent Gallup sample (from 2020 to 2022) permits the identification of different LGBTI+ subgroups, the sample size is substantially smaller compared to the HPS sample. Overall, the size of the LGBTI+ sample surveyed from 2012 to 2022 is less than half that of the LGBTI+ sample surveyed by the HPS from 2021 to 2023.

Combining the HPS and Gallup surveys, this report provides a unique set of microdata that allows addressing three critical questions: i) Who are LGBTI+ Americans?; ii) Are LGBTI+ Americans discriminated against?; iii) Does LGBTI+ equality pay?. Meanwhile, this report introduces a novel analytical framework with potential for international application, detailing the data necessary for its deployment and laying the groundwork for subsequent studies on the economic returns of greater LGBTI+ inclusion.

1.2. Who are LGBTI+ Americans?

1.2.1. Americans who self-identify as LGBTI+ represent a sizeable and growing minority

According to the Household Pulse Survey (HPS), 11.2% of American adults self-identified as LGBTI+ in 2023 (Figure 1.1). This figure is equivalent to nearly 30 million American adults, which is larger than the population in New York or Florida. Specifically, nearly 9% of Americans self-identify as cisgender LGB+, the largest subgroup being cisgender bisexuals (4.5%), followed by cisgender gays/lesbians (2.9%) and then by other cisgender non-heterosexual identities (1.5%). Regarding non-cisgender individuals, only a little more than 2% of Americans self-identify as such, with 0.9% of US adults self-identifying as transgender and 1.4% as other non-cisgender identity (e.g. non-binary).

These estimates are close to those obtained for other OECD countries, based on the 2nd edition of the biennial LGBT+ Pride Global Survey that IPSOS conducted online between February and March 2023 (IPSOS, 2023^[5]). According to this survey, the average share of LGBTI+ individuals in the 19 OECD countries surveyed is approximately 10%. If this percentage is extrapolated to the total population of OECD countries, it suggests a figure of nearly 140 million people, which is more than the population of Mexico.

Consistent with the well-documented shift towards greater acceptance of homosexuality, the share of individuals who self-identify as LGBTI+ is on the rise. This is true when one focuses on HPS data from 2021 to 2023, noting that this upward trend is even more striking when one focuses on the Gallup surveys, which reveal that the share of adults who self-identify as LGBT almost doubled between 2012 and 2022, rising from 3.5% to 6.9%.

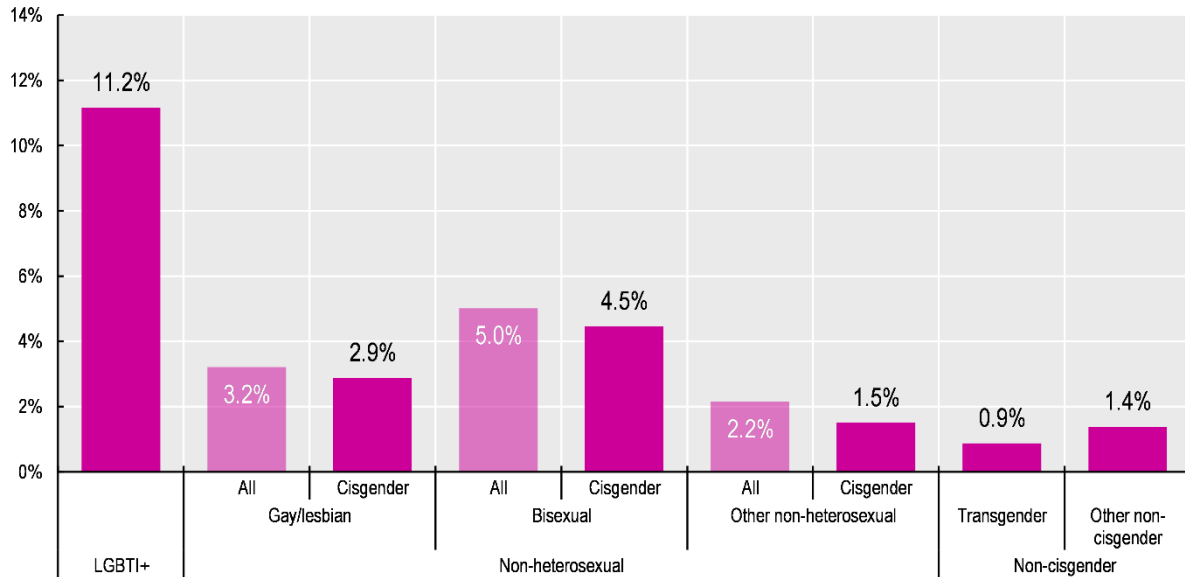
The share of self-identified LGBTI+ adults varies substantially across US regions, states, and rural/urban areas. In 2023, according to HPS data, US adults were 24% more likely to identify as LGBTI+ in the West, compared to the South. Spatial disparities also prevail across US states, including within the same region. In the West, for instance, the share of self-identified LGBTI+ US adults varies by a factor of 1.8 across states, from 8.4% in Wyoming to 15.5% in Oregon. Self-reporting of LGBTI+ status also differs between rural and urban areas, with the share being almost twice as high in the latter than in the former.

Not surprisingly, these spatial disparities are positively correlated with geographic variations in social acceptance and legal protections of LGBTI+ individuals. For instance, in a state-level analysis, an increase in the share of adults favouring LGBT non-discrimination protections by 10 percentage points is associated with an increase in LGBTI+ self-identification by 2.1 percentage points

This pattern reflects two main mechanisms. Firstly, LGBTI+ adults may be inclined to relocate to more accepting areas to escape discrimination and connect with a larger LGBTI+ community. Secondly, in more accepting areas, LGBTI+ individuals might feel more comfortable disclosing their sexual orientation and gender identity, causing a higher share of self-identified LGBTI+ individuals than in less accepting areas. While it is not possible to analyse which of the two mechanisms prevail using the HPS data, empirical evidence suggests that geographic relocation to avoid discrimination is a significant factor (Levine, 2022^[6]; Medina and Mahowald, 2023^[7]; Folch, 2022^[8]; Goldberg, 2023^[9]).

Figure 1.1. In 2023, 11.2% of US adults self-identified as LGBTI+

Share of self-identified LGBTI+ US adults (aged 18-88) in 2023



Note: LGBTI+ refers to respondents who self-identify as gay/lesbian, bisexual, transgender, other non-heterosexual, and other non-cisgender. Cisgender refers to individuals whose gender identity matches their sex assigned at birth. Person-level weights used.

Source: Household Pulse Survey (4 January 2023 – 10 July 2023).

1.2.2. Self-identified LGBTI+ Americans differ on key demographic characteristics compared to their cisgender straight peers

Factors such as sex assigned at birth, age, race/ethnicity, family structure, and educational attainment significantly influence economic and health outcomes, which, in turn, shape individual life trajectories. Exploring how LGBTI+ individuals compare with respect to these essential demographics yields five main takeaways.

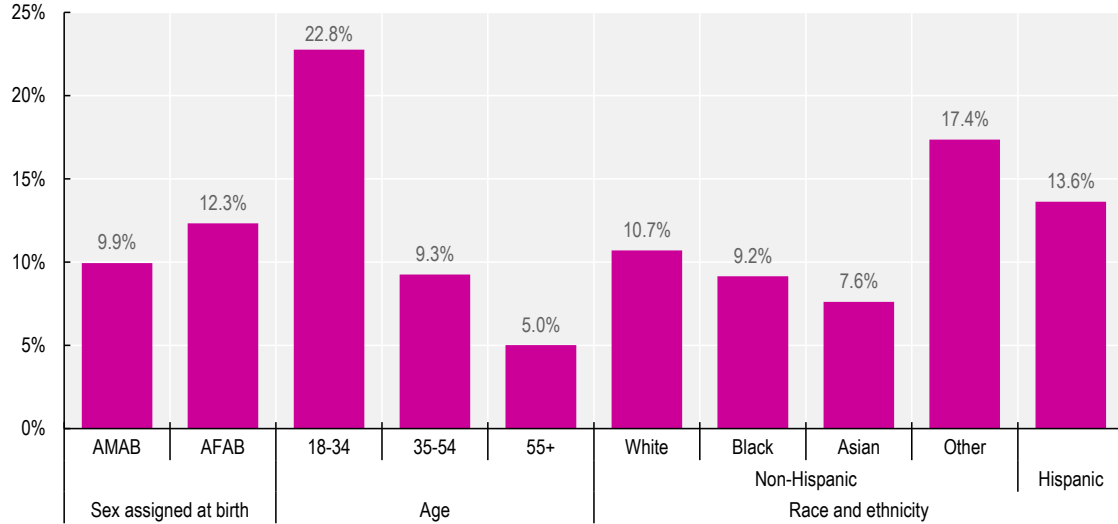
1. Individuals assigned female at birth (AFAB) are more likely to identify as LGBTI+ than individuals assigned male at birth (AMAB) (Panel A of Figure 1.2), a gap that has kept widening during the past decade. This difference is primarily driven by AFAB individuals being more than twice as likely as their AMAB counterparts to identify as cisgender bisexual. Differences in LGBTI+ self-identification between AFAB and AMAB individuals likely reflect the well-documented greater social acceptance of women expressing same-sex attraction or relationships compared to men, leading to potentially more AFABs feeling comfortable identifying as cisgender LGB+ (Herek, 2002^[10]; Bettinsoli, Suppes and Napier, 2020^[11]). Additionally, females exhibit more positive attitudes towards LGBTI+ individuals. As for the widening of this gap over time, it may reflect strides made in gender equality, hence a departure from traditional roles and expectations placed upon women, including the pressure to adhere to heteronormativity and to be feminine.
2. LGBTI+ self-identification decreases with age (Panel A of Figure 1.2) across all LGBTI+ subgroups. In 2023, nearly one in four young adults (18-34) identified as LGBTI+, compared to nearly one in ten among prime age adults (35-54) and one in 20 among mature adults (55+). Notably, the recent surge in LGBTI+ identification is driven by younger generations. Further analysis reveals that this pattern is not driven by an age effect, which would entail that, for younger generations, self-identifying as LGBTI+ is a fad that ceases once they grow older. Rather, the

Gallup data suggest that this pattern is driven by a cohort effect: among those who were aged 18 to 34 in 2012, the percentage identifying as LGBT has grown in subsequent years rather than declined. This trend indicates that, raised in more accepting environments, young LGBTI+ adults feel more comfortable disclosing who they are, a process that should intensify in the future, provided that social acceptance of LGBTI+ individuals continues to rise.

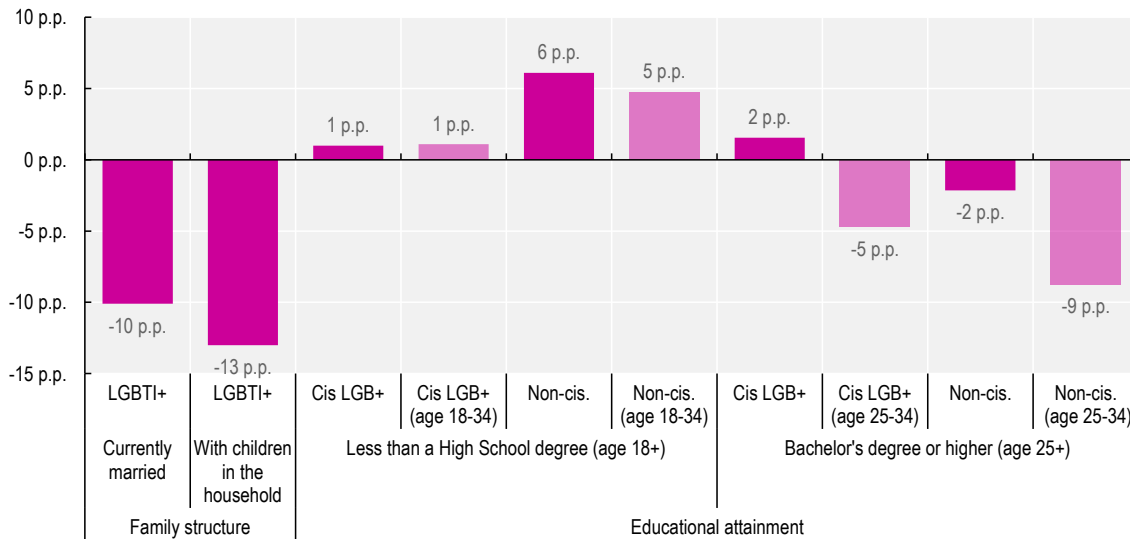
3. LGBTI+ self-identification varies by race and ethnicity. First, relative to non-Hispanic Whites, a smaller share of non-Hispanic Blacks and non-Hispanic Asians identify as LGBTI+ (Panel A of Figure 1.2). This pattern could derive from a so-called “*double minority stress*” effect (Meyer, 2010^[12]; Meyer, 2003^[13]), meaning that LGBTI+ individuals within a racial/ethnic minority group may face compounded stigma from both racism and LGBTI+-phobia, making some less inclined to openly identify as LGBTI+ (Grov et al., 2006^[14]; Rosario, Schrimshaw and Hunter, 2004^[15]; Maguen et al., 2002^[16]). This pattern could also be due to the more negative attitudes towards LGBTI+ individuals that, on average, prevail among non-Hispanic Blacks and non-Hispanic Asians, as documented in Chapter 3. Second, the greater propensity of Hispanic individuals to identify as LGBTI+ is predominantly due to their young age. Third, individuals identifying with other racial categories, including American Indian and Alaska Native, Native Hawaiian and Other Pacific Islander, and mixed-race individuals, are more likely to self-identify as LGBTI+. This finding may flow from relatively more accepting attitudes towards diverse sexual orientations and gender identities within some Indigenous groups (Jacobs, 1997^[17]; Besnier and Alexeyeff, 2014^[18]), and from a more fluid mindset among mixed-race individuals used to navigate multiple identities.
4. LGBTI+ Americans are less likely to be partnered or married, and to live in a household with children (Panel B of Figure 1.2). These disparities are consistent across all LGBTI+ subgroups, with cisgender gay men facing the most significant disparity. That said, partnership and marriage rates of LGBTI+ individuals have shown signs of convergence with those of their non-LGBTI+ peers, reflecting improved attitudes towards LGBTI+ individuals, the legalisation of same-sex marriage in 2015 by the U.S. Supreme Court, and the spread of the internet and dating apps (National Academies of Sciences, Engineering, and Medicine, 2020^[19]). Despite these recent increases in partnership and marriage rates among LGBTI+ adults, disparities in the likelihood of LGBTI+ becoming parents have remained stable. This stagnation may reflect limited access to adoption, assisted reproductive technologies, and surrogacy, stemming from a combination of legal, financial, and discriminatory barriers (Medina and Mahowald, 2023^[7]; Farr, Vazquez and Patterson, 2020^[20]). These restrictions in turn likely contribute to LGBTI+ individuals’ lower expectations about becoming parents (National Academies of Sciences, Engineering, and Medicine, 2020^[19]; Coffman, Coffman and Marzilli, 2024^[21]).
5. Non-cisgender Americans of any age face significant disadvantages in high school and college attainment, as do young cisgender non-heterosexual individuals (Panel B of Figure 1.2) – except for cisgender gay men who, irrespective of their age, experience a consistent (albeit decreasing) college attainment advantage. The fact that some cisgender non-heterosexual individuals experience a college attainment advantage presumably flows from a non-disclosure bias whereby only better educated non-heterosexual individuals disclose their sexual orientation (Valfort, 2017^[2]). This bias is especially at play among older generations who have been exposed to stronger marginalisation than younger generations. In addition, the college advantage could also flow from developing coping strategies that translate into greater perseverance in academic pursuits (Pachankis and Hatzenbuehler, 2013^[22]). Conversely, the observed educational penalties likely stem from a challenging school environment, exclusionary behaviours exhibited by parents and other family members, and mental health struggles, particularly those arising from stigma but also, in the specific case of non-cisgender individuals, from gender dysphoria (National Academies of Sciences, Engineering, and Medicine, 2020^[19]).

Figure 1.2. LGBTI+ Americans differ on key demographics, including sex assigned at birth, age, race/ethnicity, family structure, and educational attainment, relative to their cisgender straight peers

Panel A: Share of self-identified LGBTI+ US adults (aged 18-88) by sex assigned at birth, age category, and race/ ethnicity (2023)



Panel B: Percentage point difference, after demographic adjustments, in the probability: (i) of being partnered or married; (ii) of living in a household with at least one child below 18; (iii) of having less than a high school degree; and (iv) of holding a Bachelor's degree or higher, using cisgender straight US adults (aged 18-88) as the reference category (2021-23).



Note: LGBTI+ refers to respondents who self-identify as gay/lesbian, bisexual, transgender, other non-heterosexual, and other non-cisgender. Cisgender LGB+ refers to cisgender non-heterosexual individuals (gay/lesbian, bisexual and other non-heterosexual). Non-cisgender refers to individuals whose gender identity does not match their sex assigned at birth. In Panel B, the percentage point differences are estimated from a Logistic regression, controlling for key demographic characteristics (sex assigned at birth, age category, race and ethnicity, marital and parental status, number of adults in the household, educational attainment and living in one of the 15 largest metropolitan statistical areas), as well as state and survey wave fixed effects. Percentage point differences are statistically different from zero at the 95% confidence level. Person-level weights used.

Source: Household Pulse Survey (4 January 2023 – 10 July 2023) for Panel A; Gallup Poll Social Series (2020-22) and Household Pulse Survey (21 July 2021 – 10 July 2023) for Panel B.

1.3. Are LGBTI+ Americans discriminated against?

1.3.1. Anti-LGBTI+ discrimination is a reality in the United States

Evidence suggests that anti-LGBTI+ discrimination is a reality in the United States, as it is in other OECD countries. This is the conclusion that flows from three sets of evidence within the United States:

- Using the Gallup data, the first set of evidence relates to attitudes towards LGBTI+ individuals. The United States has witnessed a notable shift towards greater acceptance of homosexuality, but this has not been as widespread for other sexual orientations and non-cisgender identities. In fact, a declining minority of Americans are supportive of non-cisgender identities. For instance, only 43% of Americans consider changing one's gender to be morally acceptable in 2023, noting that this share has been decreasing in the past years. This development coincides with a rising number of anti-trans bills, with over 150 passed in various US states since 2021 (Trans Legislation Tracker, 2024^[23]).
- The second set of evidence stems from perceptions of discrimination, which continue to be widespread among LGBTI+ Americans. A survey conducted in 2022 by the Center for American Progress revealed significant disparities in experiences of discrimination between LGBTI+ and non-LGBTI+ Americans (Medina and Mahowald, 2023^[7]), with LGBTI+ Americans encountering discrimination at much higher rates than their non-LGBTI+ counterparts, especially when non-cisgender (Medina and Mahowald, 2023^[7]; Coffman, Coffman and Marzilli, 2024^[21]). Consequently, a substantial 78% of LGBTI+ respondents, including 90% of non-cisgender individuals, report taking at least one concealment action to avoid discrimination, such as hiding a personal relationship or altering one's dress style (Medina and Mahowald, 2023^[7]; Folch, 2022^[8]).
- The third set of evidence derives from field experiments, notably correspondence studies in the labour market. In the context of anti-LGBTI+ discrimination, correspondence studies involve sending out, in response to real job ads, the CVs and letters of application of fictitious candidates who are identical except for their sexual orientation or gender identity. Results from such field experiments uncover significant hiring discrimination against LGBTI+ candidates. In the United States, fictitious candidates who indicate they worked as treasurer for the gay and lesbian campus organisation have at least 30% fewer chances of being invited to a job interview than their straight peers with experience as treasurer in another progressive student association (Tilcsik, 2011^[24]; Mishel, 2016^[25]). Although correspondence studies focusing on transgender fictitious candidates are scarce, they also reveal substantial discrimination. In such studies, the transgender fictitious candidates usually add their preferred first name along their legal first name, e.g. "Anne McCarthy (Legal Name: Greg McCarthy)", while the cisgender candidates only mention their legal name (Bardales, 2013^[26]). Recently, a large-scale correspondence study conducted in the United States from 2019 to 2021 and involving over 100 Fortune 500 companies, offered additional insights (Kline, Rose and Walters, 2022^[27]; Kline, Rose and Walters, 2024^[28]). While primarily focused on assessing gender- and race-based discrimination, the study also explored other legally protected characteristics, including age and LGBTI+ identity. The findings reveal a penalty for white applicants who indicate LGBTI+ club membership on their resumes. This result emerges despite the study focusing on large firms, which are typically thought to exhibit less discriminatory behaviour due to more standardised hiring procedures. In contrast, indicating LGBTI+ club membership appears to benefit Black applicants, underscoring the intricate interplay of different identities. This complexity highlights the necessity for further research to fully understand the layered effects of possessing multiple marginalised attributes.

1.3.2. Anti-LGBTI+ discrimination is anticipated to negatively affect the labour market (and health) outcomes of LGBTI+ Americans

Anti-LGBTI+ discrimination is expected to hamper the labour market (and health) outcomes of LGBTI+ individuals via four main channels.

1. Hostile environments affect key demographic characteristics of LGBTI+ individuals which, in turn, critically shape their life trajectories. For instance, considering the positive correlation between educational attainment and employment outcomes, non-cisgender Americans, who often have fewer educational credentials presumably due to systemic bias, are likely to experience higher unemployment rates and lower labour earnings compared to their cisgender straight peers.
2. Anti-LGBTI+ discrimination in the labour market raises barriers to the hiring of LGBTI+ individuals. Consequently, some LGBTI+ individuals end up being unemployed or effectively removed from the labour force. Moreover, when LGBTI+ individuals are employed, they tend to be confined to lower-skilled positions than they would otherwise occupy in the absence of blatant or subtle anti-LGBTI+ discrimination in the workplace. This results in a misallocation of talent and an underutilisation of existing human capital, causing lower wages for LGBTI+ American workers. For instance, LGBTI+ individuals often encounter a glass ceiling that hinders their progress towards top-tier positions (McCay, 2024^[29]) and may avoid certain jobs altogether to sidestep anticipated discrimination (Medina and Mahowald, 2023^[7]). The HPS data confirm that LGBTI+ individuals often find themselves in sectors and occupations characterised by lower pay. Even those who conceal their identity to escape discrimination are adversely impacted. Concealment obstructs the formation of authentic relationships with colleagues and managers, affecting workplace collaboration, mentorship, and support. Specifically, closeted LGBTI+ individuals might avoid networking or professional development opportunities to mitigate the risk of disclosure of their identity. Furthermore, the mental burden of not being able to bring their whole self to work can significantly undermine their productivity at work (Folch, 2022^[8]; Human Rights Campaign Foundation, 2018^[30]).
3. The detrimental consequences of anti-LGBTI+ discrimination in the labour market are amplified by negative feedback effects to the extent that lower labour market prospects undermine incentives to participate in the labour force and reduce their productivity at work (Carcillo and Valfort, 2025^[31]).
4. These harmful repercussions on LGBTI+ individuals' labour supply and productivity are further compounded by the fact that anti-LGBTI+ discrimination (in and outside the labour market) undermines the mental health of LGBTI+ individuals. A rapidly growing literature is providing compelling evidence that stigmatisation does play a key role in LGBTI+ people's worse mental health, a disparity which, in turn, provides a fertile ground to other pathologies, such as cardiovascular diseases (National Academies of Sciences, Engineering, and Medicine, 2020^[19]) (Valfort, 2017^[2]).

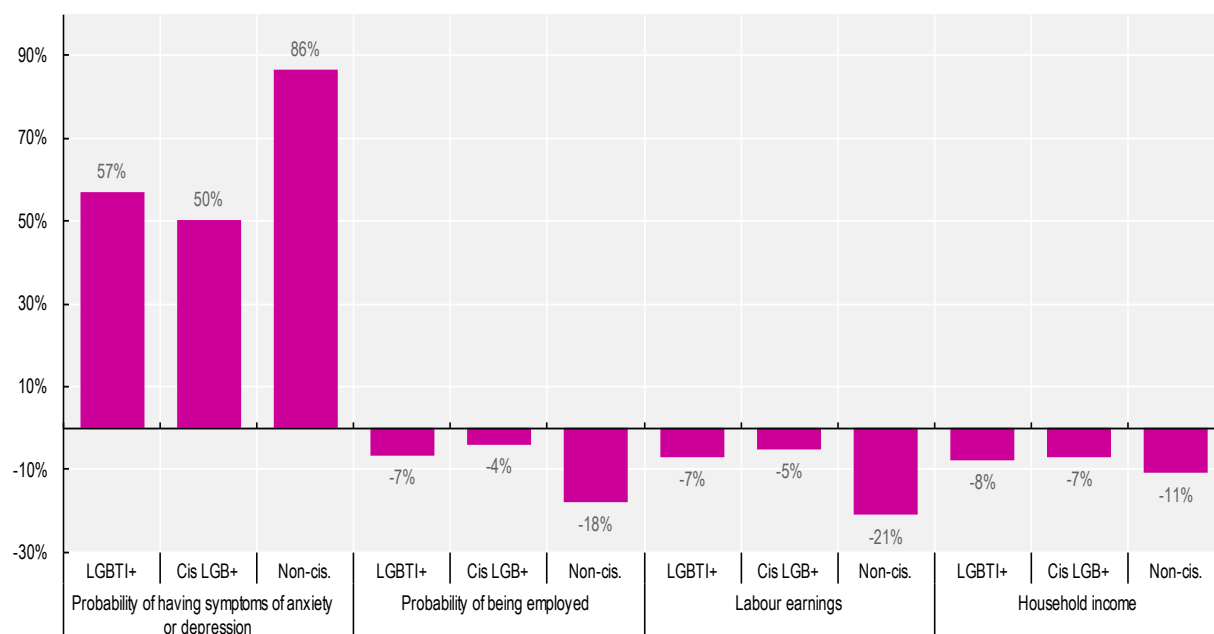
Consistent with the latter channel, our results reveal that LGBTI+ Americans suffer from major mental health unexplained gaps: they are more than 50% more likely to be at risk of generalised anxiety or major depressive disorder, compared to their cisgender straight peers (Figure 1.3). These mental health penalties are consistent across all LGBTI+ subgroups, with non-cisgender Americans facing the strongest disparities.

Importantly, comparing mental health responses from both the Gallup US Daily Survey (2012-17) and the Gallup Panel – COVID-19 Survey (2020) reveals a substantial negative impact of the COVID-19 pandemic on the mental well-being of Americans, which is particularly pronounced among LGBTI+ individuals – a trend corroborated by previous research (OECD, 2021^[32]; OECD, 2021^[33]; OECD, 2021^[34]). Several factors could explain the widening of mental health disparities between LGBTI+ individuals and their cisgender straight peers during the COVID-19 crisis. These include pre-existing higher rates of mental health issues within the LGBTI+ population, potentially making them more susceptible to pandemic-related stress. Additionally, COVID-19 containment measures, such as social distancing and lockdowns, imposed

unique challenges on LGBTI+ individuals, such as isolation from chosen families or forced cohabitation with unsupportive biological family members, further exacerbating mental health gaps.

Figure 1.3. LGBTI+ Americans suffer from unexplained mental health, labour market, and household income gaps, compared to their cisgender straight peers

Unexplained gap between LGBTI+ and cisgender straight US adults (2021-23)



Note: LGBTI+ refers to respondents who self-identify as gay/lesbian, bisexual, transgender, other non-heterosexual, and other non-cisgender. Cis. LGB+ refers to cisgender non-heterosexual individuals (gay, lesbian, bisexual and other non-heterosexual). Non-cis. refers to non-cisgender individuals (individuals whose gender identity does not match their sex assigned at birth). The percentage differences are estimated from a regression controlling for key demographic characteristics (sex assigned at birth, age groups, race and ethnicity, marital and parental status, number of adults in the household, educational attainment and living in one of the 15 largest metropolitan statistical areas), as well as state and survey wave fixed effects. Labour earnings regression also includes sectoral and occupational dummy variables. Gaps are statistically different from zero at the 95% confidence level. Person-level weights used.

Source: Household Pulse Survey (21 July 2021 – 10 July 2023) among working-age US adults (aged 18-74) living alone for probability of being employed; Household Pulse Survey (14 September 2022 – 10 July 2023) among working-age US adults (aged 18-74) living alone for labour earnings; Household Pulse Survey (21 July 2021 – 10 July 2023) among US adults (aged 18-88) for household income and probability of having symptoms of anxiety or depression.

1.3.3. LGBTI+ Americans suffer from unexplained labour market (and health) gaps compared to their cisgender straight peers

In line with the above mechanisms, extensive evidence from representative survey data reveals significant penalties for LGBTI+ individuals in terms of employment rates and labour earnings, both in the United States and in other OECD countries. This disadvantage holds after accounting for essential demographics (Badgett, 1995^[35]; Drydakis, 2022^[36]; Klawitter, 2015^[37]; Carpenter, Lee and Nettuno, 2022^[38]; Valfort, 2017^[2]; OECD, 2019^[11]).

This report distinguishes itself by dissecting the data further into specific LGBTI+ subgroups and different cohorts. Additionally, the large number of observations contained in the HPS dataset allows restricting the analysis of labour market outcomes to individuals living alone (with no other adult or children in the household). This method reduces the potential bias arising from different household specialisation across

same-sex and different-sex couples, which has historically led to overestimating the penalty for non-heterosexual men and underestimating it for non-heterosexual women (Hofmarcher and Plug, 2022^[39]; Jepsen and Jepsen, 2015^[40]; Jepsen and Jepsen, 2002^[41]).

The results confirm substantial labour market gaps, even after adjusting for essential demographics and for sector and occupation for those employed (Figure 1.3). Overall, LGBTI+ Americans are 7% less likely to be employed and, conditional on being employed, they show labour earnings which are also 7% lower. These labour market penalties translate into lower household income for LGBTI+ Americans (Figure 1.3).³ These disparities are especially strong among non-cisgender Americans, potentially due to their greater exposure to discrimination. In addition, complementary findings indicate that the labour market gaps are more pronounced among younger cohorts, presumably due to lower non-disclosure bias compared to older generations.

An analysis of income disparities for LGBT people over time indicates that the COVID-19 pandemic exacerbated their economic challenges. Between 2012 and 2017, the LGBT income penalty estimated from the Gallup US Daily was 7%, which aligns with the income penalty observed in the HPS data from 2021 to 2023. However, from April to December 2020, the Gallup COVID-19 survey recorded a significantly higher LGBT income penalty of 12.5%. These results are consistent with previous findings suggesting that LGBTI+ adults are especially vulnerable during crises (OECD, 2021^[32]; Wenham, 2020^[42]; Movement Advancement Project, 2020^[43]).

1.4. Does LGBTI+ equality pay?

Ending discrimination against LGBTI+ Americans and fostering inclusive environments and opportunities, both within and outside the labour market, ensures economic and social well-being of LGBTI+ Americans. But there is more. This shift also results in the production of more goods and services and generates greater income for spending, saving, or investing. Ultimately, these changes benefit all of society.

To quantify these economic gains, this report relies on the OECD long-term model which projects GDP up to 2060 (Guillemette and Turner, 2021^[44]; Guillemette and Turner, 2018^[45]). This model offers a framework to quantify the additional GDP that would result from a gradual convergence of the employment rate and labour productivity (proxied by labour earnings) of LGBTI+ adults to those of their cisgender straight peers, by different target years (namely 2030, 2040 and 2050). In other words, the model assumes that efforts are made every year until full labour market equality between LGBTI+ and cisgender straight Americans is achieved by the target year. This convergence scenario is compared to a baseline scenario where no such efforts are undertaken.⁴ Thus, this dynamic model helps to strengthen the economic case for LGBTI+ equality with realistic projections, acknowledging that achieving LGBTI+ equality is not an overnight process but one that requires gradual change.

Considering only the mechanisms taken into account by the OECD long term model, the economic consequences of achieving LGBTI+ equality already appear substantial. Under realistic assumptions,⁵ should the United States succeed in closing the unexplained LGBTI+ gaps in employment and labour productivity by 2050, it could expect an increase in GDP equal to 2.6% of the baseline GDP (Figure 1.4). This corresponds to a yearly increase in GDP equal to 0.1%, which amounts to 5% of the average annual US GDP growth over the past ten years. Of course, a more ambitious goal of closing the gaps by 2030 would lead to a higher yearly GDP increase, representing not 5%, but about 10% of the average annual US GDP growth observed between 2013 and 2023.

There are obviously many more benefits from LGBTI+ equality than just the economic gains derived from the OECD long-term model. For instance, LGBTI+ equality will improve public finances through increased tax revenues and reduced public expenditures. An uptick in production and labour earnings enhances public revenues from corporation tax, income tax, and social security contributions. These positive effects

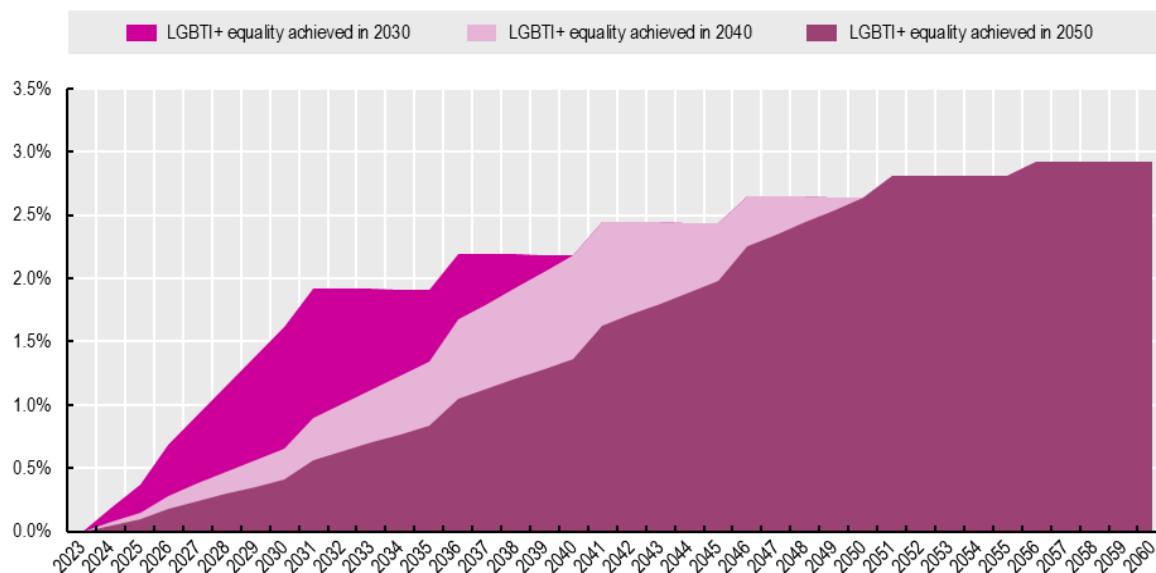
might be further magnified by increased consumption tax revenue, as LGBTI+ individuals spend more on consumer goods and services than their cisgender straight peers, a pattern confirmed by Gallup data. Moreover, eradicating anti-LGBTI+ discrimination is conducive to reduced public expenditures, as greater workforce participation reduces the need for unemployment benefits, active labour market policies, and social transfers. Greater LGBTI+ equality is also anticipated to reduce public health expenditures, as stigma is a significant driver contributing to the poorer mental, behavioural, and physical health of LGBTI+ individuals.

Beyond bolstering tax revenues and reducing public expenditures, eliminating anti-LGBTI+ discrimination might bring significant economic benefits through various additional channels, such as the potential to tap into a diversity and gender equality dividend. Specifically, LGBTI+ equality might have ripple effects on the emancipation of other groups, primarily women. LGBTI+ inclusion inherently challenges heteronormativity, which confines men and women to rigid roles that are obstructive to women's empowerment. Furthermore, embracing LGBTI+ identities means recognising the complex spectrum of gender, leading to a re-evaluation of societal expectations for both men and women. In line with this observation, countries most proactive in enacting laws that promote LGBTI+ equality also tend to have the highest support for gender equality and female labour force participation. Additionally, these countries often exhibit the narrowest gender wage gaps (OECD, 2020^[46]). Under these circumstances, the elimination of anti-LGBTI+ discrimination might bolster gender equality which, alone, has proven to generate substantial economic gains (OECD, 2023^[47]; OECD, 2022^[48]).

Finally, adapting a framework developed by the World Health Organization (World Health Organization, 2001^[49]; Badgett, 2014^[50]), this report estimates the consequences of removing the economic and well-being burden of mental health disparities for LGBTI+ Americans. In 2024, this removal would represent a benefit equivalent to a share of USD GDP ranging from 0.04% to 0.12%.

Figure 1.4. The increase in GDP from addressing the unexplained labour market disparities affecting LGBTI+ individuals is substantial

GDP gain that would be generated from efforts to achieve LGBTI+ equality by 2030, 2040, and 2050, compared to a baseline scenario where no such efforts are undertaken (2024-60)



Note: Potential GDP is defined for the total economy using a constant returns-to-scale Cobb-Douglas production function featuring total employment, labour productivity per worker, and physical capital stock.

Source: OECD estimates based on the OECD long-term model projections and Household Pulse Survey data.

References

- Badgett, M. (2014), *The Economic Cost of Stigma and the Exclusion of LGBT People: A Case Study of India*, Washington, D.C.: World Bank Group, <http://documents.worldbank.org/curated/en/527261468035379692/The-economic-cost-of-stigma-and-the-exclusion-of-LGBT-people-a-case-study-of-India>. [50]
- Badgett, M. (1995), *The wage effects of sexual-orientation discrimination*, *Industrial and Labor Relations Review*, 48(4), 726–739. [35]
- Bardales, N. (2013), “Finding a Job in a Beard and a Dress: Evaluating the Effectiveness of Transgender AntiDiscrimination Laws”, <http://www.semanticscholar.org/paper/Finding-a-Job-in-%E2%80%9C-a-Beard-and-a-Dress-%E2%80%9D-%3A-the-of-%E2%88%97-Bardales/b260619b3c78c224663689a6aae11785ed4aedde>. [26]
- Besnier, N. and K. Alexeyeff (2014), “Gender on the Edge: Transgender, Gay, and Other Pacific Islanders”, *University of Hawai’i Press*, <http://www.jstor.org/stable/j.ctt6wqhsc>. [18]
- Bettinsoli, M., A. Suppes and J. Napier (2020), *Predictors of Attitudes Toward Gay Men and Lesbian Women in 23 Countries*, *Social Psychological and Personality Science*, 11(5), 697–708, <https://doi.org/10.1177/1948550619887785>. [11]
- Carcillo, S. and M. Valfort (2025), “Discrimination in the Labor Market”, *MIT Press*. [31]
- Carpenter, C., M. Lee and L. Nettuno (2022), *Economic outcomes for transgender people and other gender minorities in the United States: First estimates from a nationally representative sample*, *Southern Economic Journal* Volume89, Issue2. [38]
- Coffman, K., L. Coffman and K. Marzilli (2024), *Non-Binary Gender Economics*, National Bureau of Economic Research. [21]
- Drydakis, N. (2022), “Sexual orientation and earnings: A meta-analysis 2012-2020”, *Journal of Population Economics* 35: 409-440. [36]
- Farr, R., C. Vazquez and C. Patterson (2020), *LGBTQ adoptive parents and their children*, In A.E. Goldberg and K.R. Allen (Eds.), *LGBT-Parent Families: Innovations in Research and Implications for Practice* (2nd ed.). New York: Springer. [20]
- Folch, M. (2022), *The LGBTQ+ Gap: Recent Estimates for Young Adults in the United States*, <https://doi.org/10.2139/ssrn.4072893>. [8]
- Goldberg, A. (2023), *Impact of HB 1557 (Florida’s Don’t Say Gay Bill) on LGBTQ+ Parents in Florida*, Williams Institute, <https://williamsinstitute.law.ucla.edu/wp-content/uploads/Dont-Say-Gay-Impact-Jan-2023.pdf>. [9]
- Grov, C. et al. (2006), “Race, ethnicity, gender, and generational factors associated with the coming-out process among gay, lesbian, and bisexual individuals”, *The Journal of Sex Research*, 43:2, 115-121, <https://doi.org/10.1080/00224490609552306>. [14]
- Guillemette, Y. and D. Turner (2021), *The long game: Fiscal outlooks to 2060 underline need for*, OECD Economic Policy Papers, No. 29, OECD Publishing, Paris, <https://doi.org/10.1787/a112307e-en>. [44]

- Guillemette, Y. and D. Turner (2018), *The Long View: Scenarios for the World Economy to 2060*, <https://doi.org/10.1787/2226583X>. [45]
- Herek, G. (2002), *Gender gaps in public opinion about lesbians and gay men*, *Public Opinion Quarterly*, 66, 40–66. [10]
- Hofmarcher, T. and E. Plug (2022), “Specialization in same-sex and different-sex couples”, *Labour Economics*, Vol. 77, p. 101995, <https://doi.org/10.1016/j.labeco.2021.101995>. [39]
- Human Rights Campaign Foundation (2018), *A workplace divided: understanding the climate for LGBTQ workers nationwide*, <https://www.hrc.org/resources/a-workplace-divided-understanding-the-climate-for-lgbtq-workers-nationwide>. [30]
- IPSOS (2023), *LGBT+ PRIDE 2023: A 30-Country Ipsos Global Advisor Survey*, <https://www.ipsos.com/sites/default/files/ct/news/documents/2023-05/ipsos%20LGBT%2B%20Pride%202023%20Global%20Survey%20Report%20-%20rev.pdf>. [5]
- Jacobs, S. (1997), “Two-Spirit people: Native American gender identity”, *Sexuality, and spirituality*. *University of Illinois Press*. [17]
- Jepsen, C. and L. Jepsen (2015), “Labor-Market Specialization within Same-Sex and Different-Sex Couples”, *Industrial Relations: A Journal of Economy and Society*, 54 (1), 2015, p. 109–130. [40]
- Jepsen, L. and C. Jepsen (2002), “An Empirical Analysis of the Matching Patterns of Same-Sex and Opposite-Sex Couples”, *Demography*, 39 (3), 2002, p. 435–453. [41]
- Jones, T. et al. (2016), *Intersex: Stories and Statistics from Australia*, Open Book Publishers, <https://doi.org/10.11647/obp.0089>. [52]
- Klawitter, M. (2015), *Meta-analysis of the effects of sexual orientation on earnings*, *Industrial Relations*, 54(1), 4–32, <https://doi.org/10.1111/irel.12075>. [37]
- Kline, P., E. Rose and C. Walters (2024), *A Discrimination Report Card*, National Bureau of Economic Research, Cambridge, MA, <https://doi.org/10.3386/w32313>. [28]
- Kline, P., E. Rose and C. Walters (2022), “Systemic Discrimination Among Large U.S. Employers”, *The Quarterly Journal of Economics*, Vol. 137/4, pp. 1963–2036, <https://doi.org/10.1093/qje/qjac024>. [27]
- Levine, A. (2022), “Sexualities and Geographic Mobility Between Childhood and Adulthood in the United States”, *Demography* 1 August 2022; 59 (4): 1541–1569, <https://doi.org/10.1215/00703370-10085223>. [6]
- Maguen, S. et al. (2002), “Developmental milestones and disclosure of sexual orientation among, gay, lesbian, and bisexual youths”, *Journal of Applied Developmental Psychology*, 2002;23(2):219–233. [16]
- McCay, L. (2024), “Breaking the Rainbow Ceiling: How LGBTQ+ people can thrive and succeed at work”, *Business, Bloomsbury*. [29]

- Medina, C. and L. Mahowald (2023), *Discrimination and Barriers to Well-Being: The State of the LGBTQI+ Community in 2022*, The Center for American Progress, <https://www.americanprogress.org/article/discrimination-and-barriers-to-well-being-the-state-of-the-lgbtqi-community-in-2022/>. [7]
- Medina, C. and L. Mahowald (2021), “Key Issues Facing People With Intersex Traits”, *The Center for American Progress*, <https://www.americanprogress.org/article/key-issues-facing-people-intersex-traits/>. [51]
- Meyer, I. (2010), “Identity, Stress, and Resilience in Lesbians, Gay Men, and Bisexuals of Color”, *The Counseling Psychologist*, 38(3), 442–454, <https://doi.org/10.1177/0011000009351601>. [12]
- Meyer, I. (2003), *Prejudice, social stress, and mental health in lesbian, gay and bisexual populations: conceptual issues and research evidence*, *Psychol Bull.* 2003;129:674-697. [13]
- Mishel, E. (2016), *Discrimination against queer women in the U.S. workforce: A résumé audit study*, *Socius: Sociological Research for a Dynamic World*, 2, 1–13. [25]
- Movement Advancement Project (2020), “The disproportionate impacts of COVID-19 on LGBTQ households in the U.S.”, <http://www.lgbtmap.org/2020-covid-lgbtq-households>. [43]
- National Academies of Sciences, Engineering, and Medicine (2020), *Understanding the Well-Being of LGBTQI+ Populations*, Washington, DC: The National Academies Press, <https://doi.org/10.17226/25877>. [19]
- OECD (2023), *Joining Forces for Gender Equality: What is Holding us Back?*, OECD Publishing, Paris, <https://doi.org/10.1787/67d48024-en>. [47]
- OECD (2023), *The Road to LGBTI+ Inclusion in Germany: Progress at the Federal and Länder Levels*, OECD Publishing, Paris, <https://doi.org/10.1787/977b463a-en>. [4]
- OECD (2022), *The Economic Case for More Gender Equality in Estonia*, Gender Equality at Work, OECD Publishing, Paris, <https://doi.org/10.1787/299d93b1-en>. [48]
- OECD (2021), “COVID-19 and Well-being: Life in the Pandemic”, *OECD Publishing, Paris*, <https://doi.org/10.1787/1e1ecb53-en>. [32]
- OECD (2021), “Supporting young people’s mental health through the COVID-19 crisis”, *OECD Policy Responses to Coronavirus (COVID-19)*, *OECD Publishing, Paris*, <https://doi.org/10.1787/84e143e5-en>. [34]
- OECD (2021), “Tackling the mental health impact of the COVID-19 crisis: An integrated, whole-of-society response”, *OECD Policy Responses to Coronavirus (COVID-19)*, *OECD Publishing, Paris*, <https://doi.org/10.1787/0cca0b0b-en>. [33]
- OECD (2020), *Over the Rainbow? The Road to LGBTI Inclusion*, OECD Publishing, Paris, <https://doi.org/10.1787/8d2fd1a8-en>. [46]
- OECD (2019), *Society at a Glance 2019: OECD Social Indicators*, OECD Publishing, Paris, https://doi.org/10.1787/soc_glance-2019-en. [1]
- Pachankis, J. and M. Hatzenbuehler (2013), *The Social Development of Contingent Self-Worth in Sexual Minority Young Men: An Empirical Investigation of the ‘Best Little Boy in the World’ Hypothesis.*, *Basic and Applied Social Psychology* 35(2):176–90. [22]

- Robertson, R. et al. (2018), *Estimates of Non-Heterosexual Prevalence: The Roles of Anonymity and Privacy in Survey Methodology*, *Archives of Sexual Behavior* 47 (4): 1069-84. [3]
- Rosario, M., E. Schrimshaw and J. Hunter (2004), "Ethnic/racial differences in the coming-out process of lesbian, gay, and bisexual youths: a comparison of sexual identity development over time.", *Cultur Divers Ethnic Minor Psychol.* 2004 Aug;10(3):215-28, <https://doi.org/10.1037/1099-9809.10.3.215>. [15]
- Rosenwohl-Mack, A. et al. (2020), "A national study on the physical and mental health of intersex adults in the U.S.", *PLoS One.* 2020 Oct 9;15(10):e0240088, <https://doi.org/10.1371/journal.pone.0240088>. [53]
- Tilcsik, A. (2011), *Pride and prejudice: Employment discrimination against openly gay men in the United States*, *American Journal of Sociology*, 117(2), 586–626. [24]
- Trans Legislation Tracker (2024), "2024 anti-trans bills tracker: Anti-trans bills under consideration and passed, 2021-2024", *Trans Legislation Tracker*, <https://translegislation.com/>. [23]
- Valfort, M. (2017), "LGBTI in OECD Countries: A Review", *OECD Social, Employment and Migration Working Papers*, No. 198, OECD Publishing, Paris, <https://doi.org/10.1787/d5d49711-en>. [2]
- Wenham, C. (2020), "The gendered impact of the COVID-19 crisis and post-crisis period", *European Parliament, Brussels*, <http://www.europarl.europa.eu/supporting-analyses>. [42]
- World Health Organization (2001), *Macroeconomic and Health: Investing in Health for Economic Development*, Commission on Macroeconomics and Health. Prepared for the World Health Organization. [49]

Notes

¹ Research has found that people with intersex traits report identifying as non-cisgender or non-heterosexual at higher rates than people who are not intersex (Medina and Mahowald, 2021^[51]; Rosenwohl-Mack et al., 2020^[53]; Jones et al., 2016^[52]). Yet, the HPS questionnaire does not allow distinguishing intersex individuals among those who self-identify as LGBTI+.

² For transgender individuals, it is possible that some respondents identify their sexual orientation based on their sex assigned at birth, while others might use their gender identity. Given that the HPS sample does not allow distinguishing between these two reporting strategies, non-cisgender heterosexual and non-cisgender non-heterosexual are combined in one group.

³ While the labour market penalties faced by LGBTI+ Americans are likely the primary factor leading to their lower household income, several other factors could also contribute to the LGBTI+ income penalty, such as lower family economic support and discrimination in the housing and financial markets.

⁴ In particular, the OECD long-term model is based on a standard Cobb-Douglas production function featuring physical capital stock, total employment of adults aged 15-74, and total factor productivity. The baseline scenario is grounded in a set of OECD long-run projections on total employment, labour productivity per worker, and capital stock in the United States (Guillemette and Turner, 2021^[44]).

⁵ We recognise in our projections that both the share of self-identified LGBTI+ individuals and the extent of their labour market disparities vary by cohort. Given that younger generations not only have a higher proportion of self-identified LGBTI+ individuals but also face, on average, greater labour market disparities, our projections yield an increase in both the proportion of self-identified LGBTI+ individuals and an increase in the extent of their labour market disparities within the total population. This assumption explains why GDP gains continue to increase even after achieving full labour market parity by the target year.

2 Who are LGBTI+ Americans?

This chapter presents the latest data on the share of US adults who identify as LGBTI+ and delves into information on their geographic location and demographics, including sex assigned at birth, age, race/ethnicity, family structure, and educational attainment. Over the past decade, the percentage of US adults who identify as LGBTI+ has nearly doubled, reaching 11.2% (or nearly 30 million) in 2023. LGBTI+ self-identification is more commonly reported in accepting neighbourhoods, as well as among individuals assigned female at birth, young people, and Whites. LGBTI+ adults are less likely to live with a partner or in a household with children. Historically, non-cisgender individuals have faced educational disadvantages, and this trend is now also emerging among younger generations of cisgender individuals who identify as non-heterosexual. This chapter explores the reasons for these disparities, beyond documenting them.

Who are LGBTI+ individuals?

LGBTI+ is the acronym for “lesbian, gay, bisexual, transgender and intersex”. LGBTI people are defined with respect to three distinct features: sexual orientation; gender identity; and sex characteristics. The “plus” (+) leaves the demographic category open ended to acknowledge additional sexual orientations and gender identities that are not explicitly present in the acronym. LGBTI+ individuals are also referred to by the umbrella term “queer people”.¹

Sexual orientation

Sexual orientation allows for differentiating between heterosexuals, lesbians, gay men and bisexuals. It is indicative of a person’s capacity for emotional and/or sexual attraction to different-gender, same-gender individuals, or both different- and same-gender individuals. In this context, the “plus” refers to additional sexual orientations, such as “asexual” (lacking emotional and/or sexual attraction to anyone), “pansexual” (considering gender as irrelevant in determining whether one will be emotionally or sexually attracted to someone), among others.

Gender identity

Gender identity refers to a person’s internal sense of being masculine, feminine, androgynous or neither, regardless of sexual orientation. For individuals for whom gender identity corresponds to their biological sex, that is individuals assigned male at birth (AMAB) or individuals assigned female at birth (AFAB), the Latin prefix *cis* (“on this side of”) is used to define them as “cisgender”. For those where this is not the case, the Latin prefix *trans* (“on the other side of”) is used to define them as “transgender”. A transgender person can be: i) a transgender man (a person who was assigned female at birth but whose gender identity is male); or ii) a transgender woman (a person who was assigned male at birth but whose gender identity is female). In this context, the “plus” refers to additional gender identities, such as “non-binary” (a person who was assigned either female or male at birth but whose gender identity refers to a combination of these genders), “agender” (not identifying as having a gender), “gender fluid” (not identifying as having a fixed gender), among others.

Sex characteristics

Sex characteristics refer to chromosomal patterns, hormonal structures, reproductive organs and sexual anatomy that determine an individual’s biological sex. Sex characteristics are sometimes ambiguous in comparison to medical standards rooted in binary concepts of “male” and “female”. An individual whose sex characteristics are neither wholly female, nor wholly male is called “intersex”. Due to this non-binary pattern, and although being intersex is distinct from a person’s sexual orientation or gender identity, intersex individuals are over-represented among the LGBT population (Medina and Mahowald, 2021^[1]; Rosenwohl-Mack et al., 2020^[2]; Jones et al., 2016^[3]) – this explains why the letter “I” is typically added to the LGBT acronym to include intersex people. Importantly, being intersex is not a pathological condition, and rarely is life-threatening (Fundamental Rights Agency, 2015^[4]).

Notes:

1. Consistent with its primary meaning to depict something strange, odd, or eccentric, the word “queer” began to be used in the early 1990s to refer to an identity that challenges normative ideas about sexuality and gender (after decades of colloquial use to refer to men who are sexually attracted to men). From the early 2000s, this word was increasingly used as an umbrella term to reflect the spectrum of sexual orientations and gender identities covered by the LGBTI+ acronym. While “queer” was frequently derogatory in the past, this word has been reclaimed by non-heterosexual and non-cisgender individuals, to describe themselves in a positive way. It is thus to be understood nowadays as an affirming word that is inclusive of all people in the rainbow acronym.

Source: (OECD, 2020^[5]; Kasprowski et al., 2021^[6])

2.1. Introduction and main findings

Although the road to LGBTI+ equality sometimes takes us backwards, there has been a well-documented overall shift towards greater acceptance of homosexuality over the past decades, both in OECD countries and beyond (OECD, 2019^[7]; Flores, 2019^[8]).¹ Consistent with this evolution, receptiveness to LGBTI+ individuals has improved across the United States (Jones et al., 2019^[9]; Flores and Barclay, 2015^[10]). This cultural change has gone hand in hand with enhanced recognition of LGBTI+ rights (OECD, 2020^[5]; Movement Advancement Project, 2020^[11]). In 2019, OECD countries were more than halfway to achieving full legal acceptance of LGBTI+ individuals – a sixfold increase from the late 1970s when less than 10% of essential laws for LGBTI+ inclusion had been enacted. Since then, there has been continued progress, with several countries passing legal measures viewed as ground-breaking, such as nationwide bans on conversion therapies that no OECD country had implemented by 2019. The United States is no exception. For instance, on 26 June 2015, the U.S. Supreme Court granted nationwide legal access to same-sex marriage, and labour market discrimination based on sexual orientation and gender identity has been illegal nationwide since 2020. However, this broadening of rights has recently sparked backlash, particularly against non-cisgender individuals. In the first eight months of 2023 alone, legislators introduced 550 bills to ban gender-affirming care for LGBTI+ youth, with over 80 of these bills becoming law (Human Rights Campaign Foundation, 2023^[12]).

While it is crucial to assess the disparities to advance LGBTI+ equality further, population-based estimates of individuals who self-identify as LGBTI+ are still rare. As of 2023, only two OECD countries have included a question about sexual orientation and/or gender identity in their censuses. For the latter category, respondents have the option to indicate a gender identity different from their sex assigned at birth (OECD, 2023^[13]). Canada introduced this question on gender identity in 2021. In the same year, the United Kingdom added questions on both sexual orientation (for Great Britain and Northern Ireland) and gender identity (for Great Britain only). Despite not being one of these countries, the United States has recently been active in bridging the data gap which has thus far hindered comprehensive estimates of the size of the LGBTI+ population and of their situation, especially with the introduction in 2021 of questions allowing respondents to self-identify as LGBTI+ in the Census Bureau's Household Pulse Survey. The United States is also distinctive by the fact that nationally representative information on self-identified LGBTI+ adults had already been collected prior to 2021, although not by the Census Bureau, but by the global analytics and advisory firm Gallup.²

Chapter 2 offers an in-depth look at the LGBTI+ population in the United States, drawing from these two sources. It begins by presenting the latest estimates of adults who self-identify as LGBTI+ and traces their evolution over the past decade. The chapter then delves into the spatial distribution of this population, followed by comparisons based on key demographic characteristics: sex assigned at birth, age, race and ethnicity, family structure, and educational attainment.

Main findings

- LGBTI+ individuals in the United States represent a sizeable and growing minority.
 - As of 2023, 11.2% of adults, or nearly 30 million people, self-identified as LGBTI+ – a number surpassing the populations of New York or Florida.
 - Among the LGBTI+ population, individuals who identify as cisgender and non-heterosexual are nearly four times more represented (8.9%) than those who identify as non-cisgender (2.3%).
 - Within the former group, the largest subgroup are bisexuals (4.5%), followed by gays/lesbians (2.9%) and by other non-heterosexuals (1.5%).

- As for non-cisgender individuals, 0.9% of US adults self-identify as transgender, and 1.4% as other non-cisgender identity (e.g. non-binary).
- Consistent with improving social and legal acceptance that encourages non-heterosexual and non-cisgender persons to “come out of the closet”, the share of adults who self-identify as LGBT has almost doubled between 2012 and 2022.
- The share of self-identified LGBTI+ adults varies substantially across US regions, states, and rural/urban areas. This pattern reflects geographic disparities in social acceptance and legal protections of LGBTI+ individuals, as LGBTI+ adults are more likely to relocate to more accepting neighbourhoods and might also feel more comfortable disclosing their LGBTI+ status in these areas.
- Individuals assigned female at birth (AFAB) are more likely to identify as LGBTI+ than those assigned male at birth (AMAB), a gap that has widened in the past decade.
 - These disparities likely flow from greater social acceptance of women expressing same-sex attraction or relationships compared to men, as well as from females’ more positive attitudes towards LGBTI+ individuals.
 - The widening of this gap over time probably mirrors strides made in gender equality, hence a departure from traditional roles and expectations placed upon women, including the pressure to adhere to heteronormativity and to be feminine.
- LGBTI+ self-identification decreases with age, with the recent surge in LGBTI+ identification largely driven by younger generations.
 - In 2023, nearly one in four young adults (18-34) identified as LGBTI+, compared to nearly one in ten among prime age adults (35-54) and one in 20 among older mature adults (55+).
 - The rising proportion of young adults identifying as LGBTI+ does not seem to be driven by an age effect, which would entail that, for younger generations, self-identifying as LGBTI+ is a fad that ceases once they grow older. Rather, it appears to flow from a cohort effect: raised in more accepting environments, young LGBTI+ adults feel more comfortable disclosing who they are.
 - Consequently, the share of self-identified LGBTI+ adults will likely continue to grow in the future, assuming social and legal acceptance further improves.
- LGBTI+ self-identification varies widely by race and ethnicity.
 - Relative to non-Hispanic Whites, a smaller share of non-Hispanic Blacks and Asians identify as LGBTI+, presumably due to the more negative attitudes towards LGBTI+ individuals that prevail within these populations as well as to a “double minority stress” effect.
 - The share of self-identified LGBTI+ adults is higher among Hispanic Americans compared to non-Hispanic Whites and has seen the greatest increase within this group. However, this pattern is predominantly driven by the younger average age of the Hispanic population.
- LGBTI+ adults are less likely to be partnered or married, and to live in a household with children.
 - These gaps are significant across all LGBTI+ subgroups, with cisgender gay males facing the most significant disparity in family structure.
 - In the past decade, partnership and marriage rates of LGBTI+ individuals have shown modest signs of convergence with those of their non-LGBTI+ peers, reflecting improved attitudes towards LGBTI+ individuals and the legalisation of same-sex marriage in 2015 by the U.S. Supreme Court, as well as the advent of the internet and dating apps.
 - Disparities in the likelihood of LGBTI+ becoming parents have remained stable, possibly indicating limited access to adoption, assisted reproductive technology, and surrogacy, due to a combination of legal, financial, and discriminatory barriers.

- Historically, non-cisgender individuals have faced educational disadvantages, and this trend is now emerging among younger generations of cisgender individuals who identify as non-heterosexual. Among the latter group, cisgender lesbian females have experienced the most significant decline in college attainment across age groups, while cisgender gay males are the only subgroup with a consistent (albeit decreasing) college attainment advantage.
 - The persistent educational penalties faced by non-cisgender individuals across age groups likely stem from a challenging school environment, exclusionary behaviours exhibited by parents and other family members, and mental health struggles, particularly those arising from stigma but also, in the specific case of non-cisgender individuals, gender dysphoria.
 - While several of these negative factors undoubtedly impact cisgender LGB+ individuals too, the educational premium observed for prime age and mature cisgender LGB+ adults may result from a non-disclosure bias that outweigh these factors' detrimental impact. Historically, lower-educated LGBTI+ adults have tended to refrain from openly identifying as LGBTI+ relative to their higher-educated peers.
 - As social and legal acceptance have steadily improved, the impetus for cisgender LGB+ individuals to invest more in education has likely diminished, thereby yielding an erosion of the educational attainment premium among younger generations. In addition, a greater number of lower-educated LGBTI+ individuals might now feel empowered to “come out of the closet”, meaning that the non-disclosure bias observed among older generations is progressively losing ground. The more pronounced decline for cisgender lesbian females could be attributed to their reduced motivation to challenge patriarchal norms as gender equality advances. Additionally, those perceived as gender atypical might face an additional obstacle in an environment where personal characteristics traditionally viewed as “feminine” are deemed more conducive to academic success than stereotypically masculine traits.

2.2. How many US adults self-identify as LGBTI+?

Large-scale, nationally representative surveys that include questions on respondents' sexual orientation and gender identity are critical to produce a reliable count of the population that self-identifies as non-heterosexual and/or non-cisgender.

2.2.1. Sources of data

Two sources allow estimating the size of the LGBTI+ adult population in the United States: the Census Bureau's Household Pulse Survey and the surveys that Gallup has been conducting since 2012.

2.2.2. The Census Bureau's Household Pulse Survey

The U.S. Census Bureau, the federal government's largest statistical agency, does not currently include questions on sexual orientation and/or gender identity in its long-standing demographic surveys, such as the Decennial Census, the American Community Survey (ACS), and the Current Population Survey (CPS).³ These surveys do contain information about marital status and living arrangements, enabling the identification of same-sex couples (Badgett, Carpenter and Sansone, 2021^[14]; Black et al., 2000^[15]). Yet, this subgroup only captures a small non-representative fraction of the LGBTI+ population (Deng and Watson, 2023^[16]). As for other mainstream nationally representative surveys conducted by public institutions, none offer comprehensive insights into the LGBTI+ population, such as its size or key characteristics. While a dozen of nationally representative surveys have collected data on sexual orientation, only a few have delved into gender identity. Yet, even when both aspects are addressed, the

focus typically narrows to specific areas like health, education, or crime. Importantly, no nationally representative surveys have ventured into collecting data on variations in sex characteristics or on intersex status (National Science and Technology Council, 2023^[17]).

In this context, the launch in April 2020 of the Census Bureau's Household Pulse Survey (HPS) constitutes a breakthrough. The HPS is an experimental probability-based nationally representative Census survey designed to provide near-real time data on how the COVID-19 pandemic has impacted households across the United States from a social and economic perspective. Beginning in July 2021, three questions related to LGBTI+ status were added: sex assignment at birth, gender identity, and sexual orientation (see Box 2.1 for more details about the HPS). Thus, the HPS data offer the first large-scale, nationally representative sample of both non-heterosexual and non-cisgender American adults, encompassing 155 447 LGBTI+ individuals between 21 July 2021 and 10 July 2023. Throughout this report, the average demographic characteristics and socio-economic outcomes of LGBTI+ adults are analysed separately for the following subgroups: i) cisgender non-heterosexuals, i.e. cisgender gays/lesbians, cisgender bisexuals, and cisgender other non-heterosexual individuals); and ii) non-cisgender adults, i.e. transgender and other non-cisgender individuals. The decision to group both non-cisgender heterosexual and non-cisgender non-heterosexual respondents together arises from challenges in categorizing non-cisgender individuals by sexual orientation. Some non-cisgender respondents may identify their sexual orientation based on their sex assigned at birth, while others might use their gender identity.⁴ Unfortunately, we cannot distinguish between these two reporting strategies.⁵

It is vital to note that, like other nationally representative US surveys with self-identified LGBTI+ questions, the HPS sample excludes institutionalised adults. This omission means that homeless individuals, those in medical institutions, incarceration facilities, or homeless shelters are not represented. Such exclusions might disproportionately affect LGBTI+ individuals, potentially underestimating the socio-economic and health challenges they face (Carpenter, Lee and Nettuno, 2022^[18]). This underestimation is likely to prevail even among the non-institutionalised population. The HPS is an online survey that necessitates to contact individuals via email or text message, thereby excluding individuals without an internet access or a phone number. Given their heightened economic vulnerability, LGBTI+ individuals might again be over-represented among these omissions (Martell and Roncolato, 2023^[19]).

2.2.3. The US Gallup data

Although not conducted under the supervision of a public institution, data collection on both sexual orientation and gender identity based on probability sampling was already ongoing prior to 2021. This was done in the framework of three polls conducted by the Gallup corporation: between 2012 and 2017 as part of the Gallup US Daily Survey (2008-17), between 2020 and 2022 in the context of the Gallup Poll Social Series (2001-ongoing), and between April and October 2020 as an addition to the Gallup Panel – COVID-19 Survey (2020-ongoing). Box 2.2, Box 2.3, and Box 2.4 provide a detailed overview of these three US Gallup surveys.

Box 2.1. Zooming in on the Census Bureau's Household Pulse Survey

The Census Bureau's Household Pulse Survey (HPS) was launched in April 2020. It is a 20-minute online survey aimed at assessing the social and economic impact of COVID-19 on American households. The HPS is based on probability sampling using the US Census Master Address File. It is designed to ensure national representation of the US population and to produce, besides, reliable state-level estimates as well as estimates for the 15 largest Metropolitan Statistical Areas (MSA) in the United States. Each randomly selected housing unit is contacted via email or text message using the provided contact information. The respondent is an adult member of the household, at least 18 years old, whose personal email or phone number was used for the outreach.

The HPS is part of the Census Bureau's Experimental Data Product series. It was designed to have a low respondent burden and to be a short-turnaround instrument that provides valuable data to aid in the pandemic recovery. That said, the online nature of HPS – as opposed to telephone-assisted or in-person methods – results in a relatively low response rate, approximately 6% between 2021 and 2023.¹ An examination of the HPS's early waves revealed that this aspect contributes to an underrepresentation of households with lower median incomes. Without correction, this selection bias could underestimate the economic challenges faced by LGBTI+ individuals, given consistent evidence pointing to a higher poverty rate within this group. However, this potential skew is addressed by the Census Bureau, which offers weights derived from the American Community Survey (ACS) demographic data for the sampling area. This adjustment compensates for the HPS's low response rate, ensuring the data accurately represent adults aged 18 and above residing in households. After this weighting adjustment, the demographic and socio-economic profiles of LGBTI+ adults align with findings from other surveys boasting higher response rates, such as Gallup, the National Health Interview Survey (NHIS), and the Behavioural Risk Factor Surveillance System (BRFSS).

Since July 2021, the survey asks three questions that permit the identification of non-heterosexual and/or non-cisgender individuals. Sexual orientation is measured using the question: "Which of the following best represents how you think of yourself?", allowing participants to answer: i) "Gay or Lesbian"; ii) "Straight, that is not gay or lesbian"; iii) "Bisexual"; iv) "Something else, please specify"; v) "I don't know." Gender identity is determined using a state-of-the-art two-step approach. The first question is, "What sex were you assigned at birth, on your original birth certificate?", with options: i) "Male"; ii) "Female". The second question is, "Do you currently describe yourself as male, female, or transgender?". The provided choices are: i) "Male"; ii) "Female"; iii) "Transgender"; iv) "None of these". Respondents whose sex at birth and gender do not align receive an automated "confirmation check" question to verify that the reported answers are correct.

This report utilises data collected in Phases 3.2-3.9, from 21 July 2021 to 10 July 2023. This dataset contains 1 648 098 respondents of which 155 447 self-identify as LGBTI+ (46 364 in 2023).

Note:

1. Low response rates are typical in online surveys, especially when compared to telephone and in-person surveys. This is particularly true when the surveys use probability sampling encompassing the entire national population, rather than targeting on a specific subgroup (Wu, Zhao and Fils-Aime, 2022^[20]). Furthermore, the response rate is likely underestimated in the HPS since all non-interviews were considered valid non-responses. Yet, non-interviews include several instances where the contact information, whether an email address or a mobile phone number for text messages, was invalid. Additionally, in cases where emails were the sole mode of contact, they might have been sent directly to the spam folder, making them unnoticed by potential respondents. Furthermore, even when the contact details were accurate and the intended recipients received the email or text message, there's a possibility that some perceived it as a scam.

Source: (United States Census Bureau, 2023^[21]; United States Census Bureau, 2021^[22]; Fields et al., 2020^[23]).

Box 2.2. Gallup US Daily Survey (2008-17) – information on LGBT status collected starting from 2012

The *Gallup US Daily Survey* was run between 2008 and 2017. It asked 1 000 US adults every day about various political, economic, and well-being topics. This survey was based on computer-assisted telephone interviews with randomly sampled respondents, aged 18 and above, including respondents from all 50 US states and the District of Columbia. Gallup conducted the survey using two different topical questionnaires: half of the respondents receive the Well-being track, while the other half receives the Politics and Economy track. The response rate averaged 9% for the Well-being track and 12% for the Politics and Economy track. Gallup provides weights to compensate for disproportionalities in selection probabilities and nonresponse. In addition, Gallup weights the data to match national and state-level targets from the U.S. Census Bureau by age, sex, region, gender, education, ethnicity, and race, as well as population density of self-reported location. Gallup also weights the data to be representative of the US adult population.

In 2012 the following question was introduced at the end of the survey: “I have one final question we are asking only for statistical purposes. Do you, personally, identify as lesbian, gay, bisexual or transgender?”, with the following options: i) “Yes”; ii) “No”; iii) “Don’t know”. Before September 2014, the interviewer coded the gender of the respondent as either “Male” or “Female” based on the voice and other cues detected during the phone interview, which entails a significant risk of misgendering. Starting in September 2014, the survey included the following question: “I am required to ask, are you male or female?”, with answer categories i) “Male”; ii) “Female”. Of those who refused to answer, the interviewer coded the gender of the respondent based on the same cues as those used before September 2014, including voice. It is important to emphasise that the various methods used to determine a respondent’s gender produced consistent estimates for both (LGBT) males and (LGBT) females. When examining the percentage of female respondents in the four months before and after September 2014, there is no notable change following the introduction of the direct question. Likewise, the proportion difference between LGBT males and LGBT females remained consistent after September 2014.

This report utilises data from the Gallup US Daily Survey collected from 2012 to 2017. This dataset contains 2 114 755 respondents, of which 62 193 self-identify as LGBT.

Source: Gallup Inc.

Box 2.3. Gallup Poll Social Series (2001-ongoing) – information on LGBTI+ status collected starting from 2020

The *Gallup Poll Social Series* (GPSS) has been conducted since 2001. It is a monthly telephone survey, with a unique set of 1 000 respondents, that also relies on probability-based sampling (random digit dialling). Each month is dedicated to a specific topic, designed to monitor US adults' views on numerous social, economic, and political topics. Gallup also provides weights to make the sample nationally representative. Indeed, response rates have recently declined significantly in the Gallup Poll Social Series (as with other telephone surveys): on average, during 2020-22, the response rate was 8%, compared with 28% in 1997 (Marken, 2018^[24]).

In 2020 the following question was introduced: "Which of the following do you consider yourself to be?". Respondents could select all options that apply, including: i) "Straight or heterosexual"; ii) "Lesbian"; iii) "Gay"; iv) "Bisexual"; v) "Transgender"; vi) "Don't know". Respondents were also allowed to volunteer another identity (e.g. queer, pansexual), which was recorded as "Other LGBT" by interviewers. Between 2020 and 2021, the gender of the respondent was identified through the following question that could only be answered in a binary way: "I am required to ask, are you male or female". In 2022, respondents were given the possibility to also self-identify as "non-binary" ("I am required to ask, are you male, female or nonbinary"). Of those who refused to answer, the interviewer coded the gender of the respondent as "Male" or "Female" based on the voice, together with other cues gathered during the phone interview.

This report utilises data from the Gallup Poll Social Series collected from 2020 to 2022. This dataset contains 42 557 respondents, of which 1967 self-identify as LGBTI+.

Source: Gallup Inc.

Box 2.4. Gallup Panel – COVID-19 Survey (2020-ongoing) – information on LGBT status collected from April to October 2020

The *Gallup Panel – COVID-19 Survey* is a web survey focusing on the impact of the COVID-19 pandemic, which was launched in March 2020. The panel was originally designed based on probability sampling. Panel members receive an average of three surveys per month, and members remain in the panel for as long as they would like. Starting in August 2020, the survey moved from daily surveying to a survey conducted one time per month. Beginning in 2022, the survey moved to quarterly data collection. The average response rate in 2020 was 46%. Gallup provides weights to make the sample nationally representative.

From April to October 2020, participants were asked the following question “Do you, personally, identify as lesbian, gay, bisexual, or transgender?”, with the following options: i) “Yes”; ii) “No”. The gender of the respondent was identified through the question “What is your gender?”, which respondents could only answer in a binary way, by selecting either “male” or “female”.

Of course, moving further away from the survey’s inception date, its representativeness diminishes due to attrition. Therefore, this survey is not used in this report to estimate the size or demographics of the LGBTI+ population, but to discuss how the pandemic impacted LGBTI+ individuals during 2020 (Chapter 3).

This report utilises data from the Gallup Panel – COVID-19 Survey collected from March to December 2020. This dataset contains 113 928 observations (42 955 respondents), of which 6 072 observations (2 364 respondents) self-identify as LGBT. Given the panel nature of the data, the LGBT status for respondents who had missing information in a specific survey wave month between March and December 2020 was imputed. This imputation was based on their responses from another month when the LGBT question was asked, and they provided a valid answer. Respondents who gave inconsistent answers to the LGBT question across months were excluded from the analysis.

Source: Gallup Inc.

2.2.4. Self-identified LGBTI+ adults

In 2023, 11.2% of American adults self-identified as LGBTI+, according to the Census Bureau’s Household Pulse Survey (Figure 1.1). This figure suggests that nearly 30 million of American adults self-identify as LGBTI+, more than the number of residents living in New York (19.6 million) or Florida (22.2 million).⁶ Among the LGBTI+ population, individuals who identify as cisgender and non-heterosexual are nearly four times more represented (8.9%) than those who identify as non-cisgender (2.3%). Within the former group, the largest subgroup are bisexuals (4.5%), followed by gays/lesbians (2.9%) and by other non-heterosexuals (1.5%). As for non-cisgender individuals, 0.9% of US adults self-identify as transgender,⁷ and 1.4% as other non-cisgender identity (e.g. non-binary).⁸ Box 2.5 provides a detailed description of how LGBTI+ respondents are identified in the data.

These estimates are close to those flowing from the 2nd edition of the biennial LGBT+ Pride Global Survey that IPSOS conducted online between February and March 2023 across 30 countries, based on opt-in panels (IPSOS, 2023_[25]).⁹ According to this survey, 3% of American adults self-identified as gay/lesbians, 5% as bisexuals, and 1% as being part of another non-heterosexual group (i.e. pansexual/omnisexual). Moreover, 2% self-defined as transgender or as having another non-cisgender identity. Using a probability-based online survey conducted in May 2022, the Pew Research Center found estimates for the share of transgender and non-binary individuals which are in line with those provided by the Household Pulse

Survey (HPS) and by the LGBT+ Pride Global Survey, although slightly lower since it didn't measure additional non-cisgender identity beyond being transgender and non-binary: 0.6% of US adults self-identified as transgender and 1% self-identified as non-binary (Pew Research Center, 2022^[26]).

Consistent with improving social and legal acceptance that encourages non-heterosexual and non-cisgender persons to “come out of the closet”, the share of individuals who self-identify as LGBTI+ is on the rise, with an increase of 0.8 percentage points (7.3%) between 2021 and 2023 according to the HPS, from 10.4% to 11.2% (Figure 2.2). This increase in self-reporting was driven by cisgender non-heterosexual adults. In contrast, the share of non-cisgender adults remained stable at 2.3%. This upward trend in self-reporting is even more striking when one focuses on a longer timeframe, hence on the Gallup survey data: the share of adults who self-identify as LGBT almost doubled between 2012 and 2022, rising from 3.5% to 6.9%. Similar increases have also been observed in other nationally representative surveys that include a question on sexual orientation (OECD, 2019^[7]; Valfort, 2017^[27]).¹⁰ Although not available for all US states, recent data from the Behaviour Risk Factor Surveillance System (BRFSS) survey also suggests that the number of adults who identify as transgender has remained roughly steady over time (Herman, Flores and O'Neill, 2022^[28]).

The comparison of HPS and Gallup survey data confirms another recurring finding: estimates of the LGBTI+ population are significantly higher in surveys where respondents, rather than interviewers, answer the question about LGBTI+ status (OECD, 2019^[7]; Valfort, 2017^[27]). Even holding the survey year constant, the share of individuals who self-identify as LGBT is on average 18% higher in the HPS, which is a self-administered online survey, than in the Gallup Poll Social Series which rely on (computer-assisted) telephone interviewing. As many LGBTI+ adults are still closeted in several areas of their life (Folch, 2022^[29]; Human Rights Campaign Foundation, 2018^[30]), they may be reluctant to come out in surveys that guarantee neither enough privacy (not being able to observe respondents while they give their responses) nor enough anonymity (not being able to link their responses to their identifier) (Robertson et al., 2018^[31]).

Yet, although it relies on a survey mode known to minimise non-disclosure by LGBTI+ individuals of who they are, estimates that flow from the HPS likely understate their actual share. Indeed, to measure sexual orientation, these estimates flow from questions that ask respondents to self-identify as heterosexual, homosexual or bisexual, not on questions that ask respondents to report their sexual behaviour (the gender of individuals with whom one has sexual intercourse) or their sexual attraction (the gender of individuals one feels sexually attracted to). However, sexual attraction is a more inclusive concept than sexual behaviour, which is itself more inclusive than sexual self-identification: not all people who feel sexually attracted to same-gender people engage in same-gender sexual behaviour, and not all people who engage in same-gender sexual behaviour view themselves as non-heterosexual. Accordingly, the size of the LGB population is 70% larger when it is calculated based on individuals' sexual behaviour (instead of individuals' sexual identification), and more than twice as large when sexual attraction is taken as a criterion (OECD, 2019^[7]). The downward bias of relying on self-identification to measure sexual orientation is compounded by the fact that LGB people are more likely to refrain from disclosing who they are if they are asked to self-identify as homosexual or bisexual, rather than specify the sex of the persons with whom they have sex or to whom they feel sexually attracted (Coffman, Coffman and Ericson, 2016^[32]).¹¹

Box 2.5. Identification of LGBTI+ respondents in our two data sources

The Census Bureau's Household Pulse Survey (HPS) allows for determining the following subgroups:

- **Heterosexual (straight):** respondents who answered "Straight, that is not gay or lesbian" on the sexual orientation question.
- **Non-heterosexual (LGB+)** adults, which includes:
 - **Gay/lesbian:** respondents who answered "Gay or lesbian" on the sexual orientation question.
 - **Bisexual (bi.):** respondents who answered "Bisexual" on the sexual orientation question.
 - **Other non-heterosexual (other non-het.):** respondents who answered "Something else, please specify" on the sexual orientation question. The specific answers to this category (e.g. pansexual, or asexual) are not available in the Household Pulse Survey Public Use File (PUF).
- **Non-cisgender (non-cis.)** adults, which includes:
 - **Transgender (trans.):** respondents who answered "Transgender" on the gender identity question, or respondents who reported a binary gender (male/female) different from their sex assigned at birth (male/female).¹
 - **Other non-cisgender (other non-cis.):** respondents who answered "None of these" on the gender identity question, hence non-binary and agender individuals, among others.

In the HPS, the label "LGBTI+" thus encompasses respondents classified as gay/lesbian, bisexual, other non-heterosexual, transgender, and other non-cisgender.² In contrast, the label "LGBT" specifically focuses on gay/lesbian, bisexual, and transgender respondents. Finally, the label "Cisgender non-heterosexual individuals (cis. LGB+)" refers to non-heterosexual respondents whose gender identity aligns with their sex assigned at birth.

Concerning the US Gallup data, they permit the identification of:

- LGBT respondents for the period 2012-17 and 2020-22: these respondents are those who chose "Yes" in the LGBT identification question in the Gallup US Daily Survey (2012-17) and "Lesbian", "Gay", "Bisexual", and/or "Transgender" in the corresponding question in the Gallup Poll Social Series (2020-22).
- LGBTI+ respondents for the period 2020-22: these respondents are those who answered "Lesbian", "Gay", "Bisexual", "Transgender", and/or "Other LGBT" in the LGBT identification survey question in the Gallup Poll Social Series (2020-22). For the year 2022, this group also encompasses respondents who selected "non-binary," an option introduced that same year to the survey's gender identity question.

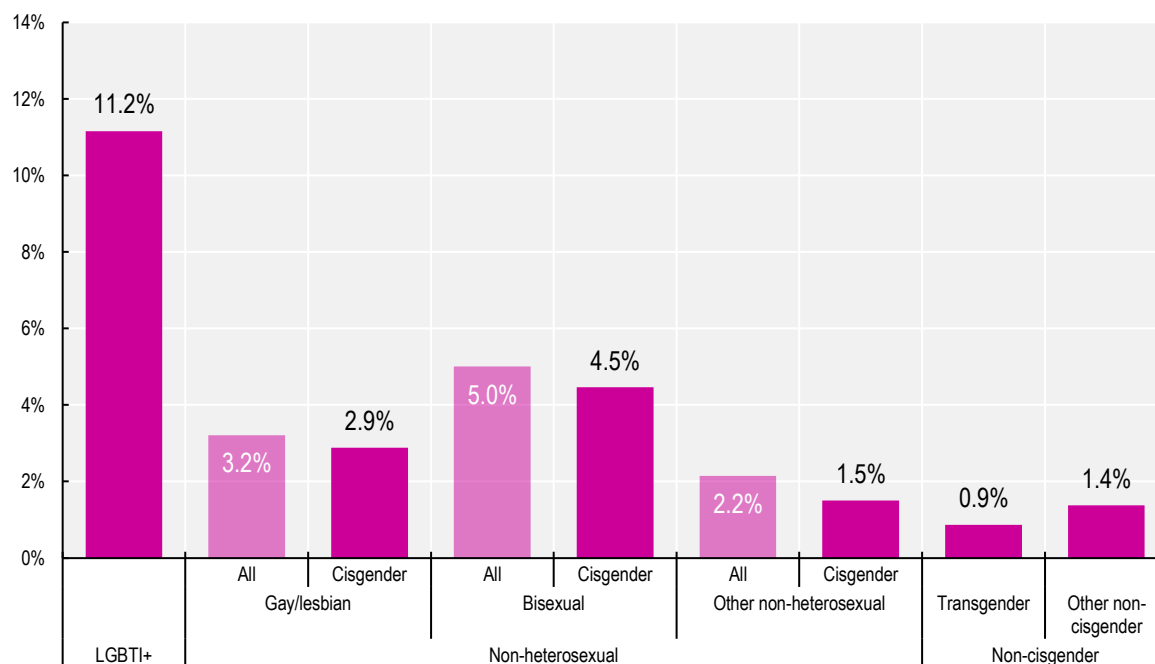
Notes:

1. During the post-data collection process, the missing information for "sex at birth" is imputed by the U.S. Census Bureau, a step essential for the creation of the survey weights. This imputation relies on a standard hot deck imputation method, which addresses missing data by replacing each missing value with an observed response from a geographically similar unit. Due to this imputation, there is a possibility that some respondents might be inaccurately categorised as transgender. Consequently, the analysis in this report is centred on the subset of respondents for whom "sex at birth" is not imputed (National Academies of Sciences, Engineering, and Medicine, 2022^[33]).

2. Although the HPS data do not collect information about people with intersex traits, recent research has found that people with intersex traits report identifying as non-cisgender or non-heterosexual at higher rates than people who are not intersex (Medina and Mahowald, 2021^[11]; Rosenwohl-Mack et al., 2020^[2]; Jones et al., 2016^[3]).

Figure 2.1. In 2023, 11.2% of US adults self-identified as LGBTI+

Share of self-identified LGBTI+ US adults (aged 18-88) in 2023

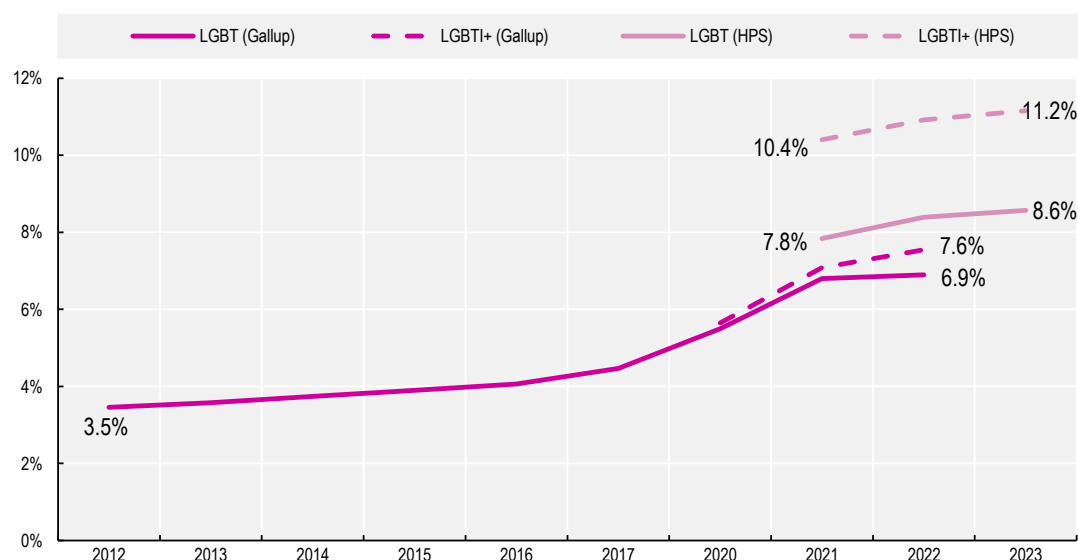


Note: LGBTI+ refers to respondents who self-identify as gay/lesbian, bisexual, transgender, other non-heterosexual, and other non-cisgender (see Box 2.5 for more details). Person-level weights used.

Source: Household Pulse Survey (4 January 2023 – 10 July 2023).

Figure 2.2. Self-reporting of LGBTI+ status is on the rise

Evolution of the share of self-identified LGBTI+ US adults between 2012 and 2023



Note: LGBT refers to respondents who self-identify as gay/lesbian, bisexual, and transgender. LGBTI+ includes those who identify as other non-heterosexual and other non-cisgender (see Box 2.5 for more details). Person-level weights used.

Source: Gallup US Daily Survey (2012-17), Gallup Poll Social Series (2020-22), and Household Pulse Survey (21 July 2021 – 10 July 2023).

2.3. Where do self-identified LGBTI+ US adults live?

The share of self-identified LGBTI+ adults varies substantially across US regions, states, and rural/urban areas. In 2023, according to HPS data, US adults were 24% more likely to self-identify as LGBTI+ in the West, compared to the South (Panel A of Figure 2.3). Spatial disparities also prevail across US states, including within the same region. In the West, for instance, the share of self-identified LGBTI+ US adults varies by a factor of 1.8 across states, from 8.4% in Wyoming to 15.5% in Oregon (Panel B of Figure 2.3). Self-reporting of LGBTI+ status also differs between rural and urban areas, with the share being almost twice as high in the latter than in the former (Panel C of Figure 2.3).

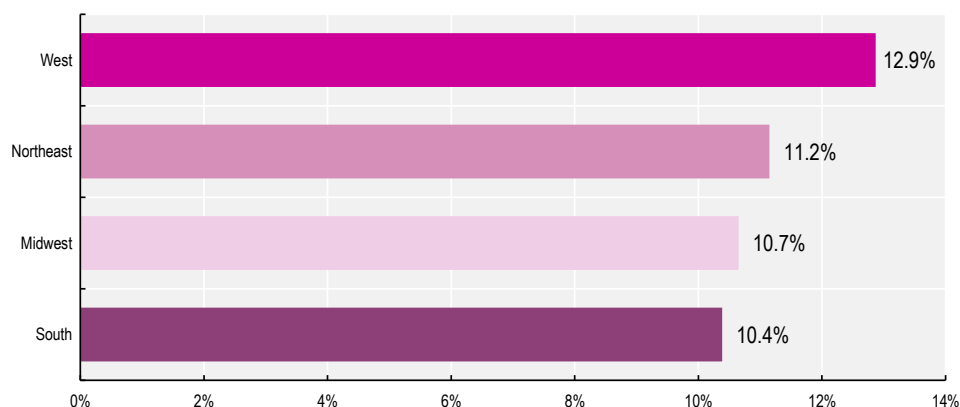
These spatial disparities could be attributed to geographic variations in social acceptance and legal protections of LGBTI+ individuals and reflect two main channels.¹² First, LGBTI+ adults may be inclined to relocate to more accepting neighbourhoods in order to escape discrimination and connect with a larger LGBTI+ community. Second, in such neighbourhoods, LGBTI+ individuals might feel more comfortable disclosing their sexual orientation and/or gender identity, both in surveys and other aspects of their lives.

Greater acceptance of LGBTI+ individuals in urban relative to rural settings has already been documented in OECD countries (OECD, 2019^[7]), including in the United States (Movement Advancement Project, 2019^[34]). Does greater acceptance of non-cisgender and/or non-heterosexual persons also characterise states and regions with a higher share of self-identified LGBTI+ adults? Figure 2.4 affirms this correlation in a state-level analysis, when both social acceptance and legal protections are considered. In 2023, an increase in the share of adults favouring LGBT non-discrimination protections by 10 percentage points is associated with an increase in LGBTI+ self-identification by 2.1 percentage points (Panel A of Figure 2.4). Similarly, states with laws (and policies) that are more supportive of LGBTI+ individuals had a higher share of self-identified LGBTI+ adults (Panel B of Figure 2.4): an increase in the inclusivity of laws and policies towards LGBTI+ individuals by 10 points is associated with an increase in LGBTI+ self-identification by 0.5 percentage points.¹³ These correlations hold true at the regional level. The US South, in addition to having the lowest share of self-identified LGBTI+ adults at 10.4% (compared to 11.7% in other regions), also has the lowest share of adults supporting LGBT non-discrimination protections at 65.2% (relative to 70.8% elsewhere). Furthermore, it ranks the lowest with respect to LGBTI+-inclusive laws and policies, with an average LGBTQ policy tally of 2.9, compared to 28.5 in other regions (see Box 2.6 for more details on the LGBTQ policy tally).

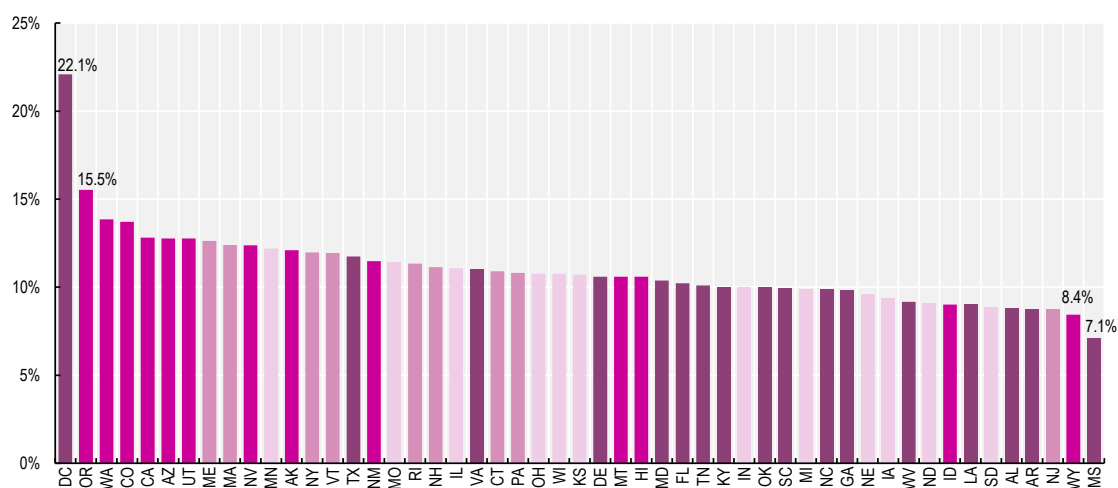
Unfortunately, since the HPS data do not collect information on the area where the respondent grew up or on any mobility patterns, we cannot investigate whether these correlations are at least partly driven by LGBTI+ individuals relocating in more accepting neighbourhoods. Yet, evidence suggests that geographic relocation of LGBTI+ individuals to avoid discrimination is a significant factor. Findings from the US Longitudinal Study of Adolescent to Adult health reveals that non-heterosexual individuals are more geographically mobile in the transition to adulthood than their non-heterosexual counterparts, a result driven by gays and lesbians – in contrast, bisexuals are indistinct from heterosexuals (Levine, 2022^[35]). Using a mix of probability-based sampling and opt-in panels, (Medina and Mahowald, 2023^[36]) found that 34% of LGBTI+ Americans moved away from their families, and 31% relocated from their residences to avoid discrimination due to their LGBTI+ identity. In addition, relying on a representative sample of recent US college graduates, (Folch, 2022^[29]) observed that LGBTI+ college graduates were more likely to earn their degrees from colleges in states different from their childhood homes. These graduates were also more likely to reside in states different from where they grew up, both one and ten years post-graduation, compared to their cisgender straight peers. Finally, a recent survey of 113 LGBTI+ adults raising children in Florida found that over half (56%) of respondents were considering moving out of Florida three months after the “Don’t Say LGBTQ+” bill was passed, and 17% had already taken steps to do so (Goldberg, 2023^[37]).¹⁴

Figure 2.3. The share of US adults who self-identify as LGBTI+ varies by a factor of 1.2, 2.2, and 1.8 across regions, states, and rural/urban areas, respectively

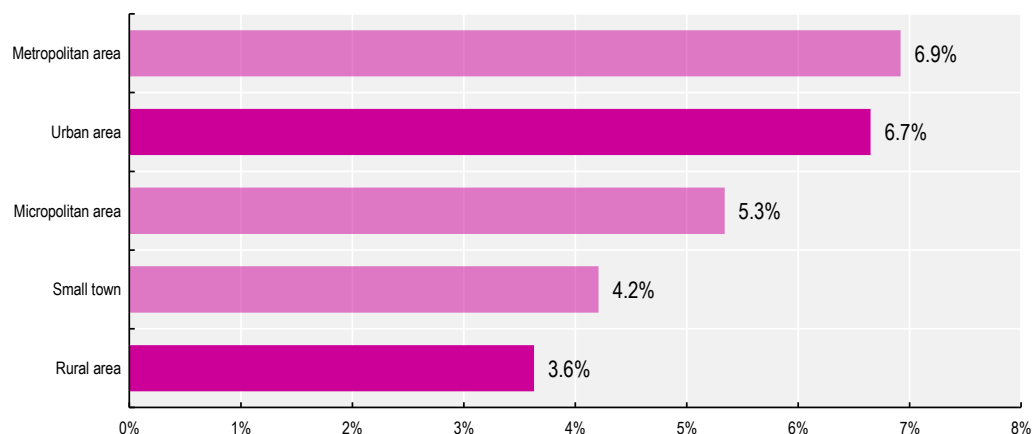
Panel A: Share of self-identified LGBTI+ US adults, by region of residence (2023)



Panel B: Share of self-identified LGBTI+ US adults, by state and D.C. residence (2023)



Panel C: Share of self-identified LGBTI+ Americans, by area of residence (2020-22)

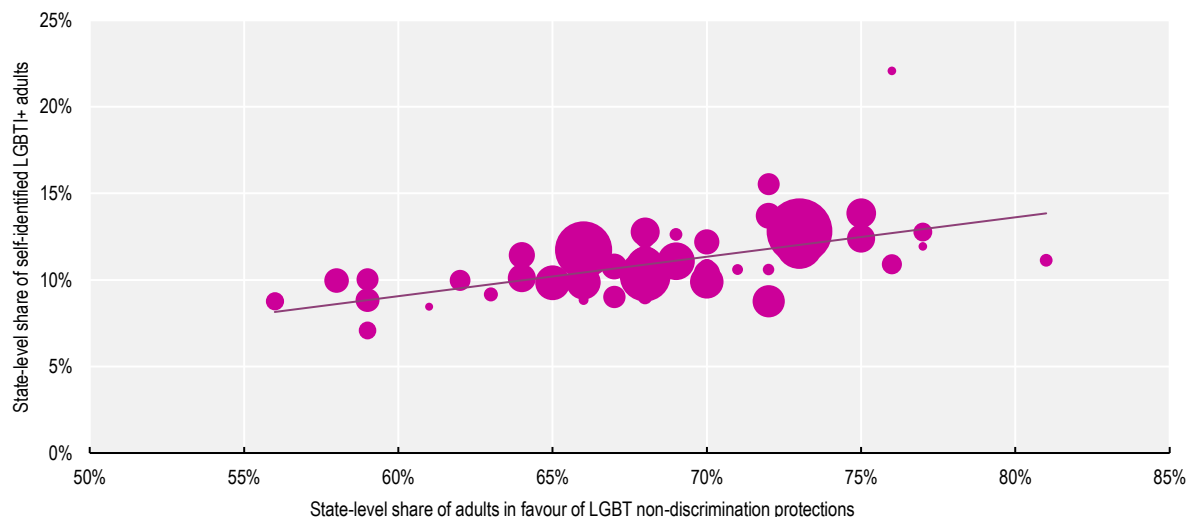


Note: LGBTI+ refers to respondents who self-identify as gay/lesbian, bisexual, transgender, other non-heterosexual, and other non-cisgender (see Box 2.5 for more details). In Panel C, the area of residence is classified using the rural-urban commuting area (RUCA) codes, which uses standard census measures of population density, levels of urbanisation and journey-to-work commuting to characterise all US census tracts and zip codes with respect to their rural/urban status. Person-level weights used.

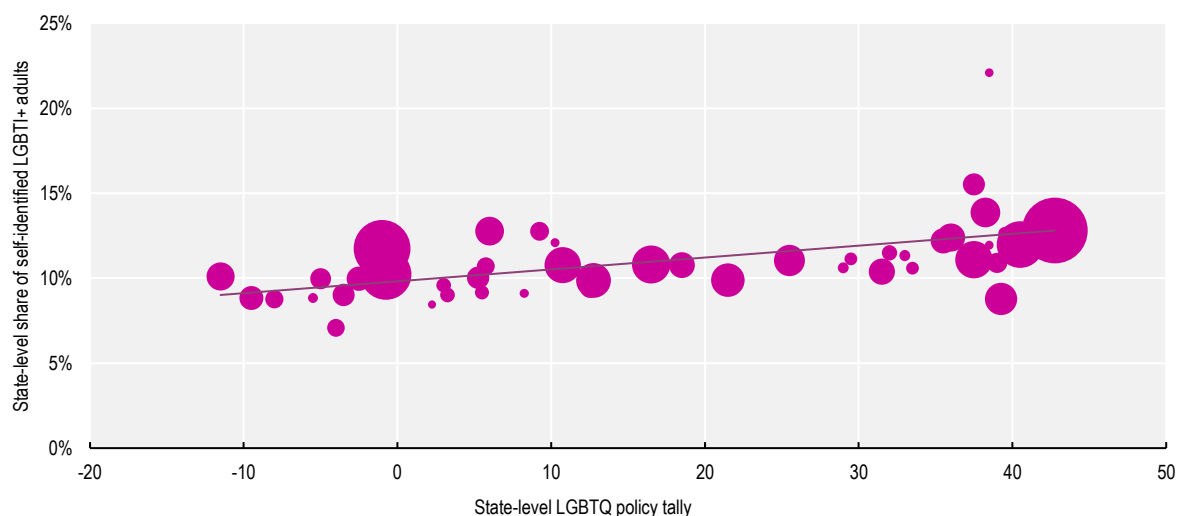
Source: Household Pulse Survey (4 January 2023 – 10 July 2023) for Panel A and Panel B; Gallup Poll Social Series (2020-22) for Panel C.

Figure 2.4. Self-reporting of LGBTI+ status is higher in US states that are more inclusive of LGBTI+ individuals

Panel A: Relationship between the share of self-identified LGBTI+ adults in 2023 and the share of adults supporting LGBT non-discrimination protections in 2018 (analysis at the US state level, including D.C.)



Panel B: Relationship between the share of self-identified LGBTI+ adults and the inclusivity of laws and policies towards LGBTI+ individuals in 2023 (analysis at the US state level, including D.C.)



Note: LGBTI+ refers to respondents who self-identify as gay/lesbian, bisexual, transgender, other non-heterosexual, and other non-cisgender (see Box 2.5 for more details). In Panel A, the share of adults supporting LGBT non-discrimination protections is derived from a 2018 telephone survey (Jones et al., 2019^[9]). This survey, based on a probability-driven, nationally representative sample of US adults, included a question asking respondents if they favoured laws protecting gay, lesbian, bisexual, and transgender individuals from discrimination in employment, public accommodations, and housing. In Panel B, the inclusivity of laws and policies towards LGBTI+ equality is measured based on the LGBTQ policy tally (see Box 2.6 for more details). Person-level weights used.

Source: Household Pulse Survey (4 January 2023 – 10 July 2023) for Panel A and Panel B; (Jones et al., 2019^[9]) for Panel A; Movement Advancement Project for Panel B.

Box 2.6. The LGBTQ “policy tally” by the Movement Advancement Project

The Movement Advancement Project (MAP) is an independent, non-profit think tank founded in 2006. MAP has tracked laws and policies supportive of LGBTI+ individuals in the United States for more than a decade.

An LGBTQ “policy tally” rates LGBT laws and policies within each state, focusing on the following areas: Relationship and Parental Recognition, Non-discrimination, Religious Exemptions, LGBTQ Youth, Health Care, Criminal Justice, and Identity Documents. Across these seven categories, each law or policy earns a score that is either positive (when the law or policy is protective of LGBTI+ individuals) or negative (when the law or policy is harmful to LGBTI+ individuals). The LGBTQ policy tally is computed as the sum of these scores and used to classify states into five groups (as of July 2023): negative (the sum of the scores is lower than 0), low (the sum of the scores is between 0 and 11), fair (the sum of the scores is between 11 and 21.5), medium (the sum of the scores is between 21.5 and 32.5), or high (the sum of the scores is greater than 32.5).

The policy tallies are updated in real time, as soon as a bill is signed into law, or a policy becomes effective. This report uses state policy tallies as of 10 July 2023. These tallies reflect significant differences in legal protections across US states. For instance, only 15 states (and D.C.) had a “high” LGBTQ policy tally (indicative of significant progress toward LGBTQ equality), while 25 states fell into either “low” or “negative” LGBTQ policy tally.

Source: (Movement Advancement Project, 2020^[11]) and (Movement Advancement Project, 2023^[38])

2.4. How do self-identified LGBTI+ US adults compare based on key demographic factors?

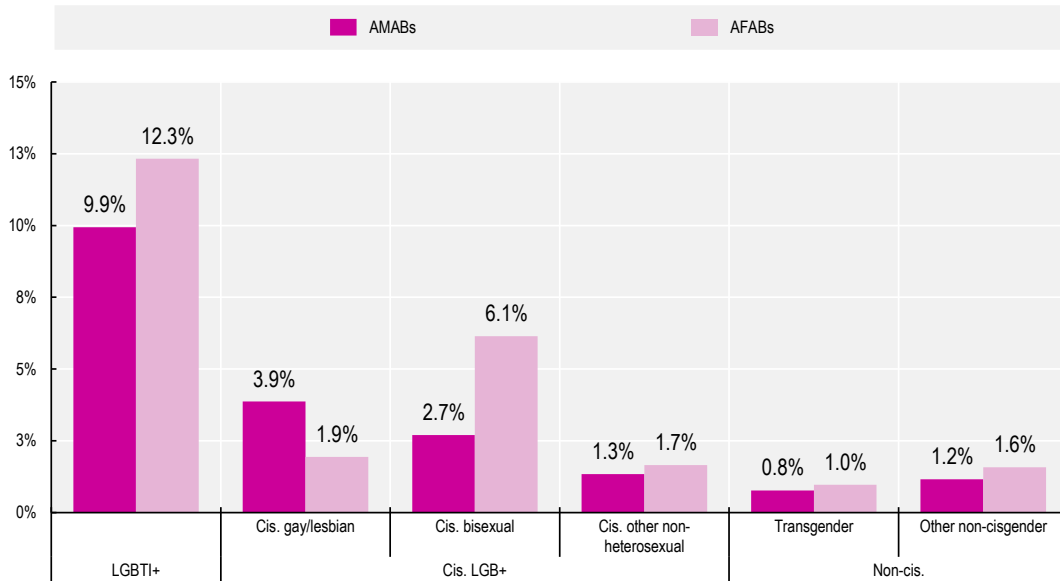
Factors such as sex assigned at birth, age, race/ethnicity, family structure, and educational attainment significantly influence economic and health outcomes, which, in turn, shape individual life trajectories. This section examines how self-identified LGBTI+ US adults fare in relation to these key demographic determinants.

2.4.1. Sex assigned at birth

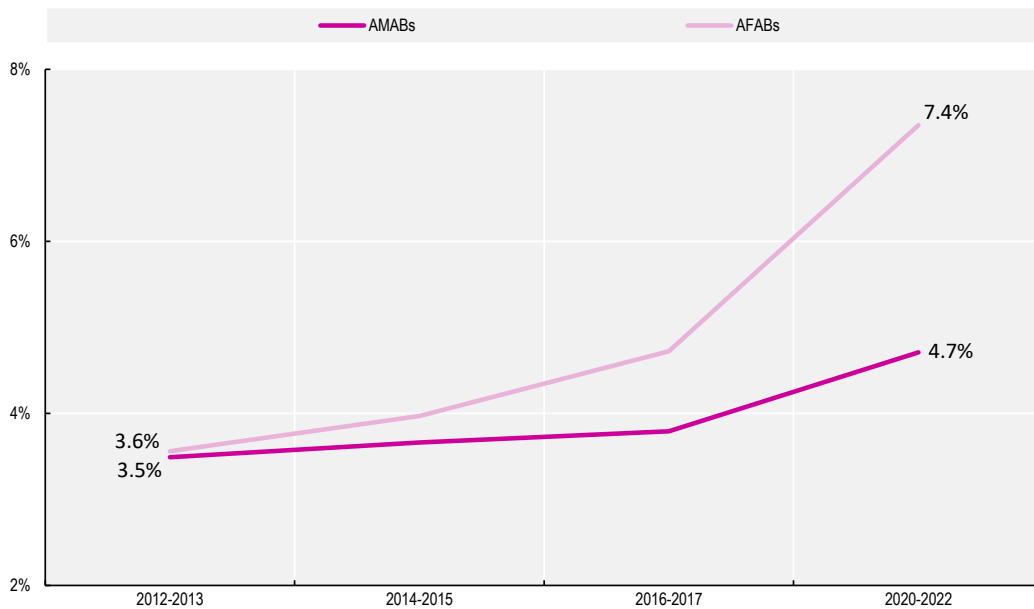
Individuals assigned female at birth (AFAB) are more likely to identify as LGBTI+ than those assigned male at birth (AMAB). In 2023, 12.3% of AFABs described themselves as non-heterosexual and/or non-cisgender, compared to 9.9% of AMABs (Panel A of Figure 2.5). This difference is primarily driven by AFAB individuals being more than twice as likely as their AMAB counterparts to identify as cisgender bisexual. This higher representation of AFABs is also observed, albeit to a lesser extent, among cisgender other non-heterosexual, and among non-cisgender individuals. A significant exception concerns the group of cisgender homosexuals where the share of AFABs is half that of AMABs (1.9% vs. 3.9%). This pattern of greater prevalence of AFABs among self-identified LGBTI+ US adults has gotten more pronounced in the past decade. In the early 2010s, they were only slightly more likely than their AMAB counterparts to report being LGBT. However, the gap has kept widening ever since (Panel B of Figure 2.5), a trend also documented in alternative datasets (National Academies of Sciences, Engineering, and Medicine, 2020^[39]).¹⁵

Figure 2.5. Individuals assigned female at birth (AFAB) are more likely to identify as LGBTI+ than those assigned male at birth (AMAB), a gap that has kept widening in the past decade

Panel A: Share of self-identified LGBTI+ US adults, by sex assigned at birth (2023)



Panel B: Evolution of the share of self-identified LGBT US adults, by sex assigned at birth (2012-22)



Note: LGBT refers to respondents who self-identify as gay/lesbian, bisexual, and transgender. LGBTI+ includes those who identify as other non-heterosexual and other non-cisgender (see Box 2.5 for more details). In Panel A, sex assigned at birth is determined based on the respondent's direct report of their sex at birth. In Panel B, for the period 2012 – August 2014, the respondent's sex at birth is inferred based on their voice and other cues detected during the interview. For the period September 2014 – 2022, it is determined by the respondent's response to the following question: "I am required to ask, are you male or female?". In Panel B, there is thus an inherent risk of misclassification: transgender men might be considered AMAB instead of AFAB, and *vice versa* for transgender women. This potential oversight could lead to an underestimation of AFABs' likelihood to identify as LGBTI+. However, the impact is likely modest due to the relatively low proportion of self-identified transgender respondents in the US population (0.9% in 2023 based on the HPS). Person-level weights used.

Source: Household Pulse Survey (4 January 2023 – 10 July 2023) for Panel A; Gallup US Daily Survey (2012-17) and Gallup Poll Social Series (2020-22) for Panel B.

Differences in LGBTI+ self-identification between AFAB and AMAB individuals likely reflect the well-documented greater social acceptance of women expressing same-sex attraction or relationships compared to men, leading to potentially more AFABs feeling comfortable identifying as cisgender LGB+ (Herek, 2002^[40]; Bettinsoli, Suppes and Napier, 2020^[41]). Additionally, females exhibit more positive attitudes towards LGBTI+ individuals, as detailed in Chapter 3. As for the widening of this gap over time, it may reflect strides made in gender equality, hence a departure from traditional roles and expectations placed upon women, including the pressure to adhere to heteronormativity and to be feminine. There is growing empirical evidence supporting enhanced fluidity in sexual orientation among women (or those AFABs). For example, from 2011 to 2019, a rising percentage of college-aged women in the United States moved away from exclusive heterosexuality, contrasting with a relatively unchanged percentage of men reporting sexual attraction and behaviour exclusively towards women (Morgan and van Dulmen, 2021^[42]).

2.4.2. Age

Self-reporting of LGBTI+ status decreases with age, for both assigned males at birth and assigned females at birth (Panel A of Figure 2.6). In 2023, nearly one in four young adults (18-34) identified as non-heterosexual and/or non-cisgender, compared to nearly one in ten among prime age adults (35-54) and one in 20 among mature adults (55+). This pattern prevails for all LGBTI+ subgroups, but especially so among cisgender bisexual individuals where the share of young adults is more than three times as high than the share of prime age adults, and more than 10 times as high as the share of mature adults (Panel B of Figure 2.6). The decrease in LGBTI+ self-identification with age is not specific to the United States. According to the 2nd edition of the biennial LGBT+ Pride Global Survey conducted by IPSOS in 2023 across 30 countries, nearly one in four Gen Zers (18-25) self-identify as LGBTI+, as opposed to less than one in 20 among Baby Boomers (60+) (IPSOS, 2023^[25]).

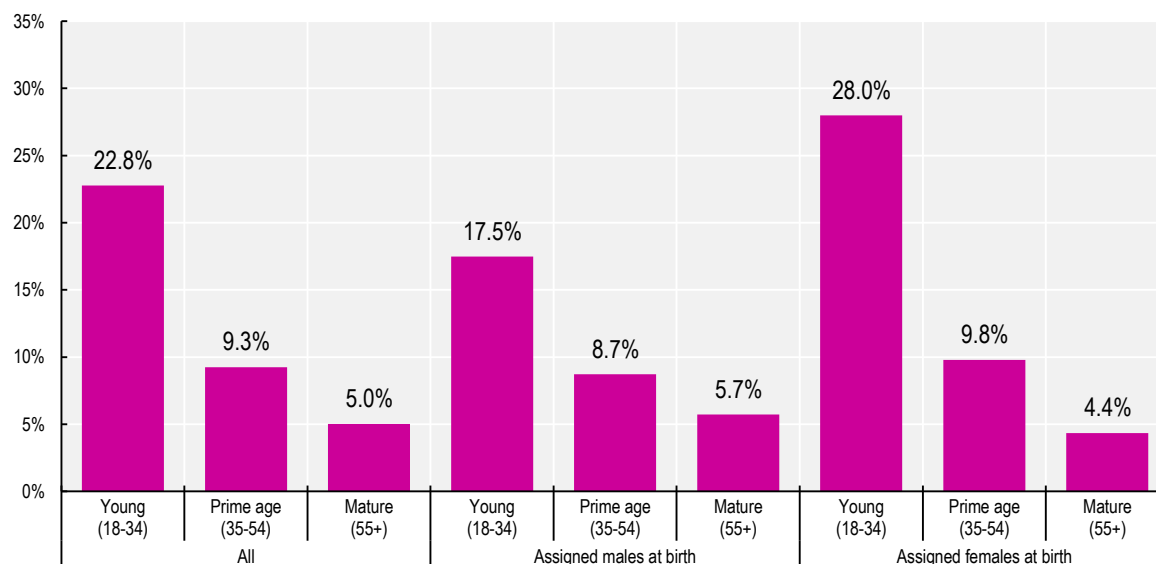
The well-documented increase in the number of individuals self-identifying as LGBT over the past decade is driven by young people (Panel A of Figure 2.7). The share of 18-34 year-olds reporting a non-heterosexual and/or non-cisgender identity more than doubled between 2012-13 and 2020-22. While the percentage of self-identified LGBT adults also rose among people aged 35-54 during the same period, its growth rate (50%) is much more modest. By contrast, the self-reporting of LGBT status among individuals above age 55 remained virtually unchanged.¹⁶

The higher and increasing share of self-identified LGBTI+ among young adults could be attributed to an age effect or to a cohort effect. An age effect posits that younger individuals are in a transitory phase of exploration regarding their sexual orientation and gender identity. This means that as today's young adults grow older, their propensity to report an LGBTI+ status might decrease. Although the majority of Americans report consistency in their sexual orientation self-identification, a minority do change their identity label over time (Hall, Dawes and Plocek, 2021^[43]). For example, some individuals who initially identify as bisexual may later identify as mostly heterosexual, or exclusively heterosexual. On the other hand, a cohort effect suggests that younger LGBTI+ generations, having been raised in more accepting environments and exposed to increased LGBTI+ visibility online and in the media, feel more comfortable with identifying as LGBTI+ throughout their lives. The increased representation of LGBTI+ individuals in mass media, coupled with social networks that connect LGBTI+ youth to their peers, has turned out instrumental in this emancipation (Ayoub and Garretson, 2017^[44]). In line with this trend, younger generations exhibit more positive attitudes towards LGBTI+ individuals, as detailed in Chapter 3. Additionally, LGBTI+ youth choose to "come out of the closet" earlier in life compared to their predecessors from older generations (Human Rights Campaign Foundation, 2023^[45]; National Academies of Sciences, Engineering, and Medicine, 2020^[39]). If the cohort effect is at play, young adults identifying as LGBTI+ today are likely to continue doing so as they age. Moreover, upcoming generations may identify at similar or even increased rates, assuming social and legal acceptance remains or further improves.

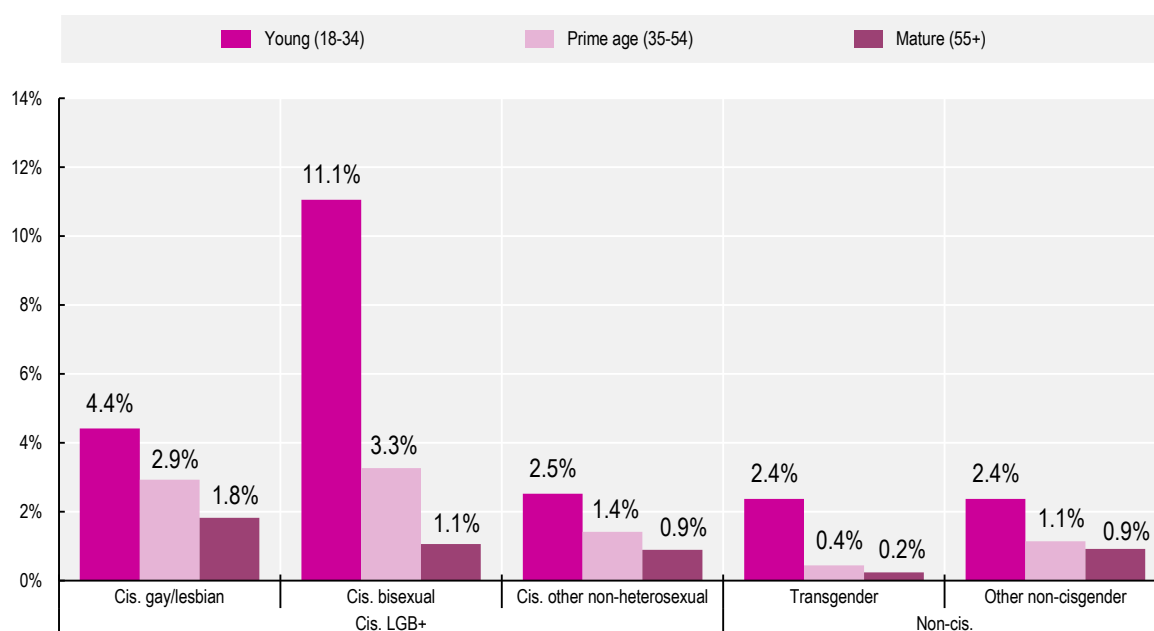
Although additional research is needed for a more in-depth understanding of the underlying mechanisms, current evidence tilts towards a cohort effect (as opposed to an age effect) when it comes to LGBT identification. Specifically, Panel B of Figure 2.7 shows that among those who were aged 18 to 34 in 2012, the percentage identifying as LGBT has grown in subsequent years rather than declined. This trend suggests that the share of LGBTI+ adults will continue to grow in the future. Concerning the decreasing trend in LGBT identification for mature adults, it could be consistent with shorter life expectancy for LGBTI+ individuals.

Figure 2.6. Self-identification as LGBTI+ declines with age, especially among cisgender bisexuals

Panel A: Share of self-identified LGBTI+ US adults, by age category and sex assigned at birth (2023)



Panel B: Distribution of self-identified LGBTI+ US adults by LGBTI+ subgroup and age category (2023)

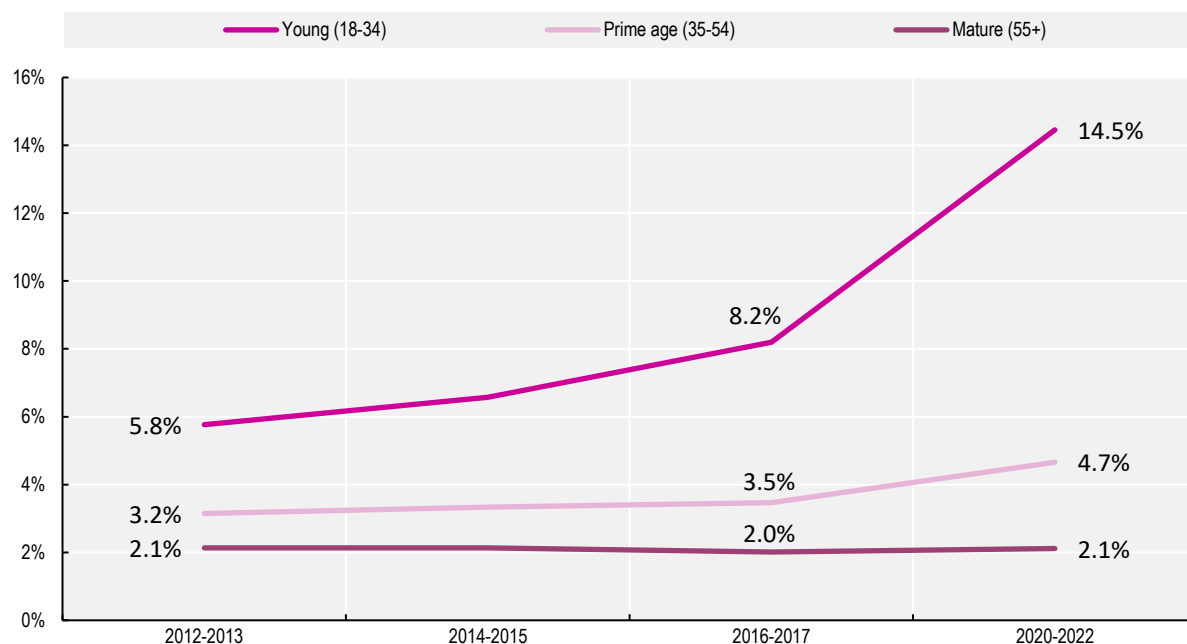


Note: LGBTI+ refers to respondents who self-identify as gay/lesbian, bisexual, transgender, other non-heterosexual, and other non-cisgender (see Box 2.5 for more details). Person-level weights used.

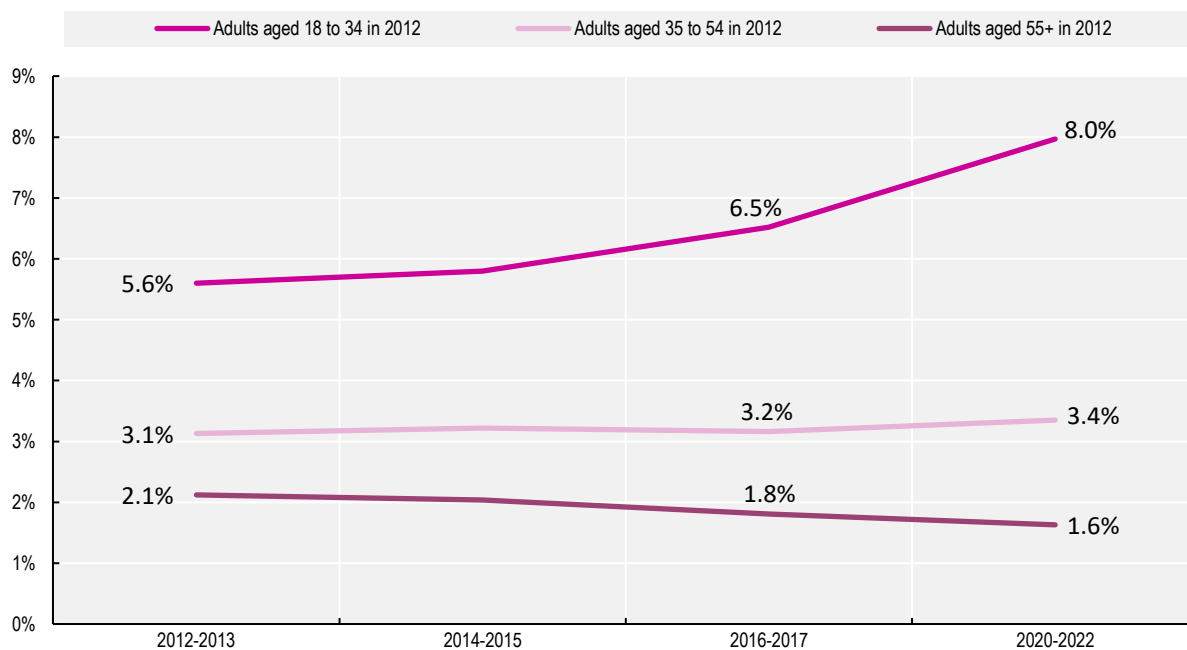
Source: Household Pulse Survey (4 January 2023 – 10 July 2023).

Figure 2.7. The increase in LGBT self-identification in the past decade is driven by young people, through a cohort effect

Panel A: Evolution of the share of self-identified LGBT US adults, by age category (2012-22)



Panel B: Evolution of the share of self-identified LGBT US adults, by age category in 2012 (2012-22)



Note: LGBT refers to respondents who self-identify as gay/lesbian, bisexual, and transgender (see Box 2.5 for more details). Person-level weights used.

Source: Gallup US Daily Survey (2012-17) and Gallup Poll Social Series (2020-22).

2.4.3. Race and ethnicity

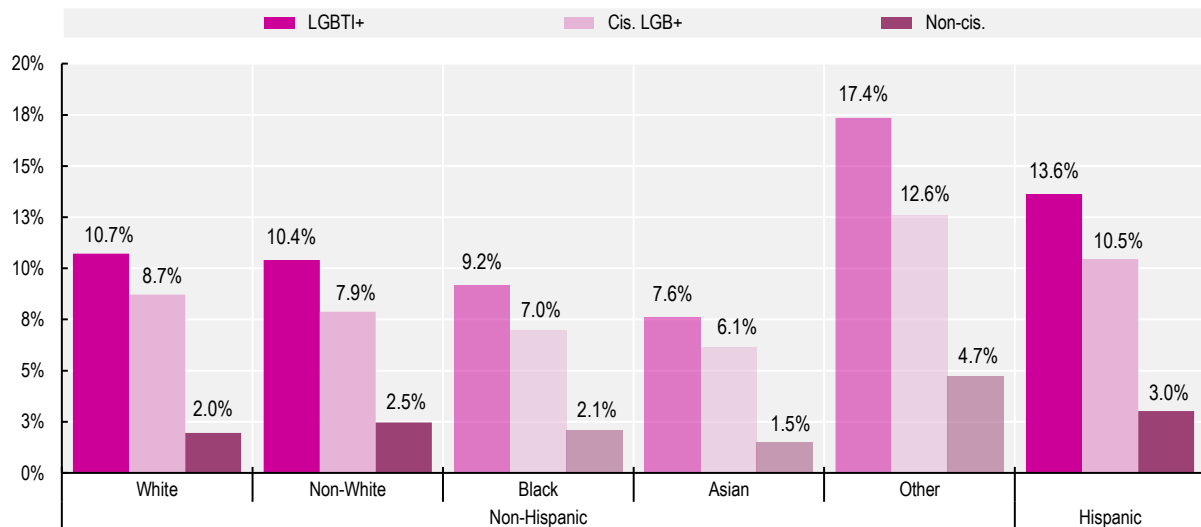
LGBTI+ self-identification among US adults varies widely by race and ethnicity. Non-Hispanic White Americans are slightly more likely than their non-White counterparts to report being LGBTI+. However, this observation conceals significant disparities. Compared to Asians and Blacks, the likelihood of self-identifying as LGBTI+ is 41% and 17% higher among Whites, respectively. Yet, it is almost 40% lower compared to individuals who identify with other racial categories, including American Indian and Alaska Native, Native Hawaiian and Other Pacific Islander, and mixed-race individuals (Panel A of Figure 2.8). This finding remains consistent even when accounting for differences across racial groups in terms of geographic location, sex assigned at birth, age, marital and parental status, number of adults in the household, and educational attainment (Annex Figure 2.A.1). However, variations in self-reporting of LGBTI+ status by ethnicity don't withstand this check: the heightened propensity of Hispanic individuals to identify as LGBTI+ that is evident in Panel A of Figure 2.8 is predominantly due to their young age (Annex Figure 2.A.1). In line with the overlap between Hispanic ethnicity and age, the most significant increase in the proportion of LGBT self-identifications between 2012 and 2022 occurred within this group. Meanwhile, the growth in LGBT self-identification was more modest among the non-Hispanic group (Panel B of Figure 2.8).

Two factors can account for the lower identification as LGBTI+ among non-Hispanic Blacks and non-Hispanic Asians. Firstly, more negative attitudes towards LGBTI+ individuals often prevail within these populations, as detailed in Chapter 3. Such attitudes might discourage individuals with non-heterosexual and/or non-cisgender identities from coming out due to fear of alienation from their community. Secondly, a “double minority stress” effect could also be at play (Meyer, 2010^[46]; Meyer, 2003^[47]): LGBTI+ individuals within a racial/ethnic minority group may face compounded stigma from both racism and LGBTI+-phobia, making some less inclined to openly identify as LGBTI+ (Groß et al., 2006^[48]; Rosario, Schrimshaw and Hunter, 2004^[49]; Maguen et al., 2002^[50]).

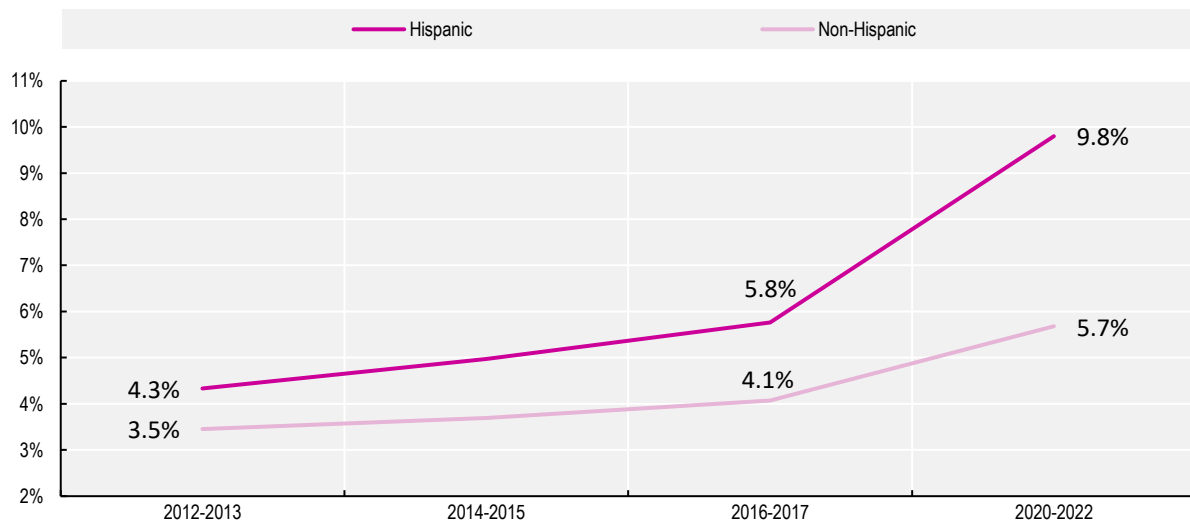
Looking further into disparities within non-Hispanic racial minorities, the fact that individuals identifying with other racial categories are more likely, rather than less, to self-identify as cisgender LGB+ and/or non-cisgender compared to non-Hispanic Whites might be linked to the composition of this group. For one, the American Indian and Alaska Native categories encompass tribes that historically recognised more than two genders. These tribes often attributed special roles or powers to “Two-Spirit” individuals, indicating a long-standing acceptance of diverse sexual orientations and gender identities (Jacobs, 1997^[51]). Similarly, Pacific Island cultures traditionally valued identities beyond the binary concept of gender. For instance, “mahu” in Hawaii or “fa’afafine” in Samoa refer to individuals who adopt a mix of male and female roles (Besnier and Alexeyeff, 2014^[52]). Finally, higher rates of LGBTI+ self-identification may be expected among mixed-race individuals. Their lived experiences of navigating multiple identities could cultivate a more fluid understanding of themselves, extending to their perceptions of sexual and gender identities.

Figure 2.8. Self-reporting of LGBTI+ status is the highest and has increased the most among Hispanic Americans

Panel A: Share of self-identified LGBTI+ US adults, by race and ethnicity (2023)



Panel B: Evolution of the share of self-identified LGBT US adults, by ethnicity (2012-22)



Note: LGBT refers to respondents who self-identify as gay/lesbian, bisexual, and transgender. LGBTI+ includes those who identify as other non-heterosexual and other non-cisgender (see Box 2.5 for more details). A reliable analysis of the evolution of the share of self-identified LGBT US adults by race (on top of ethnicity) was not possible in Panel B, due to small sample sizes attached to the following subgroups: Non-Hispanic Whites, Non-Hispanic Blacks, Non-Hispanic Asians and Non-Hispanic other racial categories. Person-level weights used.

Source: Household Pulse Survey (4 January 2023 – 10 July 2023) for Panel A; Gallup US Daily Survey (2012-17) and Gallup Poll Social Series (2020-22) for Panel B.

2.4.4. Family structure

After accounting for demographic differences between LGBTI+ and non-LGBTI+ individuals, US adults who self-identify as LGBTI+ are approximately 10 percentage points less likely to be partnered or married compared to their cisgender straight counterparts, as shown in Panel A of Figure 2.9. Additionally, they are nearly 20 percentage points less likely to be married (Panel A of Figure 2.9) and 13 percentage points less likely to be living in a household with at least one child under 18 (Panel B of Figure 2.9). These penalties are consistent across all LGBTI+ subgroups.¹⁷ Furthermore, when the sample is divided by sex assigned at birth, an additional insight emerges. Among all the categorised groups, cisgender gay males face the most significant disparity in family structure compared to their straight counterparts: their likelihood of being married is lower by 36 percentage points, while their probability of living in a household with children lies 26 percentage points below (Annex Figure 2.A.2). While the gap in marriage rates between LGBT and non-LGBT individuals has slightly narrowed over the last decade, the difference in the likelihood of being a parent to a child under age 18 has remained largely stable (Figure 2.10).

Lower partnership, marriage, and parenting rates for LGBTI+ individuals have consistently been found in US nationally representative surveys (Badgett, Carpenter and Sansone, 2021^[14]; National Academies of Sciences, Engineering, and Medicine, 2020^[39]). Historically low social and legal acceptance of same-sex couples is a major driver behind LGBTI+ individuals' greater difficulty in forming and maintaining relationships, compared to cisgender straight individuals. Fear of stigmatisation has long prevented LGBTI+ individuals from entering or publicly acknowledging relationships, as has a lack of legal recognition of same-sex partnerships and marriages. Improved attitudes towards LGBTI+ individuals and the legalisation of same-sex marriage in 2015 by the U.S. Supreme Court have likely contributed to the trend highlighted in Figure 2.10, and notably the increased number of reported same-sex cohabitating couples and the growing rate of same-sex marriage (Scherer, 2022^[53]). Similarly, the spread of the internet and dating apps has probably helped offset the traditionally lower probability of LGB+ individuals finding a romantic partner (National Academies of Sciences, Engineering, and Medicine, 2020^[39]).

Despite recent increases in partnership and marriage rates among LGBTI+ adults, many still have low expectations about becoming parents (National Academies of Sciences, Engineering, and Medicine, 2020^[39]; Coffman, Coffman and Marzilli, 2024^[54]). Naturally, same-sex couples encounter biological barriers to procreation. This is why, even today, many children raised by same-sex couples were initially born to different-sex parents, one of whom later entered a same-sex relationship (Gates, 2015^[55]; Goldberg and Conron, 2018^[56]).

LGBTI+ individuals wishing to become parents thus must turn to adoption, assisted reproductive technologies, or surrogacy. Since 2015, same-sex partners in the United States have been legally entitled to adopt (OECD, 2020^[5]). Furthermore, access to medically assisted insemination (using donor sperm) and *in vitro* fertilisation is not restricted for same-sex couples. Notably, since 2017, the United States has ensured equal treatment for “automatic co-parent recognition”. This means that when a child is born via assisted reproductive technologies, the same-sex partner of the birth parent is automatically recognised as the second legal parent, mirroring the rights of opposite-sex couples. Yet, surrogacy laws in the United States vary by state, creating challenges for male same-sex couples in regions where gestational surrogacy agreements are illegal or limited to opposite-sex couples, as seen in Florida. Such legal disparities (together with the financial challenges emphasised below) can help explain not only the reduced parenting rate among gay men but also their lower likelihood of being in partnerships or marriages compared to other LGBTI+ subgroups, as feedback effects may be in play: for many individuals, the choice to form a partnership or marry is intrinsically linked to the aspiration or anticipation of raising children.

Even where adoption, assisted reproductive technologies, or surrogacy are legally accessible, multiple barriers can still stand in the way. Potential adoptive LGBTI+ parents might confront prejudice within the adoption system. Additionally, methods like assisted reproductive technologies or surrogacy often come with high costs, rendering them unattainable for those without the necessary financial resources,

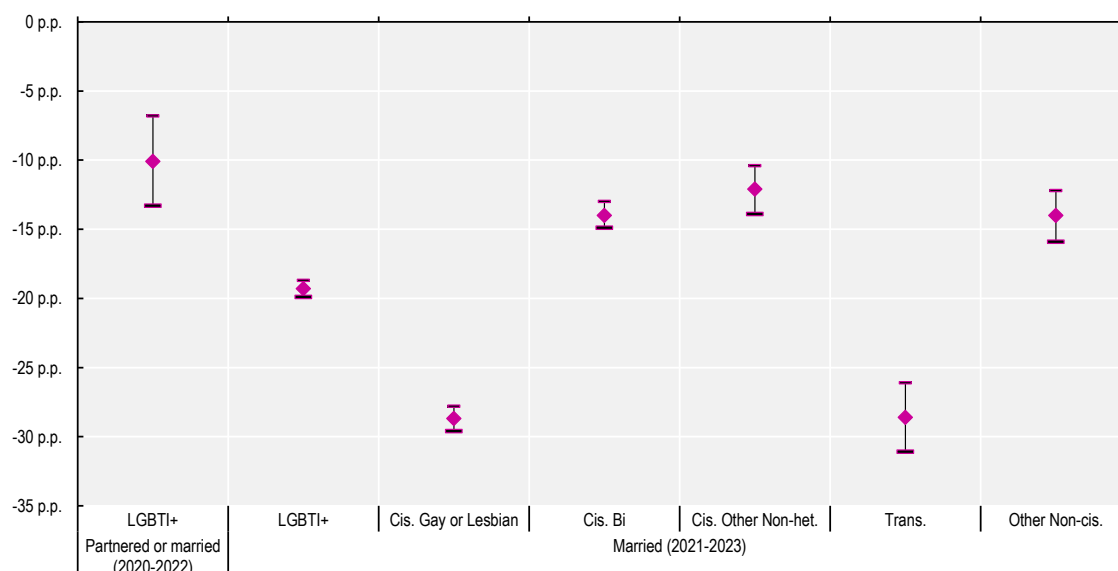
particularly as these procedures aren't usually covered for same-sex couples by health insurers and employers (Medina and Mahowald, 2023^[36]).¹⁸ It is then not surprising that same-sex couples are more likely to adopt or foster children (Taylor, 2020^[57]).

Transgender individuals, particularly those who undergo gender transitioning, may face particularly pronounced barriers to procreation. Hormone replacement therapy and specific surgical procedures can significantly impact fertility. For instance, transgender men undergoing hysterectomies (removing uterus) or oophorectomies (removing ovaries), and transgender women opting for orchiectomies (removal of the testicles) might face reproductive challenges post-surgery. While fertility preservation, such as storing eggs or sperm before medical transition, can be an option for some, it's not always feasible. The prohibitive financial costs and a lack of awareness or sensitivity from some medical professionals concerning fertility options for transgender individuals can further limit access to these reproductive choices.

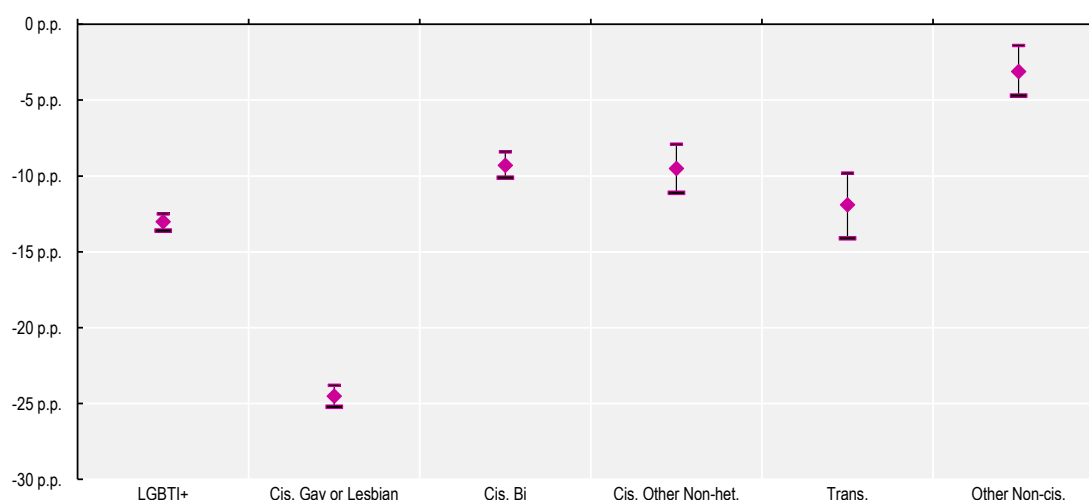
In summary, these mechanisms shed light on why the disparities in the likelihood of LGBTI+ individuals becoming parents to a child under age 18 have largely persisted, as depicted in Figure 2.10. Two opposing forces are in play. On the one hand, with the rise in social and legal acceptance, more LGBTI+ adults are "coming out". This means fewer are becoming parents within the confines of a heterosexual relationship (Tornello and Patterson, 2015^[58]; Gates and Brown, 2015^[59]). On the other hand, there is growing reliance on adoption, assisted reproductive technology, and surrogacy among LGBTI+ individuals. Yet, this phenomenon is tempered by limited access to these methods, due to a combination of legal, financial, and discriminatory barriers (Farr, Vazquez and Patterson, 2020^[60]).

Figure 2.9. After demographic adjustments, LGBTI+ adults are less likely to be partnered or married, and to live in a household with children

Panel A: Percentage point difference in the probability of being partnered or married (using cisgender straight individuals as the reference category), by sexual orientation and gender identity (2020-23)



Panel B: Percentage point difference in the probability of living in a household with at least one child below 18 (using cisgender straight individuals as the reference category), by sexual orientation and gender identity (2021-23).

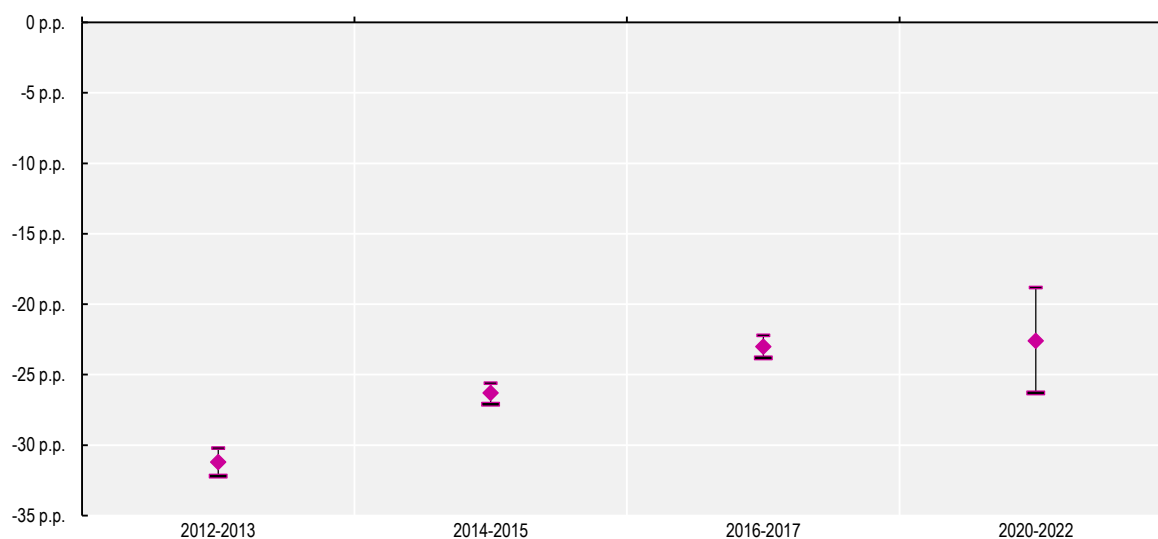


Note: LGBTI+ refers to respondents who self-identify as gay/lesbian, bisexual, transgender, other non-heterosexual, and other non-cisgender (see Box 2.5 for more details). The percentage point differences are estimated from a Logistic regression, controlling for sex assigned at birth, age groups (18-24, 25-29, 30-34, ..., 80-84, 85-88), race and ethnicity, educational attainment, living in one of the 15 largest metropolitan statistical areas (metro areas) – only in the HPS data –, as well as state and survey wave fixed effects. The percentage point differences are estimated at the average of the covariates. The error bars depict 95% confidence intervals. This means that we can be 95% confident that the true value lies within the range of the error bar. In other words, if the error bar crosses the x-axis, then the percentage point difference cannot be deemed as statistically different from 0 (at least not at this 95% confidence level). Standard errors robust to heteroscedasticity. Person-level weights used.

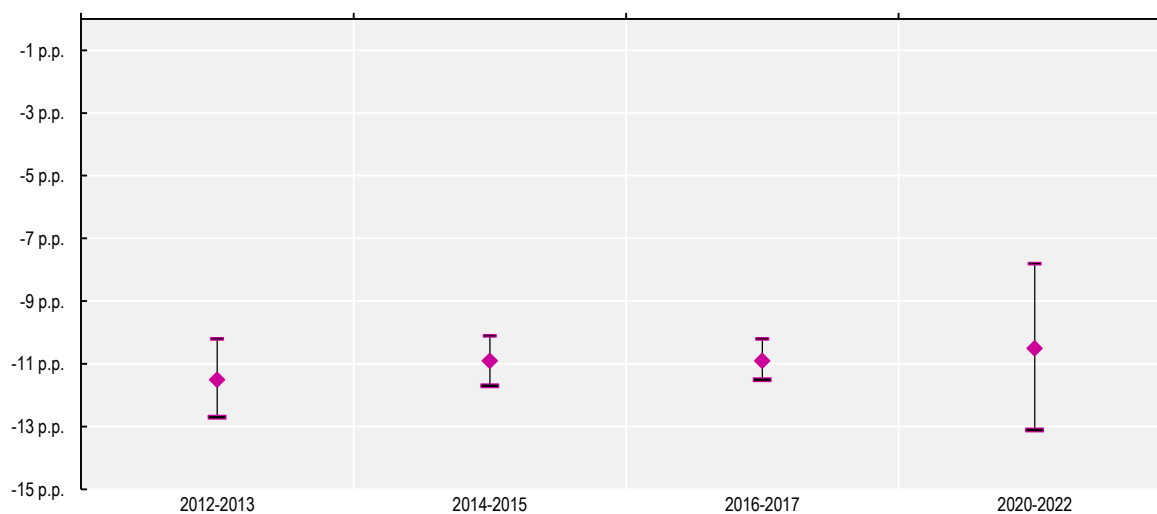
Source: Gallup Poll Social Series (2020-22) and Household Pulse Survey (21 July 2021 – 10 July 2023) for Panel A; Household Pulse Survey (21 July 2021 – 10 July 2023) for Panel B.

Figure 2.10. In the past decade, marriage rates of LGBT individuals have modestly converged with those of their non-LGBT peers, but parenting rates have remained virtually unchanged

Panel A: Over-time change in probability of being married: Comparison with non-LGBT individuals (2012-22)



Panel B: Over-time change in probability of having at least one child under 18: Comparison with non-LGBT individuals (2012-22)



Note: LGBT refers to respondents who self-identify as gay/lesbian, bisexual, and transgender (see Box 2.5 for more details). The percentage point differences are estimated from a Logistic regression, controlling for sex assigned at birth, age groups (18-24, 25-29, 30-34, ..., 80-84, 85-88), race and ethnicity, educational attainment, as well as state and year and month fixed effects. The percentage point differences are estimated at the average of the covariates. The error bars depict 95% confidence intervals. This means that we can be 95% confident that the true value lies within the range of the error bar. In other words, if the error bar crosses the x-axis, then the percentage point difference cannot be deemed as statistically different from 0 (at least not at this 95% confidence level). Standard errors robust to heteroscedasticity. Person-level weights used.

Source: Gallup US Daily Survey (2012-17) and Gallup Poll Social Series (2020-22).

2.4.5. Educational attainment

The vast majority of studies using nationally representative US data have found that LGB adults are better educated than their straight counterparts (Badgett, Carpenter and Sansone, 2021^[14]; Sansone and Carpenter, 2020^[61]; Gates, 2014^[62]; Black, Sanders and Taylor, 2007^[63]). In contrast, non-cisgender individuals show lower educational levels than cisgender individuals (Badgett, Carpenter and Sansone, 2021^[14]; Wilkinson, Shifrer and Pearson, 2021^[64]; Carpenter, Eppink and Gonzalez, 2020^[65]; Wilkinson, Pearson and Liu, 2018^[66]). However, studies focused on newer data, representing younger cohorts of LGB adults, yield a different conclusion. They indicate an educational attainment *penalty* for LGB adults, with young gay men being the only subgroup to demonstrate an educational advantage (Mittleman, 2022^[67]; Sansone, 2019^[68]; Pearson and Wilkinson, 2017^[69]).

After demographic adjustments, the Household Pulse Survey confirms these findings. The educational penalty for non-cisgender individuals remains consistent across all age groups: non-cisgender adults are significantly less likely to graduate from high school, compared to their cisgender straight counterparts (Figure 2.11). However, the nature of the educational attainment gap between cisgender LGB+ and their straight peers changes with age (Figure 2.11): there is a significant college attainment advantage for mature and prime age cisgender LGB+ adults compared to their straight counterparts, but this shifts to a disadvantage for younger LGB+ generations. Results from Gallup also corroborate this pattern, showing that young LGBT adults are less likely to hold a Bachelor's degree, while prime and mature adults are more likely to be college graduates.¹⁹ Furthermore, when the sample is divided by sex assigned at birth, an additional insight emerges. Cisgender lesbian females have experienced the most significant decline in college attainment across age groups, while cisgender gay males are the only subgroup with a consistent (albeit decreasing) college attainment advantage across those groups (Annex Figure 2.A.3).

LGBTI+ youth are evolving in environments that are much more hostile than those in which their cisgender straight peers live. This holds true both within schools and in broader societal contexts. Anti-LGBTI+ bullying at school is a worldwide problem and the United States is no exception (Unesco, 2016^[70]). The trivialisation of insults such as “fag”, “dyke”, or “tranny”, along with the rampant verbal and physical harassment of boys perceived as too feminine and girls seen as too masculine, adversely affects the mental health of LGBTI+ youth and creates barriers to their educational attainment (Human Rights Campaign Foundation, 2023^[71]; Conron, O'Neill and Vasquez, 2022^[72]; National Academies of Sciences, Engineering, and Medicine, 2020^[39]; Sansone, 2019^[68]). Although the school climate for LGBTI+ youth has improved since the early 2000s, schools remain a challenging environment (Human Rights Campaign Foundation, 2023^[45]; Kosciw, Clark and Menard, 2022^[73]). In 2021, 82% of LGBTI+ students who responded to the National School Climate Survey, conducted biennially by the Gay, Lesbian, and Straight Education Network in the United States, reported feeling unsafe in school due to at least one of their actual or perceived personal characteristics. For instance, 92% of LGBTI+ students heard negative remarks about gender expression (not acting “masculine enough” or “feminine enough”), with a majority (56%) declaring they heard these remarks frequently or often. Outside schools, the situation doesn't necessarily improve for LGBTI+ youth. Notably, the increased use of internet among youth has resulted in online bullying. In 2021, nearly half of LGBTI+ students who responded to the National School Climate Survey reported being harassed or threatened by classmates via electronic media (i.e. text messages, emails, Instagram, Twitter, Tumblr, Facebook, Snapchat) during extracurricular time. In addition, LGBTI+ youth who come out might experience exclusionary behaviours by parents and other family members (Human Rights Campaign Foundation, 2023^[45]; National Academies of Sciences, Engineering, and Medicine, 2020^[39]), which likely contribute to their elevated risk of homelessness (National Academies of Sciences, Engineering, and Medicine, 2020^[39]).

While such adversities can be immensely challenging, there is evidence that they might lead some LGBTI+ youth to develop coping strategies that translate into greater perseverance in academic pursuits (Pachankis and Hatzenbuehler, 2013^[74]). First, the anticipation that the bullying LGBTI+ individuals face in

school settings and elsewhere might continue into the labour market could encourage some to seek higher educational qualifications and other forms of human capital as a compensatory strategy. Indeed, two-thirds of LGBTI+ youth are afraid they will be discriminated against at work due to their LGBTI+ identity (Human Rights Campaign Foundation, 2023^[45]). Second, given the reduced self-esteem and increased family rejection experienced by LGBTI+ youth relative to their cisgender straight peers (National Academies of Sciences, Engineering, and Medicine, 2020^[39]), investing more in human capital can boost their self-worth and perceived acceptance. Third, LGBTI+ might invest more in human capital to ease relocation to more accepting neighbourhoods (as illustrated in Section 2.3) and to secure roles in more accepting occupations. If these coping strategies are indeed at play, the data in Figure 2.11 (which are consistent with prior research) suggest that, until recently, they have allowed LGB+ individuals to over-compensate. However, this does not seem to be the case for non-cisgender individuals. A few factors might explain these disparities. For one, non-cisgender individuals often encounter more negative attitudes than their cisgender LGB+ counterparts (OECD, 2019^[7]; Valfort, 2017^[27]), including in the school setting (Human Rights Campaign Foundation, 2023^[45]), a topic the next chapter will delve deeper into. In addition, presumably due to the enhanced stigma they face, non-cisgender individuals report higher rates of mental health disorders like depression, anxiety, and suicidal thoughts – an issue that Chapter 3 will also explore further. These mental health struggles can severely impede academic focus and performance. Lastly, the consistent educational disadvantage seen among non-cisgender individuals across all age groups might also derive from identity challenges. The process of understanding, accepting, and expressing one's gender identity might be particularly overwhelming during childhood and adolescence – a distress referred to as “gender dysphoria” –, making it challenging to balance personal growth with educational demands.

Why haven't young cisgender LGB+ individuals maintained their educational attainment premium? First, as LGBTI+ people from younger cohorts “come out of the closet” earlier (National Academies of Sciences, Engineering, and Medicine, 2020^[39]), they might be more likely to be impacted by bullying and loss of family support than their predecessors from older generations. Another plausible explanation is the continuous improvement in both social and legal acceptance, which may have reduced the need to over-compensate in terms of human capital. For example, research has shown that the legalisation of same-sex marriage across US states resulted in an increase in employment for people in same-sex couples. This change appears to be driven by a growing acceptance of homosexuality and, consequently, reduced discrimination against LGB job applicants and employees (Sansone, 2019^[75]). LGBTI+ youth might be aware of these enhanced job prospects and thus may not feel the same urgency to invest heavily in their education. Another possible explanation could be the reduction in the so-called “non-disclosure bias” (Valfort, 2017^[27]). Historically, lower-educated LGBTI+ adults might have been less inclined than their higher-educated counterparts to openly identify as LGBTI+ in surveys and other life areas. For instance, evidence has shown that, among men who report having sex with men in the United States, those of advantaged background are more likely to self-identify as homosexual or bisexual (Pathela et al., 2006^[76]; Barrett, C. and Pollack, 2005^[77]). This reluctance may stem from traditionally and persistent lower levels of acceptance of LGBTI+ individuals among populations with limited education, as illustrated in Chapter 3. However, as society shifts towards greater acceptance of non-heterosexual and non-cisgender identities, a greater number of lower-educated LGBTI+ individuals might feel empowered to “come out of the closet”.

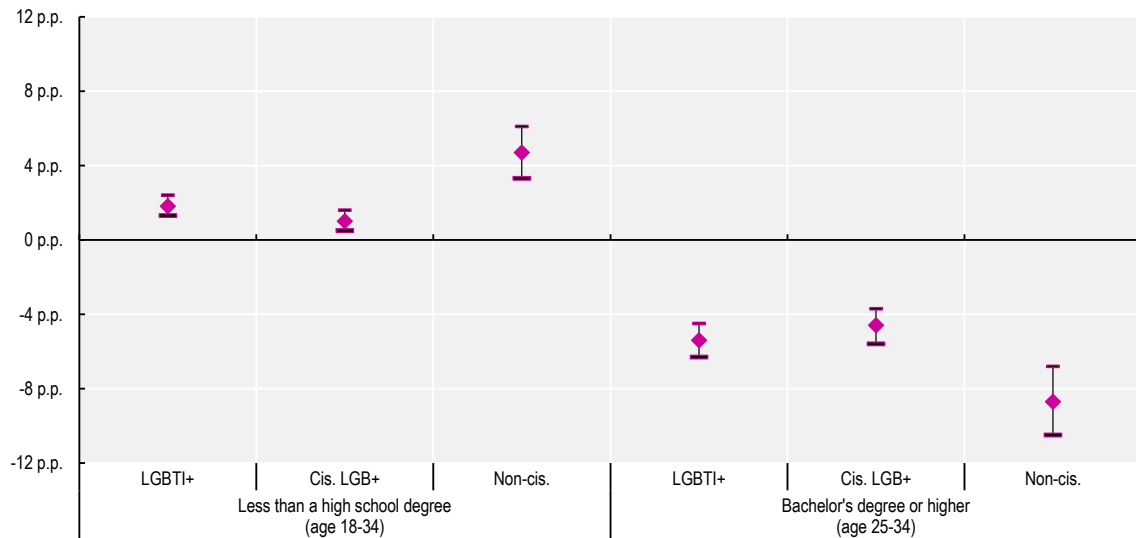
Among cisgender LGB+ individuals, lesbians have experienced the most significant decline in educational attainment across age groups (Annex Figure 2.A.3). Historically, cisgender lesbian females have surpassed their straight counterparts in education (Mittleman, 2022^[67]). One explanation is their stronger incentive to defy patriarchal norms of the era, which advocated for early marriage, childrearing, and economic dependence on a male provider. By obtaining a Bachelor's degree or higher, they could delay the traditional marital and parenting timelines. Such higher education potentially also allowed them to avoid entering a heterosexual marriage altogether by providing the prospect of financial independence. Lastly, their sexual orientation naturally protected them from unintended pregnancies, which, given the limited

access to contraceptives at the time, frequently caused several of their straight peers to not attend or drop out of college.

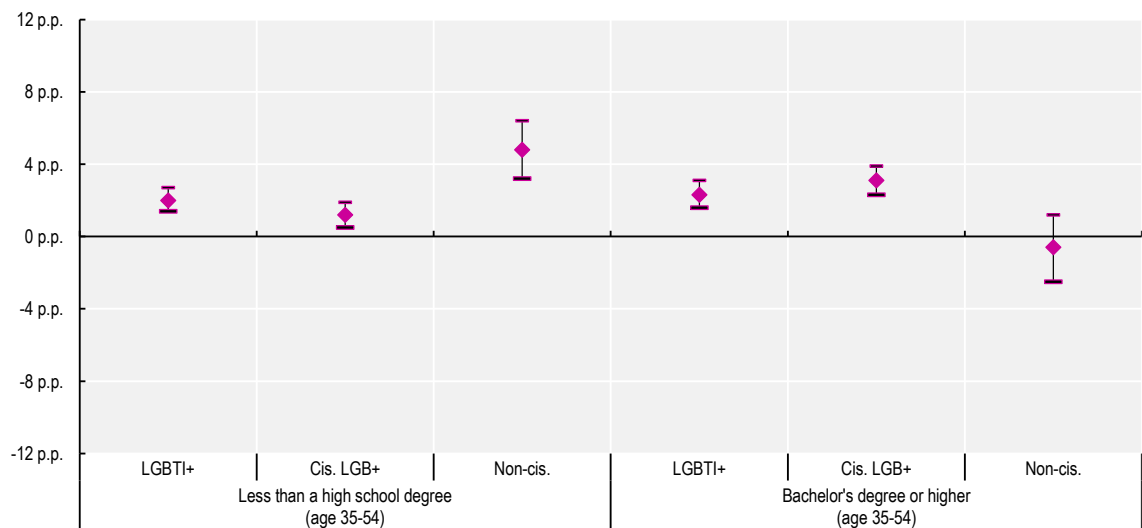
With the advancement of gender equality, many barriers that once hindered women's college completion have been removed, leading to a prevailing notion that girls should attain at least a college degree. Consequently, there has been a significant surge in girls' educational attainment, allowing them to outpace boys in many instances (OECD, 2022^[78]; England and Bearak, 2012^[79]). One explanation for this trend is that personal characteristics (including personality traits) traditionally viewed as “feminine” (which often align with societal expectations for girls) are, on average, more conducive to academic success than certain characteristics stereotypically deemed masculine, such as conscientiousness compared to excessive assertiveness. In this context, cisgender lesbian females face challenges compared to their cisgender straight counterparts (Mittleman, 2022^[67]). They have been shown to possess, on average, personal characteristics that are less typical of their gender and thus more aligned with characteristics traditionally associated with men (Burn and Martell, 2022^[80]). On the other hand, the gender atypicality observed, on average, in cisgender gay males, which has been consistently demonstrated in research, might also be a reason they surpass their straight peers academically (Mittleman, 2022^[67]). Finally, stereotypes about LGBTI+ individuals can also influence educational (and other life-related) choices of LGBTI+ individuals. For instance, stereotypes for gay men – such as that of the effeminate, rich, well-educated, urban gay male, with a successful career – are still present in contemporary representations in media (van Meer and Pollmann, 2022^[81]; McInroy and Craig, 2017^[82]; Rothmann, 2013^[83]; Meyer, 2010^[84]). All these factors could help explain why cisgender gay males remain the only LGBTI+ subgroup to consistently exhibit an educational advantage, even among younger generations (Annex Figure 2.A.3).

Figure 2.11. After demographic adjustments, an educational penalty prevails among all cohorts of non-cisgender individuals, but emerges only among younger cohorts of cisgender LGB+ individuals

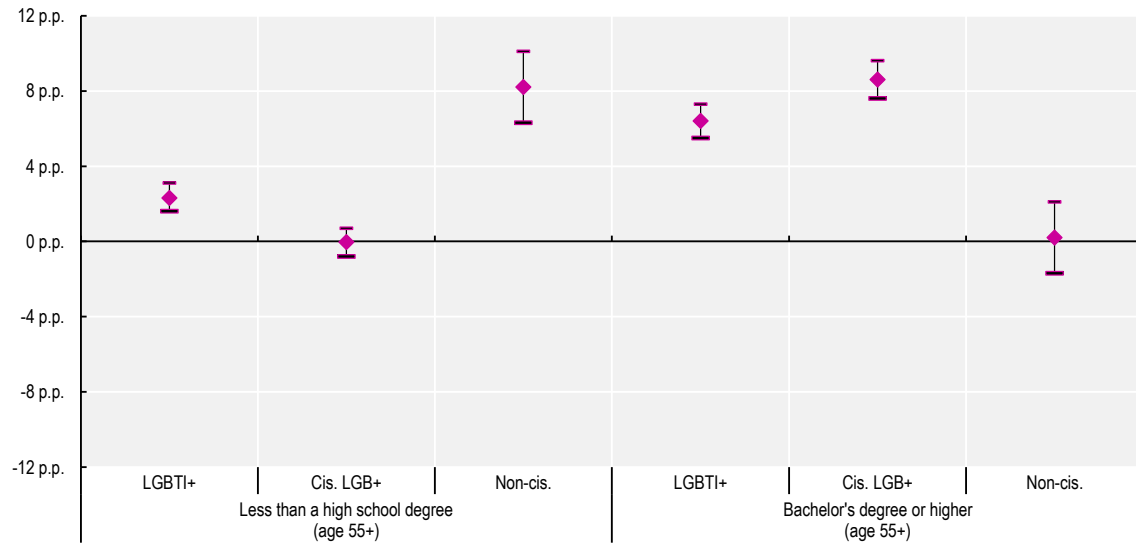
Panel A: Percentage point difference in the probability of not holding a high school degree and the probability of holding Bachelor's degree or higher (using cisgender straight individuals as the reference category) for US adults aged between 18 and 34, by sexual orientation and gender identity (2021-23)



Panel B: Percentage point difference in the probability of not holding a high school degree and the probability of holding a Bachelor's degree or higher (using cisgender straight individuals as the reference category) for US adults aged between 35 and 54, by sexual orientation and gender identity (2021-23)



Panel C: Percentage point difference in the probability of not holding a high school degree and the probability of holding a Bachelor's degree or higher (using cisgender straight individuals as the reference category) for US adults above age 55, by sexual orientation and gender identity (2021-23)



Note: LGBTI+ refers to respondents who self-identify as gay/lesbian, bisexual, transgender, other non-heterosexual, and other non-cisgender (see Box 2.5 for more details). The percentage point differences are estimated from a Logistic regression, controlling for sex assigned at birth, age groups (18-24, 25-29, 30-34, ..., 80-84, 85-88), race and ethnicity, marital and parental status, number of adults in the household, living in one of the 15 largest metropolitan statistical areas (metro areas), as well as state and survey wave fixed effects. The percentage point differences are estimated at the average of the covariates. The error bars depict 95% confidence intervals. This means that we can be 95% confident that the true value lies within the range of the error bar. In other words, if the error bar crosses the x-axis, then the percentage point difference cannot be deemed as statistically different from 0 (at least not at this 95% confidence level). Standard errors robust to heteroscedasticity. Person-level weights used.

Source: Household Pulse Survey (21 July 2021 – 10 July 2023).

References

- Aksoy, C. et al. (2020), “Do laws shape attitudes? Evidence from same-sex relationship recognition policies in Europe”, *European Economic Review*, Vol. 124, <https://doi.org/10.1016/j.euroecorev.2020.103399>. [88]
- Ayoub, P. and J. Garretson (2017), “Getting the Message Out: Media Context and Global Changes in Attitudes Toward Homosexuality”, *Comparative Political Studies*, 50(8), 1055–1085, <https://doi.org/10.1177/0010414016666836>. [44]
- Badgett, M., C. Carpenter and A. Sansone (2021), “LGBTQ Economics”, *Journal of Economic Perspectives*, 35 (2): 141-70, pp. 141-70. [14]
- Badgett, M. and N. Goldberg (2009), *Best practices for asking questions about sexual orientation on surveys*. [85]
- Barrett, D. C. and L. Pollack (2005), “Whose Gay Community? Social Class, Sexual Self-Expression, and Gay Community Involvement”, *Sociological Quarterly*, 46 (3), 2005, p. 437-456. [77]
- Besnier, N. and K. Alexeyeff (2014), “Gender on the Edge: Transgender, Gay, and Other Pacific Islanders”, *University of Hawai’i Press*, <http://www.jstor.org/stable/j.ctt6wqhsc>. [52]
- Bettinsoli, M., A. Suppes and J. Napier (2020), *Predictors of Attitudes Toward Gay Men and Lesbian Women in 23 Countries*, *Social Psychological and Personality Science*, 11(5), 697–708, <https://doi.org/10.1177/1948550619887785>. [41]
- Black, D. et al. (2000), *Demographics of the Gay and Lesbian Population in the United States: Evidence from Available Systematic Data Sources*, *Demography* 37(2): 139–54, <https://doi.org/10.1257/JEP.35.2.141>. [15]
- Black, D., S. Sanders and L. Taylor (2007), *The economics of lesbian and gay families.*, *Journal of Economic Perspectives*, 21(2), 53–70, <https://doi.org/10.1257/jep.21.2.53>. [63]
- Burn, I. and M. Martell (2022), “Gender typicality and sexual minority labour market differentials”, *British Journal of Industrial Relations*, 60(4), 784–814. [80]
- Carpenter, C., S. Eppink and G. Gonzalez (2020), *Transgender status, gender identity, and socioeconomic outcomes in the United States*, *Industrial & Labor Relations Review*, 73(3), 573-599. [65]
- Carpenter, C., M. Lee and L. Nettuno (2022), *Economic outcomes for transgender people and other gender minorities in the United States: First estimates from a nationally representative sample*, *Southern Economic Journal* Volume89, Issue2. [18]
- Coffman, K., L. Coffman and K. Ericson (2016), *The Size of the LGBT Population and the Magnitude of Antigay Sentiment Are Substantially Underestimated*, *Management Science* 63 (10): 3168-86. [32]
- Coffman, K., L. Coffman and K. Marzilli (2024), *Non-Binary Gender Economics*, National Bureau of Economic Research. [54]
- Conron, K., K. O’Neill and L. Vasquez (2022), *Educational Experiences of Transgender People*, Los Angeles, CA: The Williams Institute. [72]

- Deng, B. and T. Watson (2023), *LGBTQ+ Data Availability: What we can learn from four major surveys*, The Brookings Institution. [16]
- England, P. and J. Bearak (2012), *Women's Education and Their Likelihood of Marriage: A Historic Reversal*, Austin, TX: Council on Contemporary Families. [79]
- Farr, R., C. Vazquez and C. Patterson (2020), *LGBTQ adoptive parents and their children*, In A.E. Goldberg and K.R. Allen (Eds.), *LGBT-Parent Families: Innovations in Research and Implications for Practice* (2nd ed.). New York: Springer. [60]
- Fields, J. et al. (2020), *Design and Operation of the 2020 Household Pulse Survey*, 2020. U.S. Census Bureau. Forthcoming. [23]
- Flores, A. (2019), *Social Acceptance of LGBT People in 174 Countries, 1981 to 2017*, The Williams Institute, Los Angeles. [8]
- Flores, A. and S. Barclay (2015), *Trends in Public Support for Marriage for Same-Sex Couples by State*, Williams Institute, UCLA School of Law. [10]
- Folch, M. (2022), *The LGBTQ+ Gap: Recent Estimates for Young Adults in the United States*, <https://doi.org/10.2139/ssrn.4072893>. [29]
- Fundamental Rights Agency (2015), *The fundamental rights situation of intersex people | European Union Agency for Fundamental Rights*, <https://fra.europa.eu/en/publication/2015/fundamental-rights-situation-intersex-people> (accessed on 25 November 2019). [4]
- Gates, G. (2015), *Marriage and Family: LGBT Individuals and Same-Sex Couples.*, *The Future of Children* 25, no. 2 (2015): 67–87., <http://www.jstor.org/stable/43581973>. [55]
- Gates, G. (2014), *LGB Families and Relationships: Analyses of the 2013 National Health Interview Survey*, Los Angeles, CA: Williams Institute, UCLA School of Law. [62]
- Gates, G. and T. Brown (2015), *Marriage and Same-Sex Couples after Obergefell*, Los Angeles, CA: Williams Institute, UCLA School of Law. [59]
- Goldberg, A. (2023), *Impact of HB 1557 (Florida's Don't Say Gay Bill) on LGBTQ+ Parents in Florida*, Williams Institute, <https://williamsinstitute.law.ucla.edu/wp-content/uploads/Dont-Say-Gay-Impact-Jan-2023.pdf>. [37]
- Goldberg, S. and K. Conron (2018), *How Many Same-Sex Couples in the U.S. Are Raising Children?*, Los Angeles, CA: Williams Institute, UCLA School of Law. [56]
- Grov, C. et al. (2006), "Race, ethnicity, gender, and generational factors associated with the coming-out process among gay, lesbian, and bisexual individuals", *The Journal of Sex Research*, 43:2, 115-121, <https://doi.org/10.1080/00224490609552306>. [48]
- Hall, W., H. Dawes and N. Plocek (2021), "Sexual Orientation Identity Development Milestones Among Lesbian, Gay, Bisexual, and Queer People: A Systematic Review and Meta-Analysis", *Frontiers in Psychology*, Vol. 12, <https://doi.org/10.3389/fpsyg.2021.753954>. [43]
- Herek, G. (2002), *Gender gaps in public opinion about lesbians and gay men*, *Public Opinion Quarterly*, 66, 40–66. [40]

- Herman, J., A. Flores and K. O'Neill (2022), *How Many Adults and Youth Identify as Transgender in the United States?*, The Williams Institute, UCLA School of Law. [28]
- Human Rights Campaign Foundation (2023), "2023 LGBTQ+ Youth Report", [45]
<https://reports.hrc.org/2023-lgbtq-youth-report#>.
- Human Rights Campaign Foundation (2023), "Impact of Gender Affirming Care Bans on LGBTQ+ Adults", [12]
<https://hrc-prod-requests.s3-us-west-2.amazonaws.com/GAC-Ban-Memo-Final.pdf>.
- Human Rights Campaign Foundation (2023), *LGBTQ+ Americans Under Attack: A Report and Reflectio non the 2023 State Legislative Session*. [71]
- Human Rights Campaign Foundation (2018), *A workplace divided: understanding the climate for LGBTQ workers nationwide.*, [30]
<https://www.hrc.org/resources/a-workplace-divided-understanding-the-climate-for-lgbtq-workers-nationwide>.
- IPSOS (2023), *LGBT+ PRIDE 2023: A 30-Country Ipsos Global Advisor Survey*, [25]
<https://www.ipsos.com/sites/default/files/ct/news/documents/2023-05/ipsos%20LGBT%2B%20Pride%202023%20Global%20Survey%20Report%20-%20rev.pdf>.
- Jacobs, S. (1997), "Two-Spirit people: Native American gender identity", *Sexuality, and spirituality*. University of Illinois Press. [51]
- Jones, R. et al. (2019), *Fifty Years After Stonewall: Widespread Support for LGBT Issues – Findings from American Values Atlas 2018*, Washington, DC: Public Religion Research Institute, [9]
<https://www.prri.org/research/fifty-years-after-stonewall-widespread-support-for-lgbt-issues-findings-from-american-values-atlas-2018/>.
- Jones, T. et al. (2016), *Intersex: Stories and Statistics from Australia*, Open Book Publishers, [3]
<https://doi.org/10.11647/obp.0089>.
- Kasprowski, D. et al. (2021), "LGBTQI* people in Germany face staggering health disparities", [6]
DIW Weekly Report 5-6, pp. 42-50, https://doi.org/10.18723/diw_dwr:2021-5-1.
- Kosciw, J., C. Clark and L. Menard (2022), "The 2021 National School Climate Survey: The experiences of LGBTQ+ youth in our nation's schools.", [73]
New York: GLSEN.
- Levine, A. (2022), "Sexualities and Geographic Mobility Between Childhood and Adulthood in the United States", *Demography* 1 August 2022; 59 (4): 1541–1569, [35]
<https://doi.org/10.1215/00703370-10085223>.
- Maguen, S. et al. (2002), "Developmental milestones and disclosure of sexual orientation among, gay, lesbian, and bisexual youths", [50]
Journal of Applied Developmental Psychology. 2002;23(2):219–233.
- Marken, S. (2018), "Still Listening: The State of Telephone Surveys", [24]
<https://news.gallup.com/opinion/methodology/225143/listening-state-telephone-surveys.aspx>.
- Martell, M. and L. Roncolato (2023), *Economic Vulnerability of Sexual Minorities: Evidence from the US Household Pulse Survey.*, *Popul Res Policy Rev.* 2023;42(2):28. [19]

- McInroy, L. and S. Craig (2017), “Perspectives of LGBTQ emerging adults on the depiction and impact of LGBTQ media representation”, *Journal of Youth Studies*, Vol. 20/1, pp. 32-46, <https://doi.org/10.1080/13676261.2016.1184243>. [82]
- Medina, C. and L. Mahowald (2023), *Discrimination and Barriers to Well-Being: The State of the LGBTQI+ Community in 2022*, The Center for American Progress, <https://www.americanprogress.org/article/discrimination-and-barriers-to-well-being-the-state-of-the-lgbtqi-community-in-2022/>. [36]
- Medina, C. and L. Mahowald (2021), “Key Issues Facing People With Intersex Traits”, *The Center for American Progress*, <https://www.americanprogress.org/article/key-issues-facing-people-intersex-traits/>. [1]
- Meyer, I. (2010), “Identity, Stress, and Resilience in Lesbians, Gay Men, and Bisexuals of Color”, *The Counseling Psychologist*, 38(3), 442–454, <https://doi.org/10.1177/0011000009351601>. [46]
- Meyer, I. (2003), *Prejudice, social stress, and mental health in lesbian, gay and bisexual populations: conceptual issues and research evidence*, *Psychol Bull.* 2003;129:674-697. [47]
- Meyer, M. (2010), “Representing bisexuality on television: The case for intersectional hybrids”, *Journal of Bisexuality*, 10, 366–387. [84]
- Mittleman, J. (2022), *Intersecting the Academic Gender Gap: The Education of Lesbian, Gay, and Bisexual America.*, *American Sociological Review*, 87(2), 303–335., <https://doi.org/10.1177/00031224221075776>. [67]
- Morgan, E. and M. van Dulmen (2021), *Sexuality in Emerging Adulthood*, online edn, Oxford Academic, 23 Sept. 2021, <https://doi.org/10.1093/oso/9780190057008.001.0001>. [42]
- Movement Advancement Project (2023), *Equality Maps Snapshot: LGBTQ Equality By State*, <https://www.mapresearch.org/equality-maps/>. [38]
- Movement Advancement Project (2020), *Mapping LGBTQ Equality: 2010 to 2020*, <https://www.lgbtmap.org/file/2020-tally-report.pdf>. [11]
- Movement Advancement Project (2019), *Where We Call Home: LGBT People in Rural America.*, <http://www.lgbtmap.org/rural-lgbt>. [34]
- National Academies of Sciences, Engineering, and Medicine (2022), *Measuring Sex, Gender Identity, and Sexual Orientation*, Washington, DC: The National Academies Press, <https://doi.org/10.17226/26424>. [33]
- National Academies of Sciences, Engineering, and Medicine (2020), *Understanding the Well-Being of LGBTQI+ Populations*, Washington, DC: The National Academies Press, <https://doi.org/10.17226/25877>. [39]
- National Science and Technology Council (2023), “Federal Evidence Agenda on LGBTQI+ Equity”, *A Report by the Subcommittee on Sexual Orientation, Gender Identity, and Variations in Sex Characteristics (SOGI) Data. Subcommittee on Equitable Data.* [17]
- OECD (2023), *The Road to LGBTI+ Inclusion in Germany: Progress at the Federal and Länder Levels*, OECD Publishing, Paris, <https://doi.org/10.1787/977b463a-en>. [13]

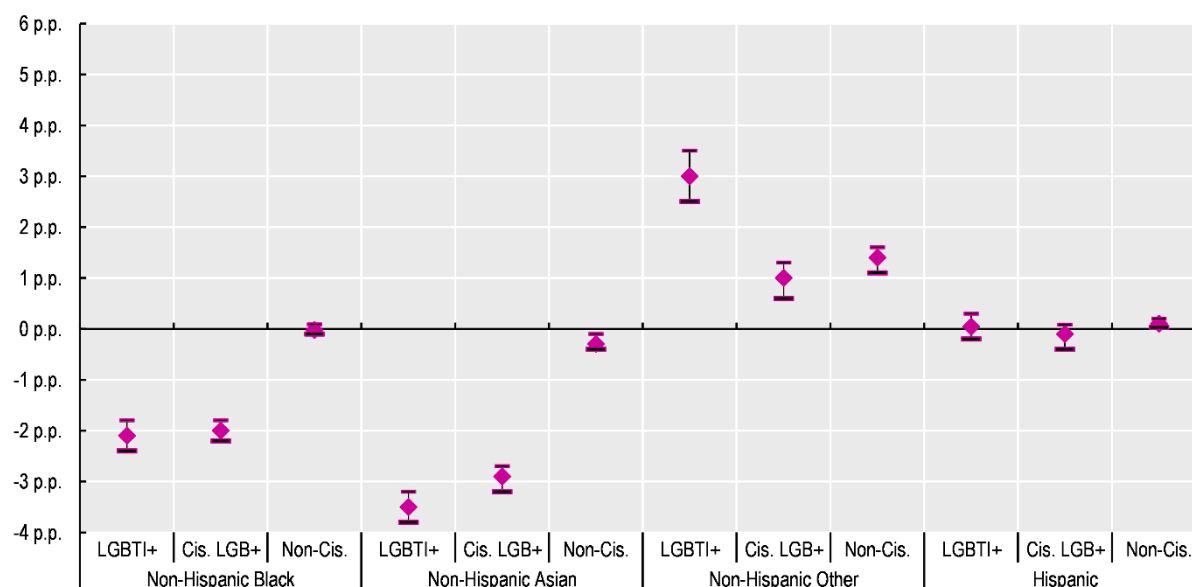
- OECD (2022), "Education at a Glance 2022: OECD Indicators", *OECD Publishing, Paris*, [78]
<https://doi.org/10.1787/3197152b-en>.
- OECD (2020), *Over the Rainbow? The Road to LGBTI Inclusion*, OECD Publishing, Paris, [5]
<https://doi.org/10.1787/8d2fd1a8-en>.
- OECD (2019), *Society at a Glance 2019: OECD Social Indicators*, OECD Publishing, Paris, [7]
https://doi.org/10.1787/soc_glance-2019-en.
- Ofosu, E. et al. (2019), "Same-sex marriage legalization associated with reduced implicit and explicit antigay bias", *Proceedings of the National Academy of Sciences*, 116(18), 8846-8851, [89]
<https://doi.org/10.1073/pnas.1806000116>.
- Pachankis, J. and M. Hatzenbuehler (2013), *The Social Development of Contingent Self-Worth in Sexual Minority Young Men: An Empirical Investigation of the 'Best Little Boy in the World Hypothesis'*, *Basic and Applied Social Psychology* 35(2):176–90. [74]
- Pathela, P. et al. (2006), "Discordance between Sexual Behavior and Self-Reported Sexual Identity: A Population-Based Survey of New York City Men", *Annals of Internal Medicine*, 145 (6), 2006, p. 416-425. [76]
- Pearson, J. and L. Wilkinson (2017), *Same-sex sexuality and educational attainment: The pathway to college*, *Journal of Homosexuality*, 64(4), 538–576. [69]
- Pew Research Center (2022), *About 5% of young adults in the U.S. say their gender is different from their sex assigned at birth*, Washington, DC: Pew Research Center. [26]
- Robertson, R. et al. (2018), *Estimates of Non-Heterosexual Prevalence: The Roles of Anonymity and Privacy in Survey Methodology*, *Archives of Sexual Behavior* 47 (4): 1069-84. [31]
- Rosario, M., E. Schrimshaw and J. Hunter (2004), "Ethnic/racial differences in the coming-out process of lesbian, gay, and bisexual youths: a comparison of sexual identity development over time.", *Cultur Divers Ethnic Minor Psychol.* 2004 Aug;10(3):215-28, [49]
<https://doi.org/10.1037/1099-9809.10.3.215>.
- Rosenwohl-Mack, A. et al. (2020), "A national study on the physical and mental health of intersex adults in the U.S.", *PLoS One*. 2020 Oct 9;15(10):e0240088, [2]
<https://doi.org/10.1371/journal.pone.0240088>.
- Rothmann, J. (2013), "Send in the (gay) clowns': Will & Grace and Modern Family as' sensibly queer", *Acta Academica*, 45, 40–83. [83]
- Sansone, D. (2019), *LGBT students: New evidence on demographics and educational outcomes*, *Economics of Education Review*, 73(April), 101933. [68]
- Sansone, D. (2019), *Pink work: Same-sex marriage, employment and discrimination*, *Journal of Public Economics*, Volume 180,, [75]
<https://doi.org/10.1016/j.jpubeco.2019.104086>.
- Sansone, D. and C. Carpenter (2020), *Turing's Children: Representation of Sexual Minorities in STEM*, *PLOS ONE* 15 (11). [61]
- Scherer, Z. (2022), "Number of Same-Sex Couple Households Exceeded 1 Million in 2021", *United States Census Bureau*, <https://www.census.gov/library/stories/2022/11/same-sex-couple-households-exceeded-one-million.html>. [53]

- Tankard, M. and E. Paluck (2017), “The Effect of a Supreme Court Decision Regarding Gay Marriage on Social Norms and Personal Attitudes”, *Psychological Science*, 28(9), 1334–1344, <https://doi.org/10.1177/0956797617709594>. [90]
- Tate, C., J. Ledbetter and C. Youssef (2013), *A two-question method for assessing gender categories in the social and medical sciences.*, *J Sex Res.* 2013;50(8):767-76. [87]
- Taylor, D. (2020), “Same-Sex Couples Are More Likely to Adopt or Foster Children”, *United States Census Bureau*. [57]
- Tordoff, D. et al. (2019), *Increased Ascertainment of Transgender and Non-binary Patients Using a 2-Step Versus 1-Step Gender Identity Intake Question in an STD Clinic Setting.*, *Sex Transm Dis.* 2019 Apr;46(4):254-259. [86]
- Tornello, S. and C. Patterson (2015), *Timing of parenthood and experiences of gay fathers: A life course perspective.*, *Journal of GLBT Family Studies*, 11, 35–56. [58]
- Unesco (2016), “Out in the open: education sector responses to violence based on sexual orientation or gender identity/expression: summary report”, <https://unesdoc.unesco.org/ark:/48223/pf0000244652>. [70]
- United States Census Bureau (2023), “Source of the Data and Accuracy of the Estimates for the Household Pulse Survey – Phase 3.9”, https://www2.census.gov/programs-surveys/demo/technical-documentation/hhp/Phase3-9_Source_and_Accuracy_Week59.pdf. [21]
- United States Census Bureau (2021), “Nonresponse Bias Report for the 2020 Household Pulse Survey”, https://www2.census.gov/programs-surveys/demo/technical-documentation/hhp/2020_HPS_NR_Bias_Report-final.pdf. [22]
- Valfort, M. (2017), “LGBTI in OECD Countries: A Review”, *OECD Social, Employment and Migration Working Papers*, No. 198, OECD Publishing, Paris, <https://doi.org/10.1787/d5d49711-en>. [27]
- van Meer, M. and M. Pollmann (2022), “Media Representations of Lesbians, Gay Men, and Bisexuals on Dutch Television and People’s Stereotypes and Attitudes About LGBs”, *Sexuality & Culture* 26, 640–664, <https://doi.org/10.1007/s12119-021-09913-x>. [81]
- Wilkinson, L., J. Pearson and H. Liu (2018), *Educational attainment of transgender adults: Does the timing of transgender identity milestones matter?*, *Social Science Research*, <https://doi.org/10.1016/j.ssresearch.2018.04.006>. [66]
- Wilkinson, L., D. Shifrer and J. Pearson (2021), *Educational Outcomes of Gender-Diverse Youth: A National Population-Based Study*, *Gender & Society* 35(5):806–37. [64]
- Wu, M., K. Zhao and F. Fils-Aime (2022), “Response rates of online surveys in published research: A meta-analysis”, *Computers in Human Behavior Reports*, Vol. 7, <https://doi.org/10.1016/j.chbr.2022.100206>. [20]

Annex 2.A. Additional Figures

Annex Figure 2.A.1. After demographic adjustments, further analysis confirms LGBTI+ self-identification's link to race but not to ethnicity

Percentage point difference in the probability of self-identifying as LGBTI+ (using non-Hispanic Whites as the reference category), by race and ethnicity (2021-23)

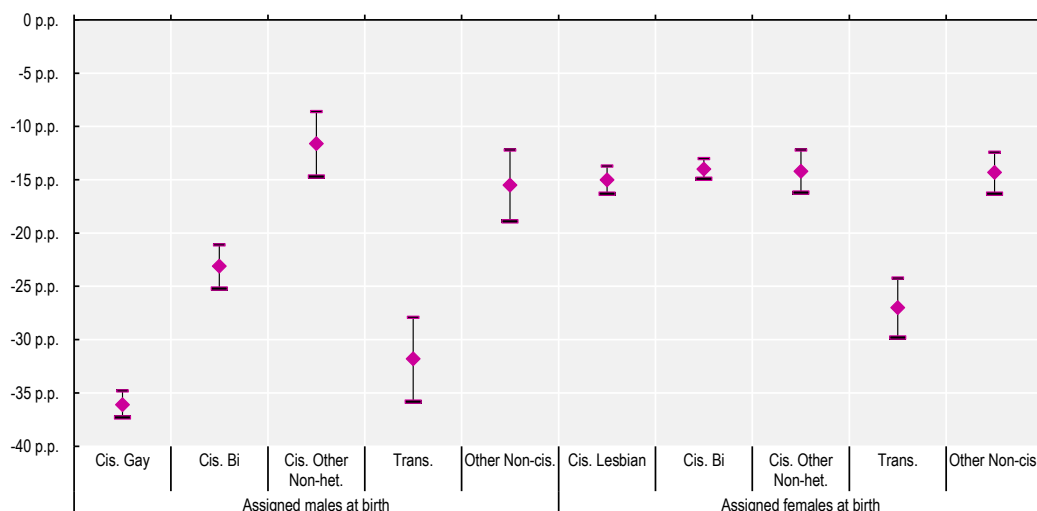


Note: LGBTI+ refers to respondents who self-identify as gay/lesbian, bisexual, transgender, other non-heterosexual, and other non-cisgender (see Box 2.5 for more details). The percentage point differences are estimated from a Logistic regression, controlling for sex assigned at birth, age groups (18-24, 25-29, 30-34, ..., 80-84, 85-88), race and ethnicity, marital and parental status, number of adults in the household, educational attainment, living in one of the 15 largest metropolitan statistical areas (metro areas), as well as state and survey wave fixed effects. The percentage point differences are estimated at the average of the covariates. The error bars depict 95% confidence intervals. This means that we can be 95% confident that the true value lies within the range of the error bar. In other words, if the error bar crosses the x-axis, then the percentage point difference cannot be deemed as statistically different from 0 (at least not at this 95% confidence level). Standard errors robust to heteroscedasticity. Person-level weights used.

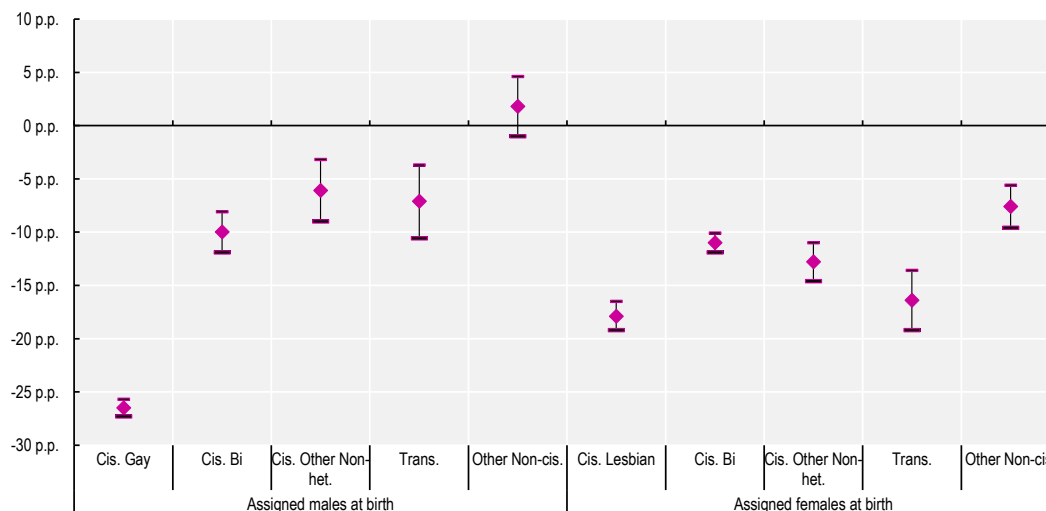
Source: Household Pulse Survey (21 July 2021 – 10 July 2023).

Annex Figure 2.A.2. Cisgender gay males face the strongest disparity in family structure

Panel A: Percentage point difference in the probability of being married (using cisgender straight individuals as the reference category), by sexual orientation, gender identity and sex assigned a birth (2021-23)



Panel B: Percentage point difference in the probability of living in a household with a least one child below 18 (using cisgender straight individuals as the reference category), by sexual orientation, gender identity and sex assigned a birth (2021-23).

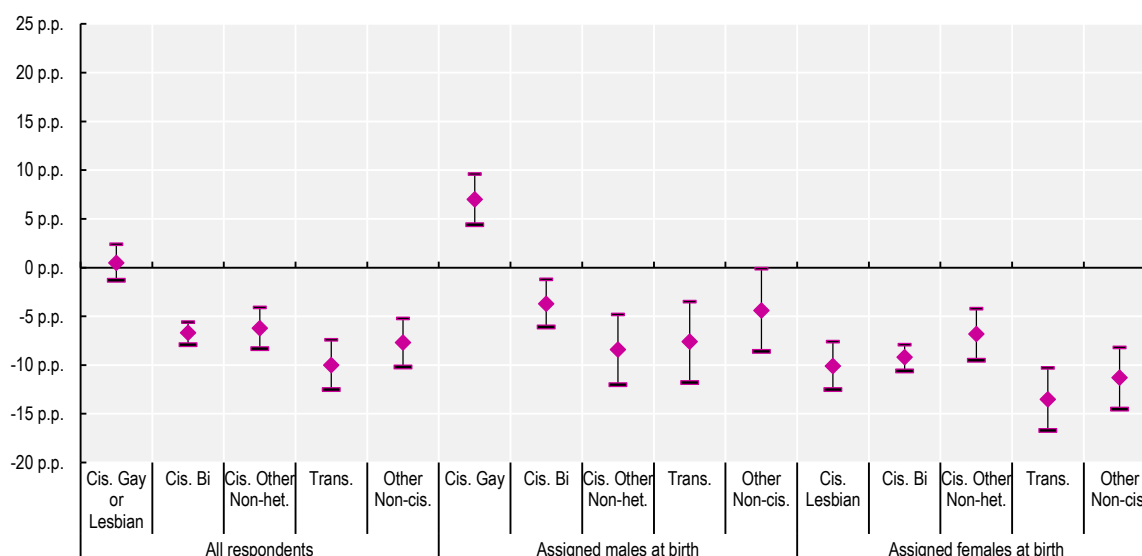


Note: The percentage point differences are estimated from a Logistic regression, controlling for age groups (18-24, 25-29, 30-34, ..., 80-84, 85-88), race and ethnicity, educational attainment, living in one of the 15 largest metropolitan statistical areas (metro areas), as well as state and survey wave fixed effects. The percentage point differences are estimated at the average of the covariates. The error bars depict 95% confidence intervals. This means that we can be 95% confident that the true value lies within the range of the error bar. In other words, if the error bar crosses the x-axis, then the percentage point difference cannot be deemed as statistically different from 0 (at least not at this 95% confidence level). Standard errors robust to heteroscedasticity. Person-level weights used.

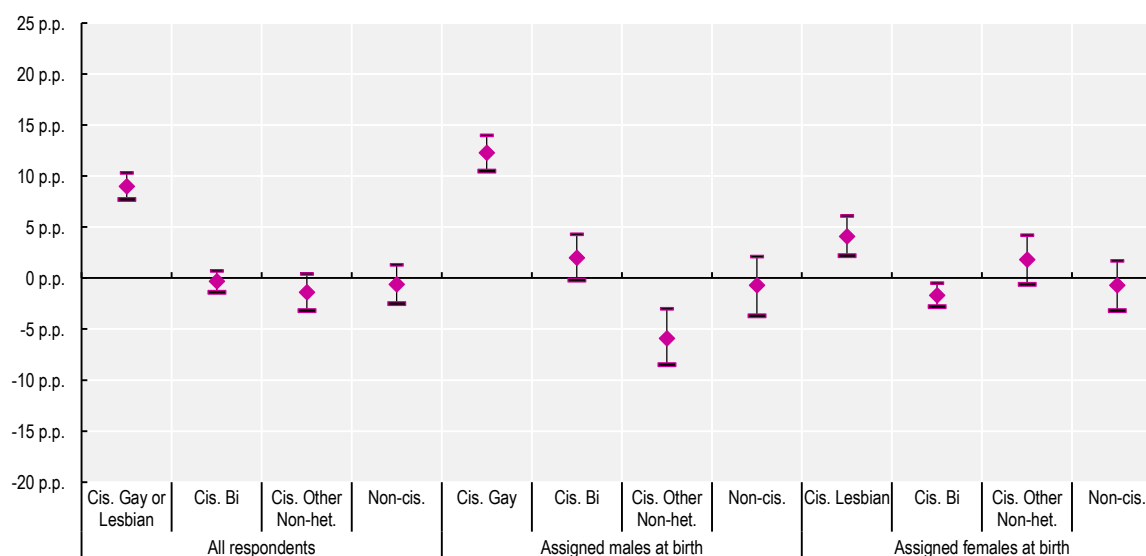
Source: Household Pulse Survey (21 July 2021 – 10 July 2023).

Annex Figure 2.A.3. College attainment declines most for cisgender lesbian females, while cisgender gay males maintain a diminishing advantage across age groups

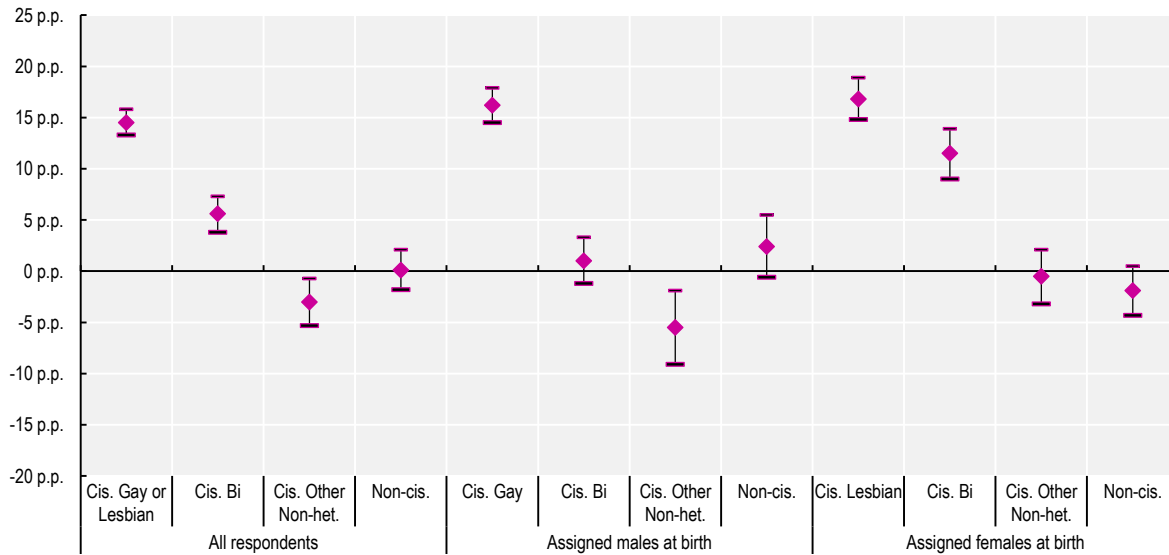
Panel A: Percentage point difference in the probability of holding a Bachelor's degree or higher (using cisgender straight individuals as the reference category) for US adults aged between 25 and 34, by sexual orientation, gender identity, and sex assigned at birth (2021-23)



Panel B: Percentage point difference in the probability of holding a Bachelor's degree or higher (using cisgender straight individuals as the reference category) for US adults aged between 35 and 54, by sexual orientation, gender identity, and sex assigned at birth (2021-23)



Panel C: Percentage point difference in the probability of holding a Bachelor's degree or higher (using cisgender straight individuals as the reference category) for US adults above age 55, by sexual orientation, gender identity, and sex assigned at birth (2021-23)



Note: The percentage point differences are estimated from a Logistic regression, controlling for sex assigned at birth, age groups (18-24, 25-29, 30-34, ..., 80-84, 85-88), race and ethnicity, marital and parental status, number of adults in the household, living in one of the 15 largest metropolitan statistical areas (metro areas), as well as state and survey wave fixed effects. The percentage point differences are estimated at the average of the covariates. The error bars depict 95% confidence intervals. This means that we can be 95% confident that the true value lies within the range of the error bar. In other words, if the error bar crosses the x-axis, then the percentage point difference cannot be deemed as statistically different from 0 (at least not at this 95% confidence level). Standard errors robust to heteroscedasticity. Transgender and other non-cisgender were not separately analysed for prime age and mature adults, due to limited sample size for transgender individuals. Person-level weights used.

Source: Household Pulse Survey (21 July 2021 – 10 July 2023).

Notes

¹ More precisely, the 2019 update of the *LGBT Global Acceptance Index* (LGBT-GAI) established by Andrew Flores shows that acceptance of LGBT people improved between 1981 and 2017 across the globe, with only limited polarisation: while 131 of 174 countries experienced increases in acceptance, only 16 are characterised by a decline (27 experienced no change). However, it is important to keep in mind that these results mainly capture levels and trends in social acceptance of homosexuality. Although the LGBT-GAI seeks to measure acceptance of lesbian, gay, bisexual and transgender people altogether, only seven of the 71 questions used to compute the index focus on acceptance of transgender individuals. Moreover, these questions are measured at only one point in time, which means that the evolution of the LGBT-GAI over time fails to capture potential improvement in attitudes towards transgender individuals.

² Numerous alternative surveys, whether nationwide or focused on specific subnational jurisdictions, have tried to distinguish self-identified LGBTI+ individuals. Some, such as the National Health Interview Survey or the General Social Survey, have included questions related to sexual orientation. Others, like the Behavioural Risk Factor Surveillance System or polls by the Pew Research Center, have addressed gender identity. Yet, none of these surveys offer a nationally representative sample of the LGBT(I+) adult population in the same way that the Gallup or the HPS data does.

³ However, it is important to note that the Census Bureau is planning to request Office of Management and Budget (OMB) approval to conduct a test of sexual orientation and gender identity questions on the ACS in 2024.

⁴ For transgender individuals, it is possible that these different ways of defining their sexual orientation are correlated with the extent of their medical and social gender transition. However, this specific information is not included in the HPS data.

⁵ While the exact implications of these numbers can be challenging to interpret, the Census Bureau's Household Pulse Survey indicates that the majority of non-cisgender adults identify as non-heterosexual. In 2023, 65% of non-cisgender adults self-identified as LGB+, 23% as straight, 11% answered "I don't know" to the sexual orientation question, and 1% did not answer the question at all.

⁶ The Census Bureau's adult population (persons 18 years and over) estimate for 2022 was 260 836 730 and Census Bureau's US state resident population totals in 2022 is obtained from: www.census.gov/newsroom/press-kits/2022/22-national-state-population-estimates.html

⁷ Of these transgender adults, 23% did not self-identify as transgender in the gender identity question but reported a binary gender (male/female) different from their sex assigned at birth (male/female). This evidence further demonstrates that surveys which identify transgender individuals through a single direct question tend to undercount the transgender population. In contrast, surveys employing a two-stage approach – first asking about sex assignment at birth and then about current gender identity – produce more accurate estimates (National Academies of Sciences, Engineering, and Medicine, 2020^[39]; Tordoff et al., 2019^[86]; Tate, Ledbetter and Youssef, 2013^[87]).

⁸ In addition, 1.9% of respondents in 2023 indicated "I don't know" in the sexual orientation question, 1.3% had a missing response in this question, and 1.2% had a missing response in the gender identity question. While some respondents who answer "I don't know" might belong to the LGBTI+ population (e.g. those questioning their sexual orientation), we did not include individuals who selected this option in the LGBTI+ estimate. We made this decision because a significant share likely chose this option due to either a lack

of understanding of the question or uncertainty about the most appropriate category to select (Badgett and Goldberg, 2009^[85]).

⁹ Reliance on opt-in panels consists in exploiting pre-existing samples, that are composed of members who have signed up to take online surveys, in exchange of small rewards. Polling companies employ a variety of statistical techniques to adjust opt-in panels to ensure they match the national population on a chosen set of dimensions. That said, the fact that respondents are self-selected raises a risk of bias that may lead to overstate the share of LGBTI+ individuals (in case those who decide to answer surveys related to LGBTI+ issues are the most open to those issues) or to understate this share (in case those who decide to answer surveys related to LGBTI+ issues are the most hostile to those issues).

¹⁰ For instance, a surge in LGB identification was documented in the General Social Survey (National Academies of Sciences, Engineering, and Medicine, 2020^[39]) and in the National Health Interview Survey (Deng and Watson, 2023^[16]), noting that this upward trend was also witnessed among self-identified cohabitating same-sex couples (Badgett, Carpenter and Sansone, 2021^[14]).

¹¹ In arriving at this conclusion, Katherine Coffman, Lucas Coffman and Keith Marzilli Ericson compared the likelihood of individuals reporting LGB status, depending on whether the sexual orientation question is asked through a self-administered online survey or through an item count technique also conducted online. (The “item count technique” is a survey method used to elicit truthful responses to sensitive questions by providing respondents with a list of items and asking them to indicate how many, but not specifically which, items apply to them, thereby preserving their anonymity.) The authors show that the proportion of LGB respondents is significantly higher when the item count technique (ICT) is used, especially if the question on sexual orientation is based on sexual self-identification. In this case, the proportion of LGB individuals increases by 65% with the ICT (compared to the self-administered online survey), noting that the impact of the ICT is smaller when sexual orientation is measured based on sexual behaviour, and almost non-existent when it is based on sexual attraction. These findings suggest that experiencing sexual attraction to a same-sex person is viewed as socially more acceptable than defining oneself as L, G, or B.

¹² Legal protections and social acceptance of LGBTI+ individuals are strongly interlinked. While countries with greater acceptance of sexual and gender minorities are more likely to pass LGBTI+-inclusive laws, evidence shows that legal changes in favour of LGBTI people in turn do cause changes in attitudes towards this population. Individuals perceive legal changes as reflections of advancements in what is socially acceptable and many are willing to conform to these shifts (Tankard and Paluck, 2017^[90]). For instance, in European countries where same-sex marriage is legal, acceptance of homosexuality increased much faster after those countries adopted same-sex relationship recognition policies (Aksoy et al., 2020^[88]). In the United States, a similar trend was observed. While both implicit and explicit antigay biases were already on the decline before state-level same-sex marriage legalization, they decreased at a sharper rate following legalisation (Ofosu et al., 2019^[89]). More generally, same-sex marriage legalisation across US states led to an increase in employment of people in same-sex couples, a change driven by improvements in attitudes towards homosexuality and, hence, lower discrimination against LGB individuals (Sansone, 2019^[75]).

¹³ Both correlations are statistically different from zero at the 99% confidence level.

¹⁴ On 28 March 2022, the Florida Legislature passed HB 1557, the “Parental Rights in Education” bill, also named the “Don’t Say Gay” bill. This bill prohibits classroom instruction on sexual orientation or gender

identity before the 4th grade and requires such instruction to be “age-appropriate or developmentally appropriate”.

¹⁵ Higher LGBTI+ self-identification among AFAB individuals and the increasing gap in LGBTI+ self-identification between AFABs and AMABs hold after accounting for differences between these groups with respect to geographic location, age, race/ethnicity, partnership and parental status, number of adults in the household, and educational attainment. Results available upon request.

¹⁶ Higher LGBTI+ self-identification among young adults and the increasing gap in LGBTI+ self-identification between them and the broader population hold after accounting for differences between these groups with respect to geographic location, age, race/ethnicity, partnership and parental status, number of adults in the household, and educational attainment. Results available upon request.

¹⁷ Even when restricting the sample to married adults living with only one other adult (likely the spouse), LGBTI+ are approximately 15 percentage points less likely to live in a household with at least one child under 18. This penalty is consistent and statistically significant at the 99% confidence level across all LGBTI+ subgroups. Results available upon request.

¹⁸ In addition, LGBTI+ adults, especially non-cisgender adults, face challenges accessing adequate health insurance: LGBTI+ adults are more likely than non-LGBTI+ adults to not be covered by a health insurance or a health coverage plan, and more likely to have insurance through Medicaid (Medina and Mahowald, 2023^[36]).

¹⁹ Logistic regressions estimated using the Gallup US Daily Survey (2017) shows a 5% penalty in college attainment for young adults, and a 5% and 10% college premium for prime and mature adults, respectively. All estimates are statistically significant at the 99% confidence level. Results available upon request.

3 Are LGBTI+ Americans discriminated against?

This chapter explores whether LGBTI+ Americans benefit from a level playing field. It begins by examining the degree of social acceptance towards LGBTI+ Americans and their perception of discrimination. The chapter then assesses their economic situation before focusing on their labour market outcomes. The analysis concludes by delving into their health and well-being status. This investigation underscores the persistent disparities faced by LGBTI+ Americans. While social acceptance towards them has increased, perceptions of discrimination remain prevalent. After adjusting for demographics, LGBTI+ Americans consistently have lower household income, which reflects the significant challenges they face in the labour market, both in terms of employment and labour earnings. These disparities, at least partly driven by societal stigmatisation, are also manifest in substantial health and well-being gaps for LGBTI+ Americans.

3.1. Introduction and main findings

Chapter 3 provides a thorough evaluation of equality for LGBTI+ Americans. It begins by exploring the levels and trends of social acceptance towards LGBTI+ Americans and their perception of discrimination. The chapter then investigates the economic situation of LGBTI+ Americans, before zooming in on their labour market outcomes. The chapter concludes by analysing their health and well-being status.

Main findings

- Although social acceptance towards LGBTI+ Americans has been increasing, their perception of discrimination remains prevalent.
 - The United States has witnessed a notable shift towards greater acceptance of homosexuality, but this has not been as widespread for other sexual orientations and non-cisgender identities.
 - The share of US adults supporting same-sex marriage has moved from a minority (42%) in 2004 to a majority (71%) in 2023, even among traditionally more conservative demographic groups.
 - Attitudes towards other sexual orientations such as bisexuality or asexuality remain negative.
 - Similarly, acceptance of gender diversity lags behind: only a minority of Americans (43%) consider changing one's gender to be morally acceptable in 2023, down from 46% in 2021.
 - The perception of discrimination continues to be widespread among LGBTI+ Americans, leading to concealment strategies that are detrimental to their lived experiences.
 - LGBTI+ Americans report nearly double the rate of discrimination in the year preceding their interview, at 36%, compared to 19% for non-LGBTI+ respondents.
 - More than three-quarters of LGBTI+ Americans, including 90% of non-cisgender individuals, report taking at least one concealment action to avoid experiencing discrimination.
 - Concealing one's identity does not prevent hampered life trajectories. For instance, closeted LGBTI+ employees are likely to still face poor career prospects as they might avoid networking and professional development opportunities. Moreover, the fear of unintended disclosure can lead to significant stress and anxiety, impacting their productivity at work.
- LGBTI+ Americans have lower household income.
 - Even after adjusting for demographics, LGBTI+ adults earn an annual household income that is 8% lower than that of cisgender straight adults, noting that this disparity was exacerbated at the onset of the COVID-19 pandemic.
 - Regardless of the sex assigned at birth, the income penalty is more pronounced for non-cisgender individuals (11%) than for cisgender LGB+ adults (7%). Within the cisgender LGB+ group, the income disparity is not statistically significant for gay men and lesbians. However, it is larger and statistically significant for bisexual men and women (8.5%). The largest disparity is observed among cisgender individuals with other non-heterosexual orientations (16.5%).
 - The income penalty for LGBTI+ individuals diminishes with age, possibly due to a non-disclosure bias. Mature adults from disadvantaged backgrounds might be less inclined to

“come out”, even in surveys. Among young adults aged 18-34, all LGBTI+ subgroups face a notable income penalty, with the exception of cisgender gay men.

- Lower household income for LGBTI+ individuals partly reflects the greater challenges they face in the labour market.
 - Both laboratory and field experiments conducted in the United States have revealed prevalent discrimination against LGBTI+ job candidates. Concurrently, extensive research using representative survey data has confirmed that either direct discrimination or the fear of such discrimination – which leads to the adoption of detrimental coping strategies – negatively impacts the labour market paths of LGBTI+ individuals.
 - We confirm that LGBTI+ Americans face substantial unexplained labour market gaps, focusing on working-age adults living alone. Restricting the analysis to adults living alone yields more accurate estimates of labour market disparities by sexual orientation and gender identity. This approach effectively neutralises the bias resulting from the lower household specialisation observed in same-sex partnerships relative to different-sex partnerships. If this bias is not addressed, it leads to overestimating the penalty for LGBTI+ men, and to underestimating this penalty for LGBTI+ women.
 - LGBTI+ Americans have a 7% unexplained lower likelihood of employment and are 24% more likely to experience job loss, compared to their cisgender straight peers. These disparities exist across all LGBTI+ subgroups and became even more pronounced during the COVID-19 pandemic, largely due to LGBTI+ individuals working in industries hard hit by the pandemic, such as restaurants, food services or art and entertainment.
 - Accounting for differences in demographics, sectors, and occupations, LGBTI+ workers have labour earnings that are 7% lower than those of their straight cisgender counterparts. It is crucial to stress that by adjusting for the tendency of LGBTI+ individuals to select certain sectors and occupations to evade discrimination, we underestimate the true labour earnings penalty caused by discrimination or the fear of it. Indeed, our data show that this avoidance strategy ultimately results in LGBTI+ individuals taking lower-paying jobs. Without considering the impact of sectoral and occupational sorting, the labour earnings penalty for LGBTI+ individuals would reach 10%.
 - The labour earnings penalty for LGBTI+ individuals hides both reduced hours worked and reduced hourly wages.
 - Regardless of the sex assigned at birth, these labour market penalties are stronger for non-cisgender individuals than for cisgender LGB+ individuals.
 - In line with results on household income disparities, unexplained labour market gaps between LGBTI+ and non-LGBTI+ Americans decrease with age, except for the employment penalty for LGBTI+ women which increases as they get older. This phenomenon could reflect a combination of sexism, ageism, and homophobia, especially given evidence of strong age discrimination against female job candidates in the United States.
- LGBTI+ Americans suffer from substantial health and well-being gaps.
 - Extensive research has documented widespread health and well-being disparities within the LGBTI+ population that flow from societal stigmatisation, discrimination from health practitioners, and economic vulnerability.
 - Even after adjusting for demographics, LGBTI+ Americans show a 50% higher risk of generalised anxiety or major depressive disorder than their cisgender straight peers. In line with prior research, this mental health penalty was most pronounced among non-cisgender

individuals. Their risk is 87% higher than that of cisgender straight American adults, largely due to the greater health challenges faced by transgender individuals.

- The mental health penalty does not differ based on sex assigned at birth. Additionally, the previously observed trend, in which the disparities for LGBTI+ individuals diminish with age, is consistent in the context of mental health. Finally, while, before the pandemic, LGBT adults were already 4% less likely to report enjoyment and 31% and 24% more likely to feel worry and stress, respectively, compared to their non-LGBT peers, all three gaps deteriorated with the onset of the COVID-19 crisis.
- Relying on the Gallup National Health and Well-Being Index which was computed between 2014 and 2017, LGBT adults had a 7% lower overall well-being score. Notably, the financial (11%) and health (7%) penalties of LGBT adults emerged as major contributors to their reduced well-being.

3.2. Social acceptance towards LGBTI+ Americans is on the rise, yet their perception of discrimination remains prevalent

3.2.1. Social acceptance of LGBTI+ Americans

The shift towards greater acceptance of homosexuality, in OECD countries and beyond, has been well documented (OECD, 2019^[1]; Flores, 2021^[2]), and the United States is no exception. Over the past two decades, the share of US adults who support same-sex marriage has moved from a minority (42%) in 2004 to a majority (71%) in 2023 (Figure 3.1). It is worth noting that this upward trend applies to all demographic subgroups, whether the analysis is performed by gender, age, race/ethnicity, educational attainment, region or political affiliation (Annex Figure 3.A.1). Consistent with previous studies (Valfort, 2017^[3]) and with key findings from Chapter 2, attitudes towards homosexuality are less positive among men, older individuals, non-Whites, the less educated, those residing in the South, and supporters of the Republican Party. However, even among these subgroups, the share of individuals who support same-sex marriage has steadily increased. Remarkably, in all these segments, a majority are now in favour of marriages between same-sex couples being recognised by the law as valid, with the same rights as traditional marriages. That said, increasing acceptance of homosexuality may conceal significant disparities, particularly when considering the gender of the homosexual population considered. A survey conducted in the United States and 22 other countries, representing both Western and non-Western societies, revealed a consistent pattern: gay men are disliked more than lesbians across all surveyed countries (Bettinsoli, Suppes and Napier, 2020^[4]). This pattern is primarily driven by male respondents. Men's acceptance of lesbians is comparable to women's acceptance of both lesbians and gay men, but men exhibit more negative attitudes towards gay men.

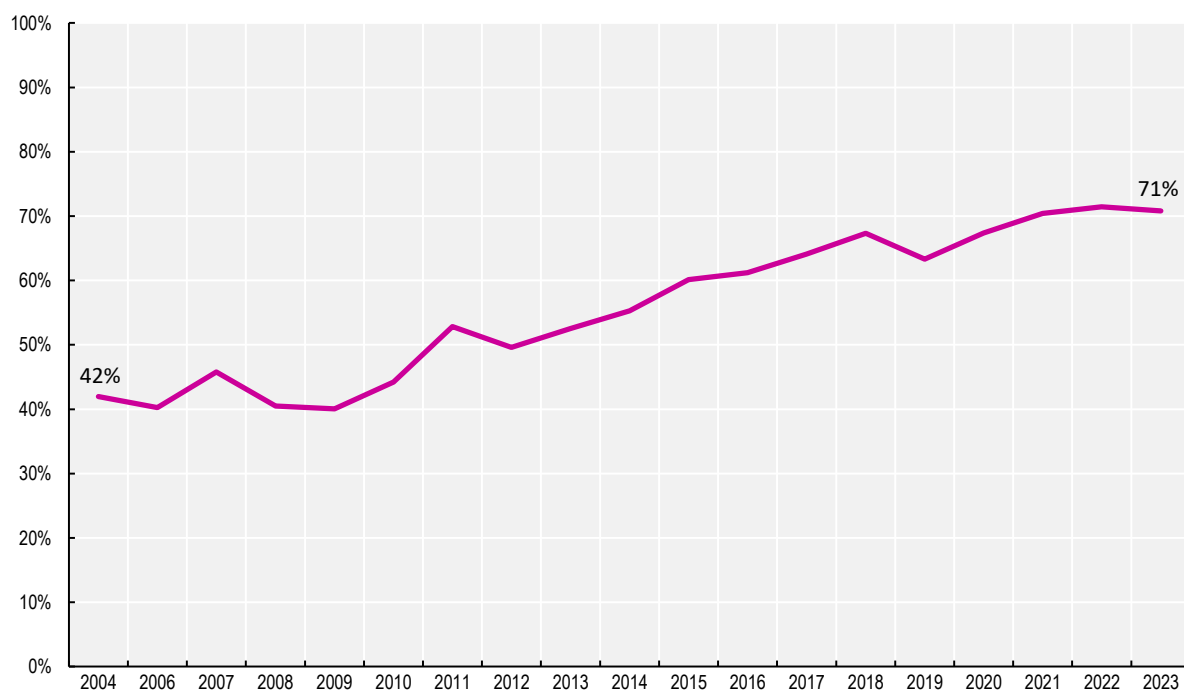
Although a significant share of individuals now supports homosexuality, even among traditionally less accepting groups, this degree of acceptance may not extend to other sexual orientations. Research has shown that bisexual individuals face unique stressors due to negative perceptions of non-monosexuality, occurring both within and outside of the LGBTI+ population (Doan Van et al., 2019^[5]; Dyar and Feinstein, 2018^[6]; Dodge et al., 2016^[7]; Pew Research Center, 2013^[8]; Herek et al., 2010^[9]). Bias against asexual individuals has also proven to be substantial, with prejudice against them being more pronounced than prejudice against homosexuals or bisexuals (Hoffarth et al., 2016^[10]; MacInnis and Hodson, 2012^[11]). Prejudice against both bisexual and asexual individuals is rooted in monosexism, which is the pervasive belief that attraction to only one gender is both normal and superior to any other pattern of attraction, whether it be to more than one gender or to none at all.

Non-cisgender individuals face lower social acceptance than do homosexual individuals. According to the 2023 round of the Gallup Poll Social Series, only a minority of US adults support transgender individuals (Figure 3.2). Notably, less than half of respondents (43%) considered changing one's gender to be morally acceptable. Alternative sources, although not based on probability sampling strictly speaking, paint a similar picture: 44% of US adults interviewed as part of the 2022 Economist/YouGov Poll agreed with the statement that "Someone can be a man or a woman even if that is different from the sex they were assigned at birth".¹ As with attitudes towards homosexuality, support for transgender individuals is greater among women, younger individuals, Whites, the better educated, residents in non-Southern states, and democrats (Annex Figure 3.A.2). In addition, hostility against non-binary individuals is stronger than anti-LGBT sentiment, and strongest among men (Coffman, Coffman and Marzilli, 2024^[12]). Although there is no empirical research on social awareness and acceptance of people with intersex variations, recent studies have found widespread reports of stigma among this group, a phenomenon that is not limited to healthcare settings (Hegarty and Smith, 2023^[13]).

Attitudes towards non-cisgender individuals are not only more negative, they have also deteriorated over the recent past. Representation of transgender and other non-cisgender identities has increased in popular media (GLAAD, 2023^[14]), and discussions on non-cisgender issues have become more prevalent in the public debate. Since 2021, the share of US adults who have any friends or relatives or co-workers who have told them, personally, that they are transgender has increased from 31% to 39% in 2023. Results from a 2021 Pew Research Center survey, a nationally representative survey of American adults, found similar increases in the number of American adults who reported personally knowing someone who is transgender or who prefers gender-neutral pronouns – 42% and 26%, respectively, up from 37% and 18% a few years earlier (Pew Research Center, 2021^[15]).² However, the study found virtually no change in the proportion of Americans who believe that gender is defined by sex assigned at birth (56%), or who report discomfort using gender-neutral pronouns to refer to someone (48%, nearly a majority). More recently, during the 2021-23 period, the share of American adults supporting transgender individuals has decreased (Figure 3.2).

Figure 3.1. Support for same-sex marriage in the United States has nearly doubled in the past 20 years

Evolution of the share of US adults in favour of same-sex marriage between 2004 and 2023

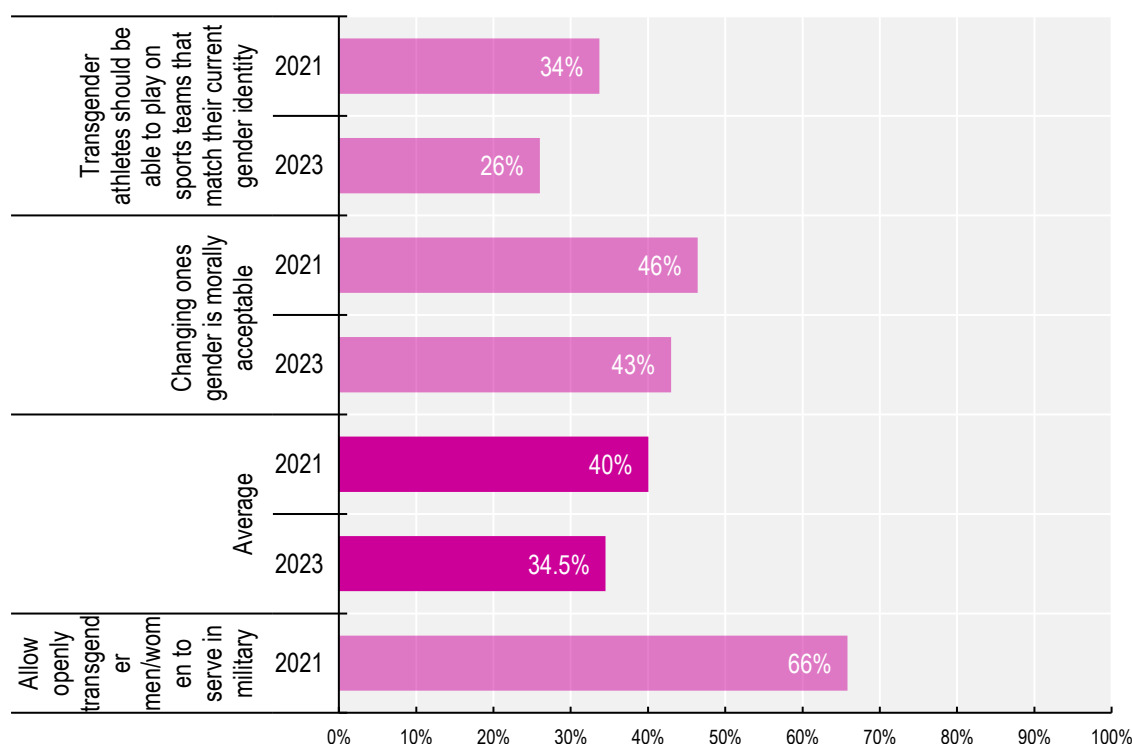


Note: The data stem from the “Values and Beliefs” poll conducted annually in May, as part of the Gallup Poll Social Series. They rely on the following question: “Do you think marriages between same-sex couples should or should not be recognised by the law as valid, with the same rights as traditional marriages?”, with answer options: “Should be valid”, “Should not be recognised as valid”, “Don’t Know”, and refused to answer.

Source: Gallup Poll Social Series (2004-23). Person-level weights used.

Figure 3.2. A minority of US adults are supportive of transgender individuals, and attitudes towards them have deteriorated since 2021

Share of US adults who are supportive of transgender individuals (2021 and 2023)



Note: The data stem from the “Values and Beliefs” poll conducted annually in May, as part of the Gallup Poll Social Series. In 2021 and 2023, this poll included the following three questions: 1) “Next, we have a question about policies for competitive sports that have separate teams for male and female athletes. Do you think transgender athletes – [rotated: should be able to play on sports teams that match their current gender identity (or) should only be allowed to play on sports teams that match their birth gender]?”, with answer options: “Transgender athletes should be able to play on sports teams that match their current gender identity”, “Transgender athletes should only be allowed to play on sports teams that match their birth gender” “Don’t Know”, and refused to answer. 2) “Next, I’m going to read you a list of issues. Regardless of whether or not you think it should be legal, for each one, please tell me whether you personally believe that in general it is morally acceptable or morally wrong. How about – Changing one’s gender”, with answer options: “Morally acceptable”, “Morally wrong”, “Depends on the situation”, “Not a moral issue”, “Don’t Know”, and refused to answer. 3) “Do you favor or oppose allowing openly transgender men and women to serve in the military?”, with answer options: “Favor”, “Oppose”, “Don’t Know”, and refused to answer. The averages for 2021 and 2023 are computed using information from 1) and 2), which were available in both years. It is worth noting that 2021 was not the first year when attitudes towards transgender individuals were assessed. This occurred first in 2016, when respondents were queried about their support for allowing transgender individuals to use restrooms corresponding to their gender identity. Although this question differs from those asked in 2021 and 2023, the results suggest that attitudes towards transgender individuals have deteriorated over time: in 2016, 43% of respondents supported policies allowing transgender individuals to use public restrooms that align with their gender identity, a higher figure than 34.5% of respondents who expressed support for transgender individuals in 2023.

Source: Gallup Poll Social Series (2021 and 2023). Person-level weights used.

3.2.2. Perception of discrimination among LGBTI+ Americans

While there have been strides in the social acceptance of LGBTI+ individuals, particularly cisgender lesbians and gay men, the perception of discrimination continues to be widespread among Americans who identify as non-heterosexual and/or non-cisgender. A US survey conducted in 2022 by the Center for American Progress revealed significant disparities in experiences of discrimination between LGBTI+ and non-LGBTI+ Americans (Medina and Mahowald, 2023^[16]). For this survey, approximately 1 500 non-

LGBTI+ adults were recruited based on probability sampling, and 1 800 LGBTI+ adults using a mix of probability sampling and opt-in panels. The findings indicate that LGBTI+ Americans encounter discrimination at much higher rates than their non-LGBTI+ counterparts. Specifically, LGBTI+ respondents reported nearly double the rate of discrimination in the year preceding the survey, at 36% compared to 19% for non-LGBTI+ respondents. This ratio increases to 3 and 3.5 for non-cisgender and intersex respondents, respectively. These disparities were consistent across various settings, including employment, healthcare, housing, and interactions with law enforcement. For example, although the survey was conducted after the 2020 U.S. Supreme Court's decision to extend non-discrimination protections to LGBT people, 23% of LGBTI+ respondents reported discrimination in the labour market, compared to 17% of non-LGBTI+ respondents. On average, 1 in 5 LGBTI+ respondents reported they had been fired or not hired (22%), denied equal pay or promotion (21%), or subjected to reduced work hours or detrimental changes in work conditions (17%) due to their sexual orientation, gender identity, or intersex status. In addition, (Coffman, Coffman and Marzilli, 2024^[12]) found that non-binary individuals report experiencing more discrimination based on their gender than do men or women across a variety of situations.

Concealment of one's LGBTI+ identity is a widespread phenomenon, often adopted to avoid discrimination. A substantial 78% of LGBTI+ respondents, including 90% of non-cisgender individuals, have reported taking at least one concealment action to avoid experiencing discrimination (Medina and Mahowald, 2023^[16]). These actions include hiding a personal relationship or altering one's dress style, strategies employed by 55% and 39% of LGBTI+ individuals, respectively. High-skilled individuals, such as college graduates, are no exception to this trend. An analysis, grounded in a representative sample of young individuals in the United States who have received a bachelor's degree, revealed that a mere 32% of those identifying as LGBTI+ were open about their identity in family, social, and work environments, with the workplace emerging as the predominant setting for concealment (Folch, 2022^[17]). This pattern is especially prevalent among bisexuals, presumably due to experiencing lower levels of social acceptance (Folch, 2022^[17]; Brown, 2019^[18]; Herek et al., 2010^[9]).

Concealment aimed at avoiding discrimination based on sexual orientation, gender identity, or intersex status is unlikely to significantly improve the lived experiences of LGBTI+ individuals. For example, even if these individuals choose not to disclose their personal lives in professional settings, their labour market outcomes may still be adversely affected. First, people might infer their sexual orientation or gender identity, even inaccurately, based on their gender atypicality – a departure, both in terms of appearance and behaviour, from the social expectations associated with one's assigned sex at birth, such as a boy who enjoys playing with dolls or a girl who is interested in mechanical work. There is evidence suggesting that gender atypical gay men, for example, are more likely to be identified as such by external observers, even those unfamiliar with them (Miller, 2018^[19]; O. Rule and Ambady, 2008^[20]). Similarly, when not outwardly apparent, a transgender identity might still be revealed, such as during employers' review of identity documents or diplomas for transgender individuals who have not undertaken the legal process of changing their gender marker. Additionally, concealing one's identity can obstruct the formation of authentic relationships with colleagues and managers, affecting workplace collaboration, mentorship and support. Specifically, closeted LGBTI+ individuals might avoid networking or professional development opportunities to mitigate the risk of disclosure of their identity (Human Rights Campaign Foundation, 2018^[21]). Furthermore, the constant management of information about their identities and fear of unintended disclosure can result in significant stress and anxiety, impacting job performance, focus, and productivity. The mental and emotional strain of maintaining separate public and private personalities can even contribute to severe mental health disorders (Folch, 2022^[17]; Pachankis et al., 2020^[22]).

3.3. LGBTI+ Americans have lower household income

The Census Bureau's Household Pulse Survey, along with the surveys that Gallup has conducted since 2012, include questions on annual household income (Box 3.1). This section first examines household

income by sexual orientation and gender identity, including through a dynamic perspective to explore the impact of the COVID-19 pandemic. It then further disaggregates the data by sex assigned at birth and age, before discussing the reasons behind the observed gaps.

Box 3.1. Turning categorical household income variable into a continuous variable

This Box presents the questions related to household income that are asked in the HPS and in the Gallup surveys and explains how we turn these categorical variables into continuous variables, through the use of interval regression.

1) Household income categorical survey questions:

Household Pulse Survey (2021-23) question: “In [previous year] what was your total household income before taxes? Select only one answer.”, with answer categories: 1) Less than \$25,000, 2) \$25,000 – \$34,999, 3) \$35,000 – \$49,999, 4) \$50,000 – \$74,999, 5) \$75,000 – \$99,999, 6) \$100,000 – \$149,999, 7) \$150,000 – \$199,999, 8) \$200,000 and more.

Gallup US Daily Survey (2012-17) question: “What is your total ANNUAL household income, before taxes? Please include income from wages and salaries, remittances from family members living elsewhere, farming, and all other sources”, with answer categories: 1) Under \$720, 2) \$720 to \$5 999, 3) \$6,000 to \$11,999, 4) \$12,000 to \$23,999, 5) \$24,000 to \$35,999, 6) \$36,000 to \$47,999, 7) \$48,000 to \$59,999, 8) \$60,000 to \$89,999, 9) \$90,000 to \$119,999, 10) \$120,000 and over.

Gallup Panel – COVID-19 Survey (2020) “What is your total annual household income before taxes?”, with answer categories: 1) Less than \$12,000, 2) \$12,000 to \$23,999, 3) \$24,000 to \$35,999, 4) \$36,000 to \$47,999, 5) \$48,000 to \$59,999, 6) \$60,000 to \$89,999, 7) \$90,000 to \$119,999, 8) \$120,000 to \$179,999, 9) \$180 000 to \$239,999, 10) \$240,000 and over.

2) Interval regression:

We then use an interval regression to turn these categorical variables into continuous variables, by making use of the “`intreg`” command in Stata.

An interval regression fits a linear model to an outcome that is unobserved but known to fall within some interval. Such censored data arise naturally in many contexts, such as income data. In this case, the exact values of household income are unobserved but known to fall within an interval with fixed endpoints (interval-censored data), unobserved but known to fall within an interval that has a fixed upper endpoint (left-censored data), and unobserved but known to fall within an interval that has a fixed lower endpoint (right-censored data).

The “`intreg`” command requires the outcome to be stored in the dataset as interval data. That is, two variables, “`income1`” and “`income2`”, are used to hold the endpoints of the interval. For the left-censored data, the lower endpoint is assumed to be minus infinity and is represented by a missing value in “`income1`”. For the right-censored data, the upper endpoint is assumed to be infinity and is represented by a missing value in “`income2`”. The other interval-censored data are represented by the two endpoints being equal.

Finally, we turn the resulting two variables (“`income1`” and “`income2`”) in logarithms. The interval regression model assumes normality, but the distribution of income is skewed and definitely non-normal. Thus, normality is more closely approximated if we model the logarithm of income.

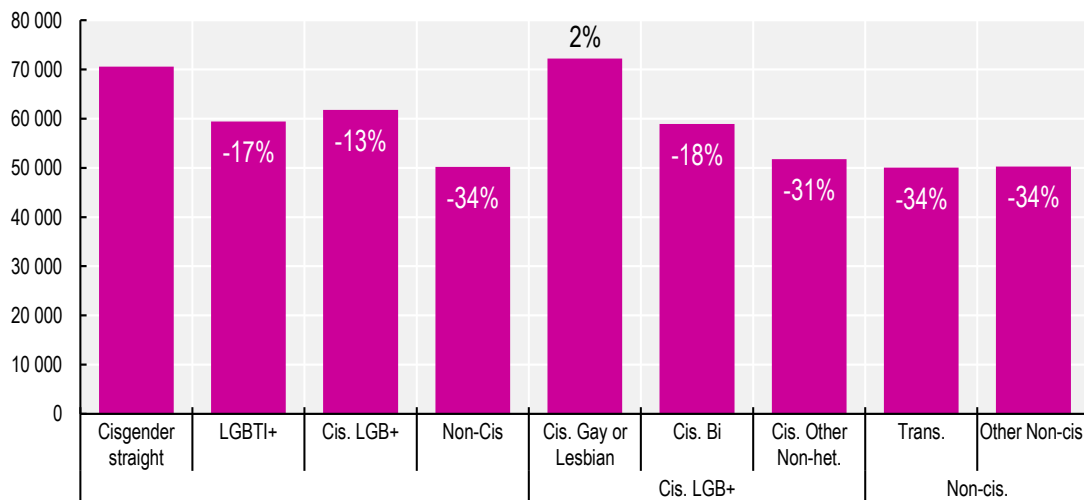
Source: Household Pulse Survey, Gallup, and Stata documentation (www.stata.com/manuals/rintreg.pdf)

3.3.1. Comparing household income by sexual orientation and gender identity

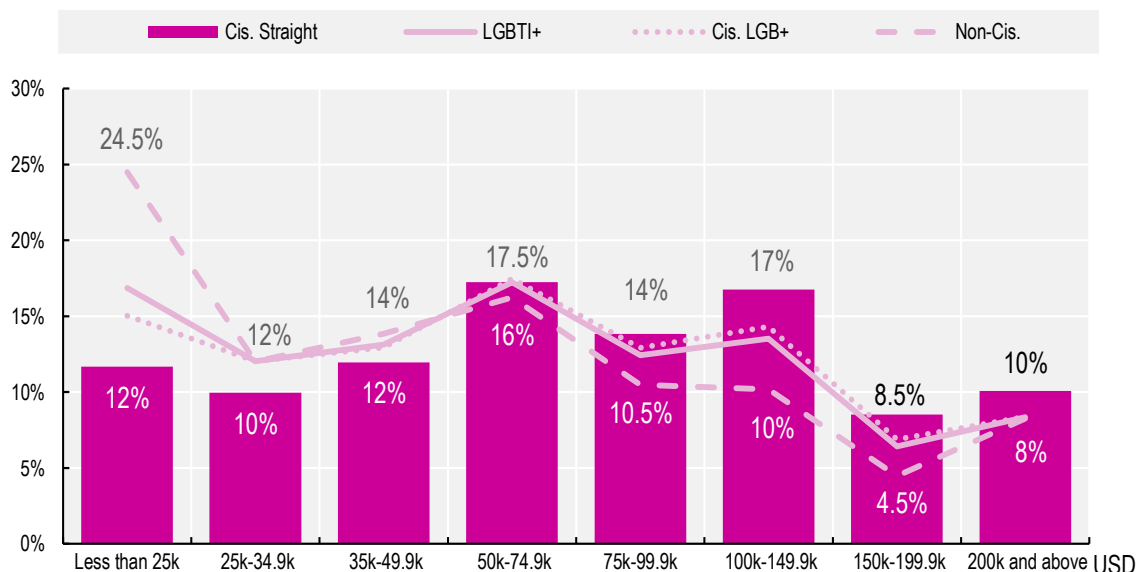
In 2023, LGBTI+ adults had an average annual household income that was 17% lower than that of cisgender straight adults, as shown in Panel A of Figure 3.3. Breaking it down further, cisgender LGB+ adults faced a 13% penalty. Notably, the subgroup of gay men and lesbians faced no penalty (indeed, they face a small premium), while penalties were 18% for bisexuals and 31% for other non-heterosexuals. Non-cisgender individuals experienced a more severe penalty at 34%, prevailing both for transgender and other non-cisgender individuals. Correspondingly, as depicted in Panel B of Figure 3.3, LGBTI+ Americans were disproportionately represented in lower income brackets and under-represented in higher ones compared to cisgender straight peers. Specifically, 17% of LGBTI+ adults lived in households earning less than USD 25 000 annually, in contrast to 12% of cisgender straight adults. On the higher end, 28% of LGBTI+ adults resided in households with incomes of USD 100 000 or above, compared to 35.5% of cisgender straight adults. The disparity was especially pronounced among non-cisgender adults. Almost one in four (24.5%) non-cisgender individuals lived in households earning less than USD 25 000 annually, while this was the case for only 15% of cisgender LGB+ adults.

Figure 3.3. LGBTI+ Americans are over-represented in low-income brackets and under-represented in high-income brackets, compared to their cisgender straight peers

Panel A: Average annual household income among US adults, by sexual orientation and gender identity (2023)



Panel B: Distribution of annual household income among US adults, by sexual orientation and gender identity (2023)



Note: Average annual household income is estimated from an Interval regression of the logarithm of household income. Person-level weights used. See Box 3.1 for more details on Interval regression methodology. In Panel A, the percentages represent the percentage difference in the estimated average annual household income between LGBTI+ (or LGBTI+ subgroups) and cisgender straight adults. In Panel B, the percentages represent the two population groups with the lowest (in white) and with the highest (in black) percentage within each income bracket. For instance, 12% of cisgender straight adults and 24.5% of non-cisgender adults have an annual household income below USD 25 000, while 8% of non-cisgender adults and 10% of cisgender straight adults have an annual household income above USD 200 000.

Source: Household Pulse Survey (4 January 2023 – 10 July 2023).

An analysis of combined 2021 and 2023 HPS data underscores that these income differences, although substantially reduced, remain significant after accounting for the demographic differences illustrated in Chapter 2 (Figure 3.4). On average, LGBTI+ adults experience an 8% lower annual household income than cisgender straight adults. The income penalty is more pronounced for non-cisgender individuals (11%) than for cisgender LGB+ adults (7%). Among LGBTI+ subgroups, only cisgender homosexuals do not face a significant income penalty. Cisgender bisexual adults have an 8.5% lower household income compared to cisgender straight adults. In contrast, cisgender adults with other non-heterosexual orientations experience an income penalty that is roughly twice as large (16.5%). For non-cisgender adults, there is no significant difference in the household income penalties between transgender individuals (10%) and those of other non-cisgender orientations (12%).

An analysis that considers how different forms of disadvantages combine shows that the household income disparities experienced by LGBTI+ individuals are more severe for those who also belong to other marginalised groups, specifically individuals assigned female at birth (see Box 3.2). This increased disadvantage, however, does not apply to LGBTI+ individuals who are non-White or Hispanic or disabled, highlighting the complex interplay of various identities.

Box 3.2. An intersectional analysis

An intersectional analysis recognises that individuals hold multiple identities and examines how different facets of their identity interact and shape individuals' experiences. The scarcity of research that combines an intersectional analysis of LGBTI+ identities with other marginalised statuses flows from the small sample size of the LGBTI+ population in representative datasets. The Household Pulse Survey, with its extensive data on LGBTI+ adults, provides a unique opportunity for such analysis. Relying on this dataset, (Carpenter, Lee and Nettuno, 2022^[23]) identified disparities in employment rates among non-cisgender Black individuals compared to their non-cisgender White counterparts, regardless of their sex assigned at birth. In a similar vein, (Martell and Roncolato, 2023^[24]) found that racial and ethnic minorities faced heightened disadvantages in some employment outcomes and some measures of economic vulnerability when also identifying as gay/lesbian or bisexual. However, for many of these outcomes, the intersectional impact of being LGBTI+ didn't exacerbate the racial/ethnic disadvantage.

Building on this method, we expand the regression used to estimate the unexplained LGBTI+ household income gap shown in Figure 3.4 and run three distinct regressions by adding: 1) an interaction term for LGBTI+ identity with sex assigned at birth, 2) an interaction term for LGBTI+ identity with race/ethnicity (defined using a binary variable that is zero for White non-Hispanic respondents and one for respondents who are non-White or Hispanic), and 3) a variable indicating whether the respondent has a disability, along with its interaction with LGBTI+ identity. Moreover, we run the regression for each intersection – race/ethnicity and disability with LGBTI+ identity – separately among AFAB and AMAB individuals.

The regression results indicate a 27% income penalty for LGBTI+ adults assigned female at birth compared to cisgender straight adults assigned male at birth. Additionally, LGBTI+ adults who are non-White or Hispanic encounter a 26% income disadvantage compared to their White non-Hispanic cisgender straight counterparts, while LGBTI+ adults with disability face a similar income reduction when compared to cisgender straight adults without disability. These penalties are consistent across individuals assigned male or female at birth. However, our analysis does not reveal a cumulative effect of race/ethnicity and disability on the LGBTI+ income gap for either AFAB or AMAB individuals. Specifically, while AFAB LGBTI+ individuals endure an income penalty relative to AFAB cisgender straight individuals, non-White or Hispanic LGBTI+ individuals do not suffer an additional income penalty relative to their non-White or Hispanic cisgender straight peers, nor do LGBTI+ individuals with disability compared to cisgender straight individuals with disability. These findings emphasise the need for further research to thoroughly comprehend the layered effects of possessing multiple marginalised identities.

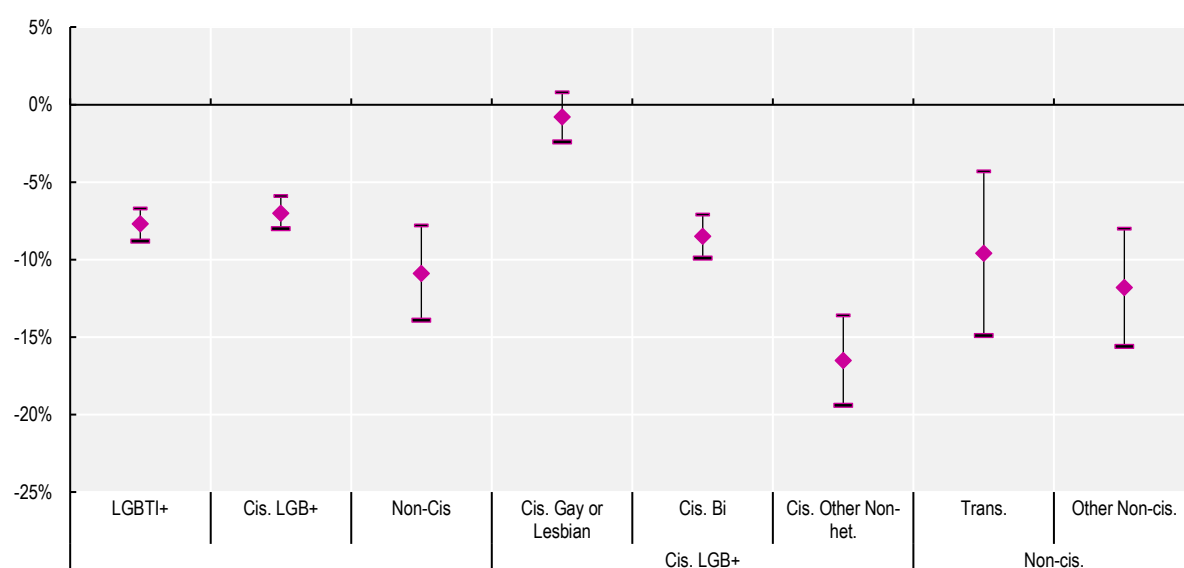
Note: The Household Pulse Survey (HPS) includes six disability-related questions: 1) "Do you have difficulty seeing, even when wearing glasses?", 2) "Do you have difficulty hearing, even when using a hearing aid?", 3) "Do you have difficulty walking or climbing stairs?", 4) "Do you have difficulty remembering or concentrating?", 5) "Do you have difficulty with self-care, such as washing all over or dressing?", 6) "Using your usual language, do you have difficulty communicating, for example understanding or being understood?". In line with the Centers for Disease Control and Prevention (CDC) guidelines, adults who report "a lot of difficulty" or an inability to perform any of these six functions (seeing, hearing, cognition, mobility, self-care, and communication) are classified as having a disability. Conversely, those who indicate "no difficulty" or "some difficulty" across all six areas are not classified as having a disability. According to this metric, it is estimated that 12% of the US adult population has a disability, with 2% identified as both LGBTI+ and disabled.

Several scholars analysing the Household Pulse Survey data from July 2021 to May 2022 have found results consistent with those reported in Figure 3.4 (Martell and Roncolato, 2023^[24]; Carpenter, Lee and Nettuno, 2022^[23]). Research from alternative datasets confirms these findings. For example, homosexual adults were found to have a similar poverty rate as straight adults. However, bisexuals and other sexual

minorities were more likely to be impoverished than their straight counterparts with the same demographic characteristics. These results are consistent across different datasets, including the National Health Interview Survey (Badgett, Carpenter and Sansone, 2021^[25]; Badgett, 2018^[26]), the General Social Survey (Chai and Maroto, 2019^[27]) and the Behavioural Risk Factor Surveillance System (Badgett, Choi and Wilson, 2019^[28]). The larger income penalty for non-cisgender individuals was also confirmed based on the latter survey: transgender adults fare worse economically, with higher poverty rates compared to both cisgender LGB and straight adults (Carpenter, Eppink and Gonzalez, 2020^[29]; Badgett, Choi and Wilson, 2019^[28]). The 2022 survey by the Center for American Progress further sheds light on this disparity. It revealed that 34% of intersex LGBTI+ respondents had a household income of less than USD 25 000 annually, compared to 27% of non-intersex LGBTI+ respondents (Medina and Mahowald, 2021^[30]).

Figure 3.4. After demographic adjustments, LGBTI+ Americans experience an 8% lower annual household income, compared to their cisgender straight peers

Percentage difference in annual household income for US adults (using cisgender straight individuals as the reference category), by sexual orientation and gender identity (2021-23)



Note: The percentage differences are estimated from an Interval regression of the logarithm of household income, controlling for sex assigned at birth, age groups (18-24, 25-29, 30-34, ..., 80-84, 85-88), race and ethnicity, marital and parental status, number of adults in the household, educational attainment, living in one of the 15 largest metropolitan statistical areas (metro areas), as well as state and survey wave fixed effects. The error bars depict 95% confidence intervals. This means that we can be 95% confident that the true value lies within the range of the error bar. In other words, if the error bar crosses the x-axis, then the percentage point difference cannot be deemed as statistically different from 0 (at least not at this 95% confidence level). Standard errors robust to heteroscedasticity. Person-level weights used. See Box 3.1 for more details on Interval regression methodology.

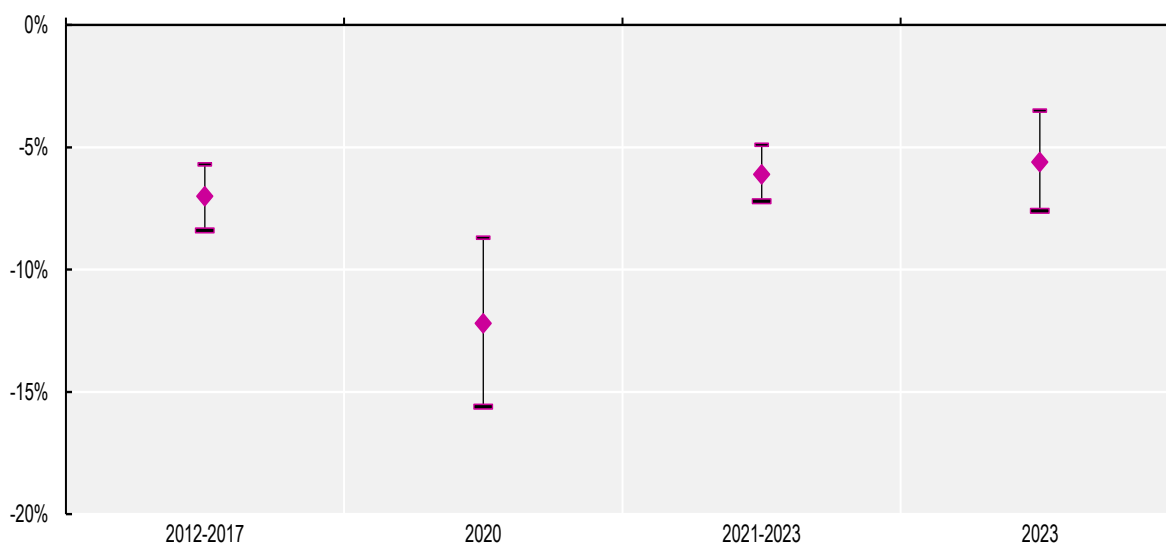
Source: Household Pulse Survey (21 July 2021 – 10 July 2023).

An analysis of income disparities for LGBT people over time indicates that the COVID-19 pandemic exacerbated their economic challenges (Figure 3.5). Between 2012 and 2017, the LGBT income penalty estimated from the Gallup US Daily was 7%, which aligns with the 6% LGBT income penalty observed in the HPS data from 2021 to 2023. However, from April to December 2020, the Gallup COVID-19 survey recorded a significantly higher LGBT income penalty of 12.5%. These results are consistent with previous findings suggesting that, notably due to their lower incomes, LGBTI+ adults are especially vulnerable during crises (OECD, 2021^[31]; Wenham, 2020^[32]; Movement Advancement Project, 2020^[33]). It is worth

noting that the 6% income penalty observed when pooling data from 2021 to 2023 remains virtually unchanged when focusing solely on 2023 (Figure 3.5). This consistency suggests that the penalty shown in Figure 3.4 does not merely reflect the impact of the COVID-19 pandemic, but a structural disadvantage for LGBTI+ individuals.

Figure 3.5. The LGBT income penalty was exacerbated at the onset of the COVID-19 pandemic

Percentage difference in annual household income for LGBT US adults, using non-LGBT individuals as the reference category (2012-23)



Note: The percentage differences are estimated from an Interval regression of the logarithm of household income, controlling for sex assigned at birth, age groups (18-24, 25-29, 30-34, ..., 80-84, 85-88), race and ethnicity, marital and parental status, number of adults in the household, educational attainment, living in one of the 15 largest metropolitan statistical areas (metro areas) – only in the HPS data –, as well as state and survey wave fixed effects. The error bars depict 95% confidence intervals. This means that we can be 95% confident that the true value lies within the range of the error bar. In other words, if the error bar crosses the x-axis, then the percentage point difference cannot be deemed as statistically different from 0 (at least not at this 95% confidence level). Standard errors robust to heteroscedasticity. Person-level weights used. See Box 3.1 for more details on Interval regression methodology.

Source: Gallup US Daily Survey (2012-17), Gallup Panel – COVID-19 Survey (March – December 2020), Household Pulse Survey (21 July 2021 – 10 July 2023).

3.3.2. Disaggregating household income gaps by sex assigned at birth and age

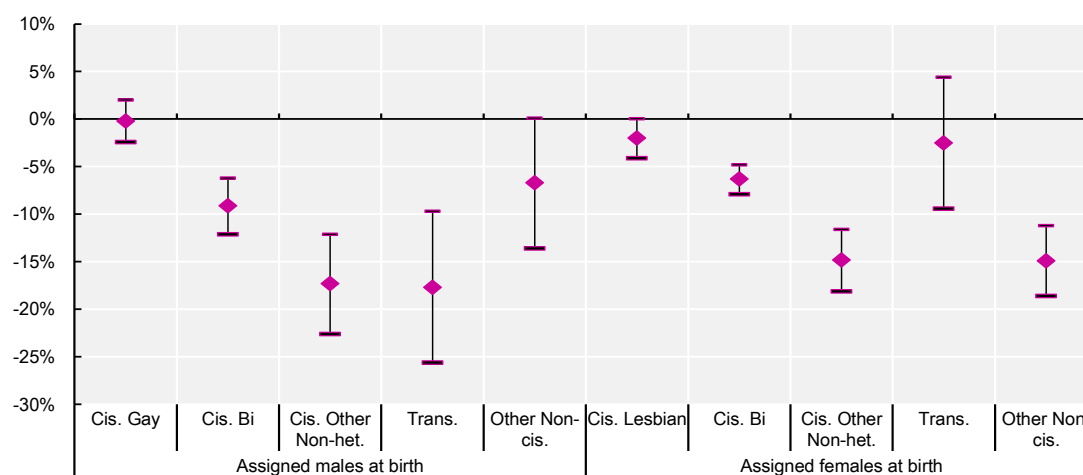
When examining household income disparities by sex assigned at birth (Figure 3.6), the patterns observed in Figure 3.4 for cisgender LGB+ adults are consistent for both those assigned males at birth (AMAB) and those assigned females at birth (AFAB). Specifically, regardless of sex assigned at birth, the income penalty is smallest for gay men and women, larger for bisexual men and women, and largest for cisgender men and women with other non-heterosexual orientations. Nevertheless, within these subgroups, the magnitude of the penalty does vary by sex assigned at birth. For example, while the penalty for cisgender gay men isn't statistically significant, cisgender lesbians face a discernible, albeit modest, significant penalty (2%). In comparison, cisgender bisexual and other non-heterosexual individuals have a more pronounced penalty if they are AMAB than if they are AFAB. It should be emphasised that the absence of a significant unexplained household income gap for cisgender gay men does not indicate they are unaffected by discrimination. In fact, in line with their educational premium highlighted in Chapter 2, this absence could mirror their effort to bolster a broad spectrum of cognitive and socioemotional skills as a compensatory strategy against discrimination. Given this extensive personal investment, the lack of an

observable income advantage for them hints at opposing forces at play, such as exclusionary attitudes and behaviours.

Unlike the patterns observed for cisgender LGB+ adults, the penalty that prevails among non-cisgender adults – suggesting a similar household income gap for transgender and other non-cisgender individuals – masks differences by sex assigned at birth. Among AMAB individuals, the gap with cisgender straight peers is most pronounced for transgender individuals. In contrast, among AFAB individuals, other non-cisgender individuals experience the strongest disadvantage. Notably, the household income difference for transgender individuals assigned female at birth is not statistically significant when compared to cisgender straight females. This finding aligns with prior research. In particular, studies have indicated that transgender men in the United States may experience a slight increase in labour earnings after transitioning, suggesting that the societal advantages associated with being recognised as a man may compensate the stigma associated with transitioning (Schilt and Wiswall, 2008^[34]). According to Figure 3.6, this increase in earnings is indeed not sufficient to translate into an improved economic situation for transgender individuals assigned female at birth in comparison to cisgender straight women. One explanation could be that only a minority of these transgender individuals are trans men who have completed their gender transition. Conversely, transitioning has been found to reduce wages for transgender women in the United States, a trend that aligns with the stigma of transitioning combining with the societal disadvantages of being recognised as a woman (Schilt and Wiswall, 2008^[34]). Consistent with this surmise, Figure 3.6 shows a 18% income penalty for transgender individuals assigned male at birth when compared to cisgender straight men.

Figure 3.6. Patterns of income disparities by sex assigned at birth are similar among cisgender LGB+ Americans but reversed among their non-cisgender peers

Percentage difference in annual household income for US adults (using cisgender straight individuals as the reference category), by sexual orientation, gender identity, and sex assigned at birth (2021-23)



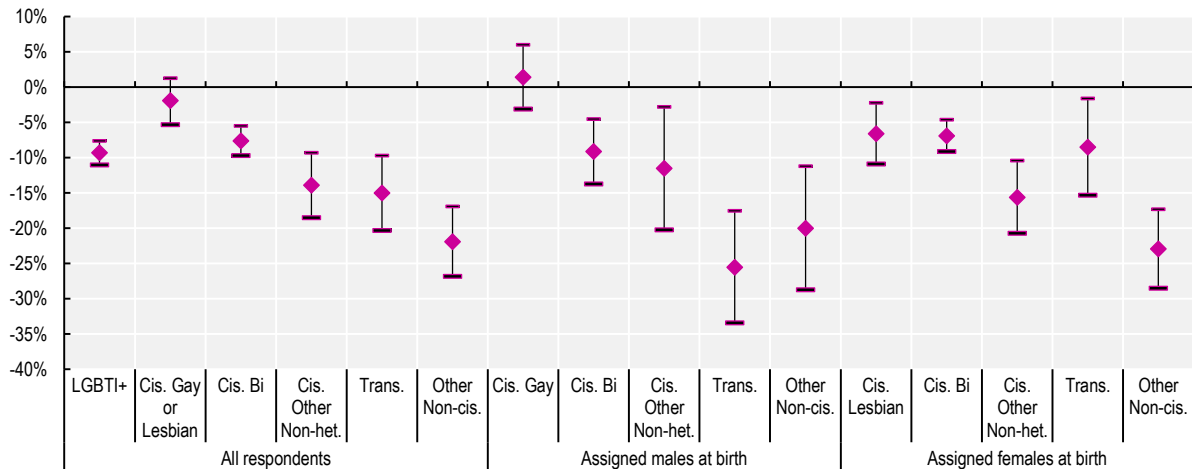
Note: The percentage differences are estimated from an Interval regression of the logarithm of household income, controlling for age groups (18-24, 25-29, 30-34, ..., 80-84, 85-88), race and ethnicity, marital and parental status, number of adults in the household, educational attainment, living in one of the 15 largest metropolitan statistical areas (metro areas), as well as state and survey wave fixed effects. The error bars depict 95% confidence intervals. This means that we can be 95% confident that the true value lies within the range of the error bar. In other words, if the error bar crosses the x-axis, then the percentage point difference cannot be deemed as statistically different from 0 (at least not at this 95% confidence level). Standard errors robust to heteroscedasticity. Person-level weights used. See Box 3.1 for more details on Interval regression methodology.

Source: Household Pulse Survey (21 July 2021 – 10 July 2023).

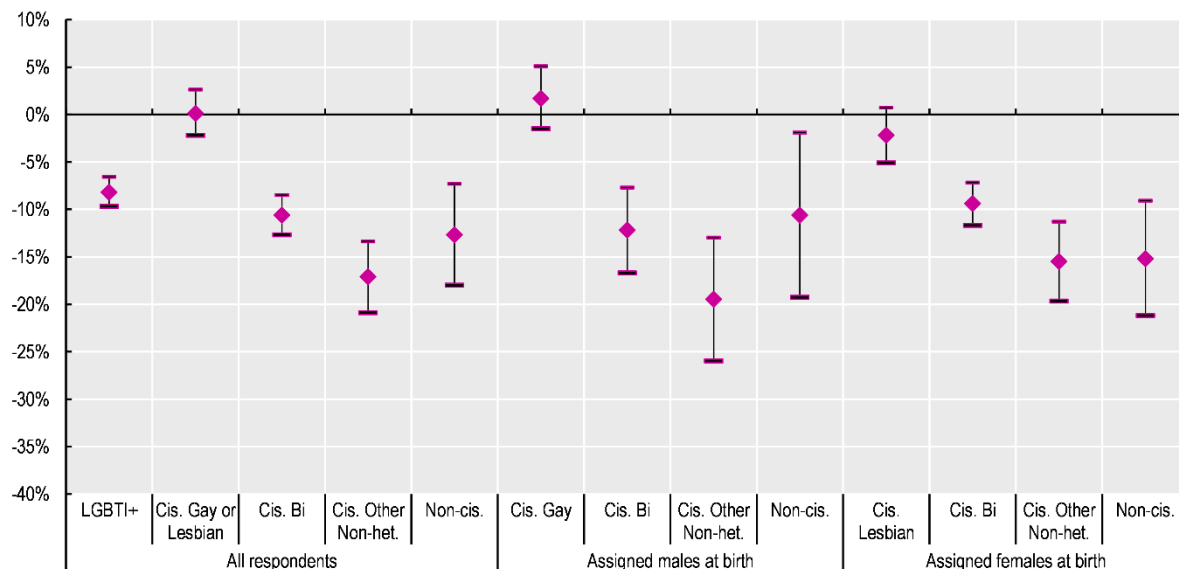
The income penalty for LGBTI+ individuals decreases with age. Among young people aged 18-34, the disparity is 9%, which changes to 8% for those in their prime age (35-54), and 3% for mature people (55+), as shown in Figure 3.7. When focusing on cisgender LGB+ adults, cisgender bisexuals and other non-heterosexual individuals face income disparities across all age groups. This pattern holds true for both AMABs and AFABs. As for cisgender homosexuals, they exhibit a different trend: while mature gay men and lesbians have a slight statistically significant income premium, this advantage disappears for younger generations, becoming a significant penalty (7%) for young cisgender lesbians. Regarding non-cisgender adults, both young people and those in their prime age experience a significant income penalty (a trend that is consistent for both AMABs and AFABs), while no disadvantage is observed among mature non-cisgender adults. Chapter 2 delved into one potential explanation behind the modest income premium for mature cisgender homosexuals and the non-significant income penalty for mature non-cisgender adults, suggesting the influence of a “non-disclosure bias” (Valfort, 2017^[3]): mature LGBTI+ adults from privileged background might be more willing to identify openly as LGBTI+ than those from less privileged backgrounds. Another perspective, developed in Section 3.5, touches upon health disparities. In a context where LGBTI+ individuals show poorer health outcomes, it is possible that self-identified LGBTI+ adults who lived up to a mature age are more likely to come from advantaged backgrounds that equipped them to more easily navigate health challenges.

Figure 3.7. The LGBTI+ income penalty in the United States decreases with age

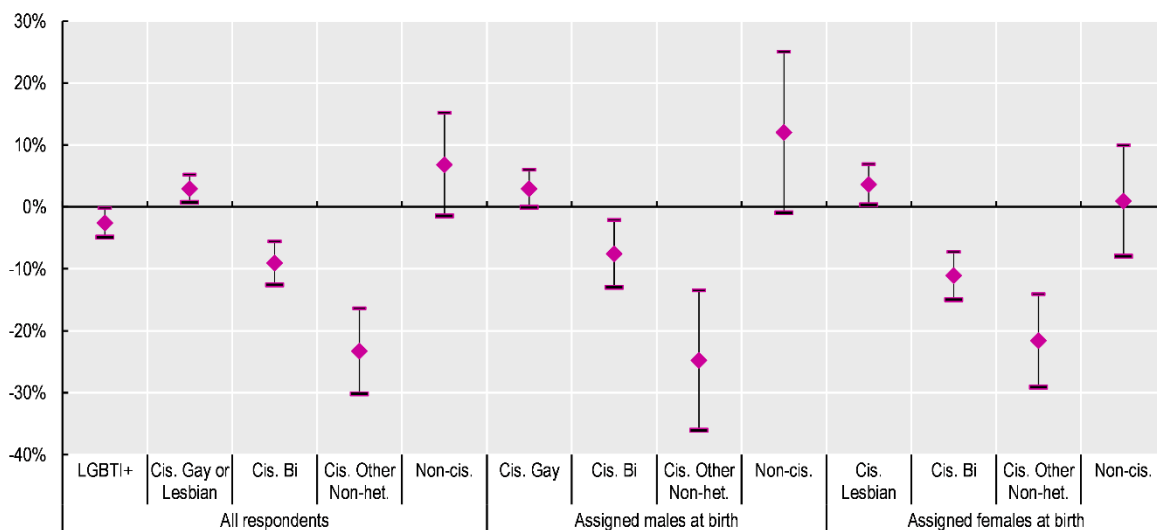
Panel A: Percentage difference in annual household income for US adults aged between 18 and 34 (using cisgender straight individuals as the reference category), by sexual orientation, gender identity, and sex assigned at birth (2021-23)



Panel B: Percentage difference in annual household income for US adults aged between 35 and 54 (using cisgender straight individuals as the reference category), by sexual orientation, gender identity, and sex assigned at birth (2021-23).



Panel C: Percentage difference in annual household income for US adults above age 55 (using cisgender straight individuals as the reference category), by sexual orientation, gender identity, and sex assigned at birth (2021-23).



Note: The percentage differences are estimated from an Interval regression of the logarithm of household income, controlling for age groups, sex assigned at birth (only when the analysis is performed over all respondents), race and ethnicity, marital and parental status, number of adults in the household, educational attainment, living in one of the 15 largest metropolitan statistical areas (metro areas), as well as state and survey wave fixed effects. The error bars depict 95% confidence intervals. This means that we can be 95% confident that the true value lies within the range of the error bar. In other words, if the error bar crosses the x-axis, then the percentage point difference cannot be deemed as statistically different from 0 (at least not at this 95% confidence level). Standard errors robust to heteroscedasticity. Transgender and other non-cisgender were not separately analysed for prime age and mature adults due to low sample size of transgender individuals. Person-level weights used. See Box 3.1 for more details on Interval regression methodology.

Source: Household Pulse Survey (21 July 2021 – 10 July 2023).

3.3.3. Accounting for the unexplained household income penalty for LGBTI+ adults

Several factors, all arising from the discriminatory contexts that LGBTI+ individuals encounter, have been identified as key drivers to the unexplained household income penalty they face. Sections 0 and 3.5 delve into the most significant of these factors. They shed light on the disparities LGBTI+ individuals experience with respect to labour market and health outcomes, bearing in mind that these disparities are not independent of each other but mutually reinforcing. On the one hand, discrimination in the labour market can adversely impact health, through different channels. For instance, a study in Italy indicates that prior experiences of workplace discrimination are closely linked to the decision of transgender individuals, who are notably overrepresented among prostitutes, to pursue sex work – a profession with a high prevalence of HIV and other sexually transmitted diseases (D'Ippoliti and Botti, 2017^[35]). Conversely, health challenges faced by LGBTI+ individuals can decrease their productivity at work, further perpetuating labour market discrimination against them, especially in places like the United States where the stigmatisation of mental health issues remains prevalent (OECD, 2021^[36]).

Beyond labour market and health gaps, several other factors impact the economic situations of LGBTI+ adults, of which three stand out. Firstly, LGBTI+ individuals are often less likely to receive family support. Some are expelled from their homes after disclosing their LGBTI+ identity, leading to housing crises for a number of LGBTI+ adolescents (National Academies of Sciences, Engineering, and Medicine, 2020^[37]; Dunne, Prendergast and Telford, 2002^[38]; Rew, Fouladi and Yockey, 2002^[39]). For instance, the 2014 LGBTQ Homeless Youth Provider Survey indicated an overrepresentation of LGBTI+ youth among homeless people (close to 30%). Among the respondents, 55% of LGB individuals and 67% of transgender individuals attributed their homelessness to either being evicted or fleeing their homes due to their sexual orientation and/or gender identity (Choi et al., 2015^[40]). In addition, according to a Pew Research Center survey conducted in 2013, about four-in-ten (39%) LGBT adults said that at some point in their lives they were rejected by a family member or close friend because of their sexual orientation or gender identity (Pew Research Center, 2013^[41]). Consistently, LGBTI+ individuals are also less likely to receive economic support from their families during challenging times or to finance their education (Badgett, 2014^[42]).

Secondly, evidence suggests that LGBTI+ individuals face discrimination when trying to access credit, which hampers their capacity to build wealth. It has been well-established that same-sex couples are less likely to own homes compared to their different-sex peers, and when they do own, they are less likely to have a mortgage (National Academies of Sciences, Engineering, and Medicine, 2020^[37]; Jepsen and Jepsen, 2009^[43]; Leppel, 2007^[44]; Leppel, 2007^[45]). After income and demographic adjustments, the Household Pulse Survey confirms these findings: self-identified LGBTI+ adults are 5 percentage points (7%) less likely to be homeowners, compared to cisgender straight adults. This penalty is significant across all LGBTI+ subgroups, with transgender facing the most significant penalty. In addition, all these penalties remain even when restricting the sample to married adults living with only another adult (likely the spouse).³ A recent study uncovers a potential driver by showing that same-sex couples experience discrimination in mortgage lending. This finding is predominantly driven by applications from male same-sex pairs – the treatment of female same-sex co-applicants, in contrast, appears indistinguishable from that of different-sex couples (Dillbary and Griffin, 2019^[46]). Specifically, an analysis of over 5 million mortgage applications to the Fair Housing Administration (FHA) reveals that mortgage applications by same-sex male co-applicants are significantly less likely to be approved than those from heterosexual couples, even though FHA-insured mortgage lenders are prohibited from discriminating based on sexual orientation. This result holds true although the same-sex male pairs were identical to their different-sex peers in all reported respects: they filed a mortgage application with the same lender, in the same county, for the same loan amount, for the same purpose, had the same income, and posed the same level of risk to the lender.

Thirdly, different expectations about future family formation may affect not only career choices, but also investment and savings decisions (Coffman, Coffman and Marzilli, 2024^[12]; Tate and Patterson, 2019^[47]). As recalled in Chapter 2, despite recent increases in partnership and marriage rates among LGBTI+ adults,

many still have low expectations about becoming parents. Growing reliance on adoption, assisted reproductive technology, and surrogacy among LGBTI+ individuals is mitigated by difficult access to these methods, due to a combination of legal, financial, and discriminatory barriers.⁴

3.4. LGBTI+ Americans face substantial hurdles in the labour market

This section first provides a brief overview of previous research on anti-LGBTI+ discrimination, relying on both experimental and observational data. It then takes advantage of HPS and Gallup data to compare labour market outcomes by sexual orientation and gender identity, focusing on unexplained gaps in both employment rates and labour earnings.

3.4.1. Anti-LGBTI+ discrimination in the labour market: an overview of previous research

Objective measures of discrimination are consistent with the subjective perceptions of discrimination reported by LGBTI+ individuals. As detailed in Section 3.2, an average of 1 in 5 LGBTI+ respondents report experiences such as being fired, not hired, denied equal pay or promotion, or subjected to reduced work hours or detrimental changes in work conditions due to their sexual orientation, gender identity, or intersex status (Medina and Mahowald, 2023^[16]). These perceptions align with employers' attitudes and behaviours, as observed in laboratory or field experiments conducted in the United States. They are also in line with unexplained labour market gaps between LGBTI+ and non-LGBTI+ individuals, based on representative survey data. Against this backdrop, it is no surprise that thousands of people in the United States filed charges of employment discrimination based on sexual orientation or gender identity between 2013 and 2016 (Baumle, Badgett and Boutcher, 2019^[48]).

Objective measures of anti-LGBTI+ discrimination in the labour market stemming from experimental data

In a recent laboratory experiment at a public Midwestern University involving introductory psychology students, participants assuming the role of employers were tasked with evaluating job applications. The results showed that, with the same CV, gay and/or lesbian applicants were rated significantly lower in competence, social skills and employability compared to heterosexual applicants. This gap widened with the degree to which the “employers” endorsed traditional gender norms (Bryant-Lees and Kite, 2021^[49]).

In the field, similar results are obtained from correspondence studies. These involve sending out, in response to real job ads, the CVs and letters of application of fictitious candidates who are identical except for their sexual orientation or gender identity. A male candidate indicating he worked as treasurer for the gay and lesbian campus organisation has 60% fewer chances of being invited to a job interview than his straight counterpart with experience as treasurer in another progressive student association (Tilcsik, 2011^[50]). Similarly, a female candidate highlighting leadership roles in a lesbian, gay, bisexual, and transgender student organisation to signal queer identity receives 30% fewer callbacks than her cisgender straight peer who, with similar experience in a non-LGBT progressive student organisation, is perceived by default by employers as a cisgender straight woman (Mishel, 2016^[51]).

Field experiments focused specifically on transgender individuals have consistently uncovered significant discrimination. In one correspondence study, researchers compared the callback rate for fictitious male-to-female transgender candidates with those of fictitious cisgender female candidates applying for high-skilled jobs in Texas (Bardales, 2013^[52]). The gender identity of the fictitious candidates was conveyed using three key pieces of information on the CV: i) the transgender woman listed her preferred name alongside her legal name, e.g. “Anne McCarthy (Legal Name: Greg McCarthy)”, while the cisgender woman only mentioned her legal name; ii) the transgender woman highlighted her membership in the “Transgender Women’s Support Group at UT San Antonio”, whereas the cisgender woman noted her involvement in the

“Women’s Health Center at UT San Antonio”; iii) the transgender woman described her role in “Male-to-Female Youth Peer Counseling”, while the cisgender woman highlighted her participation in “Young Girls Peer Counseling and Mentorship”. The study’s findings were stark: the cisgender woman’s callback rate was 50% higher than that of the transgender woman. This outcome aligns with results from a 2008 small-scale experiment in Manhattan’s retail sector. In that study, out of 24 tested employers, male-to-female, female-to-male, and gender-nonconforming transgender applicants were six times less likely to receive a job offer compared to their cisgender peers. Specifically, while 50% of cisgender applicants received job offers, only 8% of transgender applicants did (Make the Road New York, 2010^[53]).

Recently, a large-scale correspondence study conducted in the United States from 2019 to 2021 and involving over 100 Fortune 500 companies, offered additional insights (Kline, Rose and Walters, 2022^[54]; Kline, Rose and Walters, 2024^[55]). While primarily focused on assessing gender- and race-based discrimination, the study also explored other legally protected characteristics, including age and LGBTI+ identity. The findings reveal a penalty for white applicants who indicate LGBTI+ club membership on their resumes. This result emerges despite the study focusing on large firms, which are typically thought to exhibit less discriminatory behaviour due to more standardised hiring procedures. In contrast, indicating LGBTI+ club membership appears to benefit Black applicants, underscoring, following Box 3.2, the intricate interplay of different identities.

It is important to emphasise that correspondence studies likely understate the extent of hiring discrimination against LGBTI+ job candidates since they do not analyse subsequent job interview outcomes. For instance, several field experiments addressing racial and ethnic discrimination in hiring combine correspondence studies with audit studies. In these audit studies, actors represent the fictitious applicants during actual job interviews. Evidence from such combined studies reveals significant second-stage discrimination: candidates representing the racial or ethnic majority not only receive 53% more callbacks but also continue to receive 145% more job offers compared to their minority counterparts (Quillian, Lee and Oliver, 2020^[56]). This trend is likely to gain importance given the growing reliance of employers on automated CV screening tools. Such tools might push discriminatory practices further down the hiring process, making face-to-face interviews the stage where biases are most likely to emerge.

Unexplained labour market gaps between LGBTI+ and non-LGBTI+ individuals, based on representative survey data

Evidence from representative survey data indicates that discrimination – or the fear of discrimination leading to the development of detrimental coping strategies such as concealment or covering behaviours (Section 3.2.2) – adversely affects the labour market trajectories of LGBTI+ individuals. Since Badgett’s seminal 1995 article, which showed a significant individual labour earnings penalty for gay and bisexual men (Badgett, 1995^[57]), a large body of research has explored labour market disparities based on sexual orientation and gender identity. A review of these studies reveals significant penalties for LGBTI+ individuals in terms of employment rates, hours worked and hourly labour earnings, even when accounting for essential demographics – as well as, for those employed, accounting for economic sectors and occupations (Drydak, 2022^[58]; OECD, 2019^[1]; Valfort, 2017^[3]; Klawitter, 2015^[59]; Carpenter, Lee and Nettuno, 2022^[23]).

The labour market penalty found by these studies is consistent across all LGBT subgroups, with the notable exception of lesbians. This leads to contrasting results within the homosexual population: lesbians experience a labour market premium compared to heterosexual women, while gay men face a labour market penalty in comparison to heterosexual men. One likely explanation for these differences is the varying ways partners within same-sex and opposite-sex couples specialise in paid versus unpaid work, with significantly less household specialisation in same-sex partnerships (Hofmarcher and Plug, 2022^[60]; Jepsen and Jepsen, 2015^[61]; Jepsen and Jepsen, 2002^[62]). Consequently, a partnered homosexual man, on average, engages less in the labour market than a partnered heterosexual man, while a partnered

homosexual woman tends to be more involved than her heterosexual counterpart. However, it is worth noting that recent studies focusing on younger cohorts in the United States have documented a significant labour market penalty for lesbians (National Academies of Sciences, Engineering, and Medicine, 2020^[37]). This observation aligns with the trend of increasing equality in the sharing of both paid and unpaid work between men and women in heterosexual partnerships, which results in a diminishing “specialisation gap” between same-sex and different-sex couples (Giddings et al., 2014^[63]). Consequently, the effects of the household specialisation bias in latest survey data should be attenuated, unveiling the penalty also experienced by lesbians.

Lower labour earnings for LGBTI+ individuals appear to be partly driven by talent misallocation, specifically a glass ceiling effect that hinders LGBTI+ individuals from advancing beyond a certain level in organisational hierarchies (McCay, 2024^[64]). Representative data from the United Kingdom shows that LGB employees are, on average, 11% less likely to hold high managerial positions than their heterosexual peers, a trend that affects all subgroups within the LGB population, including lesbians (Aksoy et al., 2019^[65]; OECD, 2019^[1]).

The unexplained disparities for LGBTI+ individuals are associated with high levels of perceived discrimination. Data from a representative sample of US bachelor’s degree recipients interviewed in 2018 indicate that, ten years after graduation, LGBTI+ individuals earn 10% less than their non-LGBTI+ counterparts, after considering that LGBTI+ individuals choose different occupations compared to their non-LGBTI+ peers, to avoid discrimination. Meanwhile, the study reveals that almost half (48%) of LGBTI+ graduates reported experiencing workplace discrimination due to sexual orientation, gender identity or sex characteristics during the first ten years after graduation. Additionally, one-third of respondents did not find their current employer very accepting of LGBTI+ employees. In line with concealment strategies being detrimental to LGBTI+ individuals’ labour market trajectories, the study also found a more pronounced labour earnings gap for those who remain closeted in the workplace, compared to those who are open about their identities (Folch, 2022^[17]).

3.4.2. Comparing labour market outcomes by sexual orientation and gender identity

This section explores labour market gaps between LGBTI+ and non-LGBTI+ individuals. Two outcomes are analysed: employment rate and, for those employed, productivity per worker, as measured by labour earnings, obtained by multiplying hours worked by hourly wage.

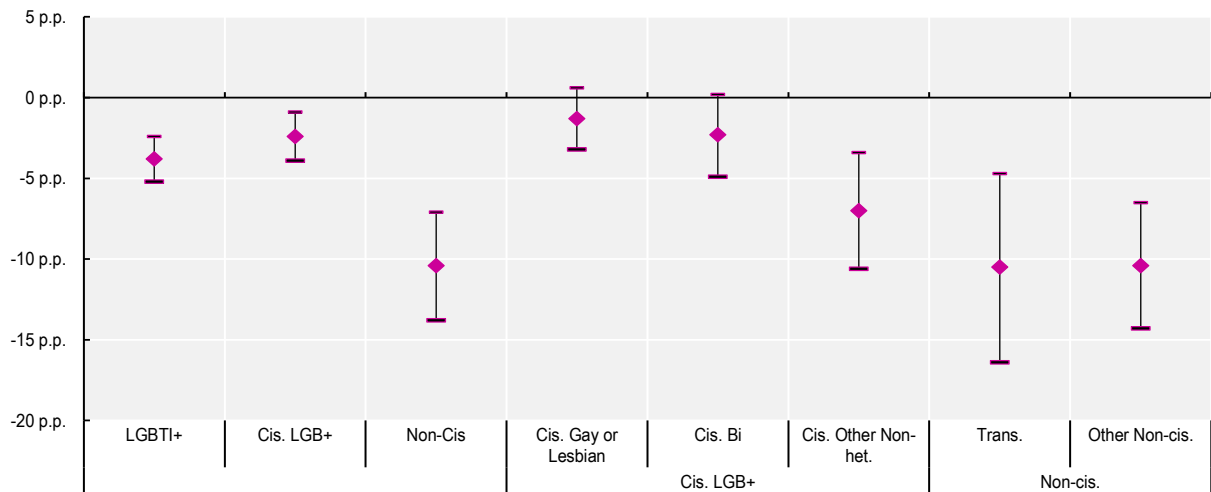
The analysis introduces two restrictions to the original sample of LGBTI+ and non-LGBTI+ respondents this report has been exploring thus far. First, we focus on working-age adults, meaning that those above 74 are excluded from the analysis. Second, we restrict our attention to individuals living alone (with no other adult or children in the household) to eliminate the potential bias from different household specialisation across same-sex and different-sex couples. This approach to obtaining more accurate estimates of labour market disparities for LGBTI+ individuals has been implemented by several researchers before us – see the pioneering work of (Aksoy, Carpenter and Frank, 2018^[66]) based on UK data – but with limited success. Indeed, focusing on singles reduced the sample size to such an extent that many of the penalties found could not be regarded as statistically different from zero. By contrast, the large number of observations contained in the HPS dataset offers a unique opportunity to run this strategy, with a lower likelihood of encountering false negatives. The number of working age adults living alone in the HPS dataset for the period between 2021 and 2023 is equal to 263 499, of which 35 174 report being LGBTI+. As expected, compared to focusing on the whole set of working-age adults, focusing on those living alone results in a lower labour market penalty for LGBTI+ individuals assigned male at birth, but a greater labour market penalty for their peers assigned female at birth. Overall, the penalty is slightly higher when computed on adults living alone.⁵

Unexplained gaps in employment rate between LGBTI+ and non-LGBTI+ individuals

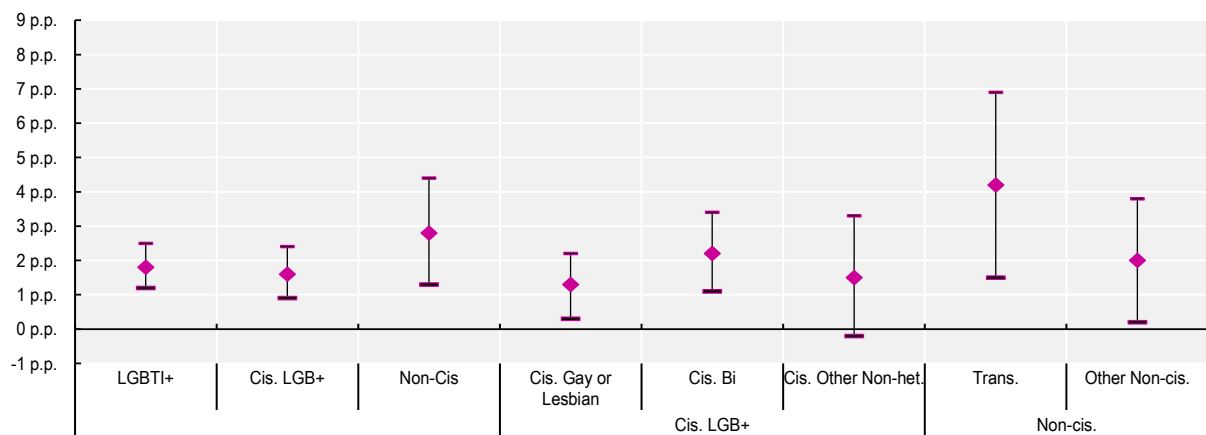
LGBTI+ US adults living alone are 4 percentage points (or 7%) less likely to be employed, compared to their cisgender straight peers, as shown in Panel A of Figure 3.8. Additionally, they are 2 percentage points (or 24%) more likely to experience job loss (Panel B of Figure 3.8). These disparities prevail among all LGBTI+ subgroups, although the gaps are more pronounced for non-cisgender individuals than for cisgender LGB+ individuals. Notably, even cisgender LGBs, who experience a non-statistically significant penalty in terms of employment, face a higher likelihood of losing employment income in the month preceding their interview.

Figure 3.8. LGBTI+ adults are less likely to be employed and more likely to experience a loss of employment income, compared to their cisgender straight peers

Panel A: Percentage point difference in the probability of being employed for pay last week among working-age US adults living alone (using cisgender straight individuals as the reference category), by sexual orientation and gender identity (2021-23)



Panel B: Percentage point difference in the probability of experiencing a loss of employment income in the last month among working-age US adults living alone (using cisgender straight individuals as the reference category), by sexual orientation and gender identity (2021-23).



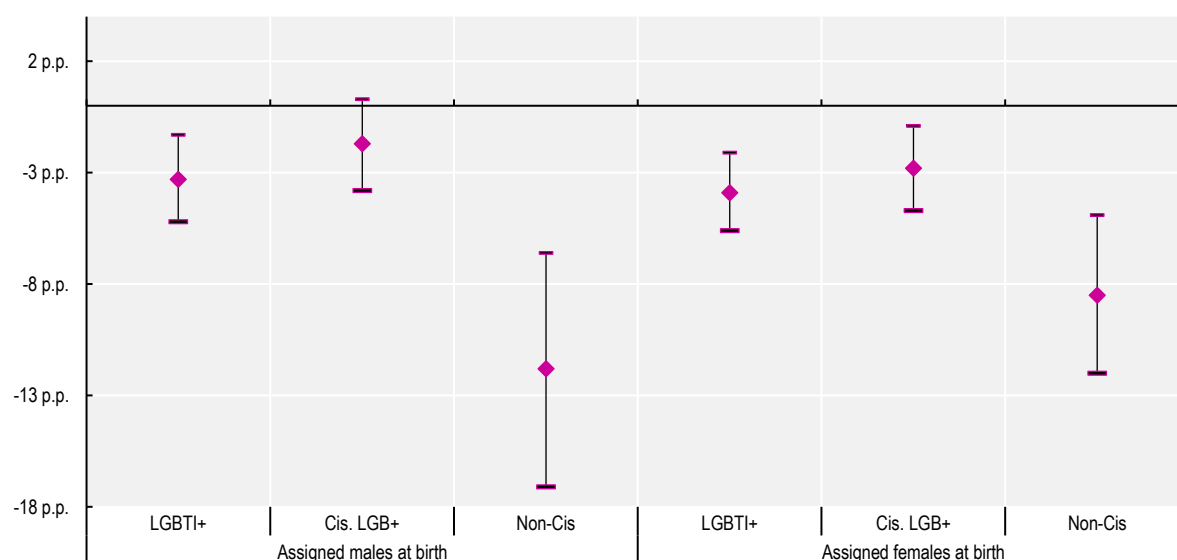
Note: The percentage point differences are estimated from a Logistic regression, controlling for sex assigned at birth, age groups (18-24, 25-29, 30-34, ..., 70-74), race and ethnicity, marital status, educational attainment, living in one of the 15 largest metropolitan statistical areas (metro areas), as well as state and survey wave fixed effects. The percentage point differences are estimated at the average of the covariates. The error bars depict 95% confidence intervals. This means that we can be 95% confident that the true value lies within the range of the error bar. In other words, if the error bar crosses the x-axis, then the percentage point difference cannot be deemed as statistically different from 0 (at least not at this 95% confidence level). Standard errors robust to heteroscedasticity. Person-level weights used.

Source: Household Pulse Survey (21 July 2021 – 10 July 2023).

While the employment penalties for LGBTI+ individuals are similar for AMAB and AFAB persons (Figure 3.9), the way this penalty evolves with age differs based on sex assigned at birth. Specifically, the employment penalty for LGBTI+ individuals assigned male at birth decreases with age (Panel A of Figure 3.10), while it increases with age for LGBTI+ individuals assigned female at birth (Panel B of Figure 3.10). This finding suggests that among AFAB individuals, other forces counteract the factors, including the non-disclosure bias, that contribute to reduce the penalty among older LGBTI+ adults. Such could be a combination of sexism, ageism and homophobia, especially given evidence of strong age discrimination against female job candidates in the United States. A large-scale correspondence study, for instance, revealed that hiring discrimination against older applicants is nearly twice as high for women compared to men, with this heightened discrimination against women occurring from the age of 50 upwards (Neumark, Burn and Button, 2019^[67]).

Figure 3.9. The employment penalty for LGBTI+ individuals are similar for AMAB and AFAB persons

Percentage point difference in the probability of being employed for pay last week among working-age US adults living alone (using cisgender straight individuals as the reference category), by sexual orientation, gender identity and sex assigned at birth (2021-23)

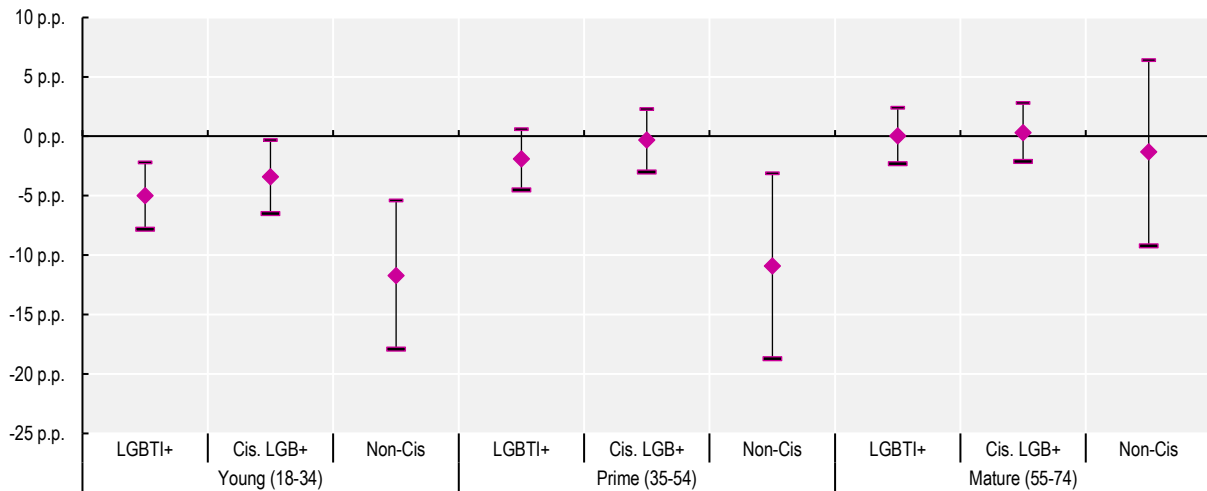


Note: The percentage point differences are estimated from a Logistic regression, controlling for age groups (18-24, 25-29, 30-34, ..., 70-74), race and ethnicity, marital status, educational attainment, living in one of the 15 largest metropolitan statistical areas (metro areas), as well as state and survey wave fixed effects. The percentage point differences are estimated at the average of the covariates. The error bars depict 95% confidence intervals. This means that we can be 95% confident that the true value lies within the range of the error bar. In other words, if the error bar crosses the x-axis, then the percentage point difference cannot be deemed as statistically different from 0 (at least not at this 95% confidence level). Standard errors robust to heteroscedasticity. Person-level weights used.

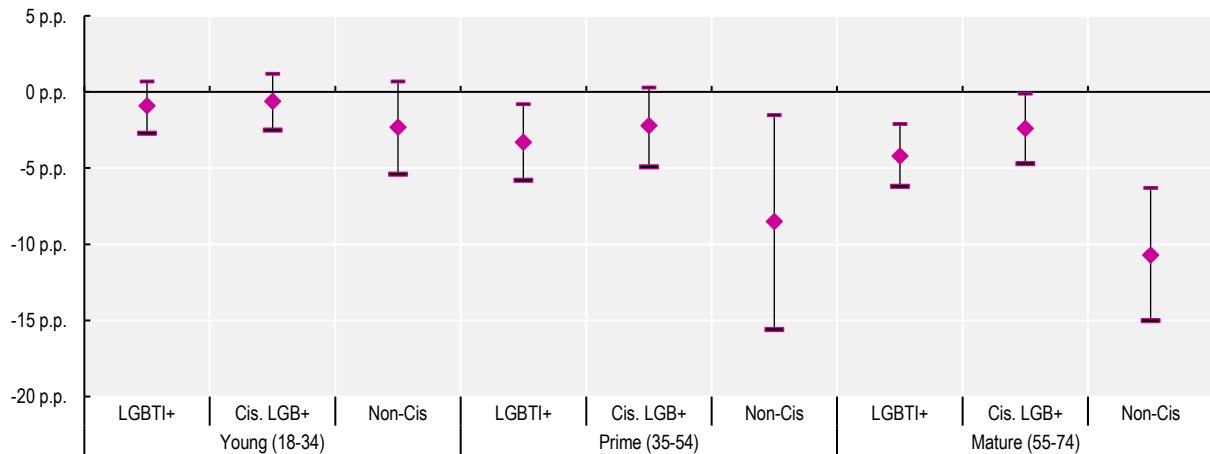
Source: Household Pulse Survey (21 July 2021 – 10 July 2023).

Figure 3.10. The employment penalty for LGBTI+ individuals assigned male at birth decreases with age, while it increases with age for LGBTI+ individuals assigned female at birth

Panel A: Percentage point difference in the probability of being employed for pay last week among working-age US adults living alone assigned male at birth (using cisgender straight individuals as the reference category), by sexual orientation, gender identity and age (2021-23)



Panel B: Percentage point difference in the probability of being employed for pay last week among working-age US adults living alone assigned female at birth (using cisgender straight individuals as the reference category), by sexual orientation, gender identity and age (2021-23)



Note: The percentage point differences are estimated from a Logistic regression, controlling for age groups, race and ethnicity, marital status, educational attainment, living in one of the 15 largest metropolitan statistical areas (metro areas), as well as state and survey wave fixed effects. The percentage point differences are estimated at the average of the covariates. The error bars depict 95% confidence intervals. This means that we can be 95% confident that the true value lies within the range of the error bar. In other words, if the error bar crosses the x-axis, then the percentage point difference cannot be deemed as statistically different from 0 (at least not at this 95% confidence level). Standard errors robust to heteroscedasticity. Person-level weights used.

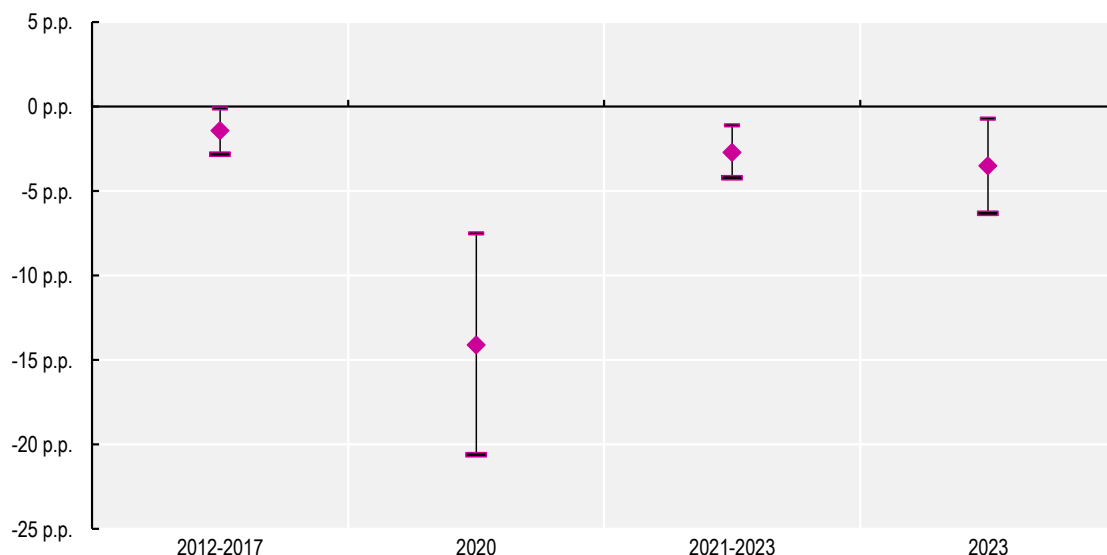
Source: Household Pulse Survey (21 July 2021 – 10 July 2023).

The COVID-19 pandemic disproportionately affected the employment of the LGBT(I+) population. Panel A of Figure 3.11 illustrates the evolution over time of the difference in employment rate between LGBT and cisgender straight working-age adults living alone. Over the period 2012-17, the employment penalty

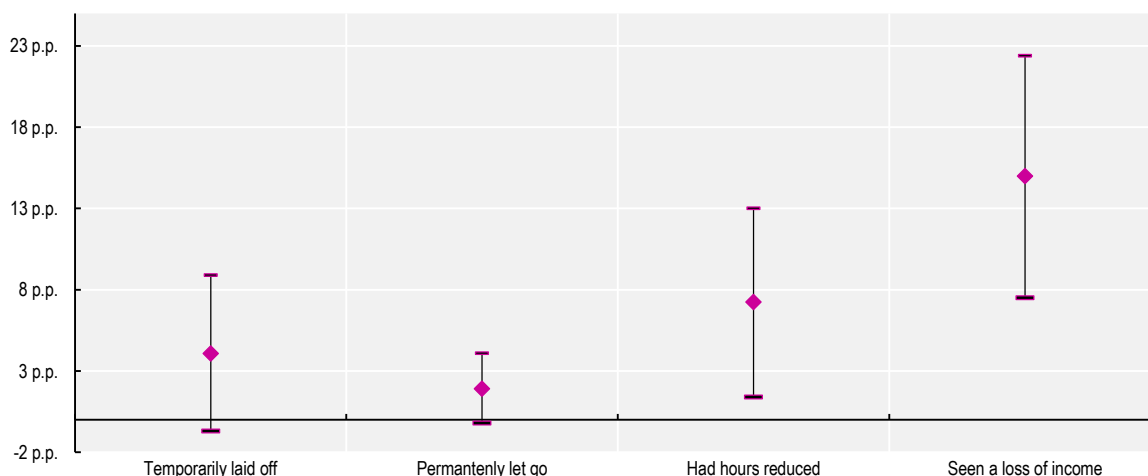
estimated from the Gallup US Daily Survey (1.5 percentage points or 2%) was lower than the 2021-23 penalty estimated from HPS (3 percentage points or 5%). However, a much more pronounced LGBT employment penalty was observed in the Gallup Panel – COVID-19 Survey from March to December 2020 (14 percentage points or 21%). Moreover, Panel B of Figure 3.11 displays responses to the survey question “Have you experienced any of the following changes to your employment as the result of the coronavirus?” from the Gallup Panel – COVID-19 Survey. On average, compared to non-LGBT adults, LGBT adults were 4 percentage points (45%) more likely to have been temporarily laid off, 2 percentage points (123%) more likely to have been permanently let go, 7 percentage points (46%) more likely to have had their hours reduced, and 15 percentage points (59%) more likely to have experienced a loss of income. These findings align with results from the KFF COVID-19 Vaccine Monitor, a US-based, nationally representative that surveyed LGBT people between December 2020 and January 2021. This investigation revealed that LGBT adults and their household members experienced COVID-era job losses at higher rates than non-LGBT adults, partly because they were more likely to work in industries hard hit by the pandemic, such as restaurants, food services or art and entertainment. Specifically, 56% of LGBT people reported that they or another adult in their household have lost a job, been placed on furlough, or had their income or hours reduced due to the coronavirus outbreak, in contrast to 44% of non-LGBT individuals (Dawson, Kirzinger and Kates, 2021^[68]).

Figure 3.11. The LGBT employment penalty was exacerbated at the onset of the COVID-19 pandemic

Panel A: Percentage point difference in the probability of being employed for pay last week for LGBT working-age US adults living alone, using non-LGBT individuals as the reference category (2012-23)



Panel B: Percentage point difference in the probability of having experienced any changes (i.e. temporarily laid off, permanently let go, had hours reduced, and seen a loss of income) as the result of the coronavirus for LGBT working-age US adults living alone, using non-LGBT individuals as the reference category (2020)



Note: The percentage point differences are estimated from a Logistic regression, controlling for sex assigned at birth, age groups (18-24, 25-29, 30-34, ..., 70-74), race and ethnicity, marital status, educational attainment, living in one of the 15 largest metropolitan statistical areas (metro areas) – only in the HPS data –, as well as state and survey wave fixed effects. The percentage point differences are estimated at the average of the covariates. The error bars depict 95% confidence intervals. This means that we can be 95% confident that the true value lies within the range of the error bar. In other words, if the error bar crosses the x-axis, then the percentage point difference cannot be deemed as statistically different from 0 (at least not at this 95% confidence level). Standard errors robust to heteroscedasticity. Person-level weights used.

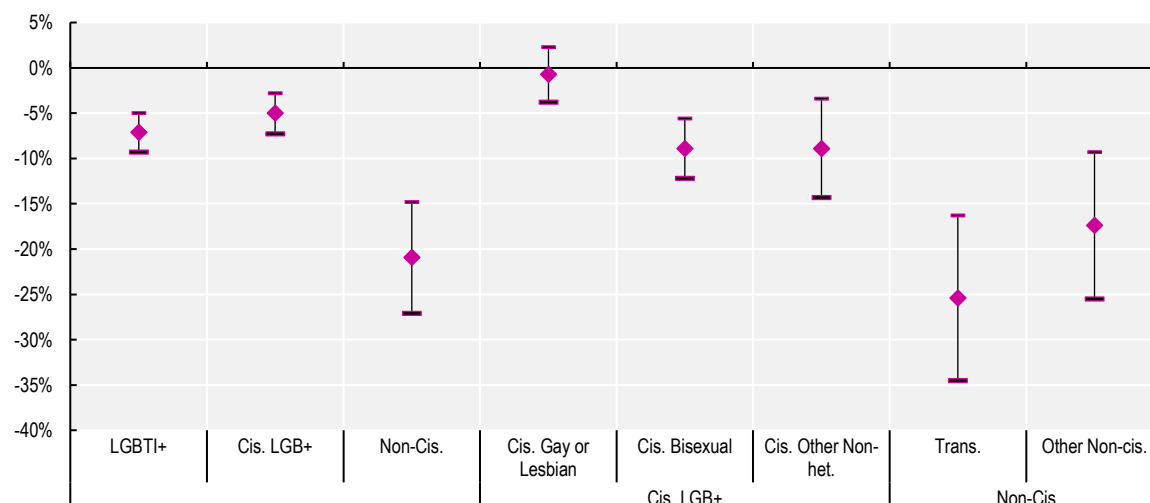
Source: Gallup US Daily Survey (2012-17), Gallup Panel – COVID-19 Survey (March – December 2020), Household Pulse Survey (21 July 2021 – 10 July 2023) for Panel A. Gallup Panel – COVID-19 Survey (March – December 2020) for Panel B.

Unexplained gaps in labour productivity (hours worked and hourly wage) between LGBTI+ and non-LGBTI+ individuals

In the absence of specific data on individual labour earnings from sources such as the HPS or Gallup surveys, we use household income of *employed* individuals living alone as a proxy for productivity per worker. On average, accounting for differences in demographics, sectors, and occupations, LGBTI+ workers have labour earnings that are 7% lower than those of their cisgender straight peers (Figure 3.12). The distribution of these labour earnings gaps varies among LGBTI+ subgroups, but the pattern is similar to that observed in Figure 3.4, which analyses household income across the entire adult population. The income penalty is more pronounced for non-cisgender workers (21%) than for their cisgender non-heterosexual counterparts (5%). Within the latter group, cisgender gay/lesbian adults experience the smallest penalty (1%, not significant at the 90% confidence level), followed by cisgender bisexual adults (9%), and finally, by cisgender other non-heterosexual adults (9%).

Figure 3.12. LGBTI+ workers experience a 7% labour earnings penalty compared to their cisgender straight peers

Percentage difference in household income among employed working-age US adults living alone (using cisgender straight individuals as the reference category), by sexual orientation and gender identity (2021-23)



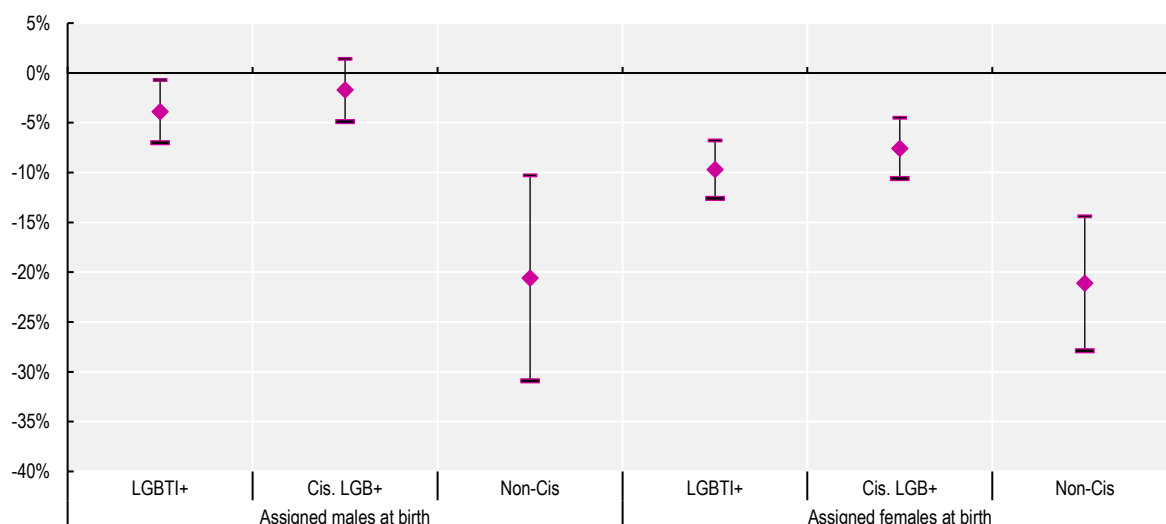
Note: The percentage differences are estimated from an Interval regression of the logarithm of household income, controlling for sex assigned at birth, age groups (18-24, 25-29, 30-34, ..., 70-74), race and ethnicity, marital status, educational attainment, sectoral and occupational dummy variables, living in one of the 15 largest metropolitan statistical areas (metro areas), as well as state and survey wave fixed effects. The error bars depict 95% confidence intervals. This means that we can be 95% confident that the true value lies within the range of the error bar. In other words, if the error bar crosses the x-axis, then the percentage point difference cannot be deemed as statistically different from 0 (at least not at this 95% confidence level). Standard errors robust to heteroscedasticity. Person-level weights used. See Box 3.1 for more details on Interval regression methodology.

Source: Household Pulse Survey (14 September 2022 – 10 July 2023).

The labour earnings penalty for LGBTI+ individuals is nearly twice as high among AFAB than AMAB individuals, as illustrated in Figure 3.13. Conversely, this penalty decreases with age, regardless of sex assigned at birth, a trend depicted in Figure 3.14. The significant labour earning disparity between AFAB and AMAB individuals can be attributed to a more pronounced penalty for cisgender LGB+ females compared to cisgender LGB+ males (who experience a non-significant penalty). This higher penalty for cisgender LGB+ females is driven by a higher representation of cisgender bisexuals and cisgender other non-heterosexuals, who, on average, experience a more pronounced income penalty than cisgender homosexuals (Figure 3.6). Additionally, the reduction of the labour earnings penalty with age aligns with the patterns previously presented in Figure 3.7, which analysed how income disparities for LGBTI+ individuals evolve across cohorts. This trend supports the hypothesis of a non-disclosure bias among employed LGBTI+ adults, suggesting that older generations from more privileged backgrounds may be more inclined to openly identify as LGBTI+ than those from less privileged backgrounds. It may also indicate a health selection process, wherein LGBTI+ adults who have reached an older age are more likely to come from advantaged backgrounds conducive to better navigating the significant health challenges that disproportionately affect the LGBTI+ population (Fredriksen-Goldsen et al., 2014^[69]). Importantly, the fact that the income penalty for employed LGBTI+ individuals living alone is maximal among younger generations (10% on average across AMAB and AFAB individuals) suggests that this penalty is mainly capturing gaps in labour earnings, not in unearned income (e.g. dividends and property gains). Indeed, the share of unearned income increases with age.⁶

Figure 3.13. The labour earnings penalty for LGBTI+ individuals is nearly twice as high among AFAB than AMAB individuals

Percentage difference in household income among employed working-age US adults living alone (using cisgender straight individuals as the reference category), by sexual orientation, gender identity and sex assigned at birth (2021-23)

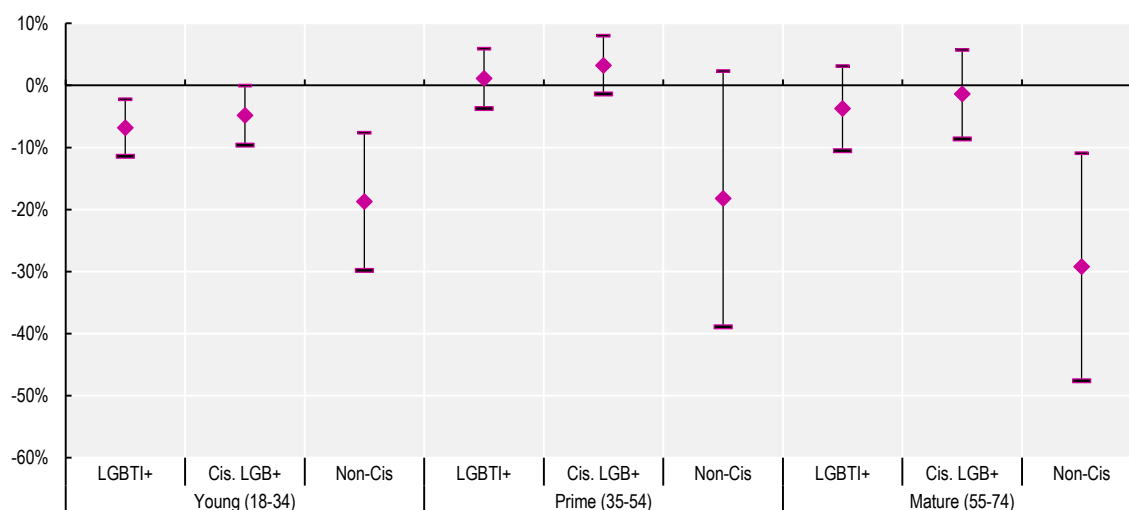


Note: The percentage differences are estimated from an Interval regression of the logarithm of household income, controlling for age groups (18-24, 25-29, 30-34, ..., 70-74), race and ethnicity, marital status, educational attainment, sectoral and occupational dummy variables, living in one of the 15 largest metropolitan statistical areas (metro areas), as well as state and survey wave fixed effects. The error bars depict 95% confidence intervals. This means that we can be 95% confident that the true value lies within the range of the error bar. In other words, if the error bar crosses the x-axis, then the percentage point difference cannot be deemed as statistically different from 0 (at least not at this 95% confidence level). Standard errors robust to heteroscedasticity. Person-level weights used. See Box 3.1 for more details on Interval regression methodology.

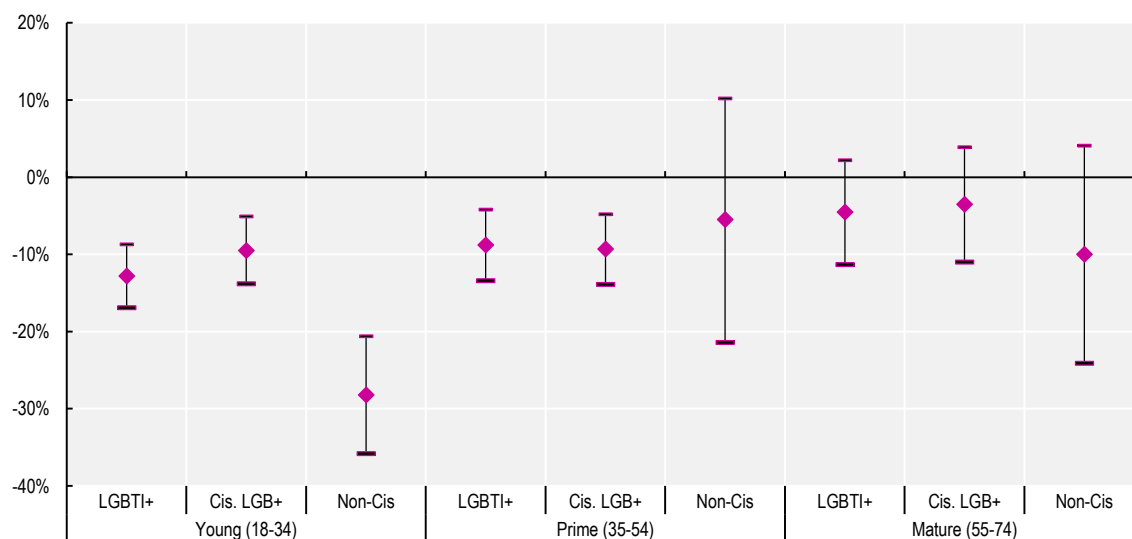
Source: Household Pulse Survey (14 September 2022 – 10 July 2023).

Figure 3.14. The labour earnings penalty for LGBTI+ individuals decreases with age

Panel A: Percentage difference in household income among employed working-age US adults living alone assigned male a birth (using cisgender straight individuals as the reference category), by sexual orientation, gender identity and age (2021-23)



Panel B: Percentage difference in household income among employed working-age US adults living alone assigned female at birth (using cisgender straight individuals as the reference category), by sexual orientation, gender identity and age (2021-23).



The percentage differences are estimated from an Interval regression of the logarithm of household income, controlling for age groups, race and ethnicity, marital status, educational attainment, sectoral and occupational dummy variables, living in one of the 15 largest metropolitan statistical areas (metro areas), as well as state and survey wave fixed effects. The error bars depict 95% confidence intervals. This means that we can be 95% confident that the true value lies within the range of the error bar. In other words, if the error bar crosses the x-axis, then the percentage point difference cannot be deemed as statistically different from 0 (at least not at this 95% confidence level). Standard errors robust to heteroscedasticity. Person-level weights used. See Box 3.1 for more details on Interval regression methodology.

Source: Household Pulse Survey (14 September 2022 – 10 July 2023).

It is important to emphasise that the unexplained labour earnings penalty we report for LGBTI+ individuals likely understates the actual impact of labour market discrimination, whether it manifests directly or indirectly, such as through strategies individuals adopt to avoid discrimination. Indeed, many of these strategies negatively impact the labour market trajectories of LGBTI+ individuals. For instance, while concealing one's identity can provide immediate protection, not being able to bring one's whole self to work can impede career advancement in various ways (Section 3.2.2). Similarly, several LGBTI+ individuals opt for sectors and occupations they perceive as less hostile to their identities, such as male-dominated occupations for gay men. On average, this sorting seems to lead both AMAB and AFAB individuals to settle for lower-paying jobs (Box 3.3). As an illustration, without accounting for this intentional job selection, the labour earnings penalty would be 7% for LGBTI+ individuals assigned male at birth and 11% for those assigned female at birth, compared to 4% and 9%, respectively, when considering the impact of sectoral and occupational sorting. Overall, the gap reported in Figure 3.12 would be 10%, as opposed to 7% currently (Annex Figure 3.A.3). In addition, the gap becomes significant for cisgender LGB+ AMAB individuals.

Finally, the labour earnings penalty for LGBTI+ individuals hides both reduced hours worked and reduced hourly wages. This insight comes from our analysis of the Gallup US Daily Survey, which, in addition to household income, captures the typical number of hours worked in a week. From this data, we conclude that employed LGBT individuals living alone typically work one hour less per week, even after accounting for differences in demographics, sectors, and occupations. However, this discrepancy in hours accounts for only 25% of the labour earnings penalty for LGBT individuals, indicating that discrimination in hourly wage is also a significant factor.⁷

Box 3.3. To avoid discrimination, LGBTI+ individuals sort into specific sectors and occupations

LGBTI+ individuals often avoid jobs where they anticipate facing discrimination. Using data from the Australian Twin Registers, a study compares gay men and lesbians with their monozygotic heterosexual twins, which allows controlling for both genetic factors as well as environmental factors common to both twins in each twin pair. The study also considers a large set of additional environmental factors that differs across twins in the same family. The results show that gay men and lesbians steer clear of occupations where they expect a high degree of hostility due to their sexual orientation, which is a pattern not observed among their heterosexual twins (Plug, Webbink and Martin, 2014^[70])

Data from the United States confirm that gay men and lesbians tend to choose gender-atypical occupations. Specifically, gay men are overrepresented in female-dominated occupations, while lesbians are overrepresented in male-dominated occupations (Ueno, Roach and Pena-Talamantes, 2013^[71]; Black, Sanders and Taylor, 2007^[72]). Supporting these previous findings, the 2022 survey by the Center for American Progress shows that 36% of LGBTI+ respondents reported making decisions about where to work to avoid discrimination (Medina and Mahowald, 2023^[16]).

Consistent with this body of research, data from the Household Pulse Survey (HPS) confirm that occupational sorting is at play among employed LGBTI+ individuals living alone. This is particularly the case for LGBTI+ assigned male at birth who are 34% less likely than their cisgender straight peers to work in occupations that are male dominated, i.e. Agriculture, Forestry, Fishing and Hunting, Mining, Quarrying, and Oil and Gas Extraction, Utilities, Construction, Manufacturing, Wholesale Trade, Transportation and Warehousing, Information Technology, and Waste Management and Remediation Services. In contrast, LGBTI+ individuals assigned female at birth are 9% less likely than cisgender straight women to work in occupations that are female dominated, i.e. Administrative and Support Services, Educational Services, Health Care, and Social Assistance.¹ The HPS data offer further insights by allowing for an exploration of sectoral sorting. Findings show that both AMAB and AFAB LGBTI+ individuals have a higher likelihood of working as self-employed – potentially to escape discrimination from employers – or working in non-profit organisations – potentially within an activist framework promoting LGBTI+ equality – or working in government sectors, in a context where public administrations in OECD countries have been more proactive in establishing comprehensive diversity strategies compared to the private sector (OECD, 2020^[73]).

Overall, the HPS data suggest that the sectoral and occupational sorting of LGBTI+ individuals result in lower pay. When accounting for this intentional job selection, the labour earnings penalty for LGBTI+ individuals living alone stands at 7%. However, when the effects of sectoral and occupational sorting are ignored, this penalty worsens to reach 10%.

Notes:

1. This is computed from 22 categories from the occupation survey question (asked only from week 49 onwards: 14 September 2022 – 10 July 2023): “What kind of business, industry, or organisation is this? That is, what do they make or do where you work?”. Occupations dominated by males are defined by having a share equal or lower of 33% of females. Occupations dominated by females are defined by having a share equal or higher than 66% of females. Regressions results are qualitatively similar when using as a threshold 25% or 50%, instead of 33%.

3.5. LGBTI+ Americans suffer from substantial health and well-being gaps

This section first provides an overview of previous research on the impact of stigma and other drivers on LGBTI+ individuals' health and well-being disparities. It then exploits HPS and Gallup data to compare health and well-being outcomes, by sexual orientation and gender identity.

3.5.1. *The impact of stigma and other related mechanisms on the health and well-being of LGBTI+ individuals*

Extensive research based on representative survey data has documented widespread health and well-being disparities within the LGBTI+ population. These disparities manifest themselves primarily through poorer mental health outcomes, such as low self-esteem, anxiety, and depression. Additionally, they often extend to behavioural health issues like sleep disorders and substance abuse, as well as physical health problems, including cardiovascular diseases and certain types of cancers. Prior research also indicates a higher risk of mortality among the LGBTI+ population, notably stemming from increased rates of suicidal ideation and suicide attempts (National Academies of Sciences, Engineering, and Medicine, 2020^[37]; OECD, 2019^[1]; Valfort, 2017^[3]).

These health disparities are suspected to arise from the pervasive exposure of LGBTI+ individuals to stigma, in a society that predominantly views heterosexuality and cisgender identity as the norm. Consequently, LGBTI+ individuals experience a unique form of stress not encountered by their heterosexual and cisgender peers, often referred to as “minority stress” (Meyer, 2003^[74]).

Several studies relying on US data confirm that a decrease in societal stigma leads to reduced psychological distress, and *vice versa*. For instance, the reduction in the gap between heterosexual and LGB youth’s suicide attempts was substantially higher in states that adopted same-sex marriage before its legalisation by the Supreme Court in 2015, than in others – a trend that was not apparent before the enactment of LGB-inclusive policies. Overall, it is estimated that same-sex marriage laws caused a reduction by nearly 15% of suicide attempts among adolescents who self-identify as gay, lesbian or bisexual (Raifman et al (2017^[75]) – see (Carpenter et al., 2021^[76]) for consistent results). Furthermore, bans on conversion therapy have been linked to improved mental health and decreased suicide rates (Harrell, 2022^[77]). On the flip side, there is emerging evidence that the backlash against the expansion of LGBTI+ rights is harming this population’s health. Starting in 2015, the United States has seen a steady increase in anti-LGBTI+ bills, from 115 bills introduced in 2015, to over 500 in 2023 (Human Rights Campaign Foundation, 2023^[78]). As of 2024, more than 150 such bills have become law (Trans Legislation Tracker, 2024^[79]). Most of these laws aim to ban gender-affirming care for LGBTI+ youth, a treatment to alleviate the distress of non-cisgender individuals, often termed “gender dysphoria”, arising from a mismatch between biological sex and gender identity (Mann, Campbell and Nguyen, 2023^[80]). These bans not only hinder the well-being of transgender and non-binary youth (Abreu et al., 2022^[81]) but also perpetuate stigmatisation, affecting the broader LGBTI+ adult population (Mann, Campbell and Nguyen, 2023^[80]). Indeed, hate crime victimisation of LGBTI+ people prevails (Flores et al., 2022^[82]), and recent FBI statistics show it has intensified over the last years (FBI, 2023^[83]). According to the 2022 survey by the Centre for American Progress, 51% of LGBTI+ adults reported that recent debates on state laws restricting LGBTI+ rights negatively impacted their mental health or safety. A staggering 86% of transgender or nonbinary individuals felt the same way. Yet, given that this latter group represents only a minority within the broader LGBTI+ population, they alone cannot account for the high share of LGBTI+ adults saying their mental health was affected. This suggests that the detrimental effects of gender-affirming care bans extend beyond just the well-being of those directly targeted by such bans (Medina and Mahowald, 2023^[16]).

In addition to stigma, discrimination from medical practitioners likely exacerbates the health disparities faced by the LGBTI+ population. In the United States, a pervasive implicit bias against lesbian women and gay men has been identified among heterosexual healthcare providers (Sabin, Riskind and Nosek, 2015^[84]). This conclusion is echoed by the first audit field experiment examining discrimination in access to mental health care, focusing on two characteristics of the patients: their race/ethnicity and their gender identity (Fumarco et al., 2020^[85]). In this “simulated patients” study, appointment requests for common mental health concerns (such as anxiety, depression, and stress) were sent by email to various mental health providers in the United States, including psychologists, counsellors, social workers, and psychiatrists, using a popular online platform. The results highlighted significant discrimination against

transgender and non-binary patients, especially when their names signalled a racial/ethnic minority background, like being African American or Hispanic. In line with these findings, the 2022 survey by the Center for American Progress showed that one-third of LGBTI+ adults experienced some form of mistreatment when interacting with a mental health professional in the past year. This rate rises to 4 in 10 for LGBTI+ people of colour and over half for transgender or nonbinary individuals. More generally, over 1 in 5 LGBTI+ adults reported either postponing or avoiding necessary medical care (23%) and preventive screenings (21%) within the past year, due to disrespect or discrimination from providers. In contrast, only 7% of non-LGBTI+ individuals reported similar experiences for both medical care and preventive screenings, which is more than three times less (Medina and Mahowald, 2023^[16]).

Poorer health outcomes for LGBTI+ individuals might also be reinforced by the more limited financial resources they can allocate to healthcare services. Consistent with their income penalty (Section 3.3), more than 1 in 3 LGBTI+ respondents to the 2022 survey by the Center for American Progress declared having postponed or not searched for necessary medical care and preventive screenings due to cost issues. In contrast, financial barriers to accessing healthcare were reported by only 1 in 5 non-LGBTI+ respondents (Medina and Mahowald, 2023^[16]). On a positive note, the negative impact of LGBTI+ individuals' economic vulnerability on their access to healthcare has likely diminished over the past decade due to two key milestones. The first was the implementation of the major provisions of the Affordable Care Act in 2014. This notably expanded Medicaid eligibility and led to a halving of the uninsured population by 2016. The second milestone was the legalisation of same-sex marriage by the Supreme Court in 2015. This allowed for the expansion of access to a spouse or partner's employer-sponsored insurance for LGBTI+ adults. LGBTI+ individuals disproportionately benefited from these changes, leading to a significant reduction in their health coverage disparities (Bolibol et al., 2023^[86]).

3.5.2. Comparing health and well-being outcomes by sexual orientation and gender identity

This section explores health and well-being gaps between LGBTI+ and non-LGBTI+ individuals. Two outcomes are analysed: mental health, as proxied in the Household Pulse Survey by the probability of showing symptoms of generalised anxiety or major depressive disorder, and a multidimensional measure of well-being stemming from the Gallup US Daily Survey.

Unexplained gaps in mental health between LGBTI+ and non-LGBTI+ individuals

In 2023, LGBTI+ Americans reported symptoms of mental health disorders at a rate of 53%, making them 80% more likely than their non-LGBTI+ peers to be at risk of generalised anxiety or major depressive disorder, as illustrated in Panel A of Figure 3.15— for a description of how this risk is measured, see Box 3.4. In line with prior research, this mental health penalty was most pronounced among non-cisgender individuals. Their risk is more than double that of cisgender straight American adults, largely due to the larger health challenges faced by transgender individuals. Although gay men and lesbians had the smallest disparity within the LGBTI+ group, they were still, on average, nearly 40% more likely to experience mental health issues than their cisgender straight peers. Adjusting for essential demographic differences slightly narrows these gaps, but the overall patterns remain unchanged (Panel B of Figure 3.15).⁸

Box 3.4. Measuring the risk of generalised anxiety or major depressive disorder with the Household Pulse Survey

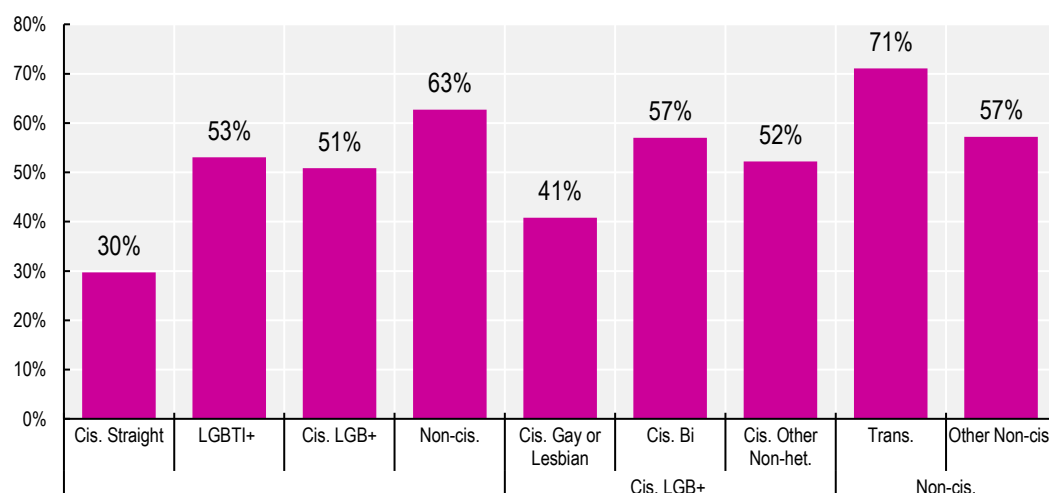
The Household Pulse Survey asks four questions to assess the mental health of the respondent, which were developed in partnership with the National Center for Health Statistics.

- Two of these questions measure symptoms of anxiety: (i) “Over the last 2 weeks, how often have you been bothered by feeling nervous, anxious, or on edge?”; and (ii) “Over the last 2 weeks, how often have you been bothered by not being able to stop or control worrying?”.
- The other two questions measure symptoms of depression: (i) “Over the last 2 weeks, how often have you been bothered by feeling down, depressed, or hopeless?”; and (ii) “Over the last 2 weeks, how often have you been bothered by having little interest or pleasure in doing things?”.

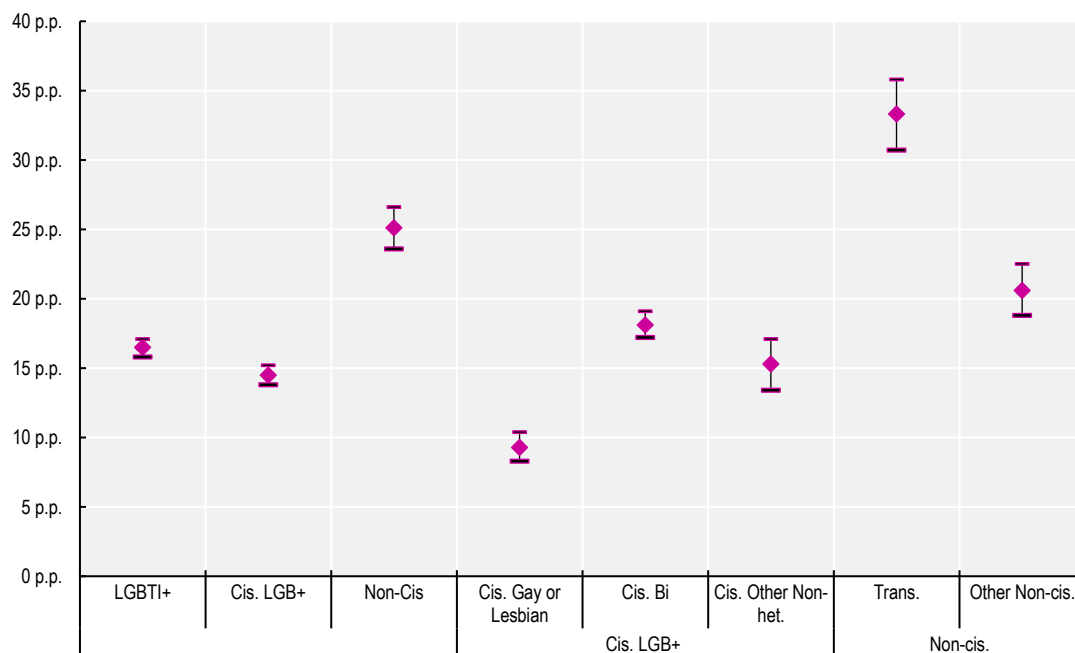
For each question, the answers are assigned a numerical value (nearly every day = 3, more than half the days = 2, several days = 1, and not at all = 0). It is considered that respondents show symptoms of generalised anxiety disorder when the sum of the numerical values associated with their answers to the two “anxiety” questions is greater than three, in which case individuals are at a heightened risk of being diagnosed with a generalised anxiety disorder. Following the same logic, it is considered that respondents show symptoms of major depressive disorder when the sum of the numerical values associated with their answers to the two “depression” questions is greater than three, in which case individuals are at a heightened risk of being diagnosed with a major depressive disorder (National Center for Health Statistics, 2023^[87]).

Figure 3.15. A majority of LGBTI+ adults, but a minority of their cisgender straight peers, are at risk of generalised anxiety or major depressive disorder

Panel A: Share of US adults with symptoms of generalised anxiety or major depressive disorder, by sexual orientation and gender identity (2023)



Panel B: Percentage point difference in the probability of having symptoms of generalised anxiety or major depressive disorder among US adults (using cisgender straight individuals as the reference category), by sexual orientation and gender identity (2021-23)



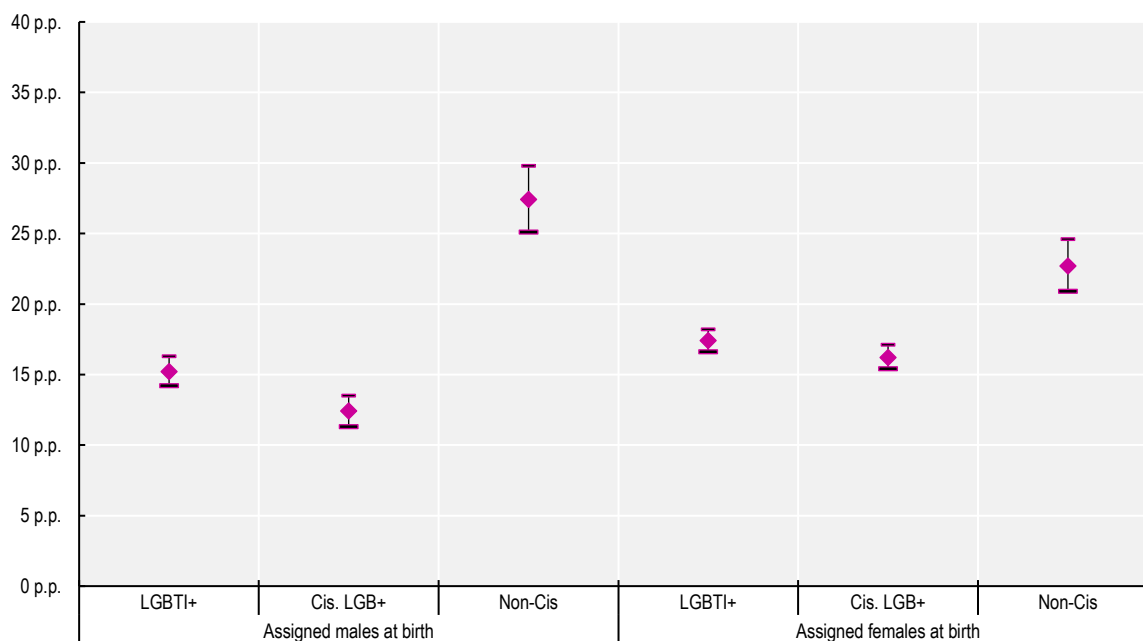
Note: The percentage differences are estimated from a Logistic regression, controlling for sex assigned at birth, age groups (18-24, 25-29, 30-34, ..., 80-84, 85-88), race and ethnicity, marital and parental status, number of adults in the household, educational attainment, living in one of the 15 largest metropolitan statistical areas (metro areas), as well as state and survey wave fixed effects. The percentage point differences are estimated at the average of the covariates. The error bars depict 95% confidence intervals. This means that we can be 95% confident that the true value lies within the range of the error bar. In other words, if the error bar crosses the x-axis, then the percentage point difference cannot be deemed as statistically different from 0 (at least not at this 95% confidence level). Standard errors robust to heteroscedasticity. Person-level weights used.

Source: Household Pulse Survey (4 January 2023 – 10 July 2023) for Panel A. Household Pulse Survey (21 July 2021 – 10 July 2023) for Panel B.

The mental health penalty does not differ based on sex assigned at birth, as illustrated in Figure 3.16. Additionally, the previously observed trend, in which the disparities for LGBTI+ individuals diminish with age, is consistent in the context of mental health, as shown in Figure 3.17. Finally, the mental health differences between LGBTI+ and cisgender straight adults remain consistent, regardless of whether an individual lives alone or with a partner, suggesting that being in a relationship does little to mitigate the negative effects of minority stress.⁹

Figure 3.16. The mental health penalty for LGBTI+ individuals does not vary with sex assigned at birth

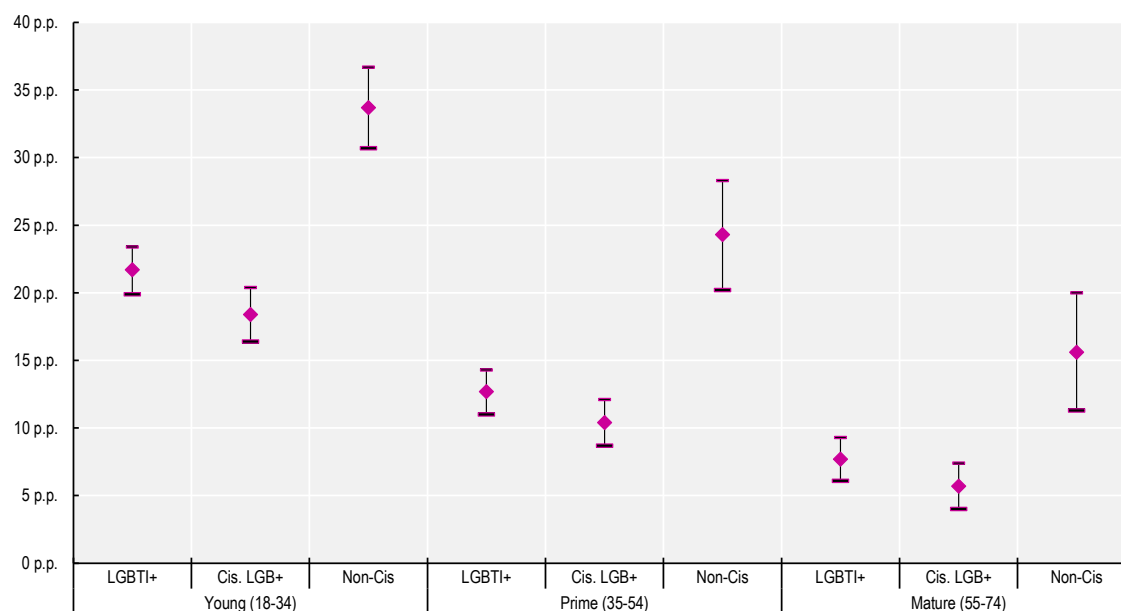
Percentage point difference in the probability of having symptoms of generalised anxiety or major depressive disorder among US adults (using cisgender straight individuals as the reference category), by sexual orientation, gender identity and sex assigned at birth (2021-23)



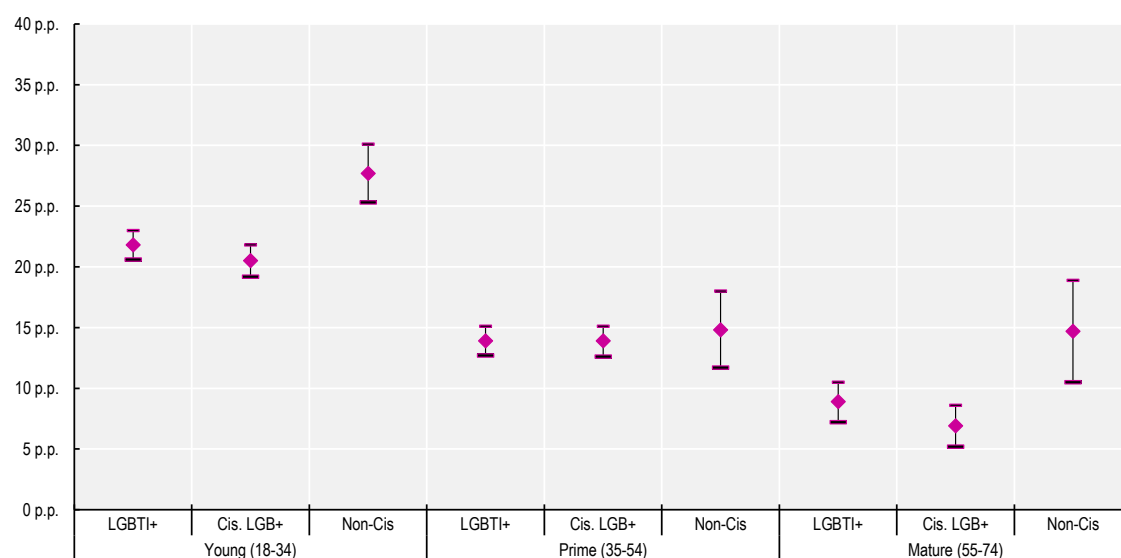
Note: The percentage differences are estimated from a Logistic regression, controlling for age groups (18-24, 25-29, 30-34, ..., 80-84, 85-88), race and ethnicity, marital and parental status, number of adults in the household, educational attainment, living in one of the 15 largest metropolitan statistical areas (metro areas), as well as state and survey wave fixed effects. The percentage point differences are estimated at the average of the covariates. The error bars depict 95% confidence intervals. This means that we can be 95% confident that the true value lies within the range of the error bar. In other words, if the error bar crosses the x-axis, then the percentage point difference cannot be deemed as statistically different from 0 (at least not at this 95% confidence level). Standard errors robust to heteroscedasticity. Person-level weights used. Source: Household Pulse Survey (21 July 2021 – 10 July 2023).

Figure 3.17. The mental health penalty for LGBTI+ individuals decreases with age

Panel A: Percentage point difference in the probability of having symptoms of generalised anxiety or major depressive disorder among US adults assigned male at birth (using cisgender straight individuals as the reference category), by sexual orientation, gender identity and age (2021-23)



Panel B: Percentage point difference in the probability of having symptoms of generalised anxiety or major depressive disorder among US adults assigned female at birth (using cisgender straight individuals as the reference category), by sexual orientation, gender identity and age (2021-23)



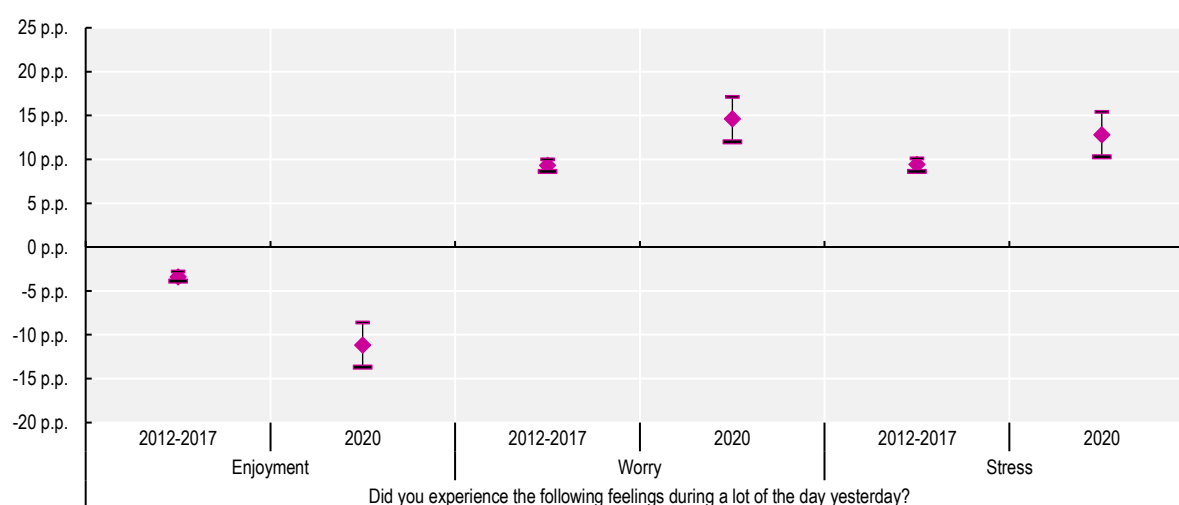
Note: The percentage differences are estimated from a Logistic regression, controlling for age groups, race and ethnicity, marital and parental status, number of adults in the household, educational attainment, living in one of the 15 largest metropolitan statistical areas (metro areas), as well as state and survey wave fixed effects. The percentage point differences are estimated at the average of the covariates. The error bars depict 95% confidence intervals. This means that we can be 95% confident that the true value lies within the range of the error bar. In other words, if the error bar crosses the x-axis, then the percentage point difference cannot be deemed as statistically different from 0 (at least not at this 95% confidence level). Standard errors robust to heteroscedasticity. Person-level weights used.

Source: Household Pulse Survey (21 July 2021 – 10 July 2023).

Comparing mental health responses from both the Gallup US Daily Survey (2012-17) and the Gallup Panel – COVID-19 Survey (2020) reveals a substantial negative impact of the COVID-19 pandemic on the mental well-being of Americans, particularly among LGBTI+ individuals – a trend corroborated by previous research (OECD, 2021^[31]; OECD, 2021^[88]; OECD, 2021^[89]). Before the pandemic, even after adjusting for essential demographic characteristics, LGBT adults were, in comparison to their non-LGBT peers, 4% less likely to experience enjoyment and 31% and 24% more likely to feel worried and stressed, respectively. However, with the onset of the pandemic, all three mental health indicators deteriorated, as depicted in Figure 3.18. Several factors could explain the widening of mental health disparities between LGBTI+ individuals and their cisgender straight peers during the COVID-19 crisis. These include pre-existing higher rates of mental health issues within the LGBTI+ population, potentially making them more susceptible to pandemic-related stress. Additionally, COVID-19 containment measures, such as social distancing and lockdowns, imposed unique challenges on LGBTI+ individuals, such as isolation from chosen families or forced cohabitation with unsupportive biological family members, further exacerbating mental health gaps.

Figure 3.18. The mental health penalty for LGBT individuals worsened at the onset of the COVID-19 pandemic

Percentage point difference in the probability of experiencing enjoyment, worry, or stress during much of the day preceding the interview with Gallup among US adults, using non-LGBT individuals as the reference category (2012-20)



Note: The percentage point differences are estimated from a Logistic regression, controlling for sex assigned at birth, age groups (18-24, 25-29, 30-34, ..., 80-84, 85-88), race and ethnicity, partnership and parental status, number of adults in the household, educational attainment, as well as state, year, and month fixed effects. The percentage point differences are estimated at the average of the covariates. The error bars depict 95% confidence intervals. This means that we can be 95% confident that the true value lies within the range of the error bar. In other words, if the error bar crosses the x-axis, then the percentage point difference cannot be deemed as statistically different from 0 (at least not at this 95% confidence level). Standard errors robust to heteroscedasticity. Person-level weights used.

Source: Gallup US Daily Survey (2012-17) and Gallup Panel – COVID-19 Survey (March – December 2020).

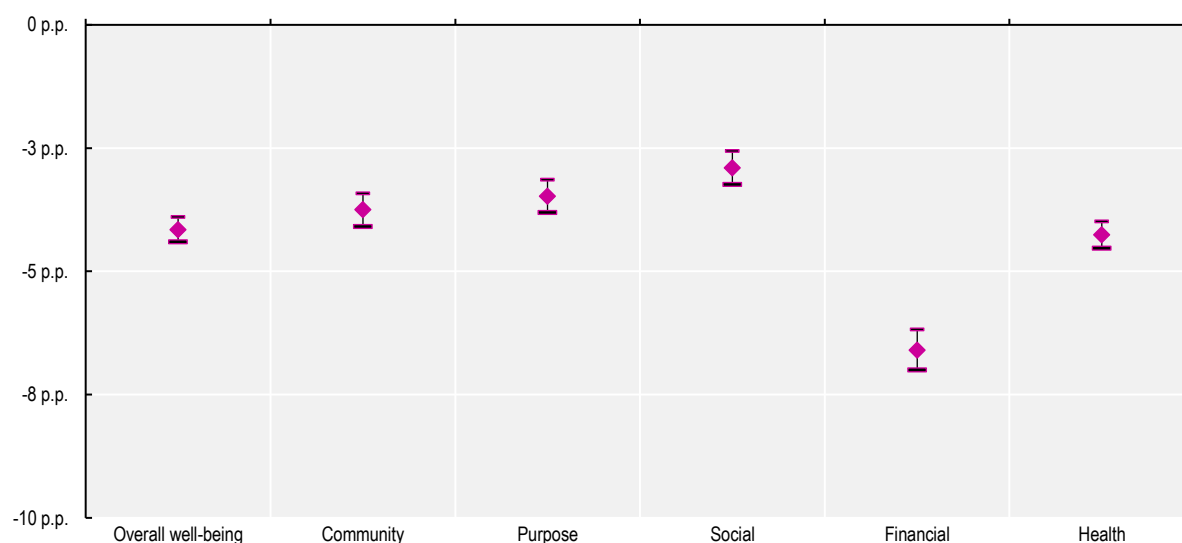
Unexplained gaps in well-being between LGBTI+ and non-LGBTI+ individuals

From 2014 to 2017, the Gallup US Daily Survey introduced several questions, as detailed in Box 2.6, to construct the multidimensional Gallup National Health and Well-Being Index. These questions were categorised into five components, subsequently aggregated into a global index, aimed at capturing key determinants of well-being: (i) the community one lives in; (ii) sense of purpose; (iii) social environment; (iv) financial situation; and (v) mental, behavioural and physical health.

Analysis of these data reveals significant unexplained well-being disparities between LGBT and non-LGBT individuals (Figure 3.19). On average, LGBT adults had a 4 percentage points (7%) lower overall well-being, with noticeable deficits across all five components of the index. Notably, the financial (7 percentage points, 11%) and health (4 percentage points, 7%) penalties of LGBT adults emerged as major contributors to their reduced well-being.

Figure 3.19. LGBT adults have lower overall well-being compared their non-LGBT peers

Percentage point difference by component of the Gallup National Health and Well-Being Index among US adults, using non-LGBT individuals as the reference category (2014-17)



Note: The percentage point differences are estimated from an OLS regression, controlling for sex assigned at birth, age groups (18-24, 25-29, 30-34, ..., 80-84, 85-88), race and ethnicity, partnership and parental status, number of adults in the household, educational attainment, as well as state, year, and month fixed effects. The error bars depict 95% confidence intervals. This means that we can be 95% confident that the true value lies within the range of the error bar. In other words, if the error bar crosses the x-axis, then the percentage point difference cannot be deemed as statistically different from 0 (at least not at this 95% confidence level). Standard errors robust to heteroscedasticity. Person-level weights used.

Source: Gallup US Daily Survey (2014-17).

Box 3.5. The questions used to construct the five components of the Gallup National Health and Well-Being Index

The questions included in each of the five components of the Gallup National Health and Well-Being Index are as follows:

The community one lives in: “I can’t imagine living in a better community than the one I live in today”, “I am satisfied with the city or area where I live”, “The city or area where I live is a perfect place for me”, “I am proud of my community (or the area where I live)”, “I always feel safe and secure”, “The house or apartment that I live in is ideal for me and my family”, “In the last 12 months, I have received recognition for helping to improve the city or area where I live”.

Sense of purpose: “There is a leader in my life who makes me enthusiastic about the future”, “I like what I do every day”, “In the past 12 months, I have reached most of my goals”, “I get to use my strengths to do what I do best every day”, “I learn or do something interesting every day”.

Social environment: “My relationship with my spouse, partner, or closest friend is stronger than ever”, “I always make time for regular trips or vacations with friends and family”, “Someone in my life always encourages me to be healthy”, “My friends and family give me positive energy every day”.

Financial situation: “There were no occasions in the past 12 months when I did not have enough money to buy food that I or my family needed”, “I have enough money to do everything I want to do”, “In the last seven days, I have not worried about money”, “There were no occasions in the past 12 months when I did not have enough money to pay for healthcare and/or medicines that I or my family needed”, “Compared to the people I spend time with, I am satisfied with my standard of living”.

Mental, behavioural and physical health: “I never use drugs or medication (including prescription drugs) which affect my mood or help me relax”, “My physical health is near-perfect”, “I have no health problems that prevent me from doing any of the things people my age normally can do”, “I didn’t experience physical pain yesterday”, “My healthcare provider didn’t tell me that I have any restrictions (on the amount or type of exercise I can do) that would limit my ability to exercise”, “Over the last two weeks, I have not been bothered by little interest or pleasure in doing things”, “In the last seven days, I have felt active and productive every day”, “I have never been told by a physician or nurse that I have any of the following (High Blood Pressure, High Cholesterol, Diabetes, Depression, Heart Attack, Asthma, Cancer)”, “I don’t have any other health problems or conditions”, “A doctor would say that I do a great job of managing my health”, “Days w/ 5 servings of fruits/veggies”, “Days w/ 30 min-plus exercise”, “I ate healthy all day yesterday”, “I always feel good about my physical appearance”, “I have little to no alcoholic drinks in a typical week (Note: One drink is equal to one beer, one glass of wine, one shot of liquor, or one mixed drink)”, “I don’t smoke”, “I don’t use any of the following tobacco products (cigarettes, cigars, pipe, smokeless tobacco, chew or snuff, other)”. In addition to these questions, the height and weight of the respondent are combined into a Body Mass Index that also enters the creation of the health component of the Gallup National Health and Well-Being Index.

Source: Gallup Inc.

References

- Abreu, R. et al. (2022), “Impact of gender-affirming care bans on transgender and gender diverse youth: Parental figures’ perspective”, *J Fam Psychol.* 2022 Aug;36(5):643-652, <https://doi.org/10.1037/fam0000987>. [81]
- Aksoy, B. and I. Chadd (2023), *Gender, Sexual Identity, and Competitiveness*, SSRN 4362469 2023. [91]
- Aksoy, C., C. Carpenter and J. Frank (2018), “Sexual Orientation and Earnings: New Evidence from the United Kingdom”, *ILR Review*, 71(1), 242-272, <https://doi.org/10.1177/0019793916687759>. [66]
- Aksoy, C. et al. (2019), “Gay glass ceilings: Sexual orientation and workplace authority in the UK”, *Journal of Economic Behavior & Organization*, Vol. 159, pp. 167-180, <https://doi.org/10.1016/j.jebo.2019.01.013>. [65]
- Badgett, M. (2018), *Left out? Lesbian, gay, and bisexual poverty in the U.S.*, Population Research and Policy Review, 37(5), 667–702. [26]
- Badgett, M. (2014), *The Economic Cost of Stigma and the Exclusion of LGBT People: A Case Study of India*, Washington, D.C. : World Bank Group, <http://documents.worldbank.org/curated/en/527261468035379692/The-economic-cost-of-stigma-and-the-exclusion-of-LGBT-people-a-case-study-of-India>. [42]
- Badgett, M. (1995), *The wage effects of sexual-orientation discrimination*, Industrial and Labor Relations Review, 48(4), 726–739. [57]
- Badgett, M., C. Carpenter and A. Sansone (2021), “LGBTQ Economics”, *Journal of Economic Perspectives*, 35 (2): 141-70, pp. 141-70. [25]
- Badgett, M., S. Choi and B. Wilson (2019), *LGBT Poverty in the United States: A Study of Differences between Sexual Orientation and Gender Identity Groups*, Los Angeles, CA: Williams Institute, UCLA School of Law. [28]
- Bardales, N. (2013), “Finding a Job in a Beard and a Dress: Evaluating the Effectiveness of Transgender AntiDiscrimination Laws”, <http://www.semanticscholar.org/paper/Finding-a-Job-in-%E2%80%9C-a-Beard-and-a-Dress-%E2%80%9D-%3A-the-of-%E2%88%97-Bardales/b260619b3c78c224663689a6aae11785ed4aedde>. [52]
- Baumle, A., M. Badgett and S. Boucher (2019), *New research on sexual orientation and gender identity discrimination: Effect of state policy on charges filed at the EEOC*, Journal of Homosexuality, 67(8), 1135–1144. [48]
- Bettinsoli, M., A. Suppes and J. Napier (2020), *Predictors of Attitudes Toward Gay Men and Lesbian Women in 23 Countries*, Social Psychological and Personality Science, 11(5), 697–708, <https://doi.org/10.1177/1948550619887785>. [4]
- Black, D., S. Sanders and L. Taylor (2007), “The economics of lesbian and gay families”, *Journal of Economic Perspectives* 21(2): 53-70. [72]
- Bolibol, A. et al. (2023), “Health Insurance Coverage And Access To Care Among LGBT Adults, 2013–19”, *Health Affairs*, Vol. 42/6, pp. 858-865, <https://doi.org/10.1377/hlthaff.2022.01493>. [86]

- Brown, A. (2019), “Bisexual adults are far less likely than gay men and lesbians to be ‘out’ to the people in their lives”, *Pew Research Center*, <https://www.pewresearch.org/short-reads/2019/06/18/bisexual-adults-are-far-less-likely-than-gay-men-and-lesbians-to-be-out-to-the-people-in-their-lives/>. [18]
- Bryant-Lees, K. and M. Kite (2021), “Evaluations of LGBT job applicants: consequences of applying “out””, *Equality, Diversity and Inclusion*, Vol. 40 No. 7, pp. 874-891. [49]
- Buser, T., L. Geijtenbeek and E. Plug (2014), *Sexual orientation, competitiveness and income*, *Journal of Economic Behavior & Organization*, 2018, 151, 191-198. [90]
- Business, B. (ed.) (2024), *Breaking the Rainbow Ceiling: How LGBTQ+ people can thrive and succeed at work*. [64]
- Carpenter, C. et al. (2021), “Effects of Access to Legal Same-Sex Marriage on Marriage and Health”, *J. Pol. Anal. Manage.*, 40: 376-411, <https://doi.org/10.1002/pam.22286>. [76]
- Carpenter, C., S. Eppink and G. Gonzalez (2020), *Transgender status, gender identity, and socioeconomic outcomes in the United States*, *Industrial & Labor Relations Review*, 73(3), 573-599. [29]
- Carpenter, C., M. Lee and L. Nettuno (2022), *Economic outcomes for transgender people and other gender minorities in the United States: First estimates from a nationally representative sample*, *Southern Economic Journal* Volume89, Issue2. [23]
- Chai, L. and M. Maroto (2019), *Economic insecurity among gay and bisexual men: Evidence from the 1991–2016 U.S. General Social Survey*, *Sociological Perspectives*, 63(1), 50–68. [27]
- Choi, S. et al. (2015), “Serving our youth 2015: The needs and experiences of Lesbian, Gay, Bisexual, Transgender, and Questioning youth experiencing homelessness”, *The Williams Institute with True Colors Fund*. [40]
- Coffman, K., L. Coffman and K. Marzilli (2024), *Non-Binary Gender Economics*, National Bureau of Economic Research. [12]
- D’Ippoliti, C. and F. Botti (2017), “Sex Work among Trans People: Evidence from Southern Italy”, *Feminist Economics*, Vol. 23/3, pp. 77-109, <https://doi.org/10.1080/13545701.2016.1177656>. [35]
- Dawson, L., A. Kirzinger and J. Kates (2021), “The Impact of the COVID-19 Pandemic on LGBT People”, *KFF*, https://www.kff.org/coronavirus-covid-19/poll-finding/the-impact-of-the-covid-19-pandemic-on-lgbt-people/?utm_campaign=KFF-2021-polling-surveys&utm_medium=email&hsmi=115337308&hsenc=p2ANqtz--3JuXAA0nFrNGilJMeNLmylXdGcM2uJ28qSTQK1PdIROsr_PSGklt1SBDp9qGoV. [68]
- Dillbary, J. and E. Griffin (2019), “An Empirical Analysis of Sexual Orientation Discrimination”, *University of Chicago Law Review: Vol. 86: Iss. 1, Article 4*, <https://chicagounbound.uchicago.edu/uclrev/vol86/iss1/4>. [46]
- Doan Van, E. et al. (2019), *Perceived Discrimination, Coping Mechanisms, and Effects on Health in Bisexual and Other Non-Monosexual Adults*, *Arch Sex Behav* 48, 159–174, <https://doi.org/10.1007/s10508-018-1254-z>. [5]

- Dodge, B. et al. (2016), *Attitudes toward bisexual men and women among a nationally representative probability sample of adults in the United States*, PLoS One, 11(10), e0164430. [7]
- Drydakis, N. (2022), "Sexual orientation and earnings: A meta-analysis 2012-2020", *Journal of Population Economics* 35: 409-440. [58]
- Dunne, G., S. Prendergast and D. Telford (2002), "Young, gay, homeless and invisible: A growing population?", *Culture, Health and Sexuality* 4(1): 103-115. [38]
- Dyar, C. and B. Feinstein (2018), *Binegativity: Attitudes toward and stereotypes about bisexual individuals*, In D. J. Swan & S. Habibi (Eds.), *Bisexuality: Theories, research, and recommendations for the invisible sexuality* (pp. 95–111). Springer International Publishing, https://doi.org/10.1007/978-3-319-71535-3_6. [6]
- FBI (2023), "Hate Crime in the United States Incidence Analysis", <https://cde.ucr.cjis.gov/LATEST/webapp/#/pages/explorer/crime/hate-crime>. [83]
- Flores, A. (2021), *Social Acceptance of LGBTI People in 175 Countries and Locations: 1981 to 2020*, UCLA Williams Institute. [2]
- Flores, A. et al. (2022), "Hate crimes against LGBT people: National Crime Victimization Survey, 2017-2019", *PLoS One*. 2022 Dec 21;17(12), <https://doi.org/10.1371/journal.pone.0279363>. [82]
- Folch, M. (2022), *The LGBTQ+ Gap: Recent Estimates for Young Adults in the United States*, <https://doi.org/10.2139/ssrn.4072893>. [17]
- Fredriksen-Goldsen, K. et al. (2014), "Successful Aging Among LGBT Older Adults: Physical and Mental Health-Related Quality of Life by Age Group", *The Gerontologist*, Vol. 55/1, pp. 154-168, <https://doi.org/10.1093/geront/gnu081>. [69]
- Fumarco, L. et al. (2020), "Gender Identity, Race, and Ethnicity-based Discrimination in Access to Mental Health Care: Evidence from an Audit Correspondence Field Experiment", *National Bureau of Economic Research, Working Paper Series* 28164, <https://doi.org/10.3386/w28164>. [85]
- Giddings, L. et al. (2014), "Birth cohort and the specialization gap between same-sex and different-sex couples", *Demography* 51(2): 509-534. [63]
- GLAAD (2023), *Where We Are on TV (2022-2023)*, Technical Report, GLAAD Media Institute, New York (2023). [14]
- Harrell, B. (2022), "Conversion Therapy Bans, Suicidality, and Mental Health", https://static1.squarespace.com/static/5f0df932fe24d51944633598/t/636a767467ca3d1c62923e1e/1667921524648/Conversion+Therapy+Bans%2C+Suicidality%2C+and+Mental+Health_10.10.22.pdf. [77]
- Hegarty, P. and A. Smith (2023), "Public understanding of intersex: an update on recent findings", *Int J Impot Res*. 2023 Feb;35(1):72-77, <https://doi.org/10.1038/s41443-021-00485-w>. [13]
- Herek, G. et al. (2010), *Demographic, Psychological, and Social Characteristics of Self-Identified Lesbian, Gay, and Bisexual Adults in a US Probability Sample*, *Sexuality Research and Social Policy* 7 (3): 176–200. [9]

- Hoffarth, M. et al. (2016), *Development and validation of the Attitudes Towards Asexuals (ATA) scale*, Psychology & Sexuality, <https://doi.org/10.1080/19419899.2015.1050446>. [10]
- Hofmarcher, T. and E. Plug (2022), “Specialization in same-sex and different-sex couples”, *Labour Economics*, Vol. 77, p. 101995, <https://doi.org/10.1016/j.labeco.2021.101995>. [60]
- Human Rights Campaign Foundation (2023), *LGBTQ+ Americans Under Attack: A Report and Reflection on the 2023 State Legislative Session*, Human Rights Campaign Foundation, <https://hrc-prod-requests.s3-us-west-2.amazonaws.com/Anti-LGBTQ-Legislation-Impact-Report.pdf>. [78]
- Human Rights Campaign Foundation (2018), *A workplace divided: understanding the climate for LGBTQ workers nationwide*, <https://www.hrc.org/resources/a-workplace-divided-understanding-the-climate-for-lgbtq-workers-nationwide>. [21]
- Jepsen, C. and L. Jepsen (2015), “Labor-Market Specialization within Same-Sex and Different-Sex Couples”, *Industrial Relations: A Journal of Economy and Society*, 54 (1), 2015, p. 109-130. [61]
- Jepsen, C. and L. Jepsen (2009), “Does home ownership vary by sexual orientation?”, *Regional Science and Urban Economics* 39(3): 307–315. [43]
- Jepsen, L. and C. Jepsen (2002), “An Empirical Analysis of the Matching Patterns of Same-Sex and Opposite-Sex Couples”, *Demography*, 39 (3), 2002, p. 435-453. [62]
- Klawitter, M. (2015), *Meta-analysis of the effects of sexual orientation on earnings*, *Industrial Relations*, 54(1), 4–32, <https://doi.org/10.1111/irel.12075>. [59]
- Kline, P., E. Rose and C. Walters (2024), *A Discrimination Report Card*, National Bureau of Economic Research, Cambridge, MA, <https://doi.org/10.3386/w32313>. [55]
- Kline, P., E. Rose and C. Walters (2022), “Systemic Discrimination Among Large U.S. Employers”, *The Quarterly Journal of Economics*, Vol. 137/4, pp. 1963-2036, <https://doi.org/10.1093/qje/qjac024>. [54]
- Leppel, K. (2007), “Home-ownership among opposite- and same-sex couples in the US”, *Feminist Economics* 13(1): 1-30. [45]
- Leppel, K. (2007), “Married and unmarried, opposite- and same-sex couples: A decomposition of homeownership differences”, *Journal of Housing Research* 16(1): 61-81. [44]
- MacInnis, C. and G. Hodson (2012), *Intergroup bias toward “Group X”: Evidence of prejudice, dehumanization, avoidance, and discrimination against asexuals*, *Group Processes & Intergroup Relations*, 15(6), 725–743. [11]
- Make the Road New York (2010), “Transgender Need Not Apply: A Report on Gender Identity Job Discrimination”, Brooklyn, NY: *Make the Road New York*. [53]
- Mann, S., T. Campbell and D. Nguyen (2023), “Access to Gender-Affirming Care and Transgender Mental Health: Evidence from Medicaid Coverage”, *American Journal of Health Economics*, <https://doi.org/10.1086/728080>. [80]
- Martell, M. and L. Roncolato (2023), *Economic Vulnerability of Sexual Minorities: Evidence from the US Household Pulse Survey*, *Popul Res Policy Rev.* 2023;42(2):28. [24]

- Medina, C. and L. Mahowald (2023), *Discrimination and Barriers to Well-Being: The State of the LGBTQI+ Community in 2022*, The Center for American Progress, <https://www.americanprogress.org/article/discrimination-and-barriers-to-well-being-the-state-of-the-lgbtqi-community-in-2022/>. [16]
- Medina, C. and L. Mahowald (2021), “Key Issues Facing People With Intersex Traits”, *The Center for American Progress*, <https://www.americanprogress.org/article/key-issues-facing-people-intersex-traits/>. [30]
- Meyer, I. (2003), *Prejudice, social stress, and mental health in lesbian, gay and bisexual populations: conceptual issues and research evidence*, *Psychol Bull.* 2003;129:674-697. [74]
- Miller, A. (2018), “Searching for gaydar: Blind spots in the study of sexual orientation perception”, *Psychology & Sexuality*, Vol. 9/3, pp. 188-203, <https://doi.org/10.1080/19419899.2018.1468353>. [19]
- Mishel, E. (2016), *Discrimination against queer women in the U.S. workforce: A résumé audit study*, *Socius: Sociological Research for a Dynamic World*, 2, 1–13. [51]
- Movement Advancement Project (2020), “The disproportionate impacts of COVID-19 on LGBTQ households in the U.S.”, <http://www.lgbtmap.org/2020-covid-lgbtq-households>. [33]
- National Academies of Sciences, Engineering, and Medicine (2020), *Understanding the Well-Being of LGBTQI+ Populations*, Washington, DC: The National Academies Press, <https://doi.org/10.17226/25877>. [37]
- National Center for Health Statistics (2023), *U.S. Census Bureau, Household Pulse Survey, 2020–2023. Anxiety and Depression*, <https://www.cdc.gov/nchs/covid19/pulse/mental-health.htm>. [87]
- Neumark, D., I. Burn and P. Button (2019), “Is It Harder for Older Workers to Find Jobs? New and Improved Evidence from a Field Experiment”, *Journal of Political Economy*, Vol. 127/2, pp. 922-970, <https://doi.org/10.1086/701029>. [67]
- O. Rule, N. and N. Ambady (2008), “Brief exposures: Male sexual orientation is accurately perceived at 50ms”, *Journal of Experimental Social Psychology*, Vol. 44/4, pp. 1100-1105, <https://doi.org/10.1016/j.jesp.2007.12.001>. [20]
- OECD (2021), “COVID-19 and Well-being: Life in the Pandemic”, *OECD Publishing, Paris*, <https://doi.org/10.1787/1e1ecb53-en>. [31]
- OECD (2021), “Fitter Minds, Fitter Jobs: From Awareness to Change in Integrated Mental Health, Skills and Work Policies”, *Mental Health and Work*, *OECD Publishing, Paris*, <https://doi.org/10.1787/a0815d0f-en>. [36]
- OECD (2021), “Supporting young people’s mental health through the COVID-19 crisis”, *OECD Policy Responses to Coronavirus (COVID-19)*, *OECD Publishing, Paris*, <https://doi.org/10.1787/84e143e5-en>. [89]
- OECD (2021), “Tackling the mental health impact of the COVID-19 crisis: An integrated, whole-of-society response”, *OECD Policy Responses to Coronavirus (COVID-19)*, *OECD Publishing, Paris*, <https://doi.org/10.1787/0ccaafa0b-en>. [88]

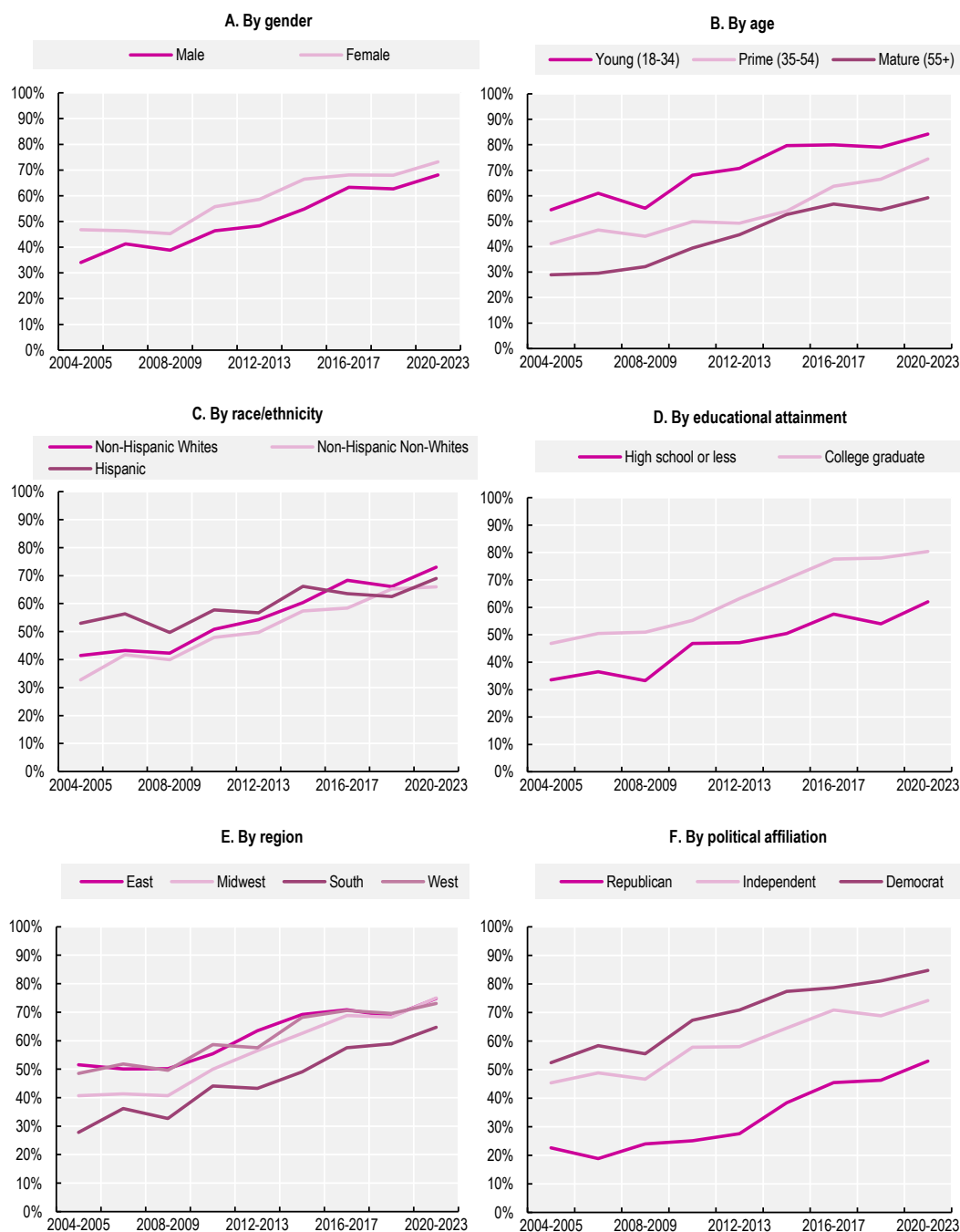
- OECD (2020), “All Hands In? Making Diversity Work for All”, *OECD Publishing, Paris*, [73]
<https://doi.org/10.1787/efb14583-en>.
- OECD (2019), *Society at a Glance 2019: OECD Social Indicators*, OECD Publishing, Paris, [1]
https://doi.org/10.1787/soc_glance-2019-en.
- Pachankis, J. et al. (2020), “Sexual Orientation Concealment and Mental Health: A Conceptual and Meta-Analytic Review”, *Psychol Bull.* 2020 Oct; 146(10): 831–871, [22]
<https://doi.org/10.1037/bul0000271>.
- Pew Research Center (2021), *Rising shares of U.S. adults know someone who is transgender or goes by gender-neutral pronouns*, Washington, DC: Pew Research Center, [15]
<https://www.pewresearch.org/short-reads/2021/07/27/rising-shares-of-u-s-adults-know-someone-who-is-transgender-or-goes-by-gender-neutral-pronouns/>.
- Pew Research Center (2013), “A Survey of LGBT Americans”, [41]
<https://www.pewresearch.org/social-trends/2013/06/13/a-survey-of-lgbt-americans/>.
- Pew Research Center (2013), “A Survey of LGBT Americans. Chapter 2: Social Acceptance”, [8]
<https://www.pewresearch.org/social-trends/2013/06/13/chapter-2-social-acceptance/>.
- Plug, E., D. Webbink and N. Martin (2014), “Sexual orientation, prejudice, and segregation”, [70]
Journal of Labor Economics 32(1): 123-159.
- Quillian, L., J. Lee and M. Oliver (2020), “Evidence from Field Experiments in Hiring Shows Substantial Additional Racial Discrimination after the Callback”, *Social Forces*, Vol. 99/2, pp. 732-759, [56]
<https://doi.org/10.1093/sf/soaa026>.
- Raifman, J. et al. (2017), *Difference-in-Differences Analysis of the Association Between State Same-Sex Marriage Policies and Adolescent Suicide Attempts*, *JAMA Pediatr.* 2017;171(4):350–356. [75]
- Rew, L., R. Fouladi and R. Yockey (2002), “Sexual health practices of homeless youth”, *Journal of Nursing Scholarship* 34(2): 139-145. [39]
- Sabin, J., R. Riskind and B. Nosek (2015), “Health Care Providers’ Implicit and Explicit Attitudes Toward Lesbian Women and Gay Men”, *Am J Public Health.* 2015 Sep;105(9):1831-41, [84]
<https://doi.org/10.2105/AJPH.2015.302631>.
- Schilt, K. and M. Wiswall (2008), “Before and after: Gender transitions, human capital, and workplace experiences”, *The B.E. Journal of Economic Analysis and Policy* 8:1 (2008): 1–28. [34]
- Tate, D. and C. Patterson (2019), “Desire for Parenthood in Context of Other Life Aspirations Among Lesbian, Gay, and Heterosexual Young Adults”, *Front Psychol.* 2019 Nov 28;10:2679, [47]
<https://doi.org/10.3389/fpsyg.2019.02679>.
- Tilcsik, A. (2011), *Pride and prejudice: Employment discrimination against openly gay men in the United States*, *American Journal of Sociology*, 117(2), 586–626. [50]
- Trans Legislation Tracker (2024), “2024 anti-trans bills tracker: Anti-trans bills under consideration and passed, 2021-2024”, *Trans Legislation Tracker*, [79]
<https://translegislation.com/>.

- Ueno, K., T. Roach and A. Pena-Talamantes (2013), “Sexual orientation and gender typicality of the occupation in young adulthood”, *Social Forces* 92(1): 81-108. [71]
- Valfort, M. (2017), “LGBTI in OECD countries: A review”, *OECD Social, Employment and Migration Working Papers*, No. 198, OECD Publishing, Paris, <https://doi.org/10.1787/d5d49711-en>. [3]
- Wenham, C. (2020), “The gendered impact of the COVID-19 crisis and post-crisis period”, *European Parliament, Brussels*, <http://www.europarl.europa.eu/supporting-analyses>. [32]

Annex 3.A. Additional Figures

Annex Figure 3.A.1. The upward trend in support for same-sex marriage applies to all demographic groups, although significant heterogeneities remain in terms of levels

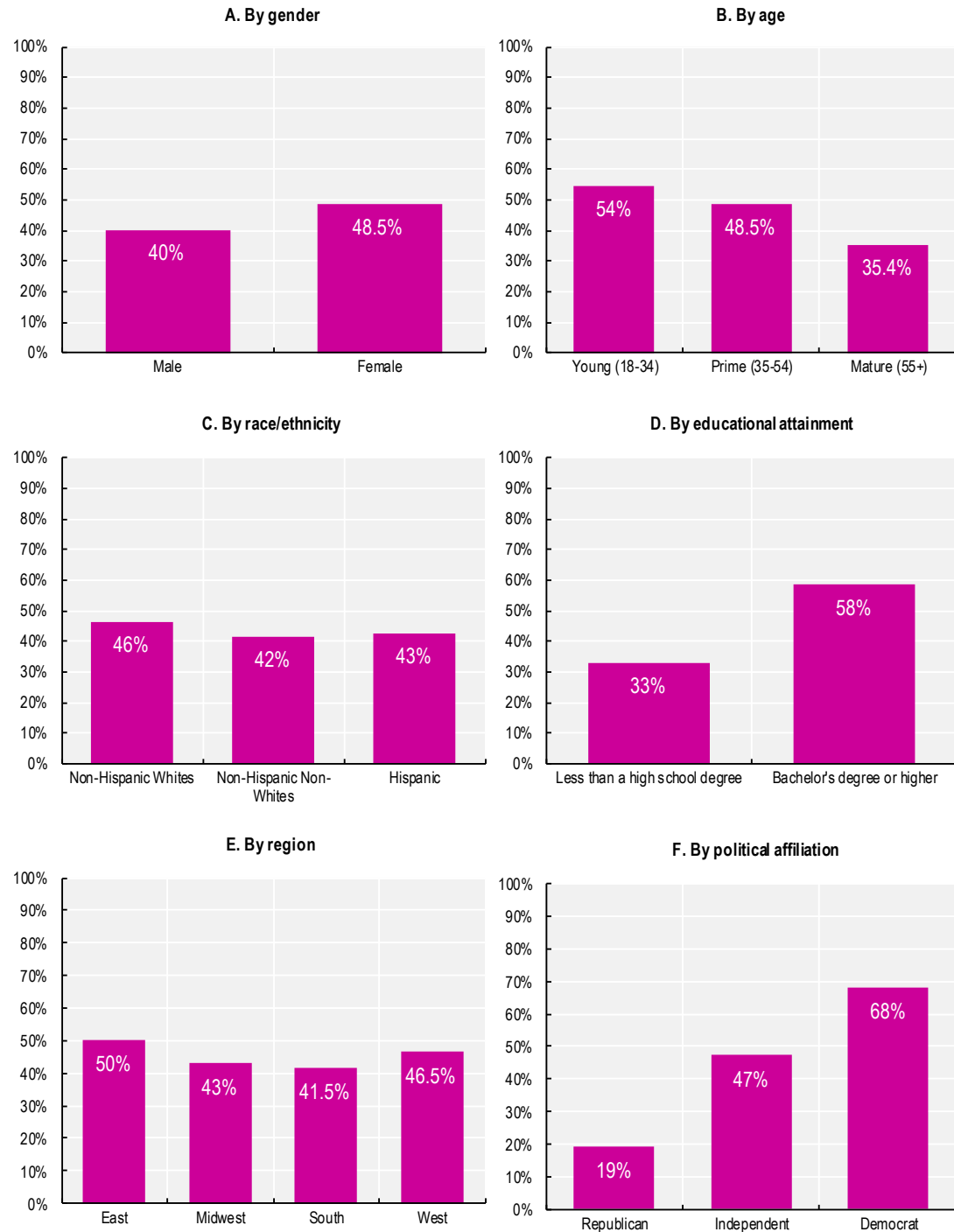
Evolution of the share of US adults in favour of same-sex marriage by key demographics (2004-23)



Source: Gallup Poll Social Series (2004-23). Person-level weights used.

Annex Figure 3.A.2. Support for transgender individuals is greater among women, younger individuals, Whites, the better educated, residents in non-Southern states, and democrats

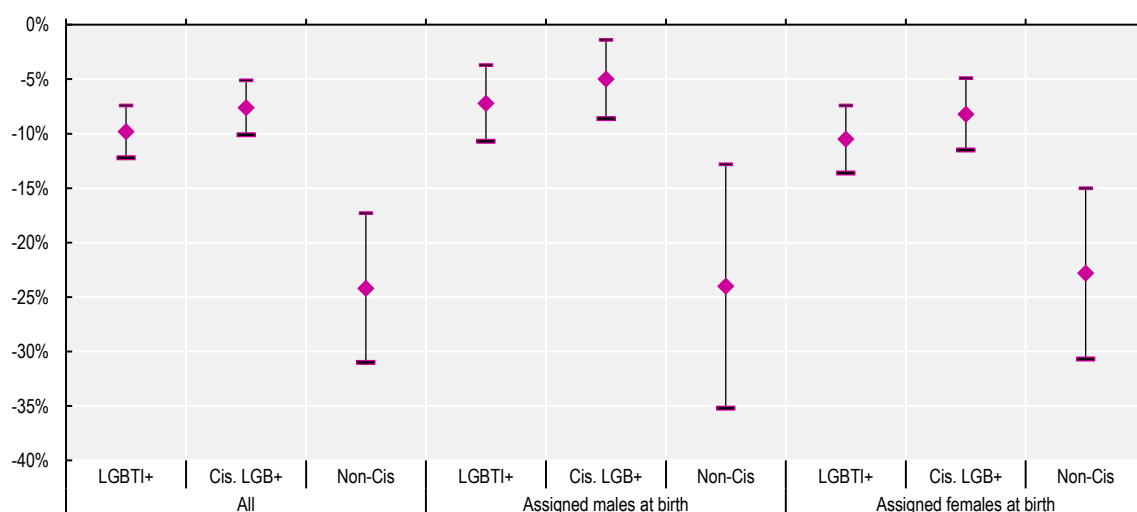
Share of US adults who considered changing one's gender to be morally acceptable (2021-23)



Source: Gallup Poll Social Series (2021 and 2023). Person-level weights used.

Annex Figure 3.A.3. When the effects of sectoral and occupational sorting are ignored, the LGBTI+ penalty worsens to reach 10%

Percentage difference in household income among employed working-age US adults living alone (using cisgender straight individuals as the reference category), by sexual orientation, gender identity and sex assigned at birth (2021-23)



Note: The percentage differences are estimated from an Interval regression of the logarithm of household income, controlling for age groups (18-24, 25-29, 30-34, ..., 70-74), race and ethnicity, marital status, educational attainment, living in one of the 15 largest metropolitan statistical areas (metro areas), as well as state and survey wave fixed effects. The error bars depict 95% confidence intervals. This means that we can be 95% confident that the true value lies within the range of the error bar. In other words, if the error bar crosses the x-axis, then the percentage point difference cannot be deemed as statistically different from 0 (at least not at this 95% confidence level). Standard errors robust to heteroscedasticity. Person-level weights used. See Box 3.1 for more details on Interval regression methodology.

Source: Household Pulse Survey (14 September 2022 – 10 July 2023).

Notes

¹ YouGov is an international Internet-based market research and data analytics firm. Its methodology involves online opt-in panels, i.e. obtaining responses from an invited group of Internet users, and then weighting these responses so that the sample becomes nationally representative with respect to essential demographics.

² Respondents are members of the Center's American Trends Panel (ATP), an online survey panel that is recruited through national, random sampling of residential addresses. Thus, nearly all US adults have a chance of selection. The survey is weighted to be representative of the US adult population by gender, race, ethnicity, partisan affiliation, education and other categories. More about the ATP's methodology: www.pewresearch.org/our-methods/u-s-surveys/the-american-trends-panel/.

³ Results available upon request.

⁴ The burgeoning field of behavioural economics focusing on LGBTI+ individuals reveals distinctive patterns in beliefs and preferences that can differentially influence their educational, labour market, financial, and health outcomes. For example, research by (Coffman, Coffman and Marzilli, 2024^[12]) indicates that non-binary individuals exhibit lower levels of competitiveness, risk tolerance, and patience compared to men or women, and show less prosocial behaviour than women. Additionally, (Buser, Geijtenbeek and Plug, 2014^[90]; Aksoy and Chadd, 2023^[91]) suggest that gay men are less competitive than their straight counterparts. Although further research is necessary to fully understand the roots of these behavioural differences, it is plausible that the persistence of anti-LGBTI+ discrimination plays a considerable role in shaping these outcomes.

⁵ Results available upon request.

⁶ Source: U.S. Bureau of Labor Statistics (<https://beta.bls.gov/dataQuery/search>).

⁷ Results available upon request.

⁸ These results are robust to analysing the risk of generalised anxiety disorder and the risk of major depressive disorder separately.

⁹ Results available upon request.

4 Does LGBTI+ equality pay?

This chapter analyses the economic returns of levelling the playing field for LGBTI+ Americans. It first explores the potential increase in GDP that could result from closing unexplained LGBTI+ labour market gaps, drawing on the OECD long-term model, before discussing several additional benefits that would emerge from eliminating anti-LGBTI+ discrimination. The chapter concludes by quantifying the economic and well-being burden of mental health disparities among LGBTI+ individuals. Considering only the mechanisms included in the OECD long term model, the economic consequences of achieving LGBTI+ equality appear substantial. Under realistic assumptions, if the United States reaches this milestone by 2050, it could expect an increase in GDP equal to 2.6% of the baseline GDP. This corresponds to a yearly increase in GDP equal to 0.1%, which amounts to 5% of the average annual US GDP growth over the past ten years. Of course, a more ambitious goal of closing the gaps by 2030 would lead to a higher yearly GDP increase, representing about 10% of the average annual US GDP growth observed between 2013 and 2023.

4.1. Introduction and main findings

LGBTI+ inclusion is a question of human rights, but it also makes a lot of economic sense. Ending discrimination against LGBTI+ Americans and creating inclusive environments and opportunities in and outside the labour market not only means that LGBTI+ Americans can get their fair share, but also that there are more goods and services being produced in the economy, and more income to spend and to save or invest, thus benefiting all of society. The lack of a strong evidence-based economic case for LGBTI+ inclusion is a barrier to further progressing in social acceptance of LGBTI+ individuals, and thus to expanding legal and lived equality for this population (OECD, 2020^[1]): informing people, companies, and policy makers about the economic gains of LGBTI+ inclusion increases their support for LGBTI+ equality, including in countries where anti-LGBTI+ prejudice is widespread (Aksoy et al., 2022^[2]).

This chapter aims to fill the gap by providing an evaluation of some of the economic returns of levelling the playing field for LGBTI+ individuals in the United States. It begins by exploring the potential increase in GDP that could result from closing LGBTI+ labour market gaps, drawing on the OECD long-term model. It then contextualises these figures by discussing several additional benefits that are not captured by this model but that could emerge from eliminating anti-LGBTI+ discrimination. The chapter concludes by quantifying the economic and well-being burden of mental health disparities among LGBTI+ individuals.

Main findings

- Anti-LGBTI+ discrimination prevents LGBTI+ individuals from reaching their full employment and labour productivity potential.
 - It raises barriers to their hiring and tends to confine them, when they are employed, to lower-skilled positions than they would otherwise be able to occupy.
 - These detrimental consequences of discrimination might be amplified by negative feedback effects from the labour supply to the extent that reduced labour market prospects undermine incentives to participate in the labour force as well as productivity at work.
 - These harmful repercussions on LGBTI+ individuals' labour supply and productivity are further compounded by anti-LGBTI+ discrimination (in and outside the labour market) undermining the health of LGBTI+ individuals.
- Closing LGBTI+ labour market gaps will substantially increase US GDP.
 - By projecting GDP up to 2060, the OECD long-term model offers a framework to quantify the additional GDP, compared to a baseline, that could result from gradually bridging the unexplained gaps in employment and labour productivity for LGBTI+ individuals, along different time horizons. This dynamic model thus helps to strengthen the economic case for LGBTI+ equality with realistic projections, acknowledging that achieving LGBTI+ equality is not an overnight process but one that requires gradual change.
 - Two different baseline scenarios are considered.
 - A sex- and cohort- blind approach: This approach is considered naïve as it fails to account for variations in both the share of self-identified LGBTI+ individuals and their unexplained labour market disparities across different sexes and cohorts. It assumes that the 2024 figures for each sex and age group are equal to their average in the total working-age population, maintaining these averages static through 2060.
 - A cohort approach: In contrast, this approach is deemed realistic as it recognises that both the share of self-identified LGBTI+ individuals and the extent of their labour market disparities differ by sex and cohort. It assumes that the 2024 figures for each sex and

age group are equal to the averages observed within each of these groups, with these averages remaining unchanged as the groups age.

- The economic benefits of achieving LGBTI+ equality are projected to be higher under the realistic method (cohort approach). This stems from the fact that younger generations not only have a higher proportion of self-identified LGBTI+ individuals but also face greater labour market disparities – likely due to a reduced incidence of non-disclosure bias compared to older generations. Consequently, the cohort approach generates an increase in both the proportion of self-identified LGBTI+ individuals and the extent of their labour market disparities within the total population.
- Specifically, should the United States achieve LGBTI+ equality by either 2030, 2040 or 2050 (meaning that the employment and labour productivity of LGBTI+ adults converge linearly with those of their cisgender straight peers by these target years), it could expect an increase in GDP ranging from 1.5% (sex- and cohort-blind approach) to 1.6% (cohort approach) of the baseline GDP by 2030 if LGBTI+ equality is achieved by 2030, from 1.5% to 2.1% of the baseline GDP by 2040 if LGBTI+ equality is achieved by 2040, and from 1.5% to 2.6% of the baseline GDP if LGBTI+ equality is achieved by 2050.
- Attaining LGBTI+ equality by 2030 would result in an average annual increase in GDP of approximately 0.2% under the two approaches, which amounts to 10% of the average annual US GDP growth observed between 2013 and 2023. However, delaying the achievement of LGBTI+ equality would yield reduced economic benefits when translated in yearly GDP gains. The average annual increase in GDP ranges between 0.09% and 0.13% if the target year is 2040, and between 0.06% and 0.10% if the target year is 2050.
- The OECD long-term model can also be used in a static capacity to estimate the cost of not closing employment and labour productivity penalties faced by LGBTI+ individuals or, conversely, the gain associated with closing those gaps immediately. When used in this way, it reveals that instantaneously achieving LGBTI+ equality results in a GDP gain equivalent to approximately 1% of the baseline GDP, irrespective of the baseline scenario considered.
- There are many more benefits from LGBTI+ equality than just the economic gains derived from the OECD long-term model.
 - For instance, LGBTI+ equality will improve public finances through increased tax revenues and reduced public expenditures. An uptick in production and labour earnings enhances public revenues from corporation tax, income tax, and social security contributions. These positive effects might be further magnified by increased consumption tax revenue, as LGBTI+ individuals spend more on consumer goods and services than their cisgender straight peers. Moreover, eradicating anti-LGBTI+ discrimination is conducive to reduced public expenditures, as greater workforce participation reduces the need for unemployment benefits, active labour market policies, and social transfers. Greater LGBTI+ equality is also anticipated to reduce public health expenditures, as stigma is a significant driver contributing to the poorer mental, behavioural, and physical health of LGBTI+ individuals.
 - Beyond bolstering tax revenues and reducing public expenditures, eliminating anti-LGBTI+ discrimination might bring significant economic benefits through various additional channels, such as the potential to tap into a diversity and gender equality dividend.
- Removing the economic and well-being burden of mental health disparities for LGBTI+ Americans will give benefits equivalent to a significant share of GDP.
 - The burden faced by LGBTI+ individuals due to their disproportionate health challenges can be thought of as comprising three main components: i) the reduction in market income

- (which is only partly accounted for in the growth model); ii) the reduction in longevity; and iii) the reduction in psychological well-being, often labelled as “pain and suffering”.
- It is possible to convert the mental health-related burdens of LGBTI+ individuals into a loss that can then be expressed as a percentage of GDP. This is done by multiplying the average number of disability-adjusted life years (DALYs) in the total population attached to mental health issues faced by LGBTI+ individuals by one time GDP per capita (lower bound representing the direct economic impact) to three times GDP per capita (upper bound reflecting the broader value of health beyond earning capacity).
- Applying this methodology to the case of anxiety and depression indicates that removing the related economic and well-being burdens for LGBTI+ individuals will represent in 2024 a benefit equivalent to a share of USD GDP ranging from 0.04% (lower bound) to 0.12% (upper bound), or an amount varying between USD 11.2 billion and USD 33.6 billion.

4.2. Closing unexplained LGBTI+ labour market gaps will increase GDP

Unequal labour market success for LGBTI+ adults not only contradicts the moral imperative of equality in society, but it also entails significant economic costs by preventing LGBTI+ adults from realising their full productive potential, as shown in the previous chapters. In particular, employment discrimination against LGBTI+ workers leads to an underutilisation of existing human capital and to a misallocation of talents as LGBTI+ individuals are passed over in hiring and in career advancement. As a result, some LGBTI+ workers end up unemployed or in lower-skilled positions than they would otherwise be able to occupy. These detrimental consequences of discrimination might be amplified by negative feedback effects from the labour supply to the extent that reduced labour market prospects undermine incentives to participate in the labour force and productivity at work. These harmful repercussions on LGBTI+ individuals' labour supply and productivity are further compounded by anti-LGBTI+ discrimination (in and outside the labour market) undermining the health of LGBTI+ individuals.

To evaluate the economic benefits of discontinuing these mechanisms, this section relies on the OECD long-term model (Guillemette and Turner, 2021^[3]; Guillemette and Turner, 2018^[4]). It begins by succinctly explaining how the model works, before outlining the features of the baseline scenario used to predict GDP. Subsequently, the section presents estimates of the average annual extra GDP that result from departing from the baseline, by closing unexplained labour market gaps between LGBTI+ and their cisgender straight peers. The section concludes with an examination of prior attempts to quantify the economic costs associated with anti-LGBTI+ discrimination.

4.2.1. The OECD long-term model

The OECD long-term model is a tool employed by the OECD not only to project GDP trends up to 2060 for its member countries but also to quantify the economic returns of greater inclusion of marginalised groups. The model has already been used by the OECD on two notable occasions: first, to assess the benefits of enhanced labour market participation among older individuals across various OECD countries (OECD, 2020^[5]), and second, to analyse the economic gains from achieving gender equality in the labour market (OECD, 2023^[6]; OECD, 2022^[7]).

In this model, economic output Y in a given year t is measured based on a Cobb-Douglas production function. This production function has the advantage of presenting a great deal of realism as it is consistent with three economic stylised facts, i.e. patterns that hold true in various economies and times (see Box 2.6 for an overview).

More precisely, the Cobb-Douglas production function on which the OECD long-term model is grounded features three main factors: physical capital stock (K_t), total employment – or, equivalently, total number of workers – denoted by N_t , and labour productivity per worker (A_t). Specifically, we adapt the model to categorise the population of workers into two groups: cisgender straight individuals (CS) and LGBTI+ individuals. In alignment with the initial version of the model, we further divide each of these groups by sex assigned at birth (AMAB or AFAB) and by age group, which includes young (15-34), prime age (35-54), and mature (55-74) individuals.

Against this backdrop, GDP in year t is given by:

$$Y_t = K_t^{\frac{1}{3}} [A_t^{CS} * N_t^{CS} + A_t^{LGBTI+} * N_t^{LGBTI+}]^{\frac{2}{3}},$$

where A_t^{CS} and N_t^{CS} is the productivity per worker and the number of workers within the cisgender straight population, while A_t^{LGBTI+} and N_t^{LGBTI+} is the productivity per worker and the number of workers within the LGBTI+ population.¹

Box 4.1. The Cobb-Douglas production function has the advantage of reflecting three economic stylised facts

The OECD long-term model is grounded in a standard Cobb-Douglas production function featuring physical capital stock (K), total employment of adults aged 15-74 (N), and total factor productivity (TFP)¹. Specifically, real GDP is given by:

$$Y = F(K, N) = TFP * K^{\frac{1}{3}} * N^{\frac{2}{3}}.$$

The Cobb-Douglas production function is consistent with three economic patterns that hold true in most economies and times:

- Constant returns to scale, which imply that $F(\lambda K, \lambda N) = \lambda F(K, N)$, where $\lambda > 0$. In other words, multiplying each input by a positive factor yields an increase of the total output by the same factor.
- Diminishing marginal product of capital stock and total employment, which implies that both the second derivative of $F(K, N)$ with respect to K and the second derivative of $F(K, N)$ with respect to N is negative. In other words, if we keep adding units of one of the two inputs (holding the other input constant), the quantity of additional output that each new unit of input produces will be smaller than that added by the previous unit of input.
- The share of the total output Y that is paid out as rents to holders of physical capital is equal to 1/3 on average, with most countries, should they be rich or poor, having physical capital shares lying fairly near this average. Conversely, the share of the total output Y that is paid out as wages to workers is equal to 2/3 on average. To the extent that, under perfect competition, inputs are paid at their marginal productivity, this stylised fact implies that:

$$(MPK * K) / Y = \frac{1}{3} \text{ and } (MPN * N) / Y = \frac{2}{3},$$

where MPK stands for the marginal productivity of capital and MPN stands for the marginal productivity of employment.

The Cobb-Douglas production function satisfies these last two conditions.

Note:

1. In the OECD long term model, total factor productivity is entirely driven by labour productivity per worker through the following relationship: $TFP = A^{2/3}$ (Guillemette et al., 2017^[9]).

4.2.2. The baseline scenarios: no progress towards greater inclusion of LGBTI+ individuals

The key assumption in the baseline scenarios is the absence of a reduction in unexplained labour market gaps between LGBTI+ individuals and their cisgender straight peers. This assumption indicates a stagnation in the progress towards greater inclusion of LGBTI+ individuals.

The OECD long-term model offers a framework to quantify the additional GDP, compared to a baseline, that could result from bridging the unexplained gaps in employment and labour productivity for LGBTI+ individuals. The underlying assumption is that anti-LGBTI+ discrimination prevents them from reaching their full employment and labour productivity potential.

Specifically, the baseline scenarios are grounded in a set of long-run projections that pertain to key model parameters, namely total employment, labour productivity per worker, and capital stock (Guillemette and Turner, 2021^[3]). The projections for total employment (N_t) are influenced by trends in the size and sex/age composition of the working-age population, according to United Nations population projections. They are also shaped by trends in employment rates across different sex/age groups, with a cohort approach applied to cyclically adjusted historical employment rates. Beyond these standard dynamics, which are integral to the initial version of the OECD long-term model, we introduce an innovation. We differentiate between total employment for cisgender straight adults and LGBTI+ adults, possibly further breaking down the data by sex and age group. This differentiation is based on the proportion of self-identified LGBTI+ adults within the total population, as well as on the demographic-adjusted employment disparities between LGBTI+ individuals and their cisgender straight peers. The results associated with these two dimensions are summarised in columns 1 and 2 of Table 4.1, based on the analysis performed in Chapters 2 and 3.

Regarding the projections for labour productivity per worker (A_t), we refine the initial version of the OECD long-term model by disaggregating labour productivity by LGBTI+ status, possibly differentiating by sex and age group. This estimation is informed by gaps in household income for employed individuals living alone, comparing LGBTI+ adults with their cisgender straight peers, and adjusting for essential demographic characteristics, as well as sector and occupation. These income disparities are presented in column 3 of Table 4.1, based on the estimates obtained in Chapter 3.

Finally, the projections for capital stock (K_t) rely on the assumption that the capital-to-output ratio will remain constant from 2024 onwards. This assumption implies that firms proportionally adapt their capital stock in response to changes in labour input. Although this approach abstracts from the dynamic adjustment of the capital stock, it is noteworthy that fluctuations in the capital-to-output ratio have had a relatively minor impact on economic growth in the United States over the past few decades (Jones, 2016^[9]). In line with this prior research, our estimates of the economic gains of greater LGBTI+ inclusion remain virtually unchanged if we allow for fluctuations in the capital-to-output ratio.²

Specifically, we consider two distinct baseline scenarios:

- **A sex- and cohort-blind approach.** This approach is considered naïve as it fails to account for variations in both the share of self-identified LGBTI+ individuals and their unexplained labour market disparities across different sexes and cohorts. It assumes that in 2024, the share of LGBTI+ individuals and their unexplained labour market disparities in each sex and age group is equal to their average in the total working-age population, as reported in Table 4.1, maintaining these averages static through 2060. With this baseline scenario, the GDP is projected to increase by 1.5% by the year 2060, assuming LGBTI+ equality is achieved by that time, which coincides with the last year of the OECD long-term model.
- **A cohort approach.** This approach is deemed realistic as it recognises that both the share of self-identified LGBTI+ individuals and the extent of their labour market disparities differ by sex and cohort. This method assumes that for each sex and age group in 2024, the percentage of LGBTI+ adults and the degree of their labour market disparities remain unchanged as individuals in this

group age. For instance, if AFAB LGBTI+ individuals aged 20 in 2024 experience a 12.8% lower labour productivity than their cisgender straight peers, as specified in Table 4.1, this gap is expected to still be equal to 12.8% as they age to 56 by the year 2060 (while the share of LGBTI+ individuals among the AFAB population aged 55-74 will then be equal to their share in the AFAB population aged 15-34 in 2024). With this baseline scenario, the GDP is projected to increase by 2.9% by the year 2060, provided that LGBTI+ equality is achieved by that time.

Table 4.1. Share of self-identified LGBTI+ individuals and unexplained employment and labour productivity disparities, in the working-age population and by sex assigned at birth and age group

	Share of self-identified LGBTI+ individuals	LGBTI+ unexplained employment gap	LGBTI+ unexplained labour productivity gap
TOTAL WORKING-AGE POPULATION	11.6%	-3.9 percentage points	-7.2%
YOUNG WORKING-AGE POPULATION (age 15-34)			
All	22.8%	-3.3 percentage points	-10.4%
AFAB (Assigned Female at Birth)	28%	0 percentage point (-1 percentage point but not statistically significant at the 90% confidence level)	-12.8%
AMAB (Assigned Male at Birth)	17.5%	-5.1 percentage points	-6.9%
PRIME AGE WORKING-AGE POPULATION (age 35-54)			
All	9.3%	-2.8 percentage points	-4%
AFAB (Assigned Female at Birth)	9.8%	-3.3 percentage points	-8.9%
AMAB (Assigned Male at Birth)	8.7%	0.0 percentage point (-1.9 percentage points but not statistically significant at the 90% confidence level)	0.0% (+1.1% but not statistically significant at the 90% confidence level)
MATURE WORKING-AGE POPULATION (age 55-74)			
All	5.1%	-1.8 percentage points	-4.9%
AFAB (Assigned Female at Birth)	4.4%	-4.2 percentage points	0.0% (-4.5% but not statistically significant at the 90% confidence level)
AMAB (Assigned Male at Birth)	5.8%	0 percentage point	0.0% (-3.7% but not statistically significant at the 90% confidence level)

Note: The LGBTI+ unexplained employment gap computed for the total working-age population is more pronounced than the LGBTI+ unexplained employment gap computed separately for the three age groups. This discrepancy arises as the gaps are estimated at the average of the covariates ("marginal effect at the mean"). Since different samples produce different average values for the covariates, the gap estimated at the average of the covariates using the total working-age population may fall outside of the range of the three gaps estimated at the average of the covariates using the three age group samples separately.

Source: Panel A of Figure 2.6. of Chapter 2 based on the Household Pulse Survey (4 January 2023 – 10 July 2023) for the share of self-identified LGBTI+ (the share of self-identified LGBTI+ individuals for working-age population and mature adults is slightly higher when excluding adults aged 75+), Figure 3.10. of Chapter 3 based on the Household Pulse Survey (21 July 2021 – 10 July 2023) for the LGBTI+ unexplained employment gap, Figure 3.14. of Chapter 3 based on the Household Pulse Survey (14 September 2022 – 10 July 2023) for the LGBTI+ unexplained labour productivity gap.

An additional baseline scenario could be envisioned that relies on a life-cycle approach. However, although this approach recognises that both the share of self-identified LGBTI+ individuals and the extent of their labour market disparities differ by sex and cohort, it is not a realistic method as it concomitantly assumes that these shares and disparities change as individuals in each group age. This assumption is at odd with Chapter 2. According to Chapter 2, the rising proportion of young adults identifying as LGBTI+ is not driven by an age effect, which would entail that, for younger generations, self-identifying as LGBTI+ is a fad that ceases once they grow older. Despite its lack of realism, we analyse the economic consequences of closing labour market gaps for LGBTI+ individuals under the life-cycle approach in 1Annex 4.A and find that closing unexplained labour market gaps for LGBTI+ individuals under this approach would still generate significant GDP gains.

4.2.3. The GDP gain from closing unexplained labour market gaps for LGBTI+ individuals by 2030, 2040 and 2050

For each of the two baseline scenarios described in the previous section, we consider that convergence towards LGBTI+ equality is achieved either in 2030, 2040 or 2050. Specifically, we assume that the employment rates and labour productivity of LGBTI+ adults converge linearly with those of their cisgender straight peers by these different time horizons. This entails that efforts are made every year until full equality between cisgender straight and LGBTI+ workers is achieved by the target year, with a particular emphasis on eradicating anti-LGBTI+ hiring discrimination and the misallocation of LGBTI+ talents. The assumption of convergence by 2030 is crafted based on the principles of the Agenda 2030 for Sustainable Development, ratified by the United Nations General Assembly in September 2015 (UN General Assembly, 2015^[10]). Although LGBTI+ individuals are not explicitly categorised as a marginalised group in Agenda 2030, many of the 17 Sustainable Development Goals (SDGs) offer avenues to ensure that they are not left behind (Open for Business, 2020^[11]; Stonewall International, 2016^[12]). For instance, SDG 8 establishes that everyone has the right to employment and decent work, in a safe environment, with equal pay for work of equal value. By 2030, its objective is to attain full and productive employment for all women and men, including for young people and persons with disabilities.

The sex- and cohort-blind approach

Panel A of Figure 4.1 illustrates the GDP gain that would be generated from efforts to achieve LGBTI+ equality by 2030, 2040, and 2050, compared to the baseline scenario, when one relies on the sex- and cohort-blind approach. This means that the share of LGBTI+ individuals and their unexplained labour market disparities does not vary by sex and age group at the beginning of the period (2024), matching the average for the total working-age population.

Under these conditions, the United States could anticipate an increase in GDP equal to approximately 1.5% of the baseline GDP by 2030. Pushing this goal further would translate into the same increase in GDP by 2040 and 2050. To put an estimate of 1.5% of the baseline GDP into perspective, this amount is 40% higher than the total US expenditure on public unemployment and labour market programmes in 2020/21. It is also equivalent to 20% of pension spending in 2020 and to more than 6% of total social spending in 2021.³

The cohort approach

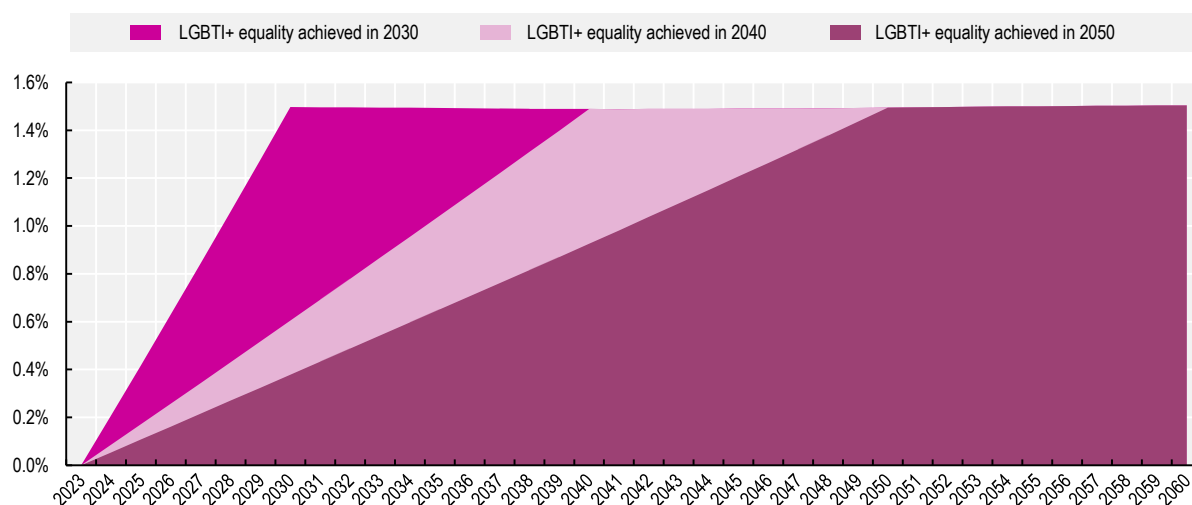
The economic benefits of achieving LGBTI+ equality would be more substantial under a cohort approach (Panel B of Figure 4.1). There are two reasons for this. First, consistent with Chapter 2's findings, such an approach accounts for the expected rise over time in the proportion of self-identified LGBTI+ individuals within the total population, as the younger cohorts, characterised by a larger share of non-heterosexual non-cisgender individuals, age. Second, it assumes that, as these younger cohorts mature, the extent of

labour market disparities, which are more pronounced within this group, presumably due to a lower tendency for non-disclosure bias, would increase in the baseline scenario.

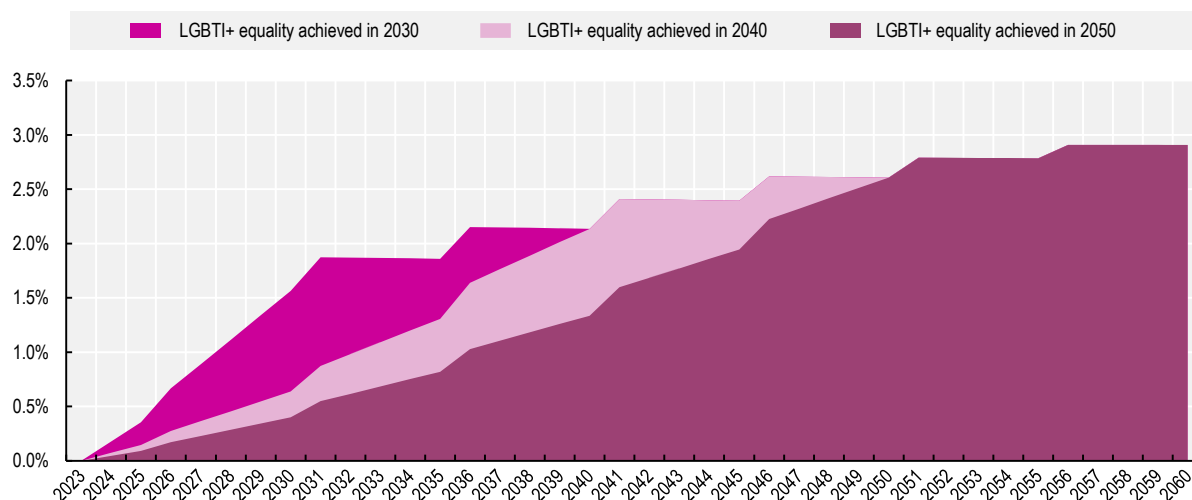
Under the cohort approach, the GDP is projected to increase by 1.6%, 2.1%, and 2.6% of the baseline GDP by 2030, 2040, and 2050, respectively. These projections assume that the United States achieves LGBTI+ equality by these respective time horizons.

Figure 4.1. The increase in GDP from addressing the unexplained labour market disparities affecting LGBTI+ individuals is substantial

Panel A: GDP gain that would be generated, under a sex- and cohort-blind approach, from efforts to achieve LGBTI+ equality by 2030, 2040, and 2050, compared to a baseline scenario where no such efforts are undertaken (2024-60)



Panel B: GDP gain that would be generated, under a cohort approach, from efforts to achieve LGBTI+ equality by 2030, 2040, and 2050, compared to a baseline scenario where no such efforts are undertaken (2024-60)



Note: Potential GDP is defined for the total economy using a constant returns-to-scale Cobb-Douglas production function featuring total employment, labour productivity per worker, and physical capital stock. The GDP gain is computed comparing the baseline scenario (unexplained LGBTI+ labour market gaps remain as in 2023), with two alternative scenarios. These scenarios assume that unexplained LGBTI+ labour market outcomes linearly converge to those of their cisgender straight peers by 2030, 2040 and 2050, respectively. In Panel A, both the baseline and the alternative scenarios rely on a sex- and age-blind approach, while they rely on a cohort approach in Panel B.

Source: OECD estimates based on the OECD long-term model projections and Household Pulse Survey data.

The yearly increase in GDP under the three different approaches

As indicated in Table 4.2, attaining LGBTI+ equality by 2030 would result in an average annual increase in GDP of approximately 0.2% under the two approaches, which amounts to 10% of the average annual US GDP growth observed between 2013 and 2023. However, as already evident in Figure 4.1, delaying the achievement of LGBTI+ equality would yield reduced economic benefits when translated in yearly GDP gains. The average annual increase in GDP ranges between 0.09% and 0.13% if the target year is 2040, and between 0.06% and 0.10% if the target year is 2050.

Table 4.2. Annual GDP gain from efforts to achieve LGBTI+ equality by 2030, 2040, and 2050, compared to a baseline scenario where no such efforts are undertaken

	2024-30	2024-40	2024-50
Sex- and cohort-blind approach	0.21% (or an increase in annual GDP growth of 0.21 percentage point)	0.09% (or an increase in annual GDP growth of 0.09 percentage point)	0.06% (or an increase in annual GDP growth of 0.06 percentage point)
Cohort approach	0.22% (or an increase in annual GDP growth of 0.22 percentage point)	0.13% (or an increase in annual GDP growth of 0.13 percentage point)	0.10% (or an increase in annual GDP growth of 0.10 percentage point)

Source: OECD estimates based on the OECD long-term model projections and Household Pulse Survey data.

4.2.4. The share of projected GDP in 2024 which is lost due to LGBTI+ individuals' unexplained labour market gaps

The OECD long-term model offers a significant advantage by extending GDP projections up to 2060. This provides a tangible forecast of the economic benefits that can be realised by narrowing labour market disparities affecting LGBTI+ individuals across different time frames. Unlike static models, which do not account for the incremental nature of such changes, this dynamic model reinforces the economic argument for LGBTI+ equality with realistic projections, acknowledging that such equality cannot be achieved overnight.

However, this model can also be used in a static capacity to estimate the cost of not closing employment and labour productivity penalties faced by LGBTI+ individuals or, conversely, the gain associated with closing those gaps immediately. When used in this way, it reveals that instantaneously achieving LGBTI+ equality results in a GDP gain equivalent to approximately 1%, irrespective of the baseline scenario considered. Here, we assume that capital does not adjust, a reasonable assumption when the increase in labour input is assumed to be instantaneous.

There have been several previous attempts to quantify the immediate economic cost of LGBTI+ labour market gaps. The first comprehensive framework for assessing this burden was developed by Lee Badgett in 2014 (Badgett, 2014^[13]) – see Box 4.2 for more details on the methodology. This framework is equivalent to a static version of the OECD long-term model, except that Badgett only considers the costs associated with lower labour productivity for employed LGBTI+ individuals, overlooking the impact of their lower employment rates. Badgett subsequently applied this framework with data from India and the Philippines (Badgett, 2020^[14]). This seminal work laid the groundwork for subsequent analyses in this field, such as in South Africa (Nyek et al., 2019^[15]), Kenya (Open for Business, 2019^[16]), Czechia (Open for Business, 2020^[17]), Hungary, Poland, Romania, and Ukraine (Open for Business, 2021^[18]), and in the English-speaking Caribbean area (Open for Business, 2021^[19]).

An interesting alternative to the static version of the OECD long-term model is a methodology developed by the International Labour Organization (ILO). This approach aims to quantify the economic output lost due to discrimination, as self-reported by affected individuals. Initially, this method was applied to study

the economic impact of discrimination against people with disabilities (Buckup, 2009^[20]) – see Box 4.3 for more details. However, this approach has its drawbacks. It is not based on a growth model, which can make it seem *ad hoc*. Additionally, using perceived discrimination as a proxy for actual discrimination can be problematic because the two can differ significantly, potentially leading to inaccurate estimates of the economic repercussions of discrimination.

The empirical application of ILOs' methodology also poses challenges. It typically requires the collection of original survey data since few nationally representative surveys provide comprehensive information on both labour market outcomes and perceptions of discrimination. Two recent World Bank reports have attempted to apply the ILO's methodology to quantify the economic cost of anti-LGBTI+ discrimination in the context of the Republic of Serbia (Flores et al., 2023^[21]) and the Republic of North Macedonia (Flores et al., 2023^[22]). However, these studies rely on data collected from a non-probability sample of LGBTI+ individuals. Specifically, these individuals were recruited through purposive methods, such as social media campaigns, and seem to come from the more privileged segments of the LGBTI+ population, leading to a selection bias that obscures any labour market disparities. Notably, they have a much higher educational attainment than the general population, which implies the presence of other unobserved traits conducive to economic success. Consequently, even when researchers control for demographic differences between the LGBTI+ sample and the general population, several of the gaps observed in terms of labour productivity, unemployment rate, or inactivity rate appear as premiums, not penalties, for LGBTI+ individuals. This selection bias leads to underestimating the economic cost of anti-LGBTI+ discrimination. Nevertheless, the World Bank provides estimates of the output lost due to the exclusion of LGBTI+ individuals which are economically significant: 0.5% of the 2021 GDP, both in Serbia and in the Republic of North Macedonia.

Box 4.2. Lee Badgett's methodology to quantify the lost economic output due to labour market disparities for LGBTI+ individuals

In 2014, Lee Badgett developed a framework for quantifying the economic cost associated with labour market disparities for LGBTI+ individuals (Badgett, 2014^[13]). Specifically, Badgett focuses on the burden resulting from lower labour productivity for employed LGBTI+ individuals. She doesn't consider the costs associated with their lower employment rates.

The starting point is the following Cobb-Douglas production function featuring physical capital stock (K), total employment (N), and total factor productivity (TFP):

$$Y = TFP * K^{\alpha} * N^{1-\alpha},$$

where α varies between 0 and 1 – while α is equal to 1/3 in most economies, as explained in Box 2.6, exceptions exist, such as in India, where α is estimated to be equal to 1/2.

Under perfect competition, the wage paid to each worker is given by $w = MPN$, where MPN stands for the marginal productivity of employment. It follows that the share of the total output Y that is paid out as wages to workers is given by:

$$(w * N) / Y = 1 - \alpha.$$

An increase in labour productivity, denoted by Δw , thus yields an increase in aggregate output given by:

$$\Delta Y = (\Delta w * N) / (1 - \alpha).$$

In other words, the economic cost of lost productivity due to LGBTI+ workers having x% lower wages than the average wage w is given by:

$$\Delta Y = (x\% * w * N^{LGBTI+}) / (1 - \alpha),$$

where $N^{\text{LGBTI+}}$ is the number of employed LGBTI+ individuals.

In empirical applications of this theoretical framework, x is typically proxied by the percentage gap in labour earnings between LGBTI+ workers and their cisgender straight peers, while $N^{\text{LGBTI+}}$ is proxied by the share of LGBTI+ individuals in the working-age population multiplied by the total number of workers in that population.

Box 4.3. ILO's methodology to quantify the economic output lost due to discrimination, as self-reported by affected individuals

This methodology comprises three steps. First, it categorises a marginalised group into three subgroups based on low, moderate, and high self-reported exposure to discrimination. The rest of the population is assumed not to experience discrimination. Second, for each subgroup, denoted by i , the methodology calculates the lost labour productivity per individual due to: (1) higher rates of inactivity; (2) higher rates of unemployment; (3) lower labour productivity, conditional on being employed.

Regarding the latter component, the labour productivity of an employed individual in group i is denoted by A_i , while the average labour productivity of an employed individual in the total working-age population is denoted by A .

It follows that the lost labour productivity for each individual in group i , denoted by P_i , is given by:

$$P_i = A \cdot (B_i - \beta_i) \cdot n_i + A \cdot (u_i - u) + A \cdot (d_i - d),$$

where:

- (1) $A \cdot (B_i - \beta_i) \cdot n_i$ reflects the lost labour productivity for employed individuals in group i . It derives from two steps. First, one computes the percentage gap in labour productivity between the average worker and an employed individual in group i . Specifically, this gap is given by the difference between the *potential* productivity of every individual in group i relative to the productivity of the average worker ($B_i = A/A = 1$) and their relative *actual* productivity ($\beta_i = A_i/A$). Second, one multiplies this difference by A (the average labour productivity of an employed individual in the total working-age population) and by n_i (the share of individuals in group i who are currently employed).
- (2) $A \cdot (u_i - u)$ captures the lost labour productivity due to higher unemployment rates in group i . It is obtained by multiplying A by the unemployment gap for group i . The latter is computed as the difference between the unemployment rate of individuals in group i (u_i) and the average unemployment rate within the working-age population (u).
- (3) $A \cdot (d_i - d)$ represents the lost labour productivity due to higher inactivity rates in group i . It is obtained by multiplying A by the inactivity gap for group i . The latter is computed as the difference between the inactivity rate of individuals in group i (d_i) and the average inactivity rate within the working-age population (d).

The economic cost C of lost labour input due to (self-reported) discrimination is then modelled as follows:

$$C = \sum_{i=1}^{i=3} (N_i \cdot P_i),$$

where N_i is the number of individuals in each group i .

4.3. There are many more benefits from LGBTI+ equality than just the economic gains derived from the OECD long-term model

Eliminating anti-LGBTI+ discrimination will lead not only to closing labour market gaps between LGBTI+ individuals and their cisgender straight peers but also to significant additional gains that the OECD long-term model does not account for. This means that the actual economic benefits of increased LGBTI+ equality are likely to be even more substantial than the previous calculations suggest. In this section, we first explore the primary channels through which ending anti-LGBTI+ discrimination can drive additional economic gains. We then discuss other potential channels which, although less apparent, can still have far-reaching effects in terms of economic growth.

4.3.1. Improved public finances through increased tax revenues and reduced public expenditures

Promoting greater LGBTI+ equality leads to production gains that subsequently bolster public finances. An uptick in production and labour earnings enhances public revenue from corporation tax, income tax, and social security contributions. These positive effects might be further magnified by increased consumption tax revenue, at least in the short run. As detailed in Box 4.4 and shown in Figure 4.2, LGBT individuals spend more on consumer goods and services than their cisgender straight peers, controlling for essential demographics, including parental status. This pattern could possibly be in response to the uneven playing field they navigate, meaning that the distinctive consumption trends of LGBT individuals might vanish as they experience a more level playing field. Moreover, eradicating anti-LGBTI+ discrimination cuts public expenditure, as greater workforce participation reduces the need for unemployment benefits, active labour market policies, and social transfers.

Box 4.4. Pink money unveiled

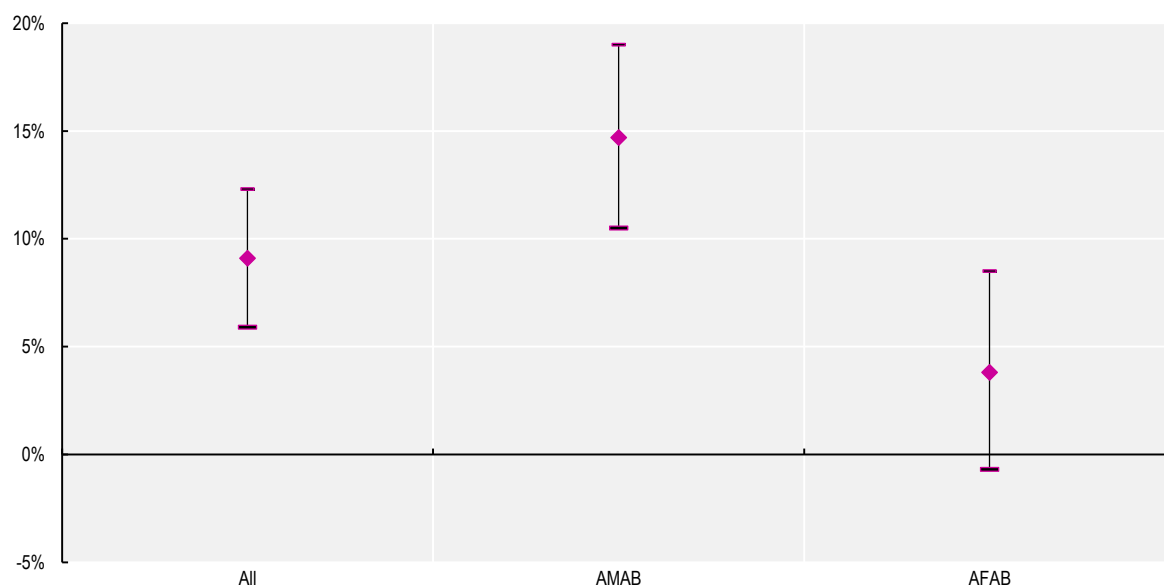
In a survey conducted in the United States in 2016, LGBT respondents were more likely to consider themselves “*spenders*”, and to spend more (and save less) compared to general population respondents (Prudential, 2017^[23]). In addition, in a recent report, LGBT Capital, an analytical firm focused on the LGBT consumer segment, estimated that the United States aggregated spending power of the LGBT population, often called “*pink money*”, is around USD 1.4 trillion per year (LGBT Capital, 2023^[24]).

These insights are confirmed by the Gallup US Daily Survey which reveals that, accounting for differences in demographics and household income, LGBT adults spend 9% more on consumer goods and services, compared to their cisgender straight peers. As shown in Figure 4.2, this pattern is mainly driven by AMAB individuals. Further analysis indicates that this spending premium prevails for all age groups within the AMAB LGBT population, although it is more pronounced among prime age individuals, aged between 35 and 54. Concerning the AFAB LGBT population, a premium emerges only among this latter age range – no statistically significant difference is observed among the other age groups. (Results available upon request.)

In this context, greater LGBTI+ equality is poised to yield increased revenue from consumption tax, at least in the short term, since the distinctive consumption patterns of LGBT individuals might vanish as they experience a more level playing field. While more research is essential, it is plausible that heightened spending on consumer goods and services among LGBT adults stems in part from their responses to the discrimination they face in various life areas, including barriers to family formation which might reduce their incentives to save.

Figure 4.2. All other things being held constant, LGBT adults spend 9% more on consumer goods and services compared to their cisgender straight peers, a result driven by AMAB individuals

Percentage difference in personal expenditure on consumer goods and services, using non-LGBT individuals as the reference category (2012-17)



Note: This figure analyses responses to the following question: “We’d like you to think about your spending yesterday, not counting the purchase of a home, motor vehicle, or your normal household bills. How much money did you spend or charge yesterday on all other types of purchases you may have made, such as at a store, restaurant, gas station, online, or elsewhere?”. It relies on an OLS regression of the logarithm of personal expenditure on consumer goods and services, controlling for sex assigned at birth, age groups (18-24, 25-29, 30-34, ..., 80-84, 85-88), race and ethnicity, marital, partnership and parental status, number of adults in the household, educational attainment, household income, as well as year and month fixed effects. The error bars depict 95% confidence intervals. This means that we can be 95% confident that the true value lies within the range of the error bar. In other words, if the error bar crosses the x-axis, then the percentage difference cannot be deemed as statistically different from 0 (at least not at this 95% confidence level). Standard errors robust to heteroscedasticity. Person-level weights used.

Source: Gallup US Daily Survey (2012-17).

Promoting greater LGBTI+ equality can also lead to economic benefits through reduced public health expenditures. As discussed in Chapter 3, stigma has been identified as a significant factor contributing to the poorer mental, behavioural, and physical health of LGBTI+ individuals. The financial implications could be substantial. In 2022, the United States spent 16.6% of its GDP to healthcare services (OECD, 2023^[25]). This spending covered personal healthcare – which includes curative, rehabilitative, long-term care, ancillary services, and medical goods – as well as collective services such as prevention, public health services, and health administration. Furthermore, reducing stigma-related health issues for LGBTI+ individuals would not only alleviate the burden on the health system but also lessen associated costs in other areas, like social security programmes, which cover paid sick leave and disability benefits. It is critical to note that these expenditure figures do not reflect the broader costs associated with the negative impact of discrimination on health, such as the loss of human capital due to the premature mortality of LGBTI+ individuals whose health suffers due to stigma, including through deaths by suicide.⁴ In Section 4.4, we aim to address this limitation by closely examining the economic and well-being burden that health disparities impose on LGBTI+ individuals.

4.3.2. Additional channels through which greater LGBTI+ equality will generate economic growth

Beyond bolstering tax revenues and reducing public expenditures, eliminating anti-LGBTI+ discrimination will bring significant economic benefits through various additional channels. These include enhanced wealth accumulation, greater opportunities for LGBTI+ individuals to start families and invest in their children, and the potential to tap into the diversity and gender equality dividend.

Enhanced wealth accumulation

Marriage equality for LGBTI+ individuals is likely to boost their capacity for wealth accumulation thanks to a stabilizing effect that facilitates long-term planning (Badgett, 2009^[26]). Evidence has confirmed this mechanism, by revealing the benefits for same-sex couples of being able to “upgrade” their civil partnership to a civil marriage, even in countries like the Netherlands where civil partnership and civil marriage are fully similar in terms of rights and obligations. More precisely, same-sex partners who transformed their civil partnership into marriage had a substantially lower separation rate following this change than similar partners who stayed in a civil partnership, which suggests that the symbolism of marriage is real and contributes to the longevity of same-sex partnerships (Chen and van Ours, 2020^[27]). In line with this finding, (Delhomme and Hamermesh, 2021^[28]) highlighted the economic benefits of marriage based on US data. Their research compared two otherwise observationally identical same-sex couples with the same duration of partnership and found that the couple who experienced a longer period of their relationship under the institution of legalised same-sex marriage had a higher household income and were more likely to own a home.

Greater equality for LGBTI+ individuals in other life areas is likely to further enhance their potential to accumulate wealth. For example, reduced barriers to parenthood can lead to increased savings. Additionally, ending the discrimination that LGBTI+ individuals face in credit and mortgage lending (see Chapter 3 for evidence of such discrimination) can significantly improve their financial health.

Increased fertility and investment in children

Lowering barriers to assisted reproductive technology for LGBTI+ individuals is likely to foster fertility. This is a promising perspective in a context where OECD countries are currently exhibiting the lowest ever recorded average fertility rate – at only 1.59 children per woman in 2020, which is far below the “replacement level” of 2.1 children per woman (Fluchtmann, van Veen and Adema, 2023^[29]). In other words, greater LGBTI+ equality could help to counteract the shrinking working-age population occurring in several OECD countries, a trend resulting from rising longevity and falling fertility rates. Simultaneously, providing LGBTI+ individuals with greater access to adoption and foster care will lead to increased investment in children, as more children without biological parents able to care for them will have the opportunity to grow up in a loving and caring family.

This increased investment in children will likely be compounded by more LGBTI+ individuals becoming biological parents through assisted reproductive technology. Indeed, compelling evidence indicates that children of same-sex parents conceived through these methods tend to have better education and health outcomes compared to the biological children of different-sex parents (OECD, 2020^[1]). This finding remains consistent even when considering differences in socio-economic status between same-sex and different-sex parents, as same-sex parents who procreate typically have higher levels of income and education than their different-sex counterparts – an expected result due to the significant time and financial costs associated with assisted reproductive technology. In line with the observed education and health advantages of children born to same-sex couples, same-sex parents tend to spend more time with their children than different-sex parents. In the United States, women (regardless of their partners’ sex) and partnered gay men engage in a similar amount of child-focused time with children (roughly 100 minutes

per day). By contrast, partnered heterosexual men dedicate less than one hour to their children, on average (Prickett, Martin-Storey and Crosnoe, 2015^[30]). The education and health premium observed in children of same-sex parents who were conceived through assisted reproductive technology may reflect that same-sex couples cannot unintentionally become parents due to birth control failure, as can happen with heterosexual couples. Instead, they deliberately choose to become parents. This dynamic may lead to a selection effect where gays and lesbians who become parents are, on average, more committed to the challenges of parenting compared to some of their heterosexual counterparts (Rosenfeld, 2010^[31]).

Tapping into the diversity and gender equality dividend

Ending anti-LGBTI+ discrimination can equip the economy to harness the economic potential of greater diversity. A work environment that is inclusive creates the condition for diversity to become a performance enhancer – while diversity can hinder performance in non-inclusive settings (OECD, 2020^[32]). An inclusive environment allows for the pooling of a richer set of perspectives, skills and experiences whose benefits can outweigh the increased communication and co-ordination costs associated with diverse backgrounds.

Dutch researchers have corroborated this mechanism, using a sample of 550 students from Amsterdam University of Applied Sciences, known for its commitment to attracting a diverse range of profiles eager to engage with peers from different backgrounds. In line with this mission, the student pool analysed has an almost equal split by gender and foreign-born/native-born status. Specifically, students of foreign origin make up 55% of the sample, with a significant proportion hailing from non-European countries (excluding North America, Australia, and New Zealand). As a component of their training, students form teams of 10 to 12 to create and manage a real company over a year. Each team handles fundraising, production, marketing, accounting, tax obligations, and more. To assess the impact of diversity, the researchers determined each team's composition, exogenously varying not only the gender proportion (from 20% to 60%) but also the mix of nationalities. Their findings indicate that both gender diversity (Hoogendoorn, Oosterbeek and van Praag, 2013^[33]) and racial/ethnic diversity (Hoogendoorn and van Praag, 2018^[34]) significantly boost the turnover and profit of the student-led start-ups, an outcome likely influenced by the inclusive environment in which students operate. Of course, further research is needed to test whether these findings apply to settings with a diversity of sexual orientations and gender identities.

The positive impacts of greater diversity in sexual orientations and gender identities, especially in environments where such diversity is valued, can be amplified by the ripple effects that greater LGBTI+ equality might have on the emancipation of other groups, primarily women. LGBTI+ inclusion inherently challenges heteronormativity, which confines men and women to rigid roles that are obstructive to women's empowerment. Furthermore, embracing LGBTI+ identities means recognising the complex spectrum of gender, leading to a re-evaluation of societal expectations for both men and women. In line with this observation, countries most proactive in enacting laws that promote LGBTI+ equality also tend to have the highest support for gender equality and female labour force participation. Additionally, these countries often exhibit the narrowest gender wage gaps (OECD, 2020^[1]). Under these circumstances, the elimination of anti-LGBTI+ discrimination might bolster gender equality which, alone, has proven to generate substantial economic gains (OECD, 2023^[6]; OECD, 2022^[7]).

4.4. Removing the economic and well-being burden of mental health disparities for LGBTI+ Americans will give benefits equivalent to a significant share of GDP

The purpose of this section is to quantify the economic and well-being burden faced by LGBTI+ individuals due to their disproportionate health challenges. This burden can be thought of as comprising three main components: i) the reduction in market income; ii) the reduction in longevity; and iii) the reduction in psychological well-being, often labelled as “pain and suffering” (World Health Organization, 2001^[35]). The reduction in market income encompasses at least four sub-components. First, the private costs of medical

treatment – public health expenditures which are discussed in Section 4.3. are not considered here as the focus is the direct burden borne by LGBTI+ individuals. Second, the loss of labour-market income due to morbidity, a factor accounted for by the OECD long-term model. Third, the diminished adult earning power resulting from diseases in childhood. And fourth, the loss of future earnings attributable to premature mortality.

This section first presents the methodological framework used to measure the economic and well-being burdens that LGBTI+ Americans face due to their health disparities. It then applies this methodology, focusing on the lower mental health of LGBTI+ individuals. In this setting, the estimates provided offer only a partial analysis of the full impact of their health disparities which often extend to behavioural health issues like sleep disorders and substance abuse, as well as physical health problems, including cardiovascular diseases and certain cancers.⁵

4.4.1. A methodological framework to quantify the economic and well-being impact of health disparities for LGBTI+ individuals

Applied to the mental health disparities of LGBTI+ individuals, the methodological framework developed by the World Health Organization entails three main steps.

1. **Estimating the additional risk for LGBTI+ individuals to exhibit generalised anxiety or major depressive disorder.** This initial step comprises three stages:
 - a. **Determining the higher prevalence of these conditions among LGBTI+ individuals** by calculating the percentage difference in the likelihood of exhibiting symptoms between LGBTI+ individuals and their cisgender straight peers, while accounting for differences in demographics and household income across these two groups, based on HPS data.
 - b. **Calculating the increased number of LGBTI+ individuals with these conditions** by applying the percentage difference obtained in the previous stage to the population of LGBTI+ individuals aged 15 or older. This figure is based on the proportion of LGBTI+ people in this age group, as derived from HPS data.
 - c. **Assessing the proportion of LGBTI+ individuals with these conditions within the overall affected population** by dividing the number from the previous stage by the total number of people affected by these conditions, as derived from HPS data.
2. **Computing the number of disability-adjusted life years (DALYs) due to the higher prevalence of generalised anxiety or major depressive disorder among LGBTI+ individuals** by multiplying the estimate from the last stage of Step 1 by the average number of DALYs in the total population attached to these conditions. Specifically, this number captures the cumulative years of life that people with at least one of these conditions lose from an early death or from poorer quality of life. It is obtained by adding the impact that these conditions have in terms of number of years of life lost (YLLs) and number of years lived with a disability (YLDs).⁶ The average number of DALYs in the total population associated with generalised anxiety and major depressive disorders is obtained from the Global Burden of Diseases, Injuries, and Risk Factors Study 2019 (GBD, 2019_[36]) which provides a 95% confidence interval for the number of DALYs related to 369 diseases and injuries, in 204 countries and territories.
3. **Converting the mental health-related burdens of LGBTI+ individuals into a GDP percentage loss.** To estimate the cost of these burdens on the economy, we draw on the World Health Organization's Commission on Macroeconomics and Health guidelines (World Health Organization, 2001_[35]). Based on estimates of individual's willingness to pay for an improvement in their health risk, these recommendations assert that the value of an extra year of healthy life – because of successfully treating a disease, for example – is worth much more than just the additional market income earned in that year. In fact, it suggests that a healthy year can be valued at up to three times an individual's annual earnings, considering not only market consumption but

also the value of leisure, the intrinsic value of longevity, and the avoidance of the sufferings associated with illness. Utilising this approach, we calculate a range for the economic cost:

- The lower bound is the total DALYs related to anxiety and depression multiplied by the GDP per capita – this represents the direct economic impact.
- The upper bound is this same figure but multiplied by three times the GDP per capita – reflecting the broader value of health beyond earning capacity.

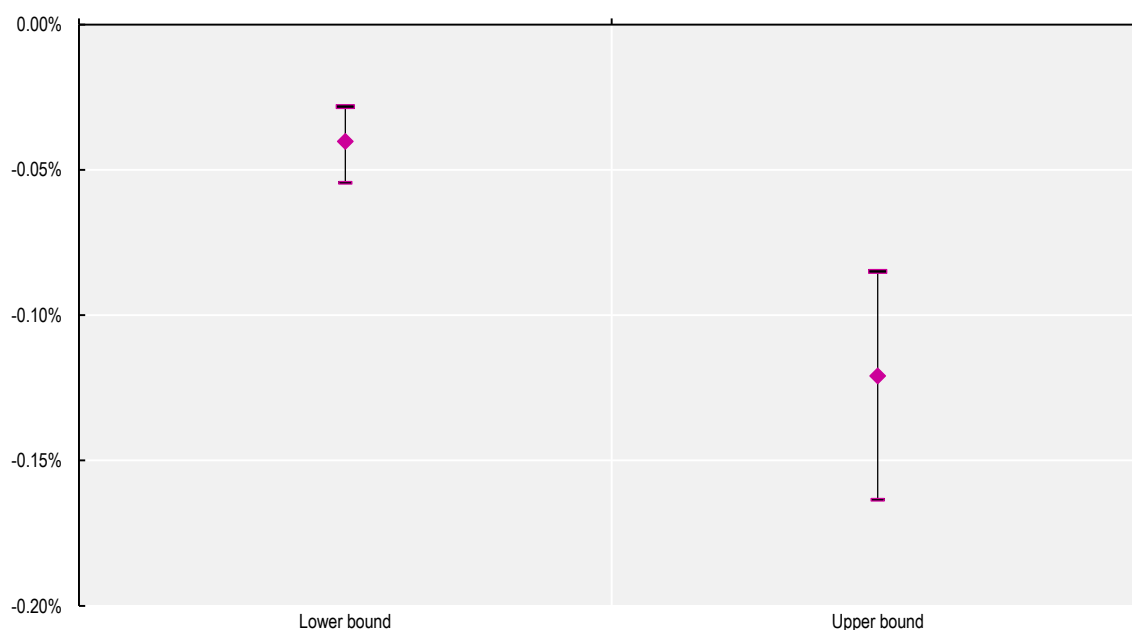
Each result is then expressed as a percentage of total GDP, providing a scale for the economic and well-being impact of mental health disparities among LGBTI+ individuals.

4.4.2. Estimating the economic and well-being burden of mental health disparities among LGBTI+ Americans

Lee Badgett was the first to adapt the methodology developed by the World Health Organization to the disproportionate health challenges for LGBTI+ individuals (Badgett, 2014^[13]). Since then, this method has been applied to the context of South Africa (Nyek et al., 2019^[15]), Kenya (Open for Business, 2019^[16]), India and the Philippines (Badgett, 2020^[14]), Czechia (Open for Business, 2020^[17]), Hungary, Poland, Romania, and Ukraine (Open for Business, 2021^[18]), the English-speaking Caribbean area (Open for Business, 2021^[19]), and Uzbekistan (Open for Business, 2021^[37]).

Figure 4.3. Removing the economic and well-being burden of mental health disparities for LGBTI+ Americans will give benefits equivalent to a significant share of GDP

Loss associated with higher incidence of anxiety and depression among LGBTI+ Americans, expressed as a percentage of GDP (2024)



Note: The economic and well-being impact of higher incidence of anxiety and depression for LGBTI+ individuals is derived from the average number of disability-adjusted life years (DALYs) in the total population attached to these conditions. Specifically, this number captures the cumulative years of life that people with at least one of these conditions lose from an early death or from poorer quality of life. It is obtained by adding the impact that these conditions have in terms of number of years of life lost (YLLs) and number of years lived with a disability (YLDs). This burden is then converted into a monetary value by multiplying the average DALYs by one (lower bound) to three (upper bound) times GDP per capita. The error bars depict 95% confidence intervals on the measure of average DALYs.

Source: 2019 Global Burden of Disease (GBD) and Household Pulse Survey (2023).

We add to this literature by focusing on the US context. The economic and well-being impact of mental health disparities among LGBTI+ Americans is substantial. In 2024, it will amount to a loss which, expressed as a share of GDP, varies between 0.04% (lower bound) and 0.12% (upper bound) of US GDP, or between USD 11.2 billion and USD 33.6 billion (Figure 4.3). In terms of magnitude, this cost is equivalent to 4%-12% of US expenditure on incapacity (spending due to sickness, disability and occupational injury) in 2020 (OECD, 2023^[38]).

References

- Aksoy, C. et al. (2022), *Reducing Sexual-Orientation Discrimination: Experimental Evidence from Basic Information Treatments*, Journal of Policy Analysis and Management. [2]
- Badgett, M. (2020), *The Economic Case for LGBT Equality: why fair and equal treatment benefits us all*, Boston, Massachusetts: Beacon Press. [14]
- Badgett, M. (2014), *The Economic Cost of Stigma and the Exclusion of LGBT People: A Case Study of India*, Washington, D.C. : World Bank Group, <http://documents.worldbank.org/curated/en/527261468035379692/The-economic-cost-of-stigma-and-the-exclusion-of-LGBT-people-a-case-study-of-India>. [13]
- Badgett, M. (2009), "The Economic Value of Marriage for Same-Sex Couples", *Drake L. Rev.*, 58, 1081. [26]
- Banks, C. (2001), *The Cost of Homophobia: Literature Review of the Economic Impact of Homophobia in Canada*, CUISR: Community - Univesity Institute for Social Research. [39]
- Buckup, S. (2009), "The Price of Exclusion: The Economic Consequences of Excluding People with Disabilities from the World of Work", *Employment Sector Working Paper 43* (Geneva: International Labour Organization, 2009).. [20]
- Chen, S. and J. van Ours (2020), "Symbolism matters: The effect of same-sex marriage legalization on partnership stability", *Journal of Economic Behavior & Organization*, Vol. 178, pp. 44-58, <https://doi.org/10.1016/j.jebo.2020.07.021>. [27]
- Delhommer, S. and D. Hamermesh (2021), "Same-Sex Couples and the Gains to Marriage: The Importance of the Legal Environment", *Journal of Policy Analysis and Management*, John Wiley & Sons, Ltd., vol. 40(4), pages 1120-1139, September., <https://doi.org/10.1002/pam.22287>. [28]
- Flores, A. et al. (2023), "The Economic Cost of Exclusion Based on Sexual Orientation, Gender Identity and Expression, and Sex Characteristics in the Labor Market in the Republic of North Macedonia", *Washington, DC: World Bank*. [22]
- Flores, A. et al. (2023), "The Economic Cost of Exclusion Based on Sexual Orientation, Gender Identity and Expression, and Sex Characteristics in the Labor Market in the Republic of Serbia", *Washington, DC: World Bank*. [21]
- Fluchtmann, J., V. van Veen and W. Adema (2023), "Fertility, employment and family policy: A cross-country panel analysis", *OECD Social, Employment and Migration Working Papers*, No. 299, OECD Publishing, Paris, <https://doi.org/10.1787/326844f0-en>. [29]

- GBD (2019), *Global Burden of Disease Collaborative Network. Global Burden of Disease Study 2019 (GBD 2019)*, Reference Life Table. Seattle, United States of America: Institute for Health Metrics and Evaluation (IHME), 2021. [36]
- Guillemette, Y. et al. (2017), *A revised approach to productivity convergence in long-term scenarios*, OECD. [8]
- Guillemette, Y. and D. Turner (2021), *The long game: Fiscal outlooks to 2060 underline need for*, OECD Economic Policy Papers, No. 29, OECD Publishing, Paris, <https://doi.org/10.1787/a112307e-en>. [3]
- Guillemette, Y. and D. Turner (2018), *The Long View: Scenarios for the World Economy to 2060*, <https://doi.org/10.1787/2226583X>. [4]
- Hoogendoorn, S., H. Oosterbeek and M. van Praag (2013), "The Impact of Gender Diversity on the Performance of Business Teams: Evidence from a Field Experiment", *Management Science*, 59 (7), p. 1514-1528. [33]
- Hoogendoorn, S. and M. van Praag (2018), "Ethnic Diversity and Team Performance: A Field Experiment", *Academy of Management Annual Meeting Proceedings*. [34]
- Jones, C. (2016), "The Facts of Economic Growth", *Handbook of Macroeconomics, Elsevier B.V.*, <https://doi.org/10.1016/bs.hesmac.2016.03.002>. [9]
- LGBT Capital (2023), "Estimated LGBT Purchasing Power: LGBT-GDP - data as of year-end 2022 -", [https://www.lgbt-capital.com/docs/Estimated_LGBT-GDP_\(table\)_-2023.pdf](https://www.lgbt-capital.com/docs/Estimated_LGBT-GDP_(table)_-2023.pdf). [24]
- Nyek, S. et al. (2019), *The economic cost of LGBT stigma and discrimination in South Africa*, UCLA Williams Institute. [15]
- OECD (2023), "Health spending (indicator)", <https://doi.org/10.1787/8643de7e-en>. [25]
- OECD (2023), *Joining Forces for Gender Equality: What is Holding us Back?*, OECD Publishing, Paris, <https://doi.org/10.1787/67d48024-en>. [6]
- OECD (2023), "Pension spending (indicator)", <https://doi.org/10.1787/a041f4ef-en>. [40]
- OECD (2023), "Public spending on incapacity (indicator)", <https://doi.org/10.1787/f35b71ed-en>. [38]
- OECD (2023), "Public spending on labour markets (indicator)", <https://doi.org/10.1787/911b8753-en>. [43]
- OECD (2023), "Public unemployment spending (indicator)", <https://doi.org/10.1787/55557fd4-en>. [42]
- OECD (2023), "Social spending (Indicator)", <https://doi.org/10.1787/7497563b-en>. [41]
- OECD (2022), *The Economic Case for More Gender Equality in Estonia*, Gender Equality at Work, OECD Publishing, Paris, <https://doi.org/10.1787/299d93b1-en>. [7]
- OECD (2021), "Preventing Harmful Alcohol Use", *OECD Health Policy Studies*, OECD Publishing, Paris, <https://doi.org/10.1787/6e4b4ffb-en>. [44]
- OECD (2020), "All Hands In? Making Diversity Work for All", <https://doi.org/10.1787/efb14583-en>. [32]

- OECD (2020), *Over the Rainbow? The Road to LGBTI Inclusion*, OECD Publishing, Paris, <https://doi.org/10.1787/8d2fd1a8-en>. [1]
- OECD (2020), *Promoting an Age-Inclusive Workforce: Living, Learning and Earning Longer*, OECD Publishing, Paris, <https://doi.org/10.1787/59752153-en>. [5]
- Open for Business (2021), *The economic case for LGBT+ inclusion in Central and Eastern Europe (CEE): Hungary, Poland, Romania, and Ukraine*. [18]
- Open for Business (2021), *The economic case for LGBT+ inclusion in the Caribbean*. [19]
- Open for Business (2021), *The Economic Case for LGBT+ Inclusion in Uzbekistan*. [37]
- Open for Business (2020), *LGBT+ Inclusion and the UN Sustainable Development Goals*. [11]
- Open for Business (2020), *The economic case for marriage equality in the Czech Republic*. [17]
- Open for Business (2019), *The economic case for LGBT+ inclusion in Kenya*. [16]
- Prickett, K., A. Martin-Storey and R. Crosnoe (2015), “A Research Note on Time With Children in Different- and Same-Sex Two-Parent Families”, *Demography*, Vol. 52/3, pp. 905-918, <https://doi.org/10.1007/s13524-015-0385-2>. [30]
- Prudential (2017), “2016/2017 LGBT Financial Experience Research Report”, <https://www.prudential.com/wps/wcm/connect/ba145fe5-88cf-48c3-ab67-6ab3cc5dba7a/PrudentialLGBT2016-2017.pdf?MOD=AJPERES&CVID=IYX8NZE&CVID=IYX8NZE>. [23]
- Rosenfeld, M. (2010), “Nontraditional families and childhood progress through school”, *Demography*, Vol. 47/3, pp. 755-775, <https://doi.org/10.1353/dem.0.0112>. [31]
- Stonewall International (2016), *LGBT Inclusion and the Sustainable Development Goals*, <https://www.stonewall.org.uk/resources/lgbt-inclusion-and-sustainable-development-goals>. [12]
- UN General Assembly (2015), *Transforming Our World: The 2030 Agenda for Sustainable Development*, A/RES/70/1. [10]
- World Health Organization (2001), *Macroeconomic and Health: Investing in Health for Economic Development*, Commission on Macroeconomics and Health. Prepared for the World Health Organization. [35]

Annex 4.A. The impact of closing unexplained labour market gaps for LGBTI+ individuals under the life-cycle approach

A life-cycle approach recognises that both the share of self-identified LGBTI+ individuals and the extent of their labour market disparities differ by sex and cohort. However, it is not a realistic method as it concomitantly assumes that these shares and disparities change as individuals in each group age. For instance, if, as detailed in Table 4.1, AFAB LGBTI+ individuals aged 20 in 2024 experience a 12.8% lower labour productivity than their cisgender straight peers, the life-cycle approach presumes that this gap will have disappeared as they age to 56 by the year 2060 (while the share of LGBTI+ individuals among the AFAB population aged 55-74 will then be equal to 4.4%, not 28% as it is the case for the AFAB population aged 15-34 in 2024). This assumption is at odd with Chapter 2. Notably, according to Chapter 2, the rising proportion of young adults identifying as LGBTI+ is not driven by an age effect which would entail that, for younger generations, self-identifying as LGBTI+ is a fad that ceases once they grow older.

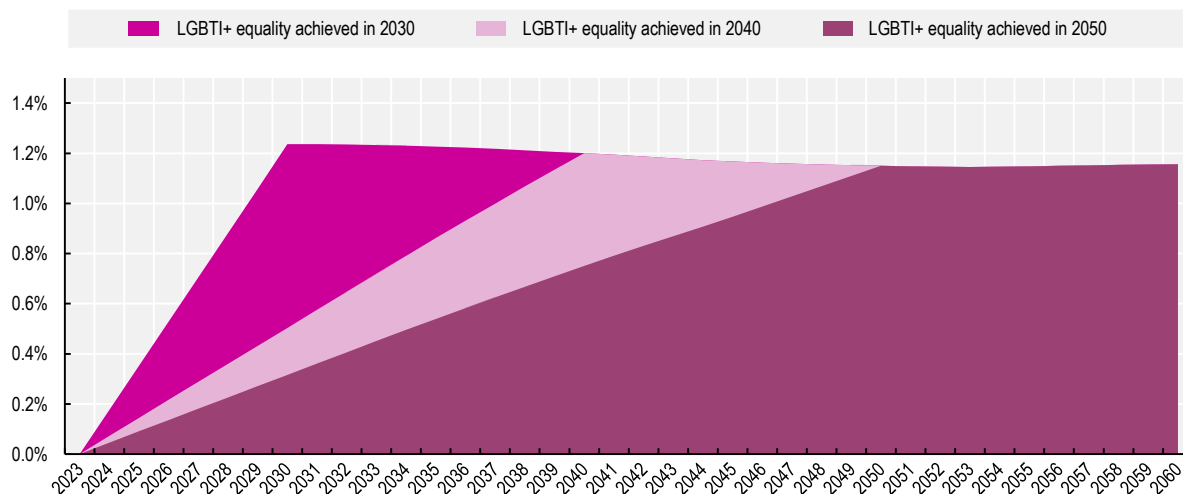
Despite its lack of realism, we analyse the economic consequences of closing labour market gaps for LGBTI+ individuals under the life-cycle approach. Not surprisingly, the economic benefits of achieving LGBTI+ equality are the lowest under this approach. Indeed, this method projects a decrease in the proportion of self-identified individuals within the total population as the younger cohorts age, leading to a decline in the extent of labour market disparities for LGBTI+ individuals in the total population.

Still the economic benefits of addressing the unexplained labour market disparities affecting LGBTI+ individuals would remain significant under a life-cycle approach (Annex Figure A.1). Under this approach, the GDP is projected to increase by 1.2%, 1.2%, and 1.1% of the baseline GDP by 2030, 2040, and 2050, respectively. These projections assume that the United States achieves LGBTI+ equality by these respective time horizons.

Under the life-cycle approach, attaining LGBTI+ equality by 2030 would result in an average annual increase in GDP of approximately 0.2%, which amounts to 10% of the average annual US GDP growth observed between 2013 and 2023. However, delaying the achievement of LGBTI+ equality would yield reduced economic benefits when translated in yearly GDP gains. The average annual increase in GDP would be equal to 0.07% if the target year is 2040, and to 0.04% if the target year is 2050.

Annex Figure A.1. The increase in GDP from addressing the unexplained labour market disparities affecting LGBTI+ individuals would still be significant under a life-cycle approach

GDP gain that would be generated, under a life-cycle approach, from efforts to achieve LGBTI+ equality by 2030, 2040, and 2050, compared to a baseline scenario where no such efforts are undertaken (2024-60)



Note: Potential GDP is defined for the total economy using a constant returns-to-scale Cobb-Douglas production function featuring total employment, labour productivity per worker, and physical capital stock. The GDP gain is computed comparing the baseline scenario (unexplained LGBTI+ labour market gaps remain as in 2023), with three alternative scenarios. These scenarios assume that unexplained LGBTI+ labour market outcomes linearly converge to those of their cisgender straight peers by 2030, 2040 and 2050, respectively.

Source: OECD estimates based on the OECD long-term model projections and Household Pulse Survey data.

Notes

¹ More precisely:

- $$A_t^{CS} * N_t^{CS} =$$

$$A_{18-34,t}^{AMAB,CS} * N_{18-34,t}^{AMAB,CS} + A_{35-54,t}^{AMAB,CS} * N_{35-54,t}^{AMAB,CS} + A_{55-74,t}^{AMAB,CS} * N_{55-74,t}^{AMAB,CS}$$

$$+ A_{18-34,t}^{AFAB,CS} * N_{18-34,t}^{AFAB,CS} + A_{35-54,t}^{AFAB,CS} * N_{35-54,t}^{AFAB,CS} + A_{55-74,t}^{AFAB,CS} * N_{55-74,t}^{AFAB,CS} ;$$
- $$A_t^{LGBTI+} * N_t^{LGBTI+} =$$

$$A_{18-34,t}^{AMAB,LGBTI+} * N_{18-34,t}^{AMAB,LGBTI+} + A_{35-54,t}^{AMAB,LGBTI+} * N_{35-54,t}^{AMAB,LGBTI+} + A_{55-74,t}^{AMAB,LGBTI+} * N_{55-74,t}^{AMAB,LGBTI+}$$

$$+ A_{18-34,t}^{AFAB,LGBTI+} * N_{18-34,t}^{AFAB,LGBTI+} + A_{35-54,t}^{AFAB,LGBTI+} * N_{35-54,t}^{AFAB,LGBTI+} + A_{55-74,t}^{AFAB,LGBTI+} * N_{55-74,t}^{AFAB,LGBTI+} .$$

² Results available upon request.

³ See: (OECD, 2023^[42]; OECD, 2023^[43]; OECD, 2023^[40]; OECD, 2023^[41])

⁴ In a 2001 study, Christopher Banks aimed to calculate the aggregate economic costs of health disparities among lesbian, gay, and bisexual individuals in Canada (Banks, 2001^[39]). His analysis focused on lost productivity from morbidity and premature mortality, direct healthcare expenditures, and other costs like prevention. Assuming that 5% to 10% of the Canadian population identified as LGB, he estimated the total economic burden of increased health conditions attributable to homophobia for five health conditions. These estimates were USD 695 million to USD 823 million for suicide, USD 281 million to USD 623 million for smoking-related issues, USD 290 million to USD 4.1 billion for alcohol abuse, USD 119 million to USD 221 million for illicit drug use, and USD 540 million to USD 2.3 billion for depression, all reported in Canadian dollars of the time (2001).

⁵ Including these additional health disparities in the analysis could significantly increase the aggregate economic and well-being costs. For instance, it is estimated that the GDP of the United States is reduced by nearly 1.9% due to the impact of diseases caused by alcohol consumption on life expectancy, health expenditure, employment, and productivity (OECD, 2021^[44]).

⁶ Years lived with a disability (YLD) are valued based on the public's perceptions about the severity of the health condition, while years of life lost (YLL) are calculated for someone with a disease by subtracting the age at death from a standard life expectancy value, defined as the lowest death rate for an age group across countries.

The Economic Case for Greater LGBTI+ Equality in the United States

Ensuring equality for LGBTI+ individuals is a human rights imperative, but it also makes a lot of economic sense. Inclusion enables LGBTI+ individuals to achieve their full employment and labour productivity potential, benefitting not only their economic and social well-being, but also society as a whole. Yet, robust evidence supporting the economic case for greater LGBTI+ equality is still scarce due to challenges in accurately measuring the size and life situation of the LGBTI+ population. This report bridges this gap by using a unique set of microdata from the United States. The report begins with an overview of the share of US adults identifying as LGBTI+, their geographic distribution and key demographics. It then evaluates the extent to which LGBTI+ Americans face discrimination, assessing how this population fares, including in the labour market. Finally, utilising the OECD long-term model, the report quantifies the potential increase in GDP resulting from closing the unexplained LGBTI+ gaps in employment and labour productivity. The findings highlight significant economic gains, although they capture only a portion of the potential benefits. Notably, the broader societal impacts, such as the advancement of women's empowerment through the disruption of heteronormative standards, are not quantified.



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